# An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields

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**Abstract**

An executable observer construction specifies a twelve-port icosahedral carrier, local seam repair, quantum instruments, memory and readback. We classify complete record laws, derive the repair gap $(5-\sqrt{5})/60$, and construct continuing observers, recovery protocols and event clocks. On a separately declared gauge carrier, protected flux classes admit a Standard Model centre-class dictionary; a declared analyzer gives charged-lepton ratios 206.7703 and 3477.47. Exact invariant sectors and positivity bounds control interacting fermionic sources without deleting their full occupation complements. One positive semileptonic source contains calibrated nucleon states and six charged/neutral Dirac branches, with an exact electron threshold and bounded nucleon energy records. Complete neutral oscillation and outgoing beta records have explicit finite error bounds. A separate autonomous matter–geometry–wave source gives bending, arrival delay and native clock records in one experiment. Its two clock factors differ at finite probe strength and share a quantitatively controlled weak-probe response. A full-carrier numerical kernel executes inside a pinned OpenWave environment. These are consequences of declared actions, preparations and observation domains. Empirical calibration, spatial refinement and physical species identification are distinguished from exact finite results; the construction supplies no common interacting continuum limit for all these source members.

## 1 The executable model and its interpretation

Observer Patch Holography proposes that observers recover the specification of their world from shared records and realize that specification within the same world. An implementation therefore accounts for its carrier, initialization, repair, controls, memory, readout and resource costs. The construction studied here specifies these objects and proves what its observers can record. A consistent implementation is not by itself an identification with nature: that identification compares complete preparation, intervention and detection laws in common physical units.

The carrier and repair law are given in Section 2; the finite observer, recovery and local clock are in Section 3. Overlapping observers read one process in Section 4. Sections 5–7 derive detector and continuum results on explicitly stated actions and refinement domains. Section 9 gives operational comparisons, and Sections 10–11 state their physical boundaries.

The mathematical roles are distinct. A finite theorem proves an implication on its stated domain; a certificate verifies a finite computation; an analytic limit theorem requires its displayed normalization and order of limits. A supplied action, state, instrument or decoder is a design input.

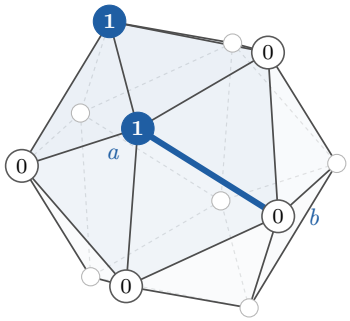
Numerical agreement with data used to choose that input is a conditional postdiction. Physical validation requires further independent observations. Host runtime, seam-attempt counts, event-clock readings, modular parameters and calibrated duration are related only by explicit maps.

The consensus paper proves quotient repair and fixed-point results [5]; the observer paper supplies the algebraic framework [6]; the carrier, gauge and spacetime papers establish their respective conditional theorems [9, 7, 8]. The flagship gives the combined mathematical argument [3]. Closure alone entails neither least-informative state selection nor inner implementation of connected response: those are separate premises in the basis registry [4].

## 2 The specification

The carrier, repair and record laws, observer algebra, services and time parameters below define the source. Each extension states its additional action, preparation and control inputs.

### 2.1 The carrier and its load configurations




twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

| $d$ | $c=0$ | $c=1$ | effect of the attempt |
|---|---|---|---|
| 2 | 1 | 1 | balance |
| 1 | 1 | 0 | swap or wait |
| 0 | 0 | 0 | no move |
| −1 | 0 | −1 | wait or swap |
| −2 | −1 | −1 | balance |

The move $j$ carries $j$ units from $a$ to $b$, and the fair coin $c$ reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps $V$ fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 1: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam $(a, b)$ has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

The specification has one carrier with twelve primitive ports. Its incidence is the oriented boundary of an icosahedron, with thirty seams and twenty faces; five seams meet at every port, the neighbours of each port form a five-cycle, and every seam lies on two oppositely oriented faces [9, 3]. Write $A$ for the adjacency matrix of the seam graph and $L = 5I - A$ for its Laplacian. The spectrum of $L$ is 0, $5 - \sqrt{5}$ with multiplicity three, 6 with multiplicity five and $5 + \sqrt{5}$ with multiplicity three, so its characteristic polynomial is $x(x-6)^5(x^2 - 10x + 20)^3$ [48, 91, 84].

A load configuration assigns a nonnegative integer $x_p$ to each port $p$, with total load $\sum_p x_p = 2$. There are 78 configurations. The 66 with two distinct occupied ports form the binary shell; the other twelve are collisions $|2u\rangle$ with load two at one port $u$. We write $\mathcal{H}_{78} = \mathbb{C}^{78}$ for the quantum load space with basis $|x\rangle$, $\mathcal{H}_{66}$ for the span of the binary shell, and $M_{78}$, $M_{66}$ for the corresponding full matrix algebras. For a binary configuration occupied at ports $u$ and $w$ we also write $|uw\rangle$ or $|\{u, w\}\rangle$.

Ports are numbered 0 to 11 by the frame vectors $u_p$: $(-1, \varphi, 0)$, $(1, \varphi, 0)$, $(-1, -\varphi, 0)$, $(1, -\varphi, 0)$, $(0, -1, \varphi)$, $(0, 1, \varphi)$, $(0, -1, -\varphi)$, $(0, 1, -\varphi)$, $(\varphi, 0, -1)$, $(\varphi, 0, 1)$, $(-\varphi, 0, -1)$, $(-\varphi, 0, 1)$, with $\varphi =$

$(1+\sqrt{5})/2$. The thirty seams are the pairs with $|u_a - u_b|^2 = 4$, each written $(a, b)$ with $a < b$, and that order is the orientation used by the repair law below [84, 85].

The construction takes this incidence as a design input. The carrier paper gives a conditional selector of the same unoriented incidence from a supplied three-dimensional quadratic readout [9].

Premise A1 of the theorem library is the layer of carriers, their federation through typed seams and coherent triple overlaps, the oriented spherical support and the complete reversible port response [4]. The construction realizes one carrier of that layer: its ports, seams and faces, its load configurations and the repair events on its seams. Coherent quantum operations run through a six-level processor whose twelve source generators, one per port, exponentiate to every executed gate on two- and four-dimensional proper codes through declared load and export services [71, 62]. Spherical-screen executions join many carriers under explicit access and refinement contracts [11]; they identify no physical current for the complete response algebra and do not discharge the full A1 attachment.

## 2.2 Native repair and its quantum record laws

Fix an oriented seam $e = (a, b)$ and a fair placement coin $c \in \{0, 1\}$. The native repair law is

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(\mathbf{e}_b - \mathbf{e}_a), \tag{1}$$

where $\mathbf{e}_p$ is the unit load vector of port $p$ and $y$ is the repaired configuration. The two endpoint loads become the floor and the ceiling of their mean; the coin assigns the ceiling to $a$ when $c = 1$; all other loads are unchanged.

**Proposition 2.1** (The classical seam rule)**.** *Let a stochastic rule on one seam (i) conserve $x_a + x_b$, (ii) end at a configuration minimizing $x_a^2 + x_b^2$, (iii) break any remaining tie by a random choice independent of the loads, and (iv) be invariant under exchange of the two endpoints. Then it is* (1) *with a fair coin.*

The carrier paper proves this characterization [9]. Without (iii), the rule that leaves the extra unit with the endpoint holding more satisfies the other three requirements. It fixes every binary configuration, so the binary shell carries no motion and the records carry no geometry. Requirement (iii) is an independent restriction on the admitted tie-breaker: it excludes retaining a preference for the endpoint holding more at the input. Internal realization alone does not impose that restriction.

The record is $r = (c, j)$, the coin and the signed current. Compatibility on a seam is specified independently of the update; it requires the two endpoint loads to differ by at most one. On the quantum load space the record defines the current instrument $\mathsf{C}$ with Kraus operators and record-discarded channel

$$L^e_{(c,j)} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \qquad \mathsf{C}_e(\rho) = \sum_r L^e_r\, \rho\, L^{e\dagger}_r. \tag{2}$$

Both maps are written in Kraus form [111]. The contrast-record variant $\mathsf{E}$ keeps $r = (c, d)$ instead, with Kraus operators $M^e_{(c,d)} = 2^{-1/2} \sum_{x:\, x_a - x_b = d} |y(x,c)\rangle\langle x|$ and record-discarded channel $\mathsf{E}_e$. Uniform independent seam attempts form the baseline schedule; on diagonal binary states the repair is then the lazy symmetric exclusion process with two particles, with generator $\frac{1}{60}\sum_e (S_e - I)$ per attempt, $S_e$ exchanging the endpoint loads [42, 48].

**Proposition 2.2** (Native repair instrument)**.** *Let $e = (a, b)$ be an oriented seam of the twelve-port carrier, let the placement coin be fair, and let $\mathsf{C}_e$ and $\mathsf{E}_e$ act on $\mathcal{H}_{78}$. Then:*

(i) *For every record value the branch map $x \mapsto y(x,c)$ is an injective translation of its support. Consequently $\sum_r L_r^{e\dagger} L_r^e = I$, and $\mathsf{C}_e$ is completely positive and trace preserving (CPTP), also when tensored with an untouched reference system. The same holds for $\mathsf{E}_e$.*

(ii) *Every branch conserves the total load, leaves spectator ports unchanged, never increases $\sum_p x_p^2$, and maps the binary shell into itself.*

(iii) *The current labels satisfy $c \in \{0,1\}$ and $j \in \{-1,0,1\}$; all six labels occur on $\mathcal{H}_{78}$ and four occur on the binary shell.*

(iv) $\mathsf{C}$ *and* $\mathsf{E}$ *induce the same classical transition on every configuration, and*

$$\mathsf{E}_e^2 = \mathsf{E}_e, \qquad \mathsf{E}_e\mathsf{C}_e = \mathsf{C}_e\mathsf{E}_e = \mathsf{E}_e, \qquad \mathsf{C}_e^n = \mathsf{E}_e + 2^{1-n}(\mathsf{C}_e - \mathsf{E}_e) \quad (n \geq 1).$$

*On the binary shell $\mathsf{E}_e$ is a trace-preserving conditional expectation.*

(v) *With the seam-total parity $p = (x_a + x_b) \bmod 2$ one has $d = 2j + (2c-1)p$, and $\mathsf{E}$ equals $\mathsf{C}$ followed by a binary record of $p$ and this relabelling, as an identity of complete unnormalized branches.*

*Proof sketch.* The inverse of a fixed-current branch adds $j$ at $a$ and subtracts it at $b$; every basis vector has one branch for each coin, so each column of the joint record/source isometry has two entries $2^{-1/2}$ with disjoint supports [46]. For fixed contrast, the unchanged spectators and the endpoint sum reconstruct the input, which gives (i) for $\mathsf{E}$. Items (ii) and (iv) are the single-seam relations of the candidate register [42]. Item (v) follows from $j = \lceil d/2 \rceil$ for $c = 0$ and $j = \lfloor d/2 \rfloor$ for $c = 1$; the parity-refined Kraus operator $\Pi_p L_{(c,j)}$ has exactly the contrast group as support [65]. The executable checks all 312 outputs of the observer's two seams against an independent floor and ceiling rule [46]. □

The current instrument is no classical coarse-graining of the contrast instrument: grouping the labels of $\mathsf{E}$ sums density operators and cannot restore the coherent Kraus sums of $\mathsf{C}$. Both belong to the family of quantum record laws classified in Section 2.3. One further law of the candidate register, the reciprocal exchange N4 that couples the native current to private six-level response factors, enters only the local-tick certificate of Section 3.10 [42].

### 2.3 Admissible record laws

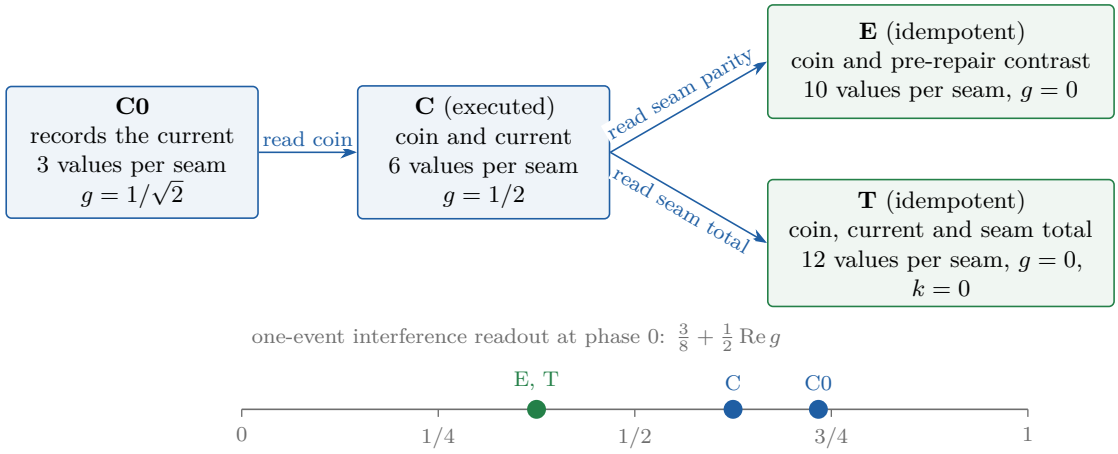
The classical rule fixes the diagonal of a record law and leaves its coherences free. For the seam $e = (a,b)$ write $\kappa(x) = (x_a, x_b)$ for the seam key and $D = \mathbf{e}_b - \mathbf{e}_a$. A channel $\Phi$ on $\mathcal{H}_{78}$ is an admissible record law for $e$ if it acts as the identity on the ten spectator ports, reproduces the classical repair distribution on every $|x\rangle\langle x|$, commutes with $U_t = \exp(i \sum_p t_p N_p)$ for every $t \in \mathbb{R}^{12}$, and is invariant under the carrier rotation that exchanges $a$ and $b$ with the current reversed. The third condition makes every local load phase a symmetry of the unread channel. It is a premise of the classification, a local charge-superselection rule for the repair channel, and it is what excludes a recorder of a pre-repair endpoint load. It guarantees that a definite-current Kraus representation exists; it does not force every recorded implementation of the channel to commit a definite current, since a change of record basis before commitment mixes the current labels and leaves the channel unchanged [41]. The classification below concerns unread channels, and statements about current histories use the compatible definite-current instrument.

**Theorem 2.3** (Coherence tables). *A channel $\Phi$ is an admissible record law if and only if*

$$\Phi(|x\rangle\langle x'|) = \sum_j G_j(\kappa(x), \kappa(x'))\, |x + jD\rangle\langle x' + jD|$$

*for endpoint-symmetric positive semidefinite matrices $G_j$ on the seam keys that can carry the current $j$, with $G_j(\kappa, \kappa) = \Pr(j \mid \kappa)$. On $\mathcal{H}_{78}$ the free entries are five multipliers: $k$ between the empty and the full seam, $g$ between the empty seam and a waiting single load, $h$ between the full seam and a waiting single load, a real $m$ between the two waiting orientations, and $n$ between a moving single load and a moving collision. Positivity gives $|k| \le 1$, $|g|, |h|, |n| \le 2^{-1/2}$ and $|m| \le \frac{1}{2}$, and $k = 1$ forces $g = h$. Repairing one seam with $(k, g, h, m, n)$ and then with $(k', g', h', m', n')$ gives $(kk', gg', hh', mm', n\bar{h}')$. Exactly two admissible laws are idempotent, those with $g = h = m = n = 0$ and $k \in \{0, 1\}$, and every law with $k = 1$ converges under repetition to the idempotent law with $k = 1$.*

The carrier paper proves this classification [9], and the exact composition checks are retained with the record-law receipt [40].




Each step records one more classical quantity after the repair and removes coherence that the previous law kept. All admissible laws share the classical current statistics, the uniform stationary state and the gap $(5 - \sqrt{5})/60$; the multiplier $g$ is what an interference experiment reads.

Figure 2: The record-law ladder of Theorem 2.3 and Table 1, and the one-event interference readout $\frac{3}{8} + \frac{1}{2}\,\mathrm{Re}\, g$ at phase zero that separates its members.

Four members form a ladder (Table 1 and Figure 2): $\mathsf{C}$ is $\mathsf{C}_0$ followed by a diagonal readout of the coin, $\mathsf{E}$ is $\mathsf{C}$ followed by a readout of the seam parity as in Proposition 2.2(v), and $\mathsf{T}$ is $\mathsf{C}$ followed by a readout of the seam total. The completion of $\mathsf{C}$ in Proposition 2.2(iv) is the case $g = h = n = \frac{1}{2}$, $m = 0$, and $\mathsf{E}$ is the idempotent law with $k = 1$. A recorder of an endpoint's pre-repair load keeps the classical law and fails covariance, because one of its branches carries two currents. $\mathsf{C}_0$ records only what conservation requires, $\mathsf{C}$ records the tie-break as an event, $\mathsf{E}$ is the coarsest law for which one repair completes the seam, and $\mathsf{T}$ records the full seam occupation; none of these principles follows from the theorem premises. The construction executes $\mathsf{C}$ as its convention and compares it with $\mathsf{E}$. Under a common definite-current record interface all members assign the same probabilities to every classical current observation; Section 9 reads $g$ in one coherent event, and Section 4.3 identifies all nine real coordinates. Every admissible table, including complex

phases and different laws on different seams, has the uniform binary density as its unique stationary source state and the same spectral gap $(5-\sqrt{5})/60$, so equilibrium selects none of the coordinates; a finite unread warm-up of 221 attempts on the binary shell, or 318 on the full carrier, brings an arbitrary initial density within 1/1000 of that state, uniformly over untouched references [40].

Table 1: Four distinguished admissible record laws [40].

| Law | Each event records | Values per seam | $k$ | $g = h$ | Idempotent |
|---|---|---|---|---|---|
| $\mathsf{C}_0$ | the current | 3 | 1 | $2^{-1/2}$ | no |
| $\mathsf{C}$ | coin and current | 6 | 1 | 1/2 | no |
| $\mathsf{E}$ | coin and pre-repair contrast | 10 | 1 | 0 | yes |
| $\mathsf{T}$ | coin and both endpoint loads | 12 | 0 | 0 | yes |

### 2.4 The finite observer algebra

An observer of length $n$ on the carrier consists of a preparation on $\mathcal{H}_{78}$, a record tape $R^{\otimes n}$, seam policies $s_k$ that map committed records and classical randomness to seams or to supplied detector outcomes, the coherent writes $W_{s_k}$, optional supplied instruments whose committed bits may enter $s_k$, and a halting rule after $n$ writes; a policy reads only committed records. Definition 2.4 is the instance $n = 2$, and Sections 3.4, 5 and 6 add declared instruments to it.

**Definition 2.4** (Two-event native observer)**.** Let $R = \mathbb{C}^6$ carry one basis vector per current label $(c, j)$. The observer carrier is

$$\mathcal{H}_{\mathrm{cap}} = \mathcal{H}_{78} \oplus (R \otimes \mathcal{H}_{78}) \oplus (R \otimes R \otimes \mathcal{H}_{78}),$$

with sectors of dimensions 78, 468 and 2808, total 3354; the sector index is the phase, the number of filled record slots.

(a) *Preparation.* The scored preparation is uniform on the 66 binary configurations; the full input family uses all 78.

(b) *Measurement and record.* At phase zero the coherent write $W_e = \sum_r |r\rangle \otimes L_r^e$ acts on the seam $e = (0, 1)$.

(c) *Record-only feedback.* At phase one the seam is $s(r) = (0, 5)$ if the stored current is positive and $(0, 1)$ otherwise, and $V_1(|r\rangle \otimes \psi) = |r\rangle \otimes W_{s(r)}\psi$.

(d) *Forecast.* Before the second event the controller reads its stored letter and issues a supplied forecast $q(r)$ of a positive second current.

(e) *Halting.* At phase two the observer acts as the identity.

The controller reads only the committed first record; it has no access to source loads or future outcomes.

**Proposition 2.5** (One CPTP map with classical-prefix preservation)**.** *Let $V_0 = W_{(0,1)}$ and $V_1$ be extended by zero to the other sectors, giving $A_0$ and $A_1$, and let $P_2$ project onto the full sector. Then:*

*(i) $V_0$ and $V_1$ are isometries on every nonfull sector, including unreachable combinations of letters and loads, and $T(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger + P_2\rho P_2$ is CPTP;*

*(ii)* *on the inhabited subspace $K = \mathcal{H}_1 \oplus \mathcal{H}_2$ every first-record projector $Q_r$ obeys $T^*(Q_r) = Q_r$, so a committed first record is reread with probability one;*

*(iii)* *for the blank input with occupations $(1, 1, 0, \ldots, 0)$, the first-record projector of the letter $(0, 0)$, extended by zero to phase zero, has expectation $0$ before the tick and $1/2$ after it, so writing a record is a transition distinct from preserving it;*

*(iv)* *for the equal superposition of $\{0, 2\}$ and $\{1, 2\}$, the reduced-density entry of the first-record pair $((0, 1), (0, 0))$ equals $1/4$ after the first write and $1/8$ after the second while the classical diagonals are unchanged, so coherence between different record values need not survive feedback.*

*Proof sketch.* Orthogonal retained labels and $W_s^\dagger W_s = I$ give $V_0^\dagger V_0 = V_1^\dagger V_1 = I$, hence $A_0^\dagger A_0 + A_1^\dagger A_1 + P_2 = I$. Every subsequent map intertwines $Q_r$ on $K$. In (iv) the working state attached to the entry has occupations $\{1, 2\}$, on which the isometries of $(0, 5)$ and $(0, 1)$ overlap by $1/2$ [46]. The checkpoint adapter verifies all $78 + 468 + 2808$ columns and both the preservation identity on the 3276-dimensional checkpoint space and the blank-sector control [71, 97]. □

**Lemma 2.6** (Feedback requires classical record preservation)**.** *Let $\Lambda$ be a CPTP map from $R \otimes W$ to $R \otimes W'$ with $\Lambda^*(B \otimes I) = B \otimes I$ for every $B$ in the full matrix algebra of $R$. Then $\Lambda^*(I \otimes O) = I \otimes \Phi^*(O)$ for a unital completely positive map $\Phi^*$, so changing only the record preparation of a product input changes no output statistic on $W'$.*

*Proof sketch.* The operators $B \otimes I$ lie in the multiplicative domain of the unital completely positive map $\Lambda^*$, because the Schwarz inequality is an equality for them [104]. Then $\Lambda^*(I \otimes O)$ commutes with every $B \otimes I$ and has the stated form [46]. □

Lemma 2.6 fixes the shape of the observer contract. A record that steers the working state is preserved as a classical record, with its projectors fixed, while coherence between different record values may change.

## 2.5 Declared services

Each component names the services it assumes; they belong to the specification and enter the resource counts where a component counts resources. The services are coherent proper-code load and export, with complementary failure outcomes retained by a complete channel definition; blank initialization, reset, storage and routing of quantum registers and classical records; Born readout of pointers and fair coins; and analogue precision of source pulses. Particular components add a calibrated external flux and probe (Section 3.4), incident-event recognition and bare-occupation readout (Section 3.10), coherent phase preparation and readout (Section 9), and, for the native process results, independent uniform seam scheduling, uniform fixed-number preparation, full initial and current readback, coherent wave processing, the return-length sampler and the response controller (Sections 7, 4 and 9). The coherent services are not products of the native grammar: committed C/E repair with diagonal accessible inputs and fresh independent diagonal ancillas cannot prepare the declared coherent reference to arbitrary accuracy, even with adaptive heralding. This coherence floor of the native interface makes the coherent services a necessary resource of the specification. The sharp joint instrument of Section 6.1 uses native integer repair as a declared primitive, and Section 6.5 attaches the compiled 78-dimensional C/E channels, routing and encoded recovery to the continuing work history. The processor algebra is $M_6$; its twelve generators are the exact source generators of the compiler, one per port [62, 71]; a proper code is a two- or

four-dimensional subspace of $\mathbb{C}^6$ with its load and export channels and the complementary failure outcome; and the service error model is the compositional contract of Section 9.3. Exact certificates exclude numerical execution error and physical service error unless a component states otherwise. Direct native acceptance effects are diagonal, so complex tomography of the binary shell, with real and imaginary analyzers on all 4,356 Hermitian coordinates, uses these coherent services [32]. A finite native warm-up replaces exact stationary input for one complete readout (Section 9.6), and Table 5 separates what repair and its counts supply from what each result declares.

### 2.6 The closure inputs $P$ and $N$

The pixel ratio $P$ is the area of one screen cell in units of $\ell_\star^2$, and $N$ is a record capacity in nats [6, 56]. The construction accepts both as closure-derived inputs and fits neither to an output. $P$ enters through a declared coupling map, the source-only structured-running branch of the fine-structure closure (one-loop running and matching conventions with the tree-level $m_Z$ closure), defined in the P-derivation code of the repository [2] and consumed unchanged; the map is a declared selection, not a result of this construction. At the pinned closure value $P = 1.630972095858897\ldots$, the midpoint of the pinned forward-closure interval, the map gives $\alpha^{-1}(P) \approx 136.9948351774$ and $g(P) = \sqrt{4\pi\alpha(P)} \approx 0.302867613117$ in Heaviside–Lorentz natural units, numerical evaluations of the executed map [54, 93]. $N$ enters only as a bound, and no component consumes its closure value. For a declared capacity carrier of dimension $D$, the record capacity $N_{\mathrm{record}} = \ln D$ bounds every correctable classical record code by $\ln M_0 \leq N_{\mathrm{record}}$ [39, 47]. The cosmological relation $N = 3\pi/(\Lambda\ell_\star^2)$ additionally requires a physical identification of capacity with horizon area, which the construction does not make.

### 2.7 Time parameters

The construction keeps the time parameters of Table 2 distinct. Conversions between them need explicit maps. The construction uses four: the fixed-reference relation $t = -\beta s$ of Section 6.6 between a model coordinate and a modular parameter, the closed-history relation $\tau = -Ls$ of the observers paper [6], the instrument-valued conversion between two local event clocks of Section 4.2, which keeps their whole count law, and the rescaling of native attempt count along the refining screens of Section 7. A calibration of native calls against a monotonic host counter is an engineered software correspondence. No preferred universal physical clock is required.

## 3 The finite observer

This section is layer three on one carrier: an observer that prepares, records, retains and acts on native records, with its capacity, memory and recovery limits.

### 3.1 The native observer

The native observer executes Definition 2.4 on the scored preparation [45, 46]. A trial is one fresh preparation followed by two repair events and a halt; running many trials repeats the preparation and does not extend one observer's memory. Event order is the time parameter. The tape, prior, forecast table, seam selector and halting rule are designed apparatus, and the forecast is a fixed table without learning.

Table 2: Time parameters and where they occur.

| **Parameter** | **Meaning** | **Occurrence** |
|---|---|---|
| Host time | Runtime of a software execution; a resource cost | A software calibration of native calls; never a physical quantity |
| Solver counter | Position in a search or repair schedule; the attempt count of a declared schedule | History repair (Section 6.7); attempt counts in the observer theorems |
| Causal order | Precedence of recorded events; counts of native operations | Observers, detectors, phase protocol, local ticks |
| Local event count | Attempted repairs incident to an observer region, or occupation changes at a port | Overlapping observers, local and active event clocks, matched screens (Section 4) |
| Closed-history length | Integer attempt count $L$ of a closed native return, the inverse-temperature multiplier of the closed-history functional | Closed-history functional and modular correlations (Section 4.5) |
| Scaled attempt count | Native attempt count $n$ at level $l$ rescaled to $\tau = n/(1260 \cdot 16^l)$ on the refining screens | Native density limits and the controlled repair response (Sections 7.3, 9) |
| Coherent wave parameter | Parameter of the admitted coherent continuation $\cos(t\sqrt{A})$ | Native waves and the return reference (Sections 7.2, 7.7) |
| Model coordinate | Parameter of a declared Hamiltonian or action | Scalar detectors, two-site apparatus, common source, compact interacting core |
| Modular parameter | Parameter of the modular automorphisms of an algebra-state pair | Modular clock and fixed-reference relation (Section 6.6), closed-history and regional references (Sections 4.5, 6.10) |
| Operational clock | Observer-accessible reading with calibrated duration | Supplied by no component |

**Proposition 3.1** (Statistics of the native observer)**.** *Prepare the 66 binary configurations uniformly, use fair coins and the controller of Definition 2.4, and let $Y$ indicate a positive second current. Then:*

*(i) the 264 histories, two coins for each of the 66 configurations at each of the two events, each have weight* $1/264$*;*

*(ii) the first record has masses* $14/33$*,* $14/33$*,* $5/66$*,* $5/66$ *for* $(c, j) = (0, 0), (1, 0), (0, 1), (1, -1)$*, the conditional probabilities* $q(r) = \Pr(Y = 1 \mid r)$ *are* $0$*,* $5/56$*,* $0$*,* $1/2$ *in the same order, and* $\Pr(Y = 1) = 5/66$*;*

*(iii) the Brier risk of forecasting $Y$ is* $395/7392$ *with* $q(r)$*,* $140/2013$ *with the best forecast based on the chosen seam alone, and* $12875/150304$ *when $q$ is evaluated on an independent record resampled within the same seam;*

*(iv) the mean final load at port* $0$ *is* $35/264$ *under record feedback and* $1/6$ *when the selector is disconnected or replaced by an independent selector with the same seam frequencies; the feedback effect is* $-3/88$*.*

*Proof sketch.* For (iii), $\mathbb{E}[(Y - q(R))^2] = \mathbb{E}[\mathrm{Var}(Y \mid R)]$. The chosen seam $S$ is a function of $R$; the best $S$-only forecast has excess risk $\mathbb{E}[\mathrm{Var}(q(R) \mid S)]$, and an independent resample within the seam adds twice that amount. For (iv), a positive first current occurs on ten binary inputs with coin zero, mass $5/66$. On that branch $x_0 = 0$, $x_1 = 1$ and $\mathbb{E}[x_5] = 1/10$; a second repair on $(0, 1)$ gives port-0 mean $1/2$ and one on $(0, 5)$ gives $1/20$. Hence the change is $(5/66)(1/20 - 1/2) = -3/88$ and the mean is $1/6 - 3/88 = 35/264$. An independent selector leaves the uniform shell uniform under either seam [46]. □

The other two current letters occur only on the full carrier, where the forecast abstains. The exact receipt retains all 264 histories [88], and a seeded execution of 20,000 fresh trials matches every trial against its exact path reference, with sample final port-0 loads 0.1327 with feedback, 0.16645 disconnected and 0.1668 with the independent selector [89]. Since the selector is a design choice and the forecast table is the exact conditional law, (iii) and (iv) are consistency checks: the Bayes forecast beats coarser forecasts and the record-wired selector moves the occupation.

### 3.2 Continuing memory and the actual stationary reference

A continuing observer [2] extends the two-event construction with an explicitly declared memory law. It keeps the same source and complete C/E instruments. A classical tag chooses C with probability $\lambda$ and E otherwise; a repair occurs with probability $r$, with an identity hold otherwise. A controller chooses a preferred seam from its retained records. The actual seam distribution is

$$p(e \mid m) = \frac{\epsilon}{30} + (1 - \epsilon)\pi(e \mid m), \qquad r, \epsilon \geq \frac{1}{16}, \quad 0 \leq \lambda \leq 1.$$

The rational parameter bound and permitted controller tables are part of the specified design, not measured physical constants. The executed controller proposes $(0, 5)$ after a positive current and $(0, 1)$ otherwise. It does not read the hidden source load. A proposal intervention preserves the uniform exploration floor. Holds are recorded with current zero.

For each classical memory input, every source branch is $K_b = \sqrt{w_b} M_b$, with $M_b$ a native partial injection or identity and $\sum_b K_b^* K_b = I_{78}$ exactly. Tensoring with the identity preserves an arbitrary untouched reference. The tests independently reconstruct the branch supports by

endpoint floor/ceiling arithmetic, including their off-diagonal action. Composition therefore defines every finite history with its unnormalized quantum output and actual probability. Exact prefix replay and retained six-attempt source/reference/last-record marginals verify C, E and the delayed tagged mixture; the complete history tree remains implicit in that channel kernel.

One to eight complete records occupy the declared buffer. Under the executed one-slot export policy, $n$ attempts leave one record and export $n-1$ records for $n \geq 1$; expected repair and hold counts are $nr$ and $n(1-r)$. Concatenating exports and the remaining buffer reconstructs the history. The halt-on-full control instead preserves its source and records with an explicit timeout outcome. These are different resource contracts. Bounded working memory with growing exports is not closed finite-resource operation for unlimited duration. No new native source-word implementation of the supplied initialization, selection, write, export or delivery services is claimed.

The stationary reference must include this controller. For $r = 1$ and $\epsilon = 1/2$, a uniform binary source with blank memory becomes nonuniform after two attempts: 30 source probabilities change, with maximum deviation $1/960$ from $1/66$. Exact rational solution of the actual source/current-sign kernel instead gives one stationary law on a 129-state recurrent subset of the 132 source/bit states. Applying the complete C/E observer instrument to that law constructs the full-record reference. Its projection recovers the bit law, and one further actual observer step preserves the entire full-record density exactly. The resulting 128 C or 192 E record blocks have 4,218 positive diagonal entries overall. Positivity establishes faithfulness only on the explicitly compressed support of each block. The growing exported tape is not a stationary object, and the delayed member's hold updates require a separate stationary solve.

This reference also makes a clock obstruction precise. In an actual nontracial block choose positive weights with ratio $q = p_i/p_j > 1$ and the dimensionless modular parameter $s_* = \pi/(2\log q)$. The state-dual modular action takes the real pair probe $(|i\rangle + |j\rangle)/\sqrt{2}$ to a density with off-diagonal entry $-i/2$. Every admitted history, with the declared real joint ancillas and classical memory, preserves real density matrices. A norm-one imaginary pair observable separates the target from every such normalized output, giving trace-distance error at least $1/2$. This witness is not a native detector, and the probe is an all-input map test, not an asserted native preparation. On the accessible commutative record algebra the modular action is trivial. Other algebras, encodings or coherent services have different contracts; no general physical-clock exclusion follows. The exact reference, implementation bound and full qualifications are retained with the executable model [2].

### 3.3 Record capacity

The retained records are evaluated with the correctable-public-record construction of the theorem library [6, 47]. There $M_0$ is the largest correctable set of reachable public records, the independence number of the compound checkpoint confusability graph; the record capacity is $N_{\mathrm{record}} = \ln M_0$ nats; and $M_0 \leq D$ for a declared carrier of dimension $D$. The counts below describe the reachable record interface; program, source, probe and service memory are additional.

**Proposition 3.2** (Record capacity of the native observer)**.** *Take halt and exact classical reread as the declared terminal continuations of the observer of Definition 2.4. Then the compound graph has no edges, and:*

*(i)* $M_0 = 11$ *on the binary preparation (four reachable first records, 264 histories) and* $M_0 = 17$ *on the full family of 78 inputs (six first records, 312 histories);*

*(ii) each terminal word projector has rank 78 in the 3354-dimensional carrier, distinct projectors are orthogonal, and the eleven reachable ones have total rank 858, so the bound is unsaturated with slack* $\ln(3354/11)$*;*

*(iii) the binary preparation reaches* $1{+}4{+}11 = 16$ *classical prefixes, so four bits encode the complete controller history; a compact controller that reads only this code reproduces the phase, action, forecast, source loads, record and halt of all 264 histories; the full family reaches 24 prefixes and needs five bits;*

*(iv) coarser retained readouts give correctable codes of size 4 for the first letter alone, 4 for the second letter alone, 3 for the forecast alone, 2 for the chosen seam alone, 4 for forecast and seam, and 1 after complete erasure.*

*Proof sketch.* Halt and reread leave the tape unchanged, so the joint kernel is the identity and the existing evaluator returns the number of reachable terminal words. The terminal projector of word $w$ occupies the basis coordinates $546 + 78w$ to $546 + 78w + 77$ of the carrier [47, 90]. □

The terminal capacity is $\ln 11$ nats, about 3.459 bits, on the binary preparation and $\ln 17$ nats on the full family; the Shannon entropy of the benchmark record law, about 2.023 nats, is a different quantity. The library's global proposal adds closure across the complete terminal fiber, $N_\star = \ln M_0(U_{N_\star})$, and a de Sitter horizon identification giving $\Lambda\ell_\star^2 = 3\pi/N_\star$ [6, 47]; the native records supply an actual finite input to that evaluator, and the two further steps are not taken here.

### 3.4 The $P$-dependent detector

The detector carries the closure input $P$ through the electromagnetic coupling into a recorded binary observation that controls the next repair [54, 55]. The charge assignment, probe, calibrated flux, readout and controller are specified apparatus (Section 9.4).

Choose the charged realization $Q_v = g(P)N_v$ of the native occupation $N_v$. A current record $j$ on the seam $(a, b)$ then transfers the charge $\Delta Q = g(P)\, j\, (\mathbf{e}_b - \mathbf{e}_a)$, and total charge is conserved. For port gauge parameters $\lambda_v$ and $U_\lambda|x\rangle = e^{ig\sum_v \lambda_v x_v}|x\rangle$, every current-grouped Kraus operator obeys $U_\lambda L_{(c,j)} U_\lambda^\dagger = e^{igj(\lambda_b-\lambda_a)} L_{(c,j)}$. The probe is a unit-charge path qubit prepared in $|{+}X\rangle$. The first native current routes it around a calibrated external loop with signed exposure $j\chi$, where $\chi = \oint A{\cdot}dx$ is prepared independently of $g$, so the relative phase is $\varphi_j = g(P)\, j\, \chi$, the Aharonov-Bohm phase [100]. A $Y$ measurement commits $y = \pm 1$ with amplitudes $m_y(j) = \frac{1}{2}[e^{-i\varphi_j/2} - iy\, e^{i\varphi_j/2}]$ and effects $E_y = \sum_r \frac{1}{2}[1 + y \sin\varphi_{j(r)}]|r\rangle\langle r|$ on the old current register. The actuator receives only $y$. It repairs $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, keeps $y$ unchanged, and the protocol halts.

**Proposition 3.3** ($P$-dependent response)**.** *Prepare the 66 binary configurations uniformly, run the protocol above, and put* $s = \sin(g(P)\chi)$*. Then:*

*(i)* $\{E_y\}$ *is a binary instrument on every input, including coherent records; each* $M_y = \sum_r m_y(j(r))|r\rangle\langle r|$ *commutes with the old record projectors, and the composed experiment is one CPTP map that preserves the committed bit;*

*(ii)* $\Pr(y \mid j_1, P, \chi) = \frac{1}{2}[1 + y \sin(g(P) j_1 \chi)]$*;*

*(iii) at* $\chi = 1$ *all 528 state, coin, detector and coin histories have positive probability, and*

$$\mathbb{E}[y] = 0, \quad \mathbb{E}[y j_1] = \tfrac{5}{33} s, \quad \mathbb{E}[j_2] = \tfrac{5}{264} s, \quad \mathbb{E}[g(P) j_2] = \tfrac{5 g(P)}{264} s;$$

*(iv) for weak exposure,* $\mathbb{E}[g\, j_2] = \frac{5\pi}{66}\alpha(P)\chi + O(\chi^3)$*;*

*(v) disconnecting the probe, or replacing $y$ by an independent fair bit, gives $\mathbb{E}[j_2] = 0$ with the same detector and seam frequencies.*

*Proof sketch.* The eigenvalues of $E_y$ lie in $[0, 1]$ and sum to one, which gives (i). For (iii), after the first repair let the source be $z$; averaging the second coin gives $\mathbb{E}[j_2 \mid z, j] = \frac{1}{4}(2z_0 - z_1 - z_5) + \frac{sj}{4}(z_1 - z_5)$. The first term averages to zero because the first repair preserves the uniform shell. Currents $\pm 1$ each have mass $5/66$, and on either branch the other load is uniform over ten ports, so $\mathbb{E}[j(z_1 - z_5)] = 5/66$. An independent floor and ceiling enumerator reproduces every coefficient over all $66 \cdot 2 \cdot 2 \cdot 2 = 528$ paths [55, 57]. □

At the computed $P$ and $\chi = 1$ the plus probability is 0.649129261647 for $j = +1$, 0.350870738353 for $j = -1$ and $1/2$ for $j = 0$; the mean second current is 0.00564883566843, the mean transferred charge 0.00171084937579 in natural units, and the final port-0 load 0.161017830998235 against $1/6$ for the controls. Off-closure controls at $0.999P$ and $1.001P$ change the detector and feedback probabilities at fixed $\chi$, and a seeded run of 200,000 trials gives mean second current 0.006390 with standard error 0.00085715 [54, 93]. The audit record $(r_1, y, r_2)$ has 24 terminal words and $M_0 = 24$ [56]. If a cut enclosing a retained record obeys the area-capacity law with area $K$ chart cells, then $N_{\mathrm{cut}} = KP/4$ and $\ln M_0 \le N_{\mathrm{cut}}$ give $K \ge 4 \ln M_0 / P$; with the pinned $P$ one detector bit needs $K \ge 2$ ($4 \ln 2 / P \approx 1.6999608573$) and the full audit $K \ge 8$ ($4 \ln 24 / P \approx 7.7942567832$).

### 3.5 Quantum mixing and memory

Discarding every seam, coin and repair record after each event turns the repairs into channels $\Phi_{\mathsf{C}}$ and $\Phi_{\mathsf{E}}$ on the source. Under independent seam attempts these channels have an exact spectral theory [48]. Theorems 3.4 and 3.5 concern this record-discarded channel under independent uniform attempts; they do not describe the retained tape or the feedback controller.

**Theorem 3.4** (Signed-difference reduction and quantum gap)**.** *Let $G$ be a finite connected graph with $n \ge 2$ vertices, positive edge weights $w_{ab}$ and $W = \sum w_{ab}$. Let each event choose the edge $ab$ with probability $w_{ab}/W$ and a fresh fair coin, and discard its record. On the binary space $\mathcal{H}_k$ of load $k$, $1 \le k \le n-1$, with $D = \binom{n}{k}$:*

*(i) for a matrix unit $|x\rangle\langle y|$ the signed difference $z = x - y$ is invariant; if $z$ has $r$ positive entries and $Z = \{v : z_v = 0\}$, then on the sector of $z$*

$$\Phi_{\mathsf{L}}\big|_z = I - \frac{L_{\mathrm{excl}}(G[Z], k - r) + K_{\mathsf{L}}(z)\, I}{2W}, \qquad \mathsf{L} \in \{\mathsf{C}, \mathsf{E}\},$$

*with the exclusion Laplacian $L_{\mathrm{excl}}$ of the induced graph, $K_{\mathsf{C}}(z) = \sum w_{ab}|z_a - z_b|$ and $K_{\mathsf{E}}(z) = 2\sum w_{ab}\mathbf{1}[z_a \ne z_b]$;*

*(ii) both channels are unital and Hilbert-Schmidt self-adjoint with spectrum in $[0, 1]$, their fixed algebra consists of the scalars, and both have gap $\gamma = \lambda_1(G)/(2W)$, where $\lambda_1(G)$ is the first positive Laplacian eigenvalue; and $\|\Phi_{\mathsf{L}}^t - \mathrm{Dep}_D\|_{2\to 2} = (1 - \gamma)^t$;*

*(iii) for every input and every untouched reference $R$, $\mathrm{dist}\big((\Phi_{\mathsf{L}}^t \otimes \mathrm{id}_R)(\rho), \tau \otimes \rho_R\big) \le \min\{1, \frac{D}{2}(1 - \gamma)^t\}$, with $\tau = I_D/D$ and $\mathrm{dist}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.*

*For the twelve-port carrier at load two ($n = 12$, $W = 30$, $\lambda_1 = 5 - \sqrt{5}$, $D = 66$) the gap is $\gamma = (5 - \sqrt{5})/60 \approx 0.046066$, and the slowest off-diagonal sectors contract by exactly $r_{\mathsf{C}} = 5/6$ and $r_{\mathsf{E}} = 7/10$ per attempt. On the full 78-state carrier the gap is the same, the full channel is block triangular with a diagonal collision block whose entries lie in $[0, 5/6]$, and the unique stationary density is the uniform binary density, of rank 66 on dimension 78.*

*Proof sketch.* A matrix unit survives a coin branch exactly when its two records agree, and then both configurations translate by the same current; comparing the coin branches gives (i). The diagonal sector is the symmetric $k$-particle exclusion chain, whose gap equals $\lambda_1(G)$ by the theorem of Caputo, Liggett and Richthammer [103]. For a nonzero sector, $K_{\mathsf{E}} \geq K_{\mathsf{C}} \geq \frac{1}{2}\sum w_{ab}(z_a - z_b)^2 \geq r\lambda_1 \geq \lambda_1$, so no off-diagonal mode is slower. Item (iii) uses Schmidt rank at most $D$ and $\|Y\|_1 \leq D\|Y\|_2$. On the carrier, $r = 1$ gives $K_{\mathsf{C}} = 10$ and $K_{\mathsf{E}} = 2(10 - \mathbf{1}[uv \text{ edge}])$, while $r = 2$ gives $K_{\mathsf{C}} \geq 16$ and $K_{\mathsf{E}} \geq 24$. The collision block has largest entry $1 - d_{\min}/W = 5/6 \leq 1 - \gamma$ [48]. The executable checks the reduction against the native primitive on all 4,356 binary matrix units and the transient structure on all 6,084 full matrix units [91]. □

Choose adjacent ports $u, v$ and the two mixtures $\rho_\pm = \frac{1}{20}\sum_{w\notin\{u,v\}}(|uw\rangle \pm |vw\rangle)(\langle uw| \pm \langle vw|)$. They have identical load populations and orthogonal supports, and

$$\operatorname{dist}(\Phi_{\mathsf{L}}^t(\rho_+), \Phi_{\mathsf{L}}^t(\rho_-)) = r_{\mathsf{L}}^t, \qquad \operatorname{dist}(\Phi_{\mathsf{C}}^t(\rho_+), \Phi_{\mathsf{E}}^t(\rho_+)) = \tfrac{1}{2}(r_{\mathsf{C}}^t - r_{\mathsf{E}}^t).$$

The second distance is $1/15$ after one attempt and peaks at four attempts at $12259/101250$; the binary effect $\frac{1}{2}(I + A_{uv} + A_{uv}^\dagger)$ with $A_{uv} = \sum_{w\notin\{u,v\}}|uw\rangle\langle vw|$ attains it. Load populations, classical load histories and diagonal geometric observables obey the same exclusion process under both laws, so only coherent preparations and readouts distinguish them under this schedule.

**Theorem 3.5** (Finite quantum memory)**.** *Under the schedule of Theorem 3.4 on the twelve-port carrier:*

*(i) on the binary shell both record-discarded channels are entanglement breaking after at most 134 attempts;*

*(ii) on the full carrier let $u, v$ be adjacent and $|\Psi\rangle = (|2u\rangle|0\rangle + |uv\rangle|1\rangle)/\sqrt{2}$ with a reference qubit; after $t$ attempts the coefficient of $|2u\rangle\langle uv|$ is $a_{\mathsf{L}}^t$, with $a_{\mathsf{C}} = 23/30$ and $a_{\mathsf{E}} = 7/10$, and the partial transpose on the reference has a principal $2 \times 2$ minor equal to $-a_{\mathsf{L}}^{2t}/4 < 0$; no finite power of either full channel is entanglement breaking;*

*(iii) every input converges to $\tau_P \otimes \rho_R$, where $\tau_P$ is the uniform binary density, with* dist $\leq \min\{1, \mu^{s/2} + \mu^s + \frac{D}{2}(1-\gamma)^{t-s}\}$ *for all $0 \leq s \leq t$, $\mu = 5/6$ and $D = 66$.*

*Proof sketch.* For the normalized Choi state $J_t$, $\|J_t - I/D^2\|_2^2 = D^{-2}\sum_{i>0}\lambda_i^{2t}$. The separable ball of Gurvits and Barnum [107] and the equivalence of separable Choi states with entanglement-breaking channels [109] reduce (i) to $65\, r_{\mathrm{mix}}^{2t} + 4290\, r_{\mathsf{L}}^{2t} \leq 1/4355$, where $r_{\mathrm{mix}} = 1 - \gamma$. The rational enclosure $559/250 < \sqrt{5} < 2237/1000$ certifies this at $t = 134$ and its failure at $t = 133$; treating all modes with $r_{\mathrm{mix}}$ would certify only 178. For (ii), 25 edges avoid $u$ and four of them touch $v$; $\mathsf{C}$ keeps one coin on each of those four and $\mathsf{E}$ keeps neither, and the triangular structure prevents any return from the binary space. Separable states have positive partial transpose [113]; the complete argument is in [48, 91]. □

Replacing the full carrier by its attracting binary shell therefore changes an exact finite-time quantum property, although both converge to the same stationary density. The number 134 is a sufficient certificate time; the exact index of entanglement breaking on the shell is not determined.

### 3.6 Recovery from complete records

Preserving committed records and recovering the unknown input from them are different questions [49]. Here a seam policy may depend on committed records and classical randomness and may not

inspect hidden loads or insert quantum corrections; a final decoder may act on the full load register and tape, keep the tape, use blank ancillas and return a state on the original carrier. That decoder access is stronger than the record-only access of the observer's controller.

**Theorem 3.6** (Optimal recovery from complete native records)**.** *Let $X$ be the 78 configurations or the 66-state shell and $D = |X|$. For a complete admitted history $h$ let $X_h$ be its compatible inputs, $P_h$ the projector onto them, $d_h = \operatorname{rank} P_h$, and $w_h$ the product of its seam and coin factors. Then:*

*(i) the history Kraus operator is $K_h = \sqrt{w_h}\, U_h P_h$ with $U_h^\dagger U_h = P_h$ and $\sum_h w_h P_h = I$; a tape-controlled inverse outputs $\sum_h w_h |h\rangle\langle h| \otimes P_h \rho P_h$, so every initial load population is recovered;*

*(ii) among all final CPTP decoders the optimal entanglement fidelity for the maximally mixed input is $F_{\mathrm{opt}} = D^{-2} \sum_h w_h d_h^2 = D^{-2} \sum_{x,y} H_{xy}$, with $H_{xy} = \sum_{h:\, x,y \in X_h} w_h$;*

*(iii) for the two-event observer, $F_{\mathrm{opt}}$ equals* 5543/8712 *(binary) and* 1301/2028 *(full) under* C*, and* 769/1452 *and* 3329/6084 *under* E *with the same current-sign selector applied to its decoded current;*

*(iv) the maximal exactly correctable algebra on the whole input is the commutant of $\{P_h\}$, and one event with full-support random seam choice reduces it to the diagonal load algebra; nevertheless the qubit code $|0_L\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $|1_L\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ satisfies the Knill-Laflamme condition $QP_hQ = \alpha_h Q$ for every seam, coin and record of either law and is recovered exactly after one event;*

*(v) under independent seam attempts $H_{xy}(t) = q_{\mathsf{L}}(x-y)^t$, with largest off-diagonal rates* 5/6 *for* C *and* 7/10 *for* E *on both carriers; $F_{\mathrm{opt}}(t) \to 1/D$, and neither retained instrument is entanglement breaking at any finite horizon;*

*(vi) a fixed* E *seam word identifies every full-carrier input exactly when at most one visited component has size one or two, and every binary input exactly when at most one component is a singleton and at most one has size two; the minimal such words have eight and seven distinct edges;*

*(vii) no bounded record-only adaptive policy makes* C *entanglement breaking; a* C *word that doubles a minimal forest, 16 or 14 events, excludes every nontrivial exact code; the policy that repeats each forest edge until its coin first differs identifies the input almost surely, with stopping time $T = m + \sum_{i=1}^{m} G_i$ for independent geometric $G_i$ of success probability one half, $\Pr(T = t) = \binom{t-m-1}{m-1} 2^{-(t-m)}$ for $t \geq 2m$, and $\mathbb{E}[T] = 3m$, namely 24 on the full carrier and 21 on the binary shell.*

*Proof sketch.* Every retained current determines a translation, so reversing them gives the input uniquely from the terminal configuration and $h$. Optimality follows from Cauchy-Schwarz on each classical branch: $\sum_a |\mathrm{Tr}(R_{ha} U_h P_h)|^2 \leq d_h^2$. For (iv), commutation is sufficient by (i), and necessity follows from equality in the Schwarz inequalities in a Stinespring dilation; the code has nonadjacent occupied pairs, equal one-port means and vanishing products $N_a N_b$ on every edge, which gives the Knill-Laflamme identity [110]. For (vi), matching E histories are those for which $z$ is constant on every visited component. For (vii), the inputs $\{u, w\}$ and $\{v, w\}$ with $u, v$ nonadjacent keep equal currents for at least one coin on every edge, so a history of length at most $T$ shares a rank-two fiber with probability at least $2^{-T}$; repeating an edge with the opposite coin reveals the initial parity through $d = 2j_1 + j_*$ [49, 92]. □

The minimal fixed $\mathsf{E}$ words are the forests $(0,1),(0,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the full carrier and $(1,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the binary shell, with component sizes $(3,3,3,3)$ and $(1,2,3,3,3)$. The full-input correctable blocks of the two-event observer have sizes $46,10,9,1$ on the binary shell and $56,10,9,1,1,1$ on the full carrier, for both laws. Existing source inverse words implement the single-seam branch reversal of (i) while keeping the classical archive [49, 92]. Native current records thus suffice to reverse classical load transport, without restoring arbitrary coherence lost across record groups. The source encoder and active recovery of Section 6.5 instead preserve the stated logical qubit between events; they do not invert the whole bare-load channel.

### 3.7 A coherent continuation and its selection boundary

An exact finite experiment attaches a reflection walk to the binary native branches, with coherent event access in place of a committed tape [50]. For each binary configuration $x$, seam $e$ and coin $c$, let $(y,j)$ be the native output and current. On the $66\cdot 60 = 3960$ arc basis vectors define

$$S|x,e,c\rangle = |y,e,c'\rangle, \qquad c' = \begin{cases} c, & j=0,\\ 1-c, & j\neq 0,\end{cases} \qquad A|x\rangle = \frac{1}{\sqrt{60}}\sum_{e,c}|x,e,c\rangle.$$

Every arc is checked: $S^2 = I$, with 300 exchanged pairs and 3360 fixed arcs, and the outgoing native record $(y,e,c,j)$ is recovered bijectively from the reversed arc. Thus $A^\dagger A = I$ and $A^\dagger SA = K$, the actual symmetric native transition kernel, and $W = S(2AA^\dagger - I)$ satisfies the hypotheses of the quantum-walk spectral mapping theorem with discriminant $K$ [121, 122]. The identity $A^\dagger W^n A = T_n(K)$, with $T_n$ the Chebyshev polynomial, gives for the centered loads $X_{x,p} = x_p - 1/6$

$$KX = X(I - L/60), \qquad U_{n+1} - 2U_n + U_{n-1} = -U_n L/30, \qquad U_n = A^\dagger W^n AX.$$

These are compressed amplitudes, and $n$ counts the supplied coherent continuation with no identification as duration. The binary domain is essential: on the full carrier $(2,0,\ldots)\mapsto(1,1,\ldots)$ has no native reverse event, so its asymmetric kernel equals $A^\dagger SA$ for no involutive unitary on the same domain. Twelve port addresses and sixty seam/coin events give a 720-arc source-compiled walk with discriminant $K_{12} = I - L/60$; the map $F_{x,p} = (x_p - 1/6)/\sqrt{10}$ satisfies $F^*F = I - \mathbf{1}\mathbf{1}^T/12$ and $K_{66}F = FK_{12}$, so it realizes the shared mean-zero density sector. Among the sixty proper carrier rotations a unique inverse-symmetric order-five class $\mathcal{C}$ of size twelve gives $L = 6I - \frac{1}{2}\sum_{g\in\mathcal{C}} U_g$, with multiplicity-free density representation $1\oplus 3\oplus 5\oplus 3$. The other order-five class is isospectral but assigns probability zero to a native seam whose actual one-attempt probability is $1/60$, so the seam records select the class.

The emitted walk word uses ten address qubits, five blank scratch qubits and 114,910 source pulses per walk with 229,820 load/export events; uniform event preparation costs a separate 36,288 pulses. The reversible circuit is checked on all 32,768 basis inputs, every clean-scratch address returns clean scratch, including 304 padding addresses, and the fifteen-qubit execution reuses raw pulse factors without target-matrix insertion or output normalization. Floating routine guards sum to $s = 829/(5\cdot 10^9)$, giving complete output trace-norm error at most $2\delta + \delta^2$ with $\delta \le s/(1-s)$; these are numerical guards, not physical precision certificates.

**A finite reference and its modular clock.** Biased native event amplitudes, the compiled reversible seam shift, inverse event preparation and a herald realize $F_p = I - pL/60$ on the protected site code. From the supplied $I_{12}/12$, $n$ successive successes give $F_p^{2n}/\operatorname{Tr} F_p^{2n}$ with success probability

at least $1/12$, every failure recorded; for $p = 30t/n \le 1$ this tends to $e^{-tL}/\operatorname{Tr} e^{-tL}$ with relative-entropy and total-variation errors at most $16t^2/[n(1-4t/n)]$. The execution at $p = 1/2$, $n = 2$, $t = 1/30$ tests all twelve input columns with success $4382551/5184000$ and bound $1/105$, and a source interference readout has contrast $260343/43825510$ against zero after dephasing. This state is central on the 44-dimensional represented group algebra, where its modular flow is trivial; nontrivial modular gaps require the full protected matrix algebra [20].

Positive subordination gives square-root weights on the same register. With $F = I - hL/2$, $h = p/30$, a Poisson number of Sibuya jumps gives positive weights with

$$\sum_n w_n F^{2n} = e^{-R\sqrt{I-F^2}} = e^{-b\sqrt{L-hL^2/4}}, \qquad b = R\sqrt{h}.$$

Finite jump and count caps and dyadic rounding retain missing mass as sampler rejection. The source instance $p = R = 1/2$ uses 64 native coin records and at most four raw filters, retains each first failure and all sixteen successful interference outcomes, and has Gibbs trace-distance bound below 0.217 and spectral diagnostic 0.01344. Its finite polynomial $M = \sum_n w_n F^{2n}$ fixes $H = -\log(M/m_0)$, $m_0 = \sum_n w_n$. Raw source words execute that clock, a current-controlled intersector kick and complete spectral-pair detectors on the same register, retaining all 3,600 two-attempt native tapes, sampler and herald failures, sixteen detector addresses and twelve reference columns. The modular identity $\sigma_s = \alpha_{-s}$ holds for the ideal successful marginal $M/\operatorname{Tr} M$ on $\mathcal{B}(\mathbb{C}^{12})$; it does not hold for every retained request block.

This density supplies eleven pairs $Q_j = X_{0j}/\sqrt{2\Delta_j}$, $P_j = Y_{0j}/\sqrt{2\Delta_j}$, $\Delta_j = p_0 - p_j > 0$, with $\langle [Q_j, P_k] \rangle = i\delta_{jk}$, symmetric covariance $\delta_{jk} \coth(g_j/2)/2$, $g_j = \log(p_0/p_j)$, and frequency $g_j$ under $H$. For chronological factors $U_N = e^{iA_m/\sqrt{N}} \cdots e^{iA_1/\sqrt{N}}$ built from these pair observables on independent identical copies, a parity commuting with the state and anticommuting with every $A_k$ gives

$$\left| z_N^N - e^{-C} \right| \le \frac{D^4/24 + |C|^2/2}{N}, \qquad z_N = \operatorname{Tr}(\rho U_N),\ D = \sum_k \|A_k\|,$$

with $C = \frac{1}{2}\sum_k \langle A_k^2 \rangle + \sum_{k>j} \langle A_k A_j \rangle$. Source pointer reads execute the one-copy factors for three band representatives with all 32 work/pointer addresses retained; a growing copy register is not executed. With physical normalization $q_j = Q_j/\sqrt{\omega_j}$, $\pi_j = \sqrt{\omega_j} P_j$ and $g_j = b\omega_j$, the clock $H/b$ gives thermal covariance $\coth(b\omega_j/2)\cos(\omega_j t)/(2\omega_j) - i\sin(\omega_j t)/(2\omega_j)$. For $m$ smeared factors with $R = \sum_k R_k$, $R_k^2 = \sum_j (f_{kj}^2/\omega_j + \omega_j h_{kj}^2)$, gaps $g_j, b\omega_j \ge \gamma > 0$, $\vartheta = \gamma/(1+\gamma)$ and $Z = 1/p_0$,

$$|z_N^N - e^{-C}| \le \frac{R^4}{\vartheta^2 N} \left( \frac{(m+1)Z}{96} + \frac{1}{8} \right),$$

so $Z/N \to 0$ suffices on a fixed positive band; the family $\rho = \operatorname{diag}(2, 1, \ldots, 1)/(d+1)$ with $N = d$ has instead the limit $e^{3(\cos(1/\sqrt{2})-1)} \ne e^{-3/4}$. On the native family $D_l = 4^l L_l$ one has $\lambda_n(D_l) \ge n/81940$, hence $\operatorname{Tr} e^{-b\sqrt{D_l}} \le 1 + 163880/b^2$, and with the lazy filter $p = 630h16^l \le 1$,

$$Z \le \frac{1 + 327760/b^2}{m_0}, \qquad P_{\text{success}} \ge \frac{m_0}{240\, 4^l};$$

both prepared and target gaps exceed $b/182$ when $(1-m_0)/m_0 \le b/182$. Preparation error $E$ with success floor $p > 2E$ gives conditional trace error at most $4E/p$, and with one-copy word error $\epsilon$ the $N$-copy amplification is at most $N\zeta/(1-N\zeta)$, $\zeta = 4E/p + \epsilon$. On a fixed band $0 \le D_0 \le \Lambda$ with $h\Lambda < 2$, retained positive weights with missing mass $\delta < a \le e^{-b\sqrt{\Lambda}}$ give

$$\| \log(m_0) I - \log M - b\sqrt{D_0} \| \le \frac{\delta}{a - \delta} + \frac{bh\Lambda^{3/2}}{4(1 + \sqrt{1 - h\Lambda/4})}.$$

A constructive growing word family sets $k = 2l + c$ with $4^c \geq b^2 \max(630, \Lambda)$, $R = 2^k$, $h = b^2 4^{-k}$, $a = 4^{-k}$, jump cap $2^{6k}$, count cap $6R + 2k$ and Taylor order $4k + 3$; the zero-request branch keeps weight at least $f = 2^{-2R}$. For $d = 240\,4^l$, request cap $K$ and word length $P$, primitive and load/export error at most $fa/(64dKP)$ gives complete stopped preparation error $E \leq fa/(32d)$ and

$$\|dS_{\mathrm{raw}} - M\| \leq 2dE, \qquad \|\log(dS_{\mathrm{raw}}) - \log M\| \leq \frac{a}{16 - a}$$

for the actual successful marginal $S_{\mathrm{raw}}$, without a reducing-band premise. The derived partition bound permits $N = 2^l$ copies with copy, preparation and detector errors tending to zero on a fixed positive band. Growing execution, spectral tails, accurate source services and a zero-energy anchor for the radial product lie outside these bounds, and fixed caps with $N_c h \to 0$ force the bounded-band generator to zero; the uncapped sampler has infinite expected work because the constant sector survives every filter [20].

**Selection controls.** Two exact controls bound what the committed records select. For $|z| = 1$, $W_z = S[AA^\dagger + z(I - AA^\dagger)]$ has the same first event $W_z A = SA$ but $A^\dagger W_z^2 A = K^2 + z(I - K^2)$; a complete added Ramsey read gives two-step plus probability 1 for $z = 1$ and $113/150$ for $z = -1$. Multiplying each coherent record branch on seam $(0, 1)$ with current $+1$ by $-1$ preserves every committed C and E branch on arbitrary references, yet the twisted embedding gives $\mathrm{Tr}[(A_\eta^\dagger S A_\eta)^3] - \mathrm{Tr}(K^3) = -1/900$. Both real continuations conserve load, respect all sixty proper rotations, fix the uniform coherent state and admit the time reversal $\Theta = S$ conjugation; under full event-permutation covariance, real orthogonal coins fixing the uniform vector are identity or reflection. A sign change on every moving arc preserves proper covariance and the committed instruments but gives $\mathrm{Tr}(D^3) - \mathrm{Tr}(K^3) = -1/90$ for $D = 2\,\mathrm{diag}(K) - K$. Scalar record phases cancel throughout committed-record histories. A source-executed interface test that reads the low unread record bit in the $X$ basis gives complete $X$-plus probabilities $3/4$ and $1/4$ for one preparation, against $1/2$ for both after commitment, using 5380 C or 5383 E source pulses per attempt [53]. These are restrictions on selection: coherent access is a supplied interface, and the test selects no phase.

### 3.8 Reversible seam repair, reference and apparatus

A different declared repair law retains the information removed by agreement in coherent memory [51]. On a one-load seam write $|\pm\rangle = (|10\rangle \pm |01\rangle)/\sqrt{2}$. With memory initially in 0, the repair has the action

$$|+\rangle|0\rangle \mapsto |+\rangle|0\rangle, \quad |-\rangle|0\rangle \mapsto |+\rangle|1\rangle,$$
$$|20\rangle|0\rangle \mapsto |11\rangle|2\rangle, \quad |02\rangle|0\rangle \mapsto |11\rangle|3\rangle.$$

Swapping the input and output vectors of each nontrivial arrow, and fixing their orthogonal complement, defines a load-conserving Hermitian unitary on the six local source states and four memory states. Its blank-memory blocks are the complete agreement-pump instrument, including both double-load origins. The sign in $|-\rangle$ is a seam exchange phase, not a derivation of fermionic statistics. Committing or discarding the record after every event recovers dissipative repair; coherent reuse defines a different history law.

In the one-load sector retain the parity bit and sixty oriented seam addresses $k = (a, b)$. Let $U_k$ exchange $|-\rangle_k|0\rangle$ and $|+\rangle_k|1\rangle$, let $A\psi = \psi \otimes \sum_k |k\rangle/\sqrt{60}$, and set

$$Q = \sum_k U_k \otimes |k\rangle\langle k|, \qquad R = 2AA^* - I, \qquad W = QR.$$

The reflection is a declared operation on the observer's address register. It is not generated by committed C/E histories. Both $Q$ and $R$ are involutions, so $W$ is unitary on $12 \cdot 2 \cdot 60 = 1440$ dimensions. If $B$ is the carrier adjacency matrix and $D$ its degree matrix, the exact discriminant, ordered by parity memory, is

$$A^*QA = \operatorname{diag}\left(I - \frac{D-B}{60},\, I - \frac{D+B}{60}\right). \tag{3}$$

Opposite seam orientations cancel the off-diagonal parity blocks. The first block is the native one-load kernel, and $A^*W^nA = T_n(A^*QA)$ gives $\cos\omega = \lambda$ for its spectral values. This is an exact finite wave law in the declared walk-step clock. It preserves the first-event population shadow; from a single port the second return probability is $643/900$, compared with $101/120$ for two classical attempts. Coherent information and its changed continuation are therefore observable properties of the specified program.

**Proposition 3.7** (A reference prepared by the same walk)**.** *For any finite unitary $W$ and $0 < \alpha < 1$, the complete two-outcome instrument*

$$K_+ = \frac{I + \alpha W}{1+\alpha}, \qquad K_- = \frac{\sqrt{\alpha}(I - W)}{1+\alpha}$$

*prepares from $I/d$, on the retained success outcome, the faithful stationary state*

$$\rho_W = \frac{(1+\alpha^2)I + \alpha(W + W^*)}{Z}, \qquad Z = (1+\alpha^2)d + 2\alpha \operatorname{Re}\operatorname{Tr} W.$$

*Its success probability is $Z/[d(1+\alpha)^2]$, and its smallest eigenvalue is at least $(1-\alpha)^2/Z$.*

*Proof.* Direct expansion gives $K_+^*K_+ + K_-^*K_- = I$. The successful numerator is $(I+\alpha W)(I+\alpha W^*)$; it commutes with $W$ and has eigenvalues $|1 + \alpha e^{i\omega}|^2 \geq (1-\alpha)^2$. □

A controlled walk and one prepared ancilla implement these two Kraus rows. For (3), $\operatorname{Tr} W = -1276$; choosing $\alpha = 1/2$ gives $Z = 524$, success probability $131/810$, and $\rho_W \geq I/2096$. Failures are recorded. Independent fresh restarts, capped after $k$ attempts, retain failure probability $(679/810)^k$. The reference is the reduced state on the walk-register algebra. A known preparation-basis label conditions a different state; stationarity of the joint walk and preparation archive is not asserted.

This reference has matrix entry $\langle a, 0, k|\rho_W|b, 0, k\rangle = -29/31440$ for $k = (a, b)$, so an ordinary register projector has nontrivial modular action. The modular generator is a function of $W + W^*$, which assigns equal weight to opposite eigenphases; it is not the walk quasienergy. Optional basis quarter-turns $V_j = I + (i-1)|j\rangle\langle j|$, the walk and complete register readout give history effects spanning all $1440^2$ Hermitian directions. Nonzero walk diagonals and support paths of length at most four give a depth bound of twenty-four unitary settings. This is tomography by a declared apparatus, not arbitrary positive-effect implementation or a derivation of quantum probability from classical counts. Its destructive probes are excluded from idle propagation.

The reference includes auxiliary memory modes: $W$ has 38 eigenvectors at $+1$, 1356 at $-1$ and 46 oscillatory dimensions. The cyclic subspace of blank-parity source inputs has dimension 23. These counts describe the finite program and provide no particle-species assignment. A committed phase-addressed agreement pump has instead a faithful nontracial reference and complete twelve-port tomography, but its one-load coherence contracts by at most $3/4 + \sqrt{2}/12$ per attempt. Its uniform damping cannot be transferred to the reversible walk, or removed by renaming its clock.

**Gauge transport in the same repair.** Attach a declared five-level internal code and unitary links $G_{ba} = G_{ab}^*$. Replace $|\pm\rangle$ by the isometries

$$m_{ab}h = \frac{|a\rangle h - |b\rangle G_{ba}h}{\sqrt{2}}, \qquad p_{ab}h = \frac{|a\rangle h + |b\rangle G_{ba}h}{\sqrt{2}}.$$

Exchanging $m_{ab}h|0\rangle$ and $p_{ab}h|1\rangle$ gives a unitary seam event. Its oriented-address average has the magnetic and signless discriminants $P_G = I - (D - B_G)/60$ and $I - (D + B_G)/60$ on the same code. The coupled walk acts on 7200 dimensions, has trace $-6380$, and Proposition 3.7 gives $Z = 2620$, the same success probability, and $\rho_W \geq I/10480$. Block-unitary links conserve the internal charge $\mathrm{diag}(2I_3, -3I_2)$ under this idle evolution.

The encoded controller derives 120 binary lifts from the sixty actual carrier rotations and executes their finite words with the central sign retained. Its two-chart transports are typed reversible arrows. The mixed $3 + 2$ code has central lift $\mathrm{diag}(I_3, -I_2)$; this is not scalar and hence does not define a projective representation of the sixty-element carrier group itself. The holonomy-cover interface of the gauge theorem permits this lift [7, 51]. A link with that central value gives two triangular Wilson traces 1 instead of 5 and changes $\mathrm{Tr}\, P_G^3$ by $-1/4500$. A prepared return observation therefore detects the static flux. One allowed link change removes it. Neither protected topological matter, dynamical gauge backreaction, physical spin, a relativistic continuum nor a rest mass follows from this finite response. A complete internal gauge response also needs its specified quotient: apparatus controls are not silently discarded to force twelve dimensions.

Finite observer agreement includes the apparatus. For implemented local cover frames $V_a$, let $S$ act by $V_a$ on the code at port $a$. The transformed links $G'_{ba} = V_b G_{ba} V_a^*$ execute $W_{G'} = S W_G S^*$ and prepare $\rho_{G'} = S\rho_G S^*$. A declared inverse-frame/read/restore wrapper implements $SP_iS^*$ and the transported quarter-turn. Thus every finite adaptive branch satisfies $B'_h = SB_hS^*$, including both preparation outcomes. Complete history probabilities agree, also with an untouched quantum reference. The exact executable check retains all four filter/detector outcomes. Omitting the detector frame instead changes one probability from 1/60 to 1/120. This accepted finite-chart domain does not supply a growing federation or the A3 information-projection dynamics.

**A joint movable defect and wave.** A finite extension makes the link register dynamical [52]. Fix the boundary and exterior of the five-face region around port 5. Ten thirty-bit link words carry one regional central flux and its exterior compensator. On each word $g$, all thirty link signs enter the same 7200-dimensional wave $W_g$. Let $T_e$ flip an interior edge when exactly one adjacent face has flux and otherwise hold. This permutation is an involution. If $\Pi_e$ projects the wave onto the edge's endpoints, set

$$B = \sum_g |g\rangle\langle g| \otimes W_g, \qquad C_e = I \otimes (I - \Pi_e) + T_e \otimes \Pi_e, \qquad U = C_4C_3C_2C_1C_0B.$$

The five interior edges have a fixed order. Each $C_e$ is a Hermitian involution, so this repeated tick is unitary on 72000 dimensions, with inverse $B^*C_0C_1C_2C_3C_4$. The link information remains coherent during idle evolution; committing a moved/held flag defines a different channel. Both regional boundary flux and $Q_5$ are conserved.

Two complete one-tick readouts establish reciprocal coupling. With the same wave preparation $A(|0,3\rangle + |5,3\rangle)/\sqrt{2}$, changing the regional defect from face $(0,1,5)$ to $(0,5,11)$ changes one wave-basis outcome from 1/120 to 0. With the initial link word fixed, preparing $W_g^*A|5,3\rangle$ or $W_g^*A|6,3\rangle$ makes the final face-$(4,5,11)$ detector return yes with probabilities 1 or 0. Complements are retained. The coherent superposition of those two preparations gives Schmidt weights $1/2, 1/2$

in the declared link/wave factorization; the invariant marginal readouts establish interaction independently of that factorization's physical interpretation.

The same joint tick has $\operatorname{Tr} U = -47850$. Its complete filter $K_+ = (2I+U)/3$, $K_- = \sqrt{2}(I-U)/3$ on $I/72000$ prepares

$$\rho_U = \frac{(5/4)I + (U+U^*)/2}{42150} \geq \frac{I}{168600}, \qquad p_+ = \frac{281}{1080}.$$

This reduced reference is stationary under this U, rather than an imported static-background state. Flipping all five interior links together with $Z$ on the doublet at port 5 defines an involution $\Gamma$ commuting with U. Its positive sector has dimension 36000 and reference normalization 21075; the complete Gauss projector retains both outcomes. Face and wave-register readouts preserve this constraint. This supplies one finite interior central Gauss law, without full continuous gauge dynamics. The boundary, move grammar, reflection and preparation are declared.

There is also a pure fixed background: uniform amplitudes on the ten links, twelve ports and sixty addresses, blank parity memory and any triplet component give $U\Omega_j = \Gamma\Omega_j = \Omega_j$. A complete Gauss filter followed by controlled native move words and a ten-level Fourier ancilla prepares it with probability 1/10 per fresh trial, retaining all failures. It is a stationary background, not a selected physical vacuum. Measuring its five faces and applying both filters $(I \pm U^2)/2$, then all four Fourier-clock bins, gives forty complete records. The minus-filter outputs are orthogonal to $\Omega_j$ and have total weight 283/7200 conditional on background preparation. This is a populated finite frequency response.

Local Gauss-dressed exchanges connect cells with a strict finite circuit cone. The tested gapless spectator branch is quadratic at small momentum. A distinct shared-link occupation law second-quantizes each $W_g$ and uses endpoint number parity for its link controls. It agrees with U at degree one and retains all occupations. Exact cubic executions resolve eight and sixty-four momenta with complete 32- and 256-outcome instruments. Their first moments equal $(-29/30)\prod_i[(1-i)/2+(1+i)\cos q_i/2]$; the tick changes a defect face with probability 1/450. A fixed-link control has the same moments and zero face movement, so moments alone do not identify interacting spectral poles.

The specified positive clock $H = \sum_x[I - \operatorname{Re} U_{F,x}] + (\pi/4)\sum_{\langle xy\rangle} d\Gamma(I - S_{xy})$ has constrained empty-wave grounds and $H \leq (2+3\pi)N$. Its uniform-link triplet energies are $0, (5-\sqrt{5})/60, 1/10, (5+\sqrt{5})/60$, with localized-port weights $1/12, 1/4, 5/12, 1/4$. The branches $E_m(k) = m + (\pi/2)\sum_i(1-\cos k_i)$ require different leading squared relativistic speeds $\pi m/2$; their diffusive limits leave a quartic mass-shell residual. An exact two-wave contact collision is available on this H, but these energies do not identify a relativistic particle spectrum.

**A positive free field of actual source pairs.** Adding the flat gauge orbit to the five odd orbits permits local exchange of the whole inside/exterior flux pair. Five face labels describe locations of one pair type. The complete receiver probabilities are $119/240, 1/240, 1/2, 0$, conditional on recorded Gauss success. A two-lane Clifford route retains all 121 cell states and has one-object limit $\omega = \pm\sqrt{|p|^2+4}$, with error $ta(3P^2+6P)$ on $|p_i| \leq P$. Its mass coin uses the literal face count $D = 2$ with coefficient one. Positive many-body energy and statistics require an occupation law and reference in addition to this frequency equation.

A separate complete graded source assigns twenty carrier slots per cell, one for each face and Clifford component. Its vacant slot encodes the flat Gauss orbit; its occupied slot encodes the actual paired-flux orbit. Every local occupation is retained. Declare fermionic grading and the

local Wilson Hamiltonian with one-body symbol

$$h_a(p) = \sum_i \alpha_i \frac{\sin(ap_i)}{a} + \beta \left[ 2 + \sum_i \frac{1 - \cos(ap_i)}{a} \right],$$
$$E_a(p)^2 = \sum_i \frac{\sin^2(ap_i)}{a^2} + \left[ 2 + \sum_i \frac{1 - \cos(ap_i)}{a} \right]^2.$$

The grading, regulator and face-count coefficient are declared architecture; ordinary qubit realizations require parity strings. This Hamiltonian is not a logarithm identified with the routing tick. Filling the negative bands gives a unique finite Slater ground with $10V$ source pairs and $E_{\text{sea}} = -10 \sum_p E_a(p)$. Relative to that same reference,

$$H_a - E_{\text{sea}} = \sum E_a(p)(b^\dagger b + d^\dagger d) \geq 0.$$

The full gap is two and the nonvacuum neutral gap is four. A particle adds one source pair; a hole removes one from the reference. Their opposite relative occupation charges are not electric charges. Native gauge currents vanish on these native-empty slots.

On $|p_i| \leq P$, the bound $\|h_a - (\alpha \cdot p + 2\beta)\| \leq b = a^2P^3/2 + 3aP^2/2$ controls dynamics by $|t|b$ and the filled-band projector by $b/2$. Wick expansion controls all finite free-field words. The resulting effective field has positive particle/antiparticle energies, spin one half, causal anticommutators and $E^2 - |p|^2 = 4$. The Wilson corner energies $2 + 2n/a$ exclude additional finite-energy valleys. The five face channels remain degenerate neutral copies. This is a controlled free field, without an interacting chiral or Yang–Mills mass-gap claim.

An actual four-mode block supplies a complete energy readout. A fermionic reservoir SWAP, with both initial settings and final flags recorded, gives four preparation alternatives. Nine particle/hole occupation projectors on the entire block give 36 joint records. At rest the six populated weights are $1/4, 1/8, 1/8, 1/4, 1/8, 1/8$. Addition and removal each give energy two; failed preparation has ground and energy-four neutral-pair components. A nine-level work battery completes this operation to an exactly energy-conserving unitary by extending each total-energy shell, including edge shells. Equality with the driven preparation holds for energy-resolved records; tracing the battery can dephase cross-energy coherence. A nonzero momentum control gives $E = \sqrt{10}$ at $a = 1, p = (0, 0, \pi/2)$. The dimensional rest-energy relation is $mc^2 = 2u_E\,\text{GeV}$, with $u_E$ unspecified.

**Native currents and chiral propagation.** Bare flux has trivial native gauge charge, and the pure-face commutant of the literal interaction is scalar. Five dark native orbitals instead carry $R = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$, with actual-current weights $(\bar{3}, 1, -2/3) \oplus (3, 2, 1/6) \oplus (1, 1, 1) \oplus (1, 2, -1/2) \oplus (\bar{3}, 1, 1/3)$. The additional central degree current is separate. On this even module $U_F = I$ and $K = I - \text{Re}\, U_F = 0$. A declared cell encoding places each occupied spin–native object in a literal compensated-flux slot. Antisymmetrizing within the cell and taking the graded tensor product between cells embeds every occupation of $\mathcal{F}(\mathbb{C}^2 \otimes R)$, preserving actual currents and the cell support of even operators. Grading, binding and transport are architecture choices; the idle law supplies no binding energy.

A complete antiperiodic cubic momentum menu supports $h_L(p) = -\boldsymbol{\sigma} \cdot p \otimes I_R$. All regulator momenta are retained. The negative-band sea has positive particle and hole excitations with opposite helicities and conjugate charges; no independent mirror family is inserted. Fourier projection gives the free Weyl limit with Sobolev tail control. Its finite position kernel has algebraic long-range hopping, so this proof supplies neither a finite causal cone nor an interacting local gauge limit.

**Quantum Higgs coupling and the field dictionary.** The inherited symmetric tensors $C_{c,k}$, for up, down and charged-lepton channels, give the spin-antisymmetric pair interaction $\frac{1}{2}c^T(\epsilon \otimes C_{c,k})c\,\Phi_k + \text{h.c.}$, with $\epsilon = i\sigma_2$. Here $\Phi_k$ is the scalar field for the up channel and its adjoint for the down and lepton channels. A five-state scalar memory retains its vacuum, two particle states and two antiparticle states. The quantum field $\Phi_k = a_k + b_k^\dagger$ satisfies the exact finite gauge Ward identities; replacing it by a number is a different background calculation. In the full 320-state active lepton–scalar block with unit lepton Yukawa, unit scalar-number energy and empty quark spectators, the ground energy is $-2$ with multiplicity two and the spectral gap is $(5-\sqrt{21})/2$. Conditioned on ground preparation, the reservoir and energy instrument has 56 records. Ground preparation succeeds with probability 1/160; the full preparation/readout law has 69 records, including thirteen preparation failures. A retained compensator balances the reservoir charge; the source plus apparatus obey all local Gauss constraints. Within the prepared-ground experiment, each reservoir setting is selected with probability 1/2. Successful addition or removal has joint weights 9/40 and 1/40 at excitation energies 1 and 3, respectively. The lower global gap has zero weight for this probe. Failed preparations are retained. These are finite onsite energies, without a relativistic mass assignment.

For the separately specified neutral background $H = (0, v)$, set $M = v\sum_c y_c C_{c,1}$ and $\Delta = \epsilon \otimes M$. The Nambu matrix is

$$B(p) = \begin{pmatrix} h_L(p) & -\Delta^* \\ \Delta & -h_L(-p)^T \end{pmatrix}.$$

For each tensor pair $(l, r)$ with sign $s_{lr}$, the canonical variables $\Psi_L = c_l$ and $\Psi_R = -s_{lr}\epsilon c_r^\dagger$ reduce this matrix to seven Dirac blocks and one neutral left-Weyl block. All occupations survive this invertible change of variables. The paired blocks obey $E_c^2 = |p|^2 + |vy_c|^2$. Their actual electric charges are $2/3, -1/3, -1$; color acts on both quark chiralities and weak currents only on the left. This determines conditional up-type, down-type and charged-lepton roles for one family. It selects no generation, Yukawa coefficient, physical Higgs vacuum or measured mass. The finite antiperiodic menu excludes $p = 0$; $|vy_c|$ is the rest parameter of the background continuum formula.

**Complete constrained two-cell spectroscopy.** A uniform open-edge member retains twelve fermion modes, one six-state quantum link and two five-state quantum scalars: $2^{12}6\cdot5^2 = 614400$ active states. Both local charged-lepton Yukawa coefficients, scalar number energy and link-excitation energy equal one; quark Yukawa coefficients vanish. Its hopping is $ic_0^\dagger\sigma_3 U_{01}c_1/2$ plus its adjoint. Exact projection onto every local Gauss generator gives 1053 neutral physical states. Complete rational eigenbasis certificates across every lepton-number and spin-projection sector place the unique ground in $L = 2$, $2S_z = 0$, with

$$-4.554826639 < E_0 < -4.554826631, \qquad E_1 - E_0 > 0.394804412.$$

These intervals bound the full constrained Hamiltonian, including states outside the empty-input sector. A binary filter on the maximally mixed physical state prepares this ground with probability 1/1053 and retains failure with probability 1052/1053, together with the compensating preparation reference.

An actual fermionic swap with a reservoir and compensator preserves the local Gauss charges and total lepton number. Its four branches are $(a^\dagger, -n, -a, I-n)/\sqrt{2}$ on this even ground. Joint conserved-sector and energy-window measurements give 460 conditional records, or 461 records including failed preparation. For the right-cell spin-up probe, the first addition and removal transfers are respectively $1.052043848 \pm 8\times10^{-9}$ and $0.909798632 \pm 8\times10^{-9}$ source units; their conditional record probabilities exceed 0.21838924 and 0.23089293. The right-positive sector's absolute ground

has $L = 4$ and is dark to this single removal, which reaches $L = 3$. Charge conjugation reflects the boundary: it pairs right addition with left removal, rather than asserting equality of the two right-cell thresholds. Odd sectors have exact Kramers doublets, while the open source has no translation dispersion or full rotation symmetry. A separate two-link periodic construction has an exact translation and complete phase measurement. Joint source/reference translation-phase records translate a retained compensator with the source. At two cells, odd-sector momenta are $\pm\pi/(2a)$, with equal squared momentum, so their energies cannot separately determine both a mass and a propagation speed. Appending a blank return link does not transfer the open dynamics: an actual physical pair preparation has full intertwining residual squared 3/16, entirely outside that embedding. An independent complete calculation on the periodic source gives 3424 neutral and 2456 states in each compensated charged Gauss sector. The unique neutral ground has $(L, 2S_z, T) = (2, 0, +1)$, energy $-4.33817023295 \pm 1.5 \times 10^{-8}$ and gap greater than 0.25770063535. Translation composed with conjugation pairs addition and removal at the same boundary. Both first transfers lie at $0.97009612853 \pm 3 \times 10^{-8}$ source units, with original-swap probability in $[0.2224969640, 0.2224969715]$. The complete adaptive energy and joint reference-phase instrument has 2627 original-attempt records, retaining ground-preparation success 1/3424 and its failure. The equal opposite phase weights at each energy follow from the retained reference position; they supply only one squared momentum.

The quantum Higgs field provides an exact operator connection. With its actual finite scalar matrices, the charged contraction $J_e = \phi_0^\dagger c_{L,10} + \phi_1^\dagger c_{L,11}$ has the gauge currents of the conjugate-singlet creator. The up and down contractions obey their native tensor identities on every local fermion occupation. For a charged creator $T$, the same-H equation is $H_t T - T H_0 = [H_{\mathrm{hop}}, T] + [H_Y, T]$ on the complete physical coordinate blocks. The charged Higgs-composite residue in the first periodic line is rigorously in $[0.4230841174, 0.4230841250]$. Its interference with hopping is retained; no scalar background or free-field equation is substituted. The gauge-neutral odd contraction $\phi_0 c_{L,11} - \phi_1 c_{L,10}$ has nonzero interacting evolution. Conserved local matter/link parities force its separated bare correlator to vanish in a symmetry-invariant reference; a transporting dressed field requires its own construction.

A full periodic cubic family retains all thirty native modes per cell, quantum links, both scalar conjugates and three commuting translations. Its exact operator identities do not transfer the reduced lepton ground. A separate large-link-penalty member admits a complete invariant low-energy graph: on an actual fifteen-state Gauss block at penalty 128, a six-state dressed generator has full evolution error below $10^{-12}|t|$. A controlled connected energy is strictly positive at penalty 65536. These finite interaction estimates supply no nonzero interacting continuum. Raw raising-link paths have norm amplification $d^{r/2}$ on a blank reference, so a classical-link comparison needs additional control. Contractive link matrices are a declared alternative with their own reference and spectrum. The computed finite spectral records identify neither relativistic masses nor observed generations (`model/coherent_defect_wave/chiral_joint/periodic_spectrum`, `model/coherent_defect_wave/chiral_joint/higgs_composites`, `model/coherent_defect_wave/chiral_joint/field_limit`, [2]).

The same periodic ground supports complete neutral and bosonic instruments. For $J_\nu = \phi_0 c_{L,11} - \phi_1 c_{L,10}$, the actual success operator $J_\nu/2$ and its adjoint setting retain their positive square-root failures and all 298 target states in each setting. Their first transfer lies at $0.62435981 \pm 3 \times 10^{-8}$ source units. The local-star symmetry pairs the translation phases and forbids separated bare transport. A scalar-number probe and two closed-link traces retain all 502 target states, elastic events and failures. They have positive weights at a common inelastic line $1.70011247 \pm 4 \times 10^{-8}$ source units. Ground-filter failure is retained throughout. Neither line is identified with an observed neutral lepton or boson (`model/coherent_defect_wave/chiral_joint/neutral_composite`,

`model/coherent_defect_wave/chiral_joint/periodic_boson_records`, [2]).

**Physical local reference and finite quantum transport.** All three native Yukawa channels and the quantum scalar act on the full $5\,2^{30}$-dimensional local source. Exact cyclic reductions give 36 positive spectral weights from the empty preparation and 49 from a neutral scalar–lepton preparation. The complete physical ground follows from a separate comparison of all zero-boundary-charge sectors. A fourteen-mode particle–hole transformation gives a collective $\mathfrak{sp}(4) \oplus \mathfrak{sl}(3)$ action. Its full quark character inventory excludes every physical quark module except one from energies below $-11.13$. The exceptional module couples to all lepton and scalar states in a complete 104-dimensional physical space, including 56 dark states at energy one. Exact root isolation proves a ground doublet near $-11.23058529425$ and the full physical gap

$$\frac{25207584560067}{281474976710656} \;\le\; \Delta_{\mathrm{loc}} \;\le\; \frac{25207584560069}{281474976710656}.$$

The ground has spin one-half, raw $L = 1$ and zero centered charge $L-I$. Its stationary spin-invariant density is $P_g/2$. Exact charge-conjugation and spin identities fix its mean native occupations to $(3, 6, 1, 2, 3)$, with total fifteen. These are reference occupations; the local gap is not identified with a particle rest mass (`model/coherent_defect_wave/chiral_joint/local_reference`, [2]).

Literal compressed Peter–Weyl multiplication on 119 quantum link states retains all 37 allowed coordinate channels, including 27 transitions between excited sectors. The full internal transporters are contractions, all endpoint currents are preserved, and all 24 proper cubic rotations act on the actual source Hamiltonian. These coordinate transitions break the minimal link law's local-star obstruction. With all native couplings one, the first neighboring neutral scalar–lepton transfer moment is $-151i/32$ at fifth order. Every native matter block contributes. A complete physical preparation and full-H remainder certify positive finite arrival without an added neutral hopping term. Separately, all five native matter representations share a dual-native reference and common observation time $1/1508$. Complete second powers of all thirty source components give a prepared-minus-unprepared local detector contrast at least $1/81866304$. The arriving charged objects retain their compensating flux and reference (`model/coherent_defect_wave/chiral_joint/peter_weyl_links`, `model/coherent_defect_wave/chiral_joint/family_transport`, [2]).

Four actual hops around a contractible square return the matter excitation and leave a gauge-invariant link trace. The exact fourth moments are $-1/256$ for the singlet and $-1/768$ for the colored up-conjugate block. All chronological orders, native currents and spin signs are included. Full-state bounds give nonzero energy-dependent eliminated loop kernels and complete loop/no-loop records under the original Hamiltonian. A magnetic term is not inserted. An exactly unitary rational clock of the unchanged H has operator error bounded by $|tB|^7/(100800n^6)$ for $n$ substeps, where $\|H\| \le B$. It preserves the certified neutral arrival and all-family detector contrast. A separate exact 72-state rational execution retains both loop outcomes and their complement. These finite results supply no many-cell Lorentz vacuum, physical photon or gluon phase, observed species or GeV mass (`model/coherent_defect_wave/chiral_joint/matter_induced_plaquette`, `model/coherent_defect_wave/chiral_joint/finite_clock`, [2]).

**A controlled interacting ground and its detector line.** The physical local reference admits a controlled finite interacting extension. Let $H_\lambda = H_0 + \lambda T$ on the side-two periodic source, with all onsite Yukawa, scalar and link coefficients one. The full kinematic dimension is $D = (5\,2^{30})^8 119^{24}$. Exact local charged-sector bounds give the atomic physical gap $g = 144856130473581/3518437208883200$ and $\|T\| \le 360$. Every omitted local occupation, scalar

and link state belongs to the complementary-space bound. A canonical graph isometry maps the 256 atomic ground states into the isolated interacting band. Its second-order generator is

$$K_2 = -24A_{\mathrm{edge}}I + 2J_{\mathrm{edge}}UH_{\mathrm{cube}}U^{\dagger}, \qquad 0.14771920681900 < J_{\mathrm{edge}} < 0.14771920681901.$$

Here $H_{\mathrm{cube}}$ is the twelve-edge spin-half Heisenberg cube; $U$ is an explicitly verified local spin change of basis. The exchange follows from the complete charged pair resolvent. All scalar shifts are retained. The actual side-two sublattice parity removes odd orders, and the full remainder is bounded by $25(360\lambda)^4/g^3$. This parity argument does not apply to the side-four source, whose longer hopping paths can join equal sublattices (`model/coherent_defect_wave/chiral_joint/atomic_reference_control`, `model/coherent_defect_wave/chiral_joint/atomic_even_remainder`, `model/coherent_defect_wave/chiral_joint/charged_ground_exchange`, [2]).

Exact diagonalization by polynomial factorization covers all 256 spin states. The cube ground is a unique singlet; its lowest energy is the least root of $\epsilon^3 + 5\epsilon^2 - \epsilon - 9 = 0$. The first excitation is a triplet at energy $-4$. Actual source translations put its three components at $(\pi, 0, 0), (0, \pi, 0), (0, 0, \pi)$, while the ground has zero momentum. At $\lambda = 2^{-29}$, the full physical Hamiltonian has a unique ground and gap in $[6.95741157484, 9.85455023443]10^{-19}$ source units. Full cubic symmetry preserves the threefold first-cluster degeneracy. No continuous spin symmetry is assumed for the full hopping Hamiltonian (`model/coherent_defect_wave/chiral_joint/atomic_spin_geometry`, `model/coherent_defect_wave/chiral_joint/atomic_momentum_response`, [2]).

A stricter coupling $\lambda = 2^{-34}$ resolves a complete detector on that same interacting reference. With the actual full-source spin current, put $O_a = \sum_x (-1)^{x_a} S_x^a$ and $B = \sum_a O_a \otimes |a\rangle/(60\sqrt{3})$. A retained vector reference carries the inverse momentum; a scalar reference records pump failure. Since $\|O_a\| = 60$, the full Halmos dilation of $B$ is unitary and preserves Gauss constraints, fermion parity and the joint cubic symmetries. Exact atomic residues and full-H projector bounds give first-line probability in $[0.00204894419, 0.00211090162]$ and outside-line probability in $[0.00001386725, 0.00001938783]$, conditioned on ground preparation. The excitation energy lies in $[8.20758426381, 8.21034719056]10^{-22}$ source units. Each source momentum retains its reference and one third of the corresponding total probability. The kinematic preparation from $I/D$ through the physical ground filter succeeds with probability $1/D$; its retained failure need not satisfy source-only Gauss. The physical protocol below preserves Gauss on all branches. Together with pump failure and all line/outside outcomes, the complete probability is one. These spectral filters are declared finite algebraic services, with no efficient native implementation asserted (`model/coherent_defect_wave/chiral_joint/collective_spin_response`, [2]).

On the physical local ground, the charged composite apparatus separately has 44 complete records, 40 positive. Scalar number and Yukawa probes share the exact partner energy $1 - 2e_g \simeq 23.4611705885$ and have 22 complete records, including all zero and elastic outcomes. Their reference and boundary charges are explicit. These local responses and the finite interacting collective line are distinct observables. The nonzero-momentum collective gap supplies neither a zero-momentum rest mass nor a growing relativistic limit, measured species identification or GeV scale (`model/coherent_defect_wave/chiral_joint/ground_composite_response`, `model/coherent_defect_wave/chiral_joint/ground_scalar_response`, [2]).

**Physical preparation and odd source response.** A finite algebraic controller starts from physical vacua, constructs the actual local ground doublets by polynomial projection, and prepares the native cube trial with retained fermionic spin references. A Householder unitary and a two-copy energy-agreement filter use the actual $H_0 + 2^{-34}T$. The Cayley filter needs no exact interacting

ground vector or ground-energy input. With 119772148350245317615208466 9 terms, preparation succeeds with probability at least 0.999999942661984; the conditional source trace distance from its true ground is at most $2.355922487\,10^{-7}$. Every failure branch remains physical. Exact algebraic synthesis, controlled matrix inversion and the finite schedule are declared services; the construction supplies no efficient native circuit (`model/coherent_defect_wave/chiral_joint/interacting_ground_preparation`, [2]).

The separate finite native service compiler executes a two-state singlet code of the actual 320-state lepton/scalar source. Its exact $HV = Vh$ has $h = \left(\begin{smallmatrix} 0 & -2 \\ -2 & 1 \end{smallmatrix}\right)$, so one simultaneous source pulse gives the full continuous code evolution; six pointer pulses supply the complete native C/E readout. A compensated $U(1)$ code also checks 491520 CAR/Gauss columns and all 32768 amplitudes. Rational control-error bounds retain export failures and reset timeouts. Coherent load/export and signed analog port control are explicit services; the passive classical repair run does not implement them. The finite general compiler includes padding and an explicit product-formula cost, without executing the full non-Abelian cube preparation (`model/native_quantum_service_bridge`, [2]).

On the thirty-mode reference vacuum, the fermion transfer is

$$B_r = (2d_r)^{-1/2} \sum_{s,a\in r} r^{\dagger}_{sa} c_{0sa}, \qquad B_r^{\dagger} B_r = N_r/(2d_r) \le I.$$

Its full dilation preserves joint Gauss charges and total fermion parity. The reference energy is $4N_{\mathrm{ref}}$; a bound over every reference occupation and source link sector proves that the neutral ground with blank reference is the unique physical ground of this enlarged Hamiltonian. Complete positive local resolvents have nine energies for $e^c$ removal and seventeen for $L$ removal. Residual bounds transfer their weights to bins of the actual full interacting Hamiltonian without assuming that unseen eigenstates are absent from those bins:

| native removal | source energy bin | selected probability |
|---|---|---|
| $e^c$ | $[0.6906800974, 0.6926332644]$ | $[0.4197041521, 0.4201989707]$ |
| $L$ | $[0.0959119144, 0.0978650814]$ | $[0.3210643365, 0.3214971263]$. |

The complete outside probabilities are enclosed respectively by $[0.0799404942, 0.0801565287]$ and $[0.1785579463, 0.1788807366]$; pump failure is about one half. The exact probabilities sum to one. These ground-reference intervals acquire the stated preparation error under the constructive protocol. Total source/reference excitation energy adds the declared reference cost four. A localized charged response does not identify an isolated propagating pole, a rest mass or an empirical lepton; the static compensator is retained in that distinction (`model/coherent_defect_wave/chiral_joint/interacting_charged_response`, [2]).

**The other three fermion-current probe families.** Literal $u^c, d^c, Q$ removal uses the actual native quark ground module, not an empty Fock reference. The normalized six-factor tensor embedding maps its 140 states into a verified 224-state CAR image; its target metric contains the required factor 1/6. All scalar and lepton factors are retained. Exact number-field resolvents give thirteen visible local energies for each probe. The $Q$ target has twenty-five distinct roots, twelve with exactly zero residue. Every complete probe measure sums to 1/2, including the normalized color/spin contraction with the static quantum reference. The equal-coupling local $u^c$ and $d^c$ measures coincide; exact degeneracy after lattice hopping is not assumed.

Full-source residual bounds give strong excitation bins $[0.7815710013291, 0.7815763657472]$ for both singlets and $[0.7819863054154, 0.7819916698335]$ for $Q$. Their original-attempt probabilities exceed 0.4953979660650 and 0.4957017902099, respectively. All other bins, preparation failures,

pump failures and outside records are retained. The reference adds four on success. This completes the five internal fermion-current probe families, without turning energy lines into species or supplying asymptotic colored states, quark propagation or GeV masses (`model/native_quark_response`, [2]).

The same quark operators also have a quantum-Higgs spectral connection. For every original annihilator $A_i = c_i$, the literal composite $C_i = [H_{\mathrm{local}}, c_i]$ obeys the full gauge, spin and raw-charge covariance identities on all occupations. Its actual target image is $C = H_{\mathrm{target}}A - AH_{\mathrm{source}}$; no scalar expectation is inserted. Contracting all indices with the complete thirty-mode reference gives $\|A\| \le 1$, $\|C\| \le 2$ and $[h, A] = C + [2^{-34}T, A]$. For the same full-source bin $P$,

$$\mathrm{Re}\langle PAg, PCg\rangle \ge Lw_{\mathrm{lower}} - 720\,2^{-34} = \gamma > 0.$$

The bounds are $\gamma > 0.3871888488$ for $u^c, d^c$ and $\gamma > 0.3876321760$ for $Q$. Fair-setting pumps $(A \pm C/2)/2$, followed by the complete hit/miss measurement, give original signed hit contrasts above 0.0967969710 and 0.0969078028. Every pump, spectral and preparation failure remains. The actual sixty-mode source/reference commutators and full target images are checked. Positive total cross weight proves a shared contribution in the bin, without isolating a pole, replacing the composite by a canonical fermion or identifying a physical quark mass (`model/native_quark_higgs_join`, [2]).

For the separately selected elementary and composite pumps, put $a = PAg$, $b = PCg$, and let $e, d$ be the bin center and half-width. Then $\|b - ea\| \le r\|a\|$, where $r = d + 720\,2^{-34}/\sqrt{w_{\mathrm{lower}}}$. The original two-source preparation's certified pair fidelity bounds its purification error $\epsilon$. With $s = \sqrt{w_{\mathrm{lower}}}$, the normalized admitted states, including the same retained environment, satisfy

$$D(\rho_A, \rho_C) \le \frac{r + (r + 2 + e)\epsilon/(s - \epsilon)}{e} < 5.193\,10^{-6}.$$

Every subsequent common finite detector history inherits this bound. The different initial setting labels are excluded from the comparison; original pump/bin success rates remain distinct and every failure is retained. No later measurement of the differing setting label is claimed to have close statistics.

Sharper windows use the same full source. With $v = 360\,2^{-34}$ and atomic gap $g_{\mathrm{at}}$, $PTP = 0$ gives $8E_0 - v^2/(g_{\mathrm{at}} - v) \le E_g \le 8E_0$. Orthogonal spectral projection retains more than 0.4197064484 unnormalized probe weight in the source-excitation enclosure $[0.6916539987293, 0.6916593631475]$. A narrower window $[0.6916566390288, 0.6916567228480]$ retains weight above 0.31477359. These bounds retain every unseen charged state. Almost all weight in a nested bin need not place the largest individual residue there: an exact finite perturbation counterexample separates those properties. A refined spectral selector therefore cannot silently replace the wider bin's selected line in another construction (`model/native_charge_resolution`, [2]).

The actual two-copy preparation also supports a complete twenty-bit energy detector without another ground projection. Its conditional source-energy mean lies between 0.6916566792649 and 0.6916566826119, with original bin admission above 0.419706188794. The blank-link identity $P_{\mathrm{blank}}TP_{\mathrm{blank}} = 0$ fixes the selected atomic first moment; the complete second moment bounds its lost tail. Preparation error is multiplied by the admitted bin width, and the rational phase clock contributes less than $1.04\,10^{-10}$. Every pointer label, preparation failure and reference state remains. A known rational neutral-energy baseline replaces an unknown exact ground-energy phase. Bin admission is a declared exact spectral service; the calculated mean is neither a single eigenvalue nor a GeV mass (`model/native_charge_centroid`, [2]).

**An elementary probe and a quantum Higgs composite at one energy.** For $A_\alpha = c^\dagger_{\alpha,e^c}$ and the actual five-state-scalar composite $B_\alpha = -\epsilon_{\alpha\beta}\phi^\dagger c_{\beta,L}$, the full operator identity is $[h, A_\alpha] =$

$B_\alpha + \lambda[T, A_\alpha]$. Both probes are odd color/weak singlets with $Y = 1$ and raw lepton increment $-1$. On the actual ground and the charged bin projector $P$,

$$\mathrm{Re}\langle PA_\alpha g, PB_\alpha g\rangle \geq Lw_{\mathrm{lower}} - 720\lambda > 0.28988126.$$

At least one actual energy consequently has nonzero elementary and composite residue. The native cubic symmetry gives a common $G_1$ doublet there. Complete interference pumps $(A \pm B/2)/2$, with retained charge/spin references and physical preparation, have original setting-hit contrast above 0.07247007595. All sixteen apparatus labels and preparation failures remain. This proves a same-Hamiltonian spectral connection without a classical scalar substitution. The positive-cross-residue selector differs from the maximum elementary residue selector; the composite is not a canonical fermion on every occupation sector, and no physical weak-scattering law follows (`model/native_higgs_spectral_join`, [2]).

The same equation bounds the normalized filtered states. Put $a = PAg$, $b = PBg$, let $e, d$ denote the bin center and half-width, and write $w_{\mathrm{lower}} \leq \|a\|^2$. Then

$$\|b - ea\| \leq q\|a\|, \qquad q = d + 720\lambda/\sqrt{w_{\mathrm{lower}}}, \qquad D(\widehat{a}, \widehat{b}) \leq q/e.$$

Here $D$ is half the trace norm on the corresponding pure densities. A depth-three finite preparation retains eight trials and seven filters. After including its error, the narrow-bin successful elementary and composite states have trace distance below $1.68597220\,10^{-7}$. Every common subsequent complete detector history has the same total-variation bound by channel contractivity. Their original pump/bin success bounds, 0.3147735248783 and 0.0376460324440, are distinct. All sixteen apparatus labels, preparation failures, references and control work remain. Selecting the actual positive-cross-residue doublet in this bin gives a rest-frequency input for the parametric free Dirac construction below, with explicitly redeclared spectral couplers. No interacting field law transfers by changing that selector (`model/native_higgs_filtered_state`, [2]).

**The quantum weak current.** The canonical particle–hole map preserves all twelve native gauge currents independently of the scalar state. Its Hamiltonian identity with a prescribed scalar background is a separate conditional statement. For the actual five-state operator, $\phi_k^3 = 0$ and $\|\phi_k\| = 1$ imply

$$\|(\phi_k - v)\eta\| \geq \frac{|v|^3}{1 + |v| + |v|^2}$$

on every normalized state, including entangled references. A nonzero exact scalar eigenbackground is excluded at this cap. The explicit product-state average over $n$ copies has variance proportional to $1/n$, so this obstruction does not exclude collective scalar limits.

The same source supplies the physical dressed density $W = \sum_a \mathrm{tr}(\sigma_a \phi\phi^\dagger) j_a$. Its full active spectrum is $\{-1, -1/2, 0, 3/2, 2\}$ and all Gauss commutators vanish. Exact atomic contractions with full-source errors give $\langle W\rangle \in [0.0495974, 0.0506134]$ and $\mathrm{Var}(W) \in [0.1517363, 0.1531928]$ on the actual ground. A current pump followed by the finite two-source Cayley detector has original nonzero-phase probability above 0.0379338, including imperfect preparation. The compensated charged pump, energy admission and subsequent current readout give conditional expectation in $[0.0903329, 0.0928015]$, including both control intervals. The complete five-outcome current law retains all failures. Current measurement can change energy, so this is an ordered instrument rather than a commuting energy/current measure. It establishes a dressed weak response, without a W-boson pole or a relativistic weak cross section (`model/native_weak_current_interface`, `model/native_weak_current_response`, [2]).

The elementary/Higgs addition branch has its own calculated response. The source antiunitary sends $\phi_i \mapsto \phi_i^\dagger$ and $j_a \mapsto -j_a$, so the same original detector cannot inherit the removal-channel mean. Its transformed operator is

$$\Theta W \Theta^{-1} = -\sum_{a,i,j} \overline{\sigma_a[j,i]}\phi_i^\dagger \phi_j \, j_a \neq W.$$

In all three complete charged blocks, exact atomic contractions give mean $-0.0376996702797\ldots$. Full-source projection and evolution bounds, with the actual eight-trial preparation, give $\langle W\rangle_A \in [-0.07004020, -0.00539719]$ and $\langle W\rangle_{B/2} \in [-0.07005609, -0.00538131]$ in the radius-$128v$ bin. Both preparations therefore have a nonzero signed current response on the unchanged interacting source. Their conditional five-label laws differ by less than $3.972\,10^{-6}$; their original success weights remain unequal. Every one of the eighty setting/pump/energy/current labels and all preparation failures is retained. Antiunitarity reverses the source time; the bound covers either sign. This does not infer CP violation from comparing two different transformed observables (`model/native_higgs_current_response`, [2]).

A local alternative retains the first three preparation/pump/bin services and replaces only the last drive. The exact current projectors give

$$f(W) = \frac{4W^4 - 4W^3 - 13W^2 + 25W}{15}, \qquad f(\{-1, -1/2, 0, 3/2, 2\}) = \{-2, -1, 0, 1, 2\}.$$

Couple $f(W)$ to the Hermitian generator of a neutral five-state cyclic pointer. Its rational 40-bit approximation $P_{40}$ has norm below $4\pi/5 + 9.078\,10^{-13}$. All 400 local source/pointer states are retained. For $G = f(W) \otimes P_{40}$ and $\tau = 2^{-20}$, the actual pulse is $\exp[-i(\tau H_{\mathrm{ext}} + G)]$, without subtracting $H_{\mathrm{ext}}$ from the drive. Double Duhamel gives

$$\|e^{-i(\tau H_{\mathrm{ext}}+G)} - e^{-i\tau H_{\mathrm{ext}}} e^{-iG}\| \leq \tfrac{\tau}{2}\|[H_{\mathrm{ext}}, G]\|.$$

The full onsite Yukawa terms, including both quark terms, have norm at most 56; all incident hops contribute at most $90\,2^{-34}$. Thus $\|[H_{\mathrm{ext}}, f(W)]\| \leq 224 + 360\,2^{-34}$. Including pointer quantization, complete-history total variation is below 0.000268446640. Both decoded current means remain below $-0.0043$ at total clock $3 + 2^{-20}$, with every earlier failure retained. The polynomial control preserves Gauss constraints throughout. Its final drive norm is below 5,270,718 and gross switching work below 10,541,436. It is a declared local apparatus primitive; the earlier full-matrix preparation and spectral services are unchanged (`model/native_local_current_readout`, [2]).

Gauge-invariant composite spectroscopy is also used in the independent lattice study of Maas, Martins and Wieland. Its scalar–fermion correlation matrices provide a useful comparison method. That study uses vectorial fermions without Yukawa couplings on a different prescribed gauge–scalar action. Its numerical spectra therefore supply neither a chiral realization nor missing premises for the native source above.

**Charge transport and its energy cost.** The same prepared source and elementary removal have six orthogonal neighbor-arrival routes. At source time $2^{-24}$ their unconditional union probabilities exceed $3.8580\,10^{-37}$ for $e^c$ and $6.2900\,10^{-37}$ for $L$. The amplitude estimate includes imperfect preparation before squaring and retains every complementary outcome. It does not identify those arrivals with the selected low energy bins.

For that distinction, let $\Pi_y$ require gauge-singlet matter at every cell except $y$, retaining all scalar, fermion, link and reference states. Gauss' law forces a nonblank link path from the origin reference to $y$. With $\ell$ its graph distance, the complete local sector bounds imply

$$\Pi_y H_0 \Pi_y \geq (8E_0 + \ell + \delta_r)\Pi_y, \qquad \Pi_y T \Pi_y = 0,$$

where $\delta_{e^c} > 0.10058529$ and $\delta_L > 0.09688849$. If $P_I$ is a selected full-Hamiltonian energy bin, with upper energy $b < a_y = 8E_0 + \ell + \delta_r + 4$, the separated-spectrum Sylvester identity gives

$$\|\Pi_y P_I\| \le \frac{360\,2^{-34}}{a_y - b}.$$

Thus at every source time the conditional probability of any of the seven remote single-charge locations is below $8.656\,10^{-15}$ for $e^c$ and $1.699\,10^{-15}$ for $L$. The eighth location and the multi-charge cloud complete the location measurement; preparation, pump and energy-selection failures give thirteen sequential records. The cloud can contain screening. An explicit positive-norm state has a locally compensated reference and a remote charge pair without a flux path back to the origin. The bound therefore concerns the unscreened single-charge domain, rather than unrestricted confinement (`model/coherent_defect_wave/chiral_joint/interacting_charged_transport`, `model/coherent_defect_wave/chiral_joint/interacting_charge_localization`, [2]).

The charged-particle criterion also depends on the intended field phase. Under the relativistic Gauss-law and asymptotic-field hypotheses of Buchholz's infraparticle theorem, a charged Coulomb sector has no sharp mass eigenstate. Its physical mass target is a dressed or inclusive energy–momentum threshold with specified detector resolution. The finite matrix bins above establish neither that Coulomb phase nor a relativistic threshold. Demanding an isolated charged pole in every continuum phase would impose an inappropriate acceptance criterion.

**A complete charged field and its native attachment.** A declared electric/magnetic architecture on the thirteen-vertex icosahedral cone retains thirteen fermionic modes, a charge reference, longitudinal Gauss constraints and 5456 transverse states at total occupation at most three. Exact reduction gives electric energy $E_C = \rho^T (D^T D)^+ \rho/2$. Literal finite Peierls hopping of strength $1/24$ preserves the constraints and couples charge motion to the field. A full-Hamiltonian residual certificate, including arithmetic error, locates the unique ground of the complete 70928-state one-charge sector:

$$E_1 \in [1.8738097549651658, 1.8738097549671659].$$

The reference contributes one unit. Transverse dressing has probability between 0.00007224653680 and 0.00007224655239, and lowers the energy by about 0.0001750058 relative to the zero-photon compression. All thirteen position probabilities are retained; these spatial levels are not thirteen particle species.

A compensated fermionic rotation prepares the specified packet with probability $1/4$. A twenty-qubit phase pointer measures the actual H, retaining every number and phase outcome. Its neutral-energy reference phase and rational controlled-evolution gates are explicit. Including preparation, acceptance lies in $[0.2499183323303489, 0.2499819397665590]$ and the accepted ground fidelity exceeds 0.99999999439512. The source evolves throughout the apparatus schedule. Exact coefficient and linear-solve services are declared; the finite specification is not an efficient local circuit (`model/affine_gauss_source`, `model/affine_gauss_mobile`, `model/affine_gauss_full_spectrum`, `model/affine_gauss_charged_clock`, [2]).

The subsequent complete position/field instrument keeps all $13 \times 2$ outcomes and uses the actual accepted state. Its conditional transverse excitation probability lies between $7.09795005\,10^{-5}$ and $7.35247970\,10^{-5}$. For ground fidelity at least $f$ and ground projector probability $p$, the lower record bound is $[\sqrt{fp} - \sqrt{(1-f)(1-p)}]^2$ when $p > 1 - f$, and zero otherwise. This fidelity bound avoids substituting a perfect ground projection. A finite-duration interaction-picture apparatus records position at the start of its readout interval while the source keeps evolving (`model/affine_gauss_clock_observables`, [2]).

The native attachment places a complete interacting eight-cell block at each vertex. Its local reference adds a conjugate of the thirty-mode reference, sharing the vacuum, and two charge ports. This enlargement is essential: particle-hole conjugation sends the original $4N$ to $120 - 4N$, rather than preserving its energy. Every native H and Gauss sector is retained. A selected actual $e^c$ energy $E_*$ in its certified bin supplies odd cubic-spin $G_1$ matrix units; the selected moving sector obeys

$$H_{\text{join}} - (13E_g + E_{\text{vac}}) = [H_{\text{affine},1} + (E_* + 2)I] \otimes I_{G_1}.$$

Its reference costs four. This selected hard-core channel is not a full spinor Fock algebra or the unpenalized Hamiltonian's full charged ground. A separate declared port penalty has its own sector-exclusion proof.

The whole-bin alternative couples normalized $P_J c_s g$ and keeps the actual native H on all of $J$. The bin half-width gives a finite-time transfer error $\epsilon|t|$. At source time $1/4$, unconditional radial and boundary arrival each exceed $3.06\,10^{-5}$; arrival together with nonvacuum transverse field exceeds $1.67\,10^{-6}$ and $2.26\,10^{-6}$ respectively. Six records retain preparation and exact ground-admission failure, pump failure, outside-bin weight and both arrival outcomes. Ground admission is an additional spectral service. A unitary field-parity map relates opposite-charge positive-time records; the native antiunitary alone reverses time. These results establish a finite native-current/field coupling. Cubic $G_1$ symmetry, a boundary energy and finite arrival do not identify a relativistic electron (`model/native_abelian_field_join`, `model/coherent_defect_wave/chiral_joint/interacting_charge_conjugation`, [2]).

Successive position and energy measurements retain their backaction on that same native attachment. A separated spectral cluster controls the actual whole-bin Hamiltonian directly; its long energy clock is not approximated by a short-time transfer estimate. At time $1/4$, the original-attempt probability of arrival followed by acceptance at the admitted charged ground exceeds $4.50\,10^{-6}$ at the apex and $1.13\,10^{-6}$ at an adjacent boundary site. The accepted infidelities are below 0.001041 and 0.004354, respectively. A coherent spatial pump gives accepted infidelity below $2.14\,10^{-8}$. Preparation, ground admission, pump, spatial complement and all phase labels remain in the complete law. The ground belongs to the admitted native bin; the full unpenalized charged-sector ground is a different question (`model/native_abelian_energy_records`, [2]).

**A complete charged occupation code and free Dirac records.** Two complete native constituents encode a spin doublet's vacuum, two one-particle states and its occupied spin singlet. Their actual energies above the product ground are $0, E_*, E_*, 2E_*$. An independent conjugate pair supplies the antiparticle modes. The sixteen occupations form a reducing CAR code inside the full native carrier; every complement is retained. With $m = E_*$, declare on a periodic cubic macro lattice

$$H_a = \sum_{x,j}(h_{xj} + H_{\text{ref},xj} - E_g) + \sum_{q,s,x,y} a^\dagger_{xqs}[\sqrt{m^2 I + L_a} - mI]_{xy} a_{yqs}.$$

The kinetic term is positive on the entire carrier, so the unique native product ground remains the actual ground. The code has exactly $H_a = d\Gamma(\Omega_a)$, where $\Omega_a(p)^2 = m^2 + 4a^{-2}\sum_i \sin^2(ap_i/2)$. The square-root couplers have full spatial range; they are declared architecture, without a microscopic finite-range light cone.

Normalized positive and negative Dirac frames $U(p), V(p)$ give the same-clock field $\Psi_a = U a\, e^{-i\Omega_a t} + V b^\dagger(-p) e^{i\Omega_a t}$. Both terms have the same boundary charge. The source excitations have positive energy, and the opposite phase follows from creation. On $|p| \le P$, $0 \le \sqrt{m^2 + |p|^2} - \Omega_a \le a^2 P^4/(24m)$. Explicit momentum, volume and tail bounds transfer complete bounded apparatus

laws to the free Dirac field, including its induced spin-half Lorentz action and causal anticommutator. A finite momentum projection is not strictly local.

Charge-conserving swaps into retained fermionic apparatus doublets give a literal sixteen-state active circuit. At $p/m = 3/4$, two swaps of angle $\pi/4$ separated by $T = \pi/(2\Omega_a)$ have probabilities

$$(p_{00}, p_{01}, p_{10}, p_{11}) = \left( \frac{313}{500} + \frac{12\sqrt{2}}{125}, \frac{137}{500} - \frac{12\sqrt{2}}{125}, \frac{1}{10}, 0 \right).$$

The other twelve spin-partner records are retained. Finite native preparation with six retained attempts bounds its complete record error below $10^{-9}$, using about $2.40\,10^{9}$ nats of displayed capacity. An explicit programmable matrix drive implements preparation and finite-duration detector pulses under the common raw Hamiltonian; additional synthesis workspace and passive local compilation are separate resources. This free vectorlike $e^c$ pair supplies neither the electron's weak chirality nor the affine field's interaction, and no measured mass or GeV scale selects $m$ (`model/native_block_dirac`, [2]).

**One finite spinor current coupled to its transverse field.** A declared open-cube Wilson–Maxwell member uses thirty-two complete native constituents. The positive elementary/Higgs source selector fixes its rest-frequency parameter $m$, and the specified spacing is $a = 1/m$. The open Wilson matrix is $H_W = A + \beta M$, where $M \geq mI$ and $\{A, \beta\} = 0$. Thus $\mathrm{Re}\langle \beta v, H_W v \rangle \geq m\|v\|^2$ proves $|H_W| \geq m$ without assuming that $A$ and $M$ commute. Its actual positive/negative frames give $\Psi = Ua + Vb^\dagger$, including pair terms in the local charge density. The Fock unitary uses the explicit grouped-mode to site-coordinate permutation. All native complements remain, with a declared global spectral-code compression of the new kinetic and current couplers.

The same cube cochains supply five transverse modes of frequencies $(2, 2, 2, \sqrt{6}, \sqrt{6})/a$, retaining all fifty-six states of total occupation at most three. The positive top-boundary completion is part of $H_{\mathrm{ph}}$. Finite coordinate commutators at that boundary remain; no unrestricted Weyl identity is imported. Seven longitudinal labels in $[-4, 4]$ and a compensator in $[-32, 32]$ enforce the full source Gauss constraints by an exact isometry. The common Hamiltonian uses the same coupling in $e^{igQ_e}$ and $g^2 \rho_{\mathrm{ref}}^T G \rho_{\mathrm{ref}}/(2a)$.

For its free matter/photon input the complete first pair-plus-photon coefficient has two independent certificates. An exact contraction gives

$$\frac{\sigma^2}{m^2} = \frac{965}{1456} + \frac{5\sqrt{6}}{36} - \frac{\sqrt{14}}{48} + \frac{\sqrt{21}}{336} + \frac{\sqrt{91}}{546} + \frac{7\sqrt{546}}{1872} > 1.$$

With declared $g = 1/64$ and $mT = 1/64$, the complete nonlinear Peierls, Coulomb and all-order Duhamel bounds give original pair-plus-photon probability above $4.75055\,10^{-8}$ and a matter/field connected response above $4.71131\,10^{-8}$. The deterministic native trial has an exact reducing-code decomposition: its code component is the free vacuum, with weight at least $p_0^{32}$, and all remaining weight is an explicit native-complement outcome. There is no initial ground projection or discarded preparation. The pointer retains 1160 physical counts and a distinct outer-Gauss failure label. Its displayed source/reference/pointer capacity is below 10479 nats; coefficient synthesis workspace is separate.

The input is nonstationary under the actual coupled Hamiltonian, and programmable preparation and readout pulses use its same clock. This is a controlled finite current/field interaction, not emission from an interacting vacuum or an asymptotic scattering cross section. The global spectral compression requires a spatial locality/refinement proof before an interacting Lorentz limit or empirical electromagnetic identification follows (`model/native_wilson_maxwell`, [2]).

**The full interacting ground and a positive local completion.** A comparison over the retained native complement is necessary. Let $E_{\mathrm{base}} = 32E_g + m(3+\sqrt{6})$. Exact filled-band contractions give

$$C_0 = \frac{4055}{4368} - \frac{\sqrt{21}}{14} - \frac{3\sqrt{39}}{208} - \frac{27\sqrt{91}}{728}, \qquad h = -\sqrt{3} + \frac{6\sqrt{7}}{7} + \frac{17\sqrt{13}}{13},$$

where $C_0 = \langle \rho^T G \rho \rangle/2$ and $h = -\langle H_{\mathrm{hop}}(0) \rangle/m$ in the free filled-band vacuum. A weighted Schur bound, retaining the full Coulomb operator, proves $P(H - E_{\mathrm{base}})P \geq (m/50000)P$. The full code complement instead contains a single neutral native triplet in any of the thirty-two constituents, at energy $E_{\mathrm{base}} + \Delta_{\mathrm{native}}$ with $\Delta_{\mathrm{native}} \in [8.20758, 8.21035]10^{-22}$. Its active lowest eigenspace has rank 96 and lies below the entire interacting code. Zero-energy preparation spectators multiply that rank.

The explicitly changed source

$$H_+ = H + mc(g)\sum_x (I - P_x), \quad c(g) = g^2 2^{-48}\left\lceil 2^{48}(C_0 + hv/2)\right\rceil, \quad v = \frac{1}{16} + \frac{1}{12\sqrt{6}},$$

uses each macrocell's complete occupation-code projector $P_x$. This positive completion makes no scalar energy subtraction. Its chosen coefficient uses only the source contractions, leaves all code energies unchanged and vanishes at $g = 0$. A $q$-defect complement costs an additional $qmc(g)$. Every $P_x$ reduces the full native Hamiltonian, Gauss dressing and count effects, so complete pattern-preserving count histories are invariant under the change. Cross-pattern phases and energy differences change.

The completed source has a unique active code ground, internal code gap above $1.7170018m$ and full physical gap above $2.52078609\,10^{-10}$. Four actual trials and three finite Cayley energy-agreement filters prepare it with success above 0.9995943286660 and trace error below $1.417923\,10^{-7}$. No ground projector or unknown ground-energy phase is used. All partners and filter labels remain. The registers require less than 42006.105 nats before the final 1161-label count pointer; synthesis workspace is separate. The original-attempt $(1,1,1)$ count probability exceeds $1.41692895\,10^{-6}$. This is stationary occupation in the free quasiparticle/photon frame, not emission from the ground. The completion is an architecture choice. Its macro-local penalty does not remove the kinetic/current terms' global code guard or establish a full-source spatial propagation cone (`model/native_wilson_maxwell_ground`, [2]).

**A controlled photon cutoff on the same cube.** Let $P_K = 1_{N_{\mathrm{ph}} \leq K}$. The actual finite link is $\exp(igP_K Q_e P_K)$; it is not replaced by $P_K \exp(igQ_e)P_K$. The completed free photon Hamiltonian is exactly the compression of $m[d\Gamma(\nu) + \operatorname{Tr}\nu/2]$. Each of the twenty-four oriented matter coefficients has norm at most $2m$. In the interaction picture its field factor is a displacement of radius $r = |g|\sqrt{1/16 + 1/(12\sqrt{6})}$. A product of $n$ displacements from vacuum has coherent radius at most $R = nr$. Projected Duhamel and the Poisson factorial moment give

$$b_K(R) = \min\left\{2, \frac{R^{K+1}}{\sqrt{(K+1)!}}(1 + \sqrt{K+1}R)\right\}.$$

For $x = 48mT < J + 2$, the complete embedded-state error obeys

$$\delta_K \leq \sum_{n=0}^{J} \frac{x^n}{n!} b_K(nr) + \frac{2x^{J+1}}{(J+1)!\,[1 - x/(J+2)]}.$$

The factorial tail supplies a constructive $K \to \infty$ comparison at fixed cube, coupling and time. At $K = 3$, $g = mT = 1/64$, $\delta_3 < 1.863\,10^{-9}$. The actual native preparation has comparison pair-plus-field probability above $4.7504776747\,10^{-8}$ and connected response above $4.7112356073\,10^{-8}$. Overflow probability is at most $\delta_3^2$. Complete record laws include the overflow label and every native complement; no rare-event normalization enters these estimates. The unbounded oscillator is a mathematical comparison, not a physical infinite register.

On the reducing code, let $E = P_T + gDG\rho_{\mathrm{ref}}$ denote the total electric flux and $B = CQ$ its magnetic curl. The transverse projector $\Pi_T$ and longitudinal projector satisfy $\Pi_T + DGD^T = I$. For the dressed oriented hop $F_e$, the two electric commutator contributions therefore give

$$[E_e, F_f] = g\delta_{ef}F_f, \qquad [B_f, E_e] = iC_{fe}, \qquad [\rho_x, F_e] = D_{ex}F_e.$$

These relations yield the finite-graph Maxwell/current equations and a local incidence algebra. Bounded electric/magnetic Hadamard tests have complete-history cutoff bounds, including disturbance and every control outcome. The finite cap has nonvanishing boundary commutators; no uniform operator-norm or spatial-cone conclusion is inferred. The complete ground and its explicit positive completion are treated above; growing-volume and ultraviolet limits require separate estimates (`model/native_wilson_maxwell_refinement`, [2]).

**The interacting ground under photon refinement.** The same positive completion works at every photon cap $K \geq 3$. On the complete code, in units of $m$, let $H_0 \geq 2N$ be the free operator and $A = 96g\sqrt{v}$. The literal Peierls links give

$$\|V_{\mathrm{hop}}\psi\| \leq A\|\sqrt{N+1}\psi\|, \qquad V_{\mathrm{hop}} \geq -\epsilon H_0 - \left(\frac{A^2}{8\epsilon} + 2\epsilon\right) I.$$

At $\epsilon = 1/10$, the same vacuum contractions and a weighted Schur bound give a unique code ground with gap above $1.3284353m$. The completed full physical gap stays above $2.52078609\,10^{-10}$. The uncompleted source retains its lower native-complement ground.

Recorded code admission avoids resolving the much smaller complement gap. Keep all sixteen admission flags of four actual trials and use

$$h = (H - E_{\mathrm{base}})/m, \qquad C = (I - ih)(I + ih)^{-1}$$

only on the admitted code, extending by identity on failure. The known native/free baseline is not an unknown interacting ground-energy input. The ground-to-excited Cayley chord exceeds $3/2$. Three complete 4096-label energy-agreement filters give original success above $0.9993248562$, conditional purification error below $8.235\,10^{-8}$ and original preparation-plus-$(1,1,1)$ probability above $4.5122\,10^{-7}$. A two-copy, one-filter 64-label setting gives a positive original record above $6.4978\,10^{-8}$.

A uniform resolvent estimate, with the full perturbation norm at most $48m$, controls the literal cap-dependent Cayley services. At $K = 2^{64}$ the complete four-copy preparation/count error is below $1.254\,10^{-11}$ and its two-field-history error below $7.304\,10^{-10}$. All partners, flags and overflow remain. The elapsed clocks are respectively six and four duration-one service slots for the two settings; control pulses explicitly cancel the same raw background. Their source/reference/pointer budgets are below 42817.043 and 21425.219 nats, excluding controller workspace. These matrices are bounded analytically, not enumerated. The source Hamiltonian and its positive completion are unchanged (`model/field_record_addenda/native_code_ground_preparation`, `model/native_wilson_maxwell_ground_refinement`, [2]).

**Ordered quark-type arrival and energy records.** On the original native charged source, every one of six flux routes obeys $P_0B^\dagger TP_eTBP_0 = P_0/128$. At an autonomous wait of one, a complete arrival projection followed by an active-source-plus-reference spectral measurement has original one-route/bin probability above $2.3275\,10^{-23}$ for $u^c, d^c$ and $2.3348\,10^{-23}$ for $Q$. The corresponding source excitation windows lie in $[1.7815710013, 1.7815763658]$ and $[1.7819863054, 1.7819916699]$, after the fixed reference cost four and actual neutral ground are subtracted. All 88 or 160 coarse records, dark local lines, failed preparations and outside-bin outcomes remain. The energy operator acts as identity on the retained preparation partner. Disjoint energy projectors without the intervening arrival measurement instead satisfy $\Pi_2 e^{-itH_{\mathrm{total}}}\Pi_1 = 0$ exactly. Thus the positive ordered event includes measurement backaction; it is not an autonomous energy gain or a rest-mass determination (`model/field_record_addenda/native_quark_spatial_spectrum`, [2]).

**Spatial records and an interaction witness.** Growing open boxes have $V = N^3$ vertices and retain four complete native constituents per vertex. Full-box preparation banks, all local code flags and first-success routing bound failure by $V(1 - p_0^4)^R$. The transverse photon minimum frequency is $2\sqrt{2}\sin[\pi/(2N)]$, so the cube's photon gap is not uniform in volume. Heat bounds for the actual Hodge operator, finite-range sign polynomials for the free sea, longitudinal dipole estimates and bounded-hopping Dyson tails give complete positive-time local-record convergence at fixed spacing. The preparation is a specified nonstationary sea, not an assumed coupled vacuum. Literal finite field caps and every failure outcome have separate error allowances.

A fair recorded electric kick or identity, actual coupled wait $mt = 2^{-32}$ and complete sine-Hadamard current readout give a strict interaction witness. With $J_e = i(F_e - F_e^\dagger)$ and $u = 2^{-16}$, $\Pr(+) = \frac{1}{2}[1 - \langle \sin(uJ_e)\rangle]$. A source-only integration-by-parts identity gives $-\mathrm{Re}\langle F_e\rangle > 1/345$ in an explicit interior finite box. Uniform field-moment bounds control the full nonlinear short-time drift. The complete finite law differs from the matched zero-coupling law by more than $1.7183\,10^{-10}$ after all failure and cutoff allowances. The conservative sufficient box has 8200 cells per axis and roughly $5.054\,10^{15}$ nats; no such matrix is enumerated. This is an interacting local-record limit at fixed spacing, not a relativistic ultraviolet or scattering theorem (`model/field_record_addenda/native_wilson_maxwell_spatial_family`, `model/field_record_addenda/native_wilson_maxwell_interaction_witness`, [2]).

**A revised regulator and its distinct ground-sector requirement.** The original free Wilson symbol has $E_F^2 - m^2 = (1 + ma)\Omega_\gamma^2 + 2a^{-2}\sum_{i<j} w_i w_j$, where $w_i = 1 - \cos(ap_i)$. At $a = 1/m$ its axial squared coefficient is two. The declared regulator $\Gamma = \alpha_1\alpha_2\alpha_3\beta$ gives

$$h_\perp = m\beta + a^{-1}\alpha\cdot\sin(ap) + a^{-1}\Gamma\sum_i w_i, \qquad E_\perp^2 - m^2 = \Omega_\gamma^2 + 2a^{-2}\sum_{i<j} w_i w_j.$$

It retains the native field and charge frame and a vectorlike four-component carrier; ordinary parity and time reversal fail at finite spacing. The revised cube, with its old scalar offset retained and declared complement coefficient $21/2$, has full physical gap above 0.3711801 source units. Its new vacuum and prepared interaction are recomputed. This free-cone correction does not itself establish an interacting mass shell (`model/field_record_addenda/native_orthogonal_wilson_bridge`, [2]).

The revised regulator has a separate growing-box proof in the original field and Gauss frame. With $W$ the graph Laplacian divided by two and $A_\theta = K + (\cos\theta\,\beta + \sin\theta\,\Gamma)W$,

$$h_\theta^2/m^2 = I + 2\cos\theta\, W + A_\theta^2 \geq I.$$

This identity does not assume that the derivative and Laplacian commute at an open boundary. The equal-rank endpoint projectors define a finite algebraic sea-rotation service. Charge conjugation gives zero initial current and mean density; fresh sign-projector, heat and dipole bounds then establish complete positive-time local records at fixed spacing. A fresh hopping integration by parts strengthens the complete finite interaction-law separation to more than $3.4360\,10^{-10}$.

The ground-sector extension requires another explicit source choice. The old raw vacuum offset is extensive: the cube coefficient $21/2$ fails to select the code ground on sufficiently large boxes, at zero coupling. A neutral onsite two-occupation trial with photon vacuum has energy at most $E_{\mathrm{base}} + 18mV$, for every finite cap and any coupling. Every noncode state is bounded below by $E_{\mathrm{base}} + mc$. Thus the source-defined choice $c(V) = 18V + 1$ places the global ground in the code with at least $m$ separation from the native complement. Code-admitted record laws are unchanged. This variational sector result does not determine the coupled ground vector, its internal gap or a relativistic charged threshold (`model/field_record_addenda/native_orthogonal_wilson_spatial_join`, [2]).

**Charged clusters and complete joint energy records.** The finite orthogonal cube at photon cap three admits a sharper fixed-charge calculation. With the same field, Gauss map and old raw sea offset, declare only the complement coefficient $c = 23/2$. Exact free positive levels start at $\sqrt{5}/2$, $\sqrt{17 - 4\sqrt{3}}/2$, $\sqrt{13}/2$ and $\sqrt{33 - 4\sqrt{7}}/2$, in units of $m$, with multiplicities $2, 4, 2, 2$. The uniform interaction estimate alone does not certify the first two states. It does isolate the first eight in each active charge sector $Q = \pm 1$: the separation from higher states exceeds $0.01652645m$, and the margin to the native complement exceeds $0.21532398m$. These are fixed-sector ranks, excluding zero-energy spectators, not counts of particle species. Charge conjugation pairs the two code spectra. The central $2\pi$ action is odd fermion parity, but the fixed corner reference breaks full cubic rotation symmetry; a continuous spin representation is not inferred.

Two-copy code admission and a 64-label energy-agreement filter precede an even two-mode fermionic pointer pump. It preserves $Q + R$, the distinct source/pointer charge $Q - n_{f+} + n_{f-}$, and joint parity. The actual reference $R = -Q$ retains cost four. Charged and neutral energy clocks have $2^{24}$ and $2^{18}$ labels. Five complete driven slots retain all $2^{53}$ records, including failed admissions, empty-pointer outcomes and every rejected energy result. The original joint acceptance exceeds 0.56935668; conditional fidelity to the charged cluster tensored with the actual neutral ground exceeds 0.99991834. Source, reference and pointer capacity is about 20980.4 nats, excluding coefficient workspace. The energy comparison obeys

$$0.57009819m < E_{\mathrm{charged}} - E_{\mathrm{ground}} - 4 < 2.35071144m.$$

It contains Coulomb and boundary-reference effects. Exact finite matrix services are declared and their background cancellation and work are bounded; the full huge matrix is not enumerated. This is a controlled charged energy record, not a relativistic mass or a species assignment (`model/field_record_addenda/native_orthogonal_wilson_charged_band`, [2]).

**Charged preparations with a receding reference.** The growing orthogonal family also admits the compensated local probes $D_+ = Y c_f^\dagger Y^\dagger$ and $D_- = Y c_f Y^\dagger$, with a fixed positive or negative $\beta$ spinor at an interior site. Its free band bound $m \le |h| \le 10m$ gives conditional pump success at least $11/20$. All assembly flags and nonvanishing pump failures remain. As every boundary and the static reference recede, complete local gauge-invariant records converge at fixed spacing and bounded source time. The compared alphabet is the local observer's record; remote fine flags re-

main in each finite source. Reference phases vanish by the dipole bound, while a three-dimensional heat estimate gives $G(x,x) \le 33$ for the point Green function needed by the energy comparison.

For the normalized charged Slater packet $P_q = P + quu^\dagger$, exact Wick contraction cancels the rank-one classical mean-density term against its exchange covariance. Relative to the neutral prepared sea, the actual-H expectation, after subtracting the displayed reference cost $4 + (mg^2/2)G(r,r)$, has a bulk limit in $[8081/8192, 81953/8192]m$ at $g = 1/64$. No Hamiltonian counterterm is introduced. A complete local density readout has limiting signed outcome imbalance at least $121/1600$ immediately after preparation; this bound is not asserted at arbitrary positive time. The input is nonstationary: its positive prepared energy is not a charged threshold above the interacting vacuum or a rest mass (`model/field_record_addenda/native_orthogonal_wilson_charged_spatial_records`, [2]).

**Neutral propagation and a controlled field limit.** The actual eight-cell ground $g$ and isolated triplet $e_\alpha$ supply dressed native-current matrix units $A_\alpha = P_g O_\alpha P_t/(60\sqrt{r}) = |g\rangle\langle e_\alpha|$, where $r > 0$ is the computed current residue and $\Delta \in [8.20758, 8.21035]10^{-22}$ is the source gap. A declared number-conserving exchange between complete blocks gives an isolated moving band $E(k) = \Delta - 2J\sum_i \cos k_i$ and a uniform positive gap. Its controlled spacing limit is Schrödinger evolution.

A different, explicitly paired architecture places $N$ such blocks at each cubic site. Write $A^{(N)}_{x\alpha} = N^{-1/2}\sum_j A_{xj\alpha}$ and $X = A^{(N)} + (A^{(N)})^\dagger$. On the full physical carrier declare

$$H_{a,N} = \frac{1}{a}\sum_{x,j}(h_{xj} - E_g) - t_a \sum_{\langle xy\rangle,\alpha} X_{x\alpha}X_{y\alpha}, \qquad t_a = \frac{\Delta(1-a^2)}{12a}, \quad 0 < a \le 1.$$

Its clock is $e^{-itH_{a,N}}$. On the exact permutation-symmetric sector, $(A^{(N)}_{x\alpha})^\dagger = b^\dagger_{x\alpha}\sqrt{1 - n_x/N}$, including the capacity boundary. Every higher native sector is retained. A variational argument places a full-source ground in this sector. Occupation-tail and square-root bounds compare its energy and vector with a specified Gaussian ground. In particular, $0 \le E_N - E_{\text{free}} \le \epsilon_E$ implies squared ground overlap at least $1 - \epsilon_E/\Delta$.

The quadratic comparison has

$$\omega_a(p)^2 = \Delta^2\left[1 + \frac{1-a^2}{3a^2}\sum_i(1 - \cos(ap_i))\right] \longrightarrow \Delta^2 + \frac{\Delta^2}{6}|p|^2.$$

Complete bounded field instruments retain their defect flags and a finite orientation reference. Explicit capacity choices bound a two-instrument, three-evolution record law by $19/20000$, with separate spacing and volume errors. This is a controlled free Klein–Gordon description with three internal scalar components. The component labels do not furnish spatial spin one. Replication, spectral dressing and the spacing-dependent clock are declared architecture; no empirical mass or GeV conversion selects their coefficients (`model/interacting_block_propagation`, `model/interacting_block_quadrature`, [2]).

**A vector readout of the same neutral source.** An explicitly different field instrument on those same blocks uses $\tau = \Delta t$, $\xi = \sqrt{6}x$, $\Phi_\alpha = \sqrt{\Delta}Q_\alpha$ and $\Pi_\alpha = P_\alpha/\sqrt{\Delta}$. With limiting mass $m = 1$ and $\Omega_0(p) = \sqrt{m^2 + |p|^2}$, declare

$$A_0(p) = \frac{ip_\alpha\Pi_\alpha(p)}{m\Omega_a(p)}, \qquad A_i(p) = \left(\delta_{i\alpha} + \frac{p_i p_\alpha}{m(\Omega_0 + m)}\right)\Phi_\alpha(p).$$

The momentum and factor $i$ in $A_0$ are required for Hermiticity and the Lorenz identity. The induced Lorentz action has polarization sum $-\eta_{\mu\nu} + p_\mu p_\nu/m^2$, three positive polarizations and the Proca equations. Its commutator is a differentiated Klein–Gordon commutator and vanishes at spacelike separation. The finite source has no exact boost symmetry; the controlled record limit carries that action.

Two specified finite members retain sixteen complete instrument outcomes and have total-variation error at most $9.5\,10^{-10}$ against the free vector comparison. Capacity, state, evolution and filter errors are separate. The rest frequency is the native $\Delta$ in the original clock; no W/Z identity or GeV scale is selected. This is a free massive spin-one readout of one declared source, not a deduction of spatial spin from the dimension of an unfiltered internal triplet (`model/interacting_block_proca`, [2]).

**A finite collision distinguishable from free evolution.** The same paired Hamiltonian has a finite-$N$ channel forbidden by its free quadratic comparison. Prepare two different internal components in Fourier packets with disjoint input and output momenta but equal total momentum. The free evolution preserves each component's momentum; the capacity square root produces a connected quartic vertex. For four momenta on one frequency shell,

$$\langle k', l'|V_1|k, l\rangle = \frac{t_a A(k)}{V}\left[\left(\frac{D}{\omega_k}\right)^2 + 1\right] > 0, \quad A(k) = 2\sum_i \cos k_i, \quad D = \Delta/a,$$

when $A(k) > 0$, with $H_{a,N} = H_{\text{free}} + V_1/N + R_N$. Global number-weighted bounds control $R_N$, the second Born term, packet admission and finite preparation before squaring the signal. The experiment retains its hit, complement and preparation failures. It uses explicitly prepared nonstationary states; an approximate interacting ground is not substituted in a rare-event bound.

An independently interval-certified diagnostic on a five-block cycle evolves the complete 1024-state reducing carrier, retaining all three components. With $D/\Delta = 2$, $t/\Delta = 1/8$ and $\Delta T = 1$, its preparation-inclusive final hit probability is enclosed by $[0.00296630042443, 0.00308409427253]$. The associated occupation records are quantum marginals, rather than classical particle paths. At $N = 1$, $H_N = P_N H_{\text{free}} P_N$; the admitted packet projection changes the separate momentum supports. Consequently the diagnostic's nonzero response alone does not certify a channel forbidden to the free evolution of those same admitted states. An exact Gaussian-kernel calculation of that matched free evolution, without an intermediate oscillator cutoff, gives the preparation-inclusive interval $[0.00218695122151, 0.00228825603677]$. The native-minus-free contrast is therefore at least $0.00067804438766$. Both comparisons retain the same admitted input, output projector and quantum reference; the complete native complement has an explicit common-carrier embedding. The large-$N$ theorem separately bounds that admission error below its connected interaction signal. Three-dimensional finite members also have strictly positive controlled collision channels. Their interaction vanishes as $N \to \infty$ in the established refinement. Neither the cycle nor those finite records proves a nontrivial interacting relativistic ultraviolet limit, an asymptotic scattering matrix or a Standard Model species assignment (`model/interacting_block_scattering`, [2]).

**A family effect and an energy clock.** For the declared cyclic family register, native repair columns give the one-tick amplitude $M = \sqrt{2}\,C(\theta)/3$, where $C = I/\sqrt{6} + (e^{i\theta}R + e^{-i\theta}R^{-1})/\sqrt{12}$ and $R^3 = I$. Positive eigenvalues of $C$ have squared values with Koide ratio $2/3$. The literal two-tick amplitude is instead $\sqrt{3}I/9$. An unconditional collision extension realizes $A = M^\dagger M$ as a generator: $N$ fresh retained ancillas over time $t$ give marginal-channel diamond error at most $\min(2, 4t^2/N)$

from $e^{-itA}$. The projector phase gate, inverse tick and schedule are explicit architecture inputs. The XOR probability 2/9 does not alone choose a Hamiltonian phase.

The separate scalar-memory construction contains three CAR families and $5\,2^{18}$ states. On its stationary vacuum-generated component, put $S = 4\sum_f y_f^2$, $D_0 = \sqrt{\omega^2 + 4S}$ and $D_f = \sqrt{\omega^2 + 4S - 8y_f^2}$. Its lower charged line is $m_f = 4y_f^2/(D_0 + D_f)$; a second line, nine dark states and the no-operation record are retained. For $y_h > y_l > 0$, $m_h/m_l > (y_h/y_l)^2$. Consequently this response cannot remove the +9.83354 ppm excess in the muon/electron comparison of the $\theta = 2/9$ squared-amplitude reading. Three independently occupiable families, a common three-state register and a neutral classical Higgs background are distinct source constructions; these finite identities do not transfer their energies to the spatial interacting source (`model/finite_family_clock`, [2]).

The retained finite candidate producers separate these inputs further. The optional native quarter-turn implements the oriented record phase; the family shift, joint analyzer and definite tail preparation are additional operations. Independent-object readout cannot realize that particular Schmidt-rank-two analyzer. In the tested transport dressing, retaining each virtual flag's clock energy gives the common equation $d = w/(d - \omega)$ and common residue, leaving family ratios unchanged. Omitting that energy defines a different model; its successful muon-ratio weight is fitted. A two-seam junction does generate nonzero CP violation, with one native assignment giving $(|V_{us}|, |V_{cb}|, |V_{ub}|, J) \simeq (0.07994105, 0.02311042, 0.01860003, -0.00003424005)$. None of its tested native, conditional or fitted assignments passes the registered four-observable comparison. The single-site first-order hierarchy failure does not exclude all junction strengths. These finite results are reproduced from retained original producers; no absent phase is silently replaced by zero (`model/particle_construction_scope`, [2]).

**A defect class and a particle representation.** The six-class register preserves colour triality and weak centre parity: $k = 3n_2 - 2n_3 \pmod 6$. Its 32 exterior states have class multiplicities $(4, 6, 6, 4, 6, 6)$, while the independently computed fifteen-state current representation gives the internal weights of one Standard Model fermion family. This comparison requires the full currents. For example, the centre acts trivially on both summands of $\mathbf{3} \otimes \overline{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$, whereas the full quadratic Casimir is $3(I - P_1)$. Two quarks also contain a $(\overline{\mathbf{3}}, \mathbf{1})_{1/3}$ channel with the gauge charges of $d^c$, but their even fermion parity differs from a single odd fermion. A class-conserving fusion therefore needs its complete physical reference and exchange law before receiving a particle interpretation. Auxiliary exterior degree alone is not the physical fermion grading (`model/particle_catalogue_correspondence`, [2]).

**Common interacting state and its scope.** Finite quantum-link transport, the quantum Higgs pair terms and invariant scalar and electric energies can be placed in one Hermitian Hamiltonian with exact local Gauss conservation. On its nonzero physical subspace, $\rho_\beta = P_G e^{-\beta H} P_G / \operatorname{Tr}(P_G e^{-\beta H})$ is a specified faithful stationary reference, with $\beta > 0$ a declared input. For $\|H\| \le B$, the filter $L = P_G e^{-\beta(H+BI)/2}$ on a maximally mixed physical input prepares it; $F = \sqrt{I - L^\dagger L}$ retains the failure branch. A two-cell quantum creation/transport control on 122880 active states gives $\|RH^2\Omega\|^2 = 3$, with $R\Omega = RH\Omega = 0$ and a strictly positive, error-bounded arrival probability at a specified finite time. Its empty input is distinct from the thermal reference. The free filled sea is not substituted for this interacting state. The separate shared-link collision source has a positive finite encounter response, but its density vertex fails a scalar boost test and its bounded contact vanishes in the stated one-flux/one-native refinement window. A nonzero connected interacting limit, controlled physical reference and mass renormalization do not

follow from either finite interaction. An energy conversion supplies units only: the background relation is $m_c c^2 = u_E |v y_c|$ GeV, without a source selection of $v, y_c, u_E$ [52].

### 3.9 The oriented record, the defect register and the generation register

Three elements of the specification carry the particle identification of the companion paper [118].

**The repair event records its source.** On the oriented address $(a, b)$ of a seam put $p = (|a\rangle + |b\rangle)/\sqrt{2}$ and $m = (|a\rangle - |b\rangle)/\sqrt{2}$. In the basis $(p0, m0, p1, m1)$ the coherent event $U_{(a,b)}$ of Section 3.8 is the swap of the port qubit $(p, m)$ with the record qubit, and the reversed address gives $U_{(b,a)} = Z_{\mathrm{rec}} U_{(a,b)} Z_{\mathrm{rec}}$. On all 30 seams, both orientations and both sources, $U|s, 0\rangle = |p\rangle \otimes (|0\rangle + \epsilon(o)\epsilon(s)|1\rangle)/\sqrt{2}$: the record holds the source endpoint relative to the address in its conjugate basis, while its agree/repair value is complementary to it.

**The oriented record.** The oriented record $V$ changes the sign of record one at the reversed address. The product $VQ$ equals the law with the screen address's unitary at both addresses, for every link phase. On the full 1440-state walk the plain law's consensus block from record zero to record one vanishes; with the oriented record it is $\frac{1}{30}\sum_{\mathrm{seams}} |p\rangle\langle m_{\mathrm{ref}}|$, so the repair branch is kept. The oriented record is the law used by the generation register.

**The defect register.** The universe runs carry a passive register in $\mathbb{Z}_6$ on the faces of every carrier. It starts blank; an accepted current-driven twist creates a conjugate pair of classes, with costs $(0, 5, 8, 10, 8, 5)$ for classes $0, \ldots, 5$ and acceptance scale $\kappa = 5$. Once the loads balance, moved unit currents release no mismatch work, so positive-cost proposals are rejected and surviving classes are pinned. The register does not feed back into the loads; a control run with the register switched off reproduces every load, current and depth record exactly. On the five-level code the class of a defect equals $6Y$ mod 6 of the particle family whose quanta see it. Keeping the integer current of every applied twist unreduced lifts the register to integers: the lifted value reduces to the stored class on every face of every stored cut of the runs at levels four to seven, and every carrier total is zero. The law reads the class only, so the integer is no conserved label of a member inside a class.

**The generation register.** Two seam objects on one face, a neutral register carrying the face third-turn $R$ and the clock $K_\theta = e^{i\theta J}$ on the oriented current form the tick $U(\theta) = K_\theta T V (W_C \otimes W_E)$. Its compression on a three-flag Fourier analyzer is $\frac{\sqrt{2}}{3} C(\theta)$ with $|b|/a = 1/\sqrt{2}$, and the register's exclusive read has probability $2/9$. The seam program satisfies $\langle I - \mathrm{Re}\, U \rangle = P(\text{record } 1)$ on every blank-record input with uniform address, so the energy clock reads read probabilities, and the read count becomes the phase $\theta = 2/9$. The resulting charged-lepton ratios are 206.770316 and 3477.4728 [118, 2].

**The weak round.** The register runs over many ticks through one added protocol on operations of the public program. On the sixty-address carrier two cells with independent address registers carry the pair, and the law's own records are the flags: the two signs of the native current in the defect's cell with weight $1/12$ each, and the written record of the second cell with weight $1/6$. One round is the tick of both cells, the kick $e^{-i(1/N)J \otimes X}$ on the exclusive read $X$, the inverse tick, one return test per cell on its uniform address state, and a reset of a cell after a negative test. The kick is the commutator of an oriented toggle with the tick of the pair. Every round reads the first-tick weights, so $N$ rounds per unit clock time give the phase $2/9 - 35/(2187N^2) + O(N^{-4})$. Executed

for $10^4$ to $2 \times 10^6$ rounds, the phase follows the closed form and tends to 2/9 within $10^{-13}$; with all four outcomes of the return tests kept, of weights 4/81, 11/324, 125/1296 and 55/1296, the coherence between current values $j, j'$ is multiplied per round by $1 - \frac{2}{9} + \frac{2}{9}e^{-i(j-j')/N}$. Without the inverse tick the count is not kept: the forward tick reads 2/9, 0.486, 0.536 and 0.467 at ticks one to four. Without the reset a cell that failed its test reads the complementary weight and the count changes to 7/9. The source events that create the pair and apply the kick are declared operations of this protocol [118, 2].

**Exact active-space reduction of the free mass compiler.** The separately specified neutral-background fermion compiler admits an exact reduction of its analyzer coordinates. For each of its three channels the rank-one native effect acts inside the span of the blank vector and its inverse-tick image. Its exterior-algebra isometry preserves all ninety physical fermion modes, twelve active ancillary modes per site and the complete quantum archive. The remaining ancillary coordinates stay in vacuum. The resulting 102 modes per site reproduce the original 324,000-mode source, including arbitrary physical occupations, flag-independent hopping and untouched references. Twelve ancillary modes are minimal within number-preserving fermionic embeddings that fix the physical modes; this is not a minimum over all dilations.

If $e_k$ are the three native effect eigenvalues and $L$ is the number of sites, the exact off-diagonal leakage norm satisfies

$$B^2 = L\left(6 - 2\sum_{k=0}^{2} e_k^2\right).$$

For $N$ committed rounds over dimensionless time $t$, the unhalved trace distance between complete output histories is at most

$$\min\{2, 2B^2t^2/N\} \le \min\{2, 12Lt^2/N\}.$$

The bound includes every nonvacuum flag and its quantum archive. It follows from the second-order block-Duhamel estimate for the vacuum compression, the exact leakage operator on every physical occupation, and telescoping of the committed channel. At one site and $t = 1$, a sufficient round count for error $10^{-12}$ is 11,724,207,179,903. This is a proved resource bound, not an executed precision campaign. The implementation and full-occupation controls are in `model/lepton_compiler_compression` [2].

**Native words for a bounded coherent lepton cycle.** The six-level processor also implements the padded preparation and coherent services of a specified lepton cycle at phase 2/9 and kick angle 1/7. Each sixty-address register is embedded in sixty-four coordinates, with padding retained. A compiled real preparation word $V$ and the complete blank-phase word implement the Householder $V(I - 2|0\rangle\langle 0|)V^\dagger$ on every input. The two-address analyzer preparation word $W$ acts on all 4096 coordinates. The exact identity

$$We^{-ih|0\rangle\langle 0|}W^\dagger = I + (e^{-ih} - 1)|\phi\rangle\langle\phi|, \qquad |\phi\rangle = W|0\rangle,$$

allows its execution through the actual source-prepared vector without materializing a dense kick. The amplitude-tree compiler and commuting phase expansion establish the identity on arbitrary inputs and references. Literal pulse replay on independent columns is an additional numerical control, not an all-input precision certificate.

Both return-test outcomes on each address remain in the output. A compiled SWAP moves the entire old source into a fresh archive register, preserving its correlations with the family and

reference. Up to three cycles retain all 4, 16 and 64 original histories and their full quantum archive factors. Proper-code complement failures are part of the instrument and have zero weight on the ideal invariant-code trajectory. Coherent loading and export, routing, fresh blanks and binary readout are supplied services. The target equalities use the ideal real-angle synthesis recipe. Its binary64 angle tape approximates that recipe; the reported residuals include angle representation, synthesis and matrix-execution rounding, without a certified numerical tolerance. The conjugation identity itself holds for any unitary preparation word, but its prepared vector need not equal the target analyzer after angle rounding. Hardware errors require separate complete-channel bounds. These words implement the stated cycle without identifying its clock or coefficients with interacting physical particle masses [118, 2]; the executable and complete archive are in `model/lepton_service_compiler`.

**Exact compression of the retained lepton archive.** The prepared sixty-address pair has four original return outcomes. Their family-conditioned environment ranks are $1, 2, 1, 2$. For one, two and four independent rounds, with total kick parameter $1/7$ and phase $2/9$, the numbers of rank-one, rank-two and rank-three histories are respectively $(2, 2, 0)$, $(4, 8, 4)$ and $(16, 64, 176)$. Six nonzero two-column minors have rational interval certificates. They imply that two rank-two factors give three independent family columns; additional nonzero tensor factors preserve independence. The complete decoder retains every original history amplitude and extends by identity on arbitrary quantum references. The Choi rank proves minimality at each fixed original classical history.

At four rounds the 48 padded source-archive qubits can therefore be represented by at most two quantum archive qubits, together with all eight original classical return bits. This is a reversible representation of the admitted prepared-instrument span, not erasure of arbitrary source inputs. Twenty-eight complete small orthogonal compressors and the local native address service are implemented. The worst four-round compression cost is 1,767,798 additional native pulses and 3,535,596 coherent loading/export events, beyond preparation, cycles, measurement and transport. Exact identities concern the ideal real-angle synthesis recipe; binary64 replay remains an approximation with numerical diagnostics. Coherent routing and blank preparation are supplied, and information dimension is not identified with thermodynamic work. Floating eigenvalue thresholding would incorrectly remove nonzero directions in 136 four-round histories; the construction instead uses the exact rank certificates. The complete construction is in `model/lepton_archive_compression` [2].

**Lepton cycles on the native event timeline.** The 66-dimensional native clock carrier can control the complete weak and mass cycles on one declared three-coordinate register. Its first nonzero current triggers the weak cycle; its second triggers the mass cycle. Every attempted slot, idle, current sign, source output, return and timeout is retained. Fresh address pairs are supplied coherent services, with their preparation and archive operations charged separately from native attempts. At three controller slots the complete instrument has 1,272 records. Six controls, including both native laws, half activity and deterministic idle prefixes, retain 7,632 records. The full-activity timeout probability is exactly $15806/16875$. For activity $q$ and cap $N$, the native zero-current effect gives the all-input bound

$$\Pr(\text{fewer than two currents by } N) \le (1-\delta)^N + N\delta(1-\delta)^{N-1}, \qquad \delta = \frac{2q}{15}.$$

This follows by conditional stochastic domination and extends to untouched quantum references. Source columns admit a reversible nine-coordinate representation using the actual address classes of sizes $50, 5, 5$. The decoder reconstructs all 3,600 original address amplitudes.

An event-only service policy is unchanged by a deterministic idle prefix. A declared slot-driven phase evolution changes its final Fourier readout under the same prefix. Reusing the correlated address pair after the weak cycle, instead of supplying a fresh pair, changes the complete sixteen return probabilities by total variation approximately 0.002341923935. These controls separate native counts, phase coordinates and supplied reset services. Complete finite Kraus composition establishes the reference extension; floating replay supplies numerical diagnostics without a certified hardware tolerance.

The limiting weak and mass generators have different spectra. Rational effect intervals exclude an affine identification for every permutation of their three coordinates, while allowing the common timeline above. The identity $e^{-i(cH)(t/c)} = e^{-iHt}$ preserves every dimensionless map, service angle and record. Thus the protocol supplies no absolute GeV or SI conversion. An empirical electron anchor fixes an energy unit without identifying interacting particles or the duration of a native attempt. The complete construction and controls are in `model/lepton_clock_energy` [2].

### 3.10 The local event clock

N4 is a reciprocal extension of the binary repair law in the construction's candidate register [42]. Each port carries a private six-level response factor, and one coherent exchange $W_e = \exp[\theta(Y \otimes \Gamma_e + X \otimes \Delta_e)]$, with $X = S_e(N_a - N_b)^2$, $Y = iS_e(N_a - N_b)$ and response generators $\Gamma_e, \Delta_e$ built from the endpoint response matrices, conjugates the binary repair, $K_r = W_e(L_r \otimes I)W_e^\dagger$, with supplied coupling $\theta$. Private preparations change bare load probabilities: at fixed bare load and $\theta = 1/8$ an endpoint occupation probability moves from 0.482020673 to 0.517979327. An interval certificate proves an edge-Gram margin $\mathcal{G} \geq mI$ at $\theta = 1/16$ and $\theta = 1/8$ [42, 68, 2].

Fix an observer port $v$. A tick is any attempted seam incident to $v$, including attempts that leave its load unchanged. Under independent uniform seam selection a tick has probability $p = 1/6$ per attempt, and remote attempts continue between ticks. In independent trials the full tracial state is prepared, the binary occupation $N_v$ is measured, the unmodified source runs until one or two ticks have occurred, and $N_v$ is measured again, with no intermediate measurement, reset or postselection. With $f = N_v - 1/6$ and $\|f\|_\tau^2 = 5/36$, write $c_k = \mathbb{E}[(X_0 - \frac{1}{6})(X_k - \frac{1}{6})]/(5/36)$ for the endpoint correlation after $k$ ticks [68, 69]. The numbers of Theorem 3.8(iii) are certified lower bounds; the full N4 two-tick correlation is not evaluated, and preparation, incident-event detection and bare-occupation readout are supplied operations that the detector of Section 3.1 does not implement.

**Theorem 3.8** (Memory at observer-local ticks)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be the averages over the thirty seams of the remote and incident seam channels, so that the first-hit channel is $\mathcal{K} = \mathcal{B}(I - \mathcal{A})^{-1}$. Then:*

*(i)* $c_k = \langle f, (\mathcal{K}^*)^k f\rangle / \|f\|^2$, *and* $c_1 = 1 - 6\alpha_{\mathrm{occ}}/11$ *for the occupation compression factor* $\alpha_{\mathrm{occ}}$*;*

*(ii) the statistic*

$$b_{\mathrm{tick}} = c_2 - c_1^2 - \frac{r^2(209r + 180)}{1331r^2 + 3300r + 1800}, \qquad r = 1 - c_1,$$

*vanishes for the native law and for the one-term exchange controls, which have* $c_1 = 5/11$ *and* $c_2 = 93/407$*;*

*(iii) for both N4 completions, built on* $\mathsf{C}$ *and on* $\mathsf{E}$, $b_{\mathrm{tick}} \geq 3m/50$, *and the certified lower bounds are*

$$b_{\mathrm{tick}} \geq \frac{139996257}{12500000000000} \approx 0.00001119970056 \quad (\theta = 1/16),$$
$$b_{\mathrm{tick}} \geq \frac{11335290321}{50000000000000} \approx 0.00022670580642 \quad (\theta = 1/8);$$

*(iv)* $c_{2,\mathsf{C}} \geq c_{2,\mathsf{E}}$ *and* $b_{\mathrm{tick},\mathsf{C}} \geq b_{\mathrm{tick},\mathsf{E}}$.

*Proof sketch.* Each seam channel is a positive semidefinite contraction of Hilbert-Schmidt space, with spectrum in $\{0, 1/2, 1\}$ for $\mathsf{C}$ and $\{0, 1\}$ for $\mathsf{E}$; N4 conjugation preserves this. Hence $0 \leq \mathcal{A} \leq (1-p)I$, the first-hit series converges, and the stopped maps are unital and trace preserving. Remote seams fix $f$, which gives (i). The closed comparator with seam mean maps $I - \frac{\alpha_{\mathrm{occ}}}{2} w_e w_e^{\top}$ yields the correction term in (ii) after substituting $\alpha_{\mathrm{occ}} = 11(1-c_1)/6$. With the Schur complement $S$ of $I-\mathcal{A}$ on the centered occupations, $S \leq I$ and $b_{\mathrm{tick}} = \langle z, S^{-1} z\rangle/(p\|f\|^2) \geq \|z\|^2/(p\|f\|^2)$. The hidden defect is $z = -\frac{1}{60}\sum_e t_e S_e$ with five incident coefficients $\pm 1$, so $\|z\|^2 \geq 5m/3600$ and $b_{\mathrm{tick}} \geq 3m/50$. On each seam $\mathsf{C} - \mathsf{E}$ is positive semidefinite, which gives (iv) [68]. □

The correction used for global attempts, $(1 - c_1)^2/10$, assigns $-16/2035$ to the native null at local ticks; the correction of Theorem 3.8(ii) is the one adapted to this stopping instrument. The receipt and eight independent checks bind the result, including a comparison with the full 66-state native stopping kernel and a complete dimension-six quantum model of the resolvent and Schur identities [68, 94].

# 4 Observers, clocks and screens on one process

This section completes layer three on one native process. Its components run the native C or E instrument itself, with the uniform independent seam schedule and fair placement coin as declared scheduling inputs. Their time parameters are attempt counts and local event counts. Each result states its preparation, observer access and services; none identifies a count with laboratory duration [14].

## 4.1 Two overlapping observers

On the 66 binary total-two configurations, prepare the uniform state and disclose the bits at ports $(0, 1, 5)$; all eight labels stay in the output menu. Observer A accesses $(0, 1)$ and observer B accesses $(0, 5)$, so the regions overlap at port 0. Each receives its initial bits and every incident local current, including zero, and ticks when an attempted seam meets its region. The protocol requests two incidents on the union with at most two native attempts per interval; after the first it applies one native repair on $(0, 1)$ if the reconstructed $n_0$ is one and on $(0, 5)$ otherwise, and that repair ticks both observers.

**Proposition 4.1** (Exact overlapping-observer correspondence)**.** *Normalize the native remote and incident branch counts by* 60*, write* $P = A + B$ *with* $B = \sum_m B_m$ *over incident marks* $m$*, and put* $D = I - A$*,* $p = 2/5$ *and* $G = D/p$*. With* $K_m = B_m D^{-1}$ *and* $F_m = D^{-1} B_m$*,*

*(i)* $K_m = G F_m G^{-1}$ *and* $\mathbf{1}^{\mathsf{T}} G = \mathbf{1}^{\mathsf{T}}$*; remote projectors, the selective feedback maps and the disclosed uniform preparation commute with or are fixed by* $G$*;*

*(ii) for attempt cap* $h$ *the marked and timeout maps are* $K_{m,h} = B_m(I + A + \cdots + A^{h-1})$ *and* $Q_h = A^h$*, their effective conjugates are* $F_{m,h} = G^{-1} K_{m,h} G$ *and* $A^h$*, and the terminal decoder* $G$ *returns the complete unnormalized source state on every output word of every finite adaptive protocol in this commuting intervention family;*

*(iii) the direct endpoint execution and the effective recursion agree exactly on all* 5,768 *output words,* 689 *of positive probability, including all terminal populations, and the timeout probability is* $369/625$*; and*

*(iv) the unread completed transfer $F = D^{-1}B$ is self-adjoint in the metric $G$ with $1/20 \le F \le 1$, so $H = -\log F$ is an effective Euclidean generator per completed union incident tick.*

*Proof sketch.* Remote repairs leave the union fields fixed and every remote seam is disjoint from the feedback ports, so the conjugation by $G$ cancels between successive reads and allowed feedback maps; induction over the protocol gives (ii). Every native seam count equals $I + \mathrm{SWAP}$, which makes $GF = B/p$ symmetric positive semidefinite, and the native triangle $(0, 1, 5)$ contributes at least $3I/60$ to $B$ [15]. □

At every positive-weight prefix each observer's decoder, which sees only its local initial pair and incident increments, returns fields that obey native continuity, and the two port-zero values agree. The transformed finite-cap maps preserve the transported cone $G^{-1}\mathbb{R}^{66}_{+}$; they are not entrywise-positive stochastic maps, and the comparison is of decoded full states. The quantum extension checks the native remote resolvent on all 4,356 matrix units with coherent preparations, untouched references and complete local clock records, and an ordinary completely positive effective instrument for local outputs is a positive mixture of native repairs; its one-particle coherent experiment retains all 81 outputs [16]. The compiled clock history replaces the ideal preparation, intervention and pointer of that experiment by words of the six-level source compiler. Exact rational propagation of the binary source arrays keeps all 512 code addresses and 1,656 clock, timeout and complement-abort outputs within $10^{-11}$ of the ideal history, at most 51 pulses (54 with the compiled Bell preparation), two code services per pulse and five native repair attempts [17]. The native instrument, the scheduler and the code services are specified inputs.

### 4.2 Joint and active event clocks

A local event clock ticks on actual occupation changes at an observer port, or on attempted repairs incident to a region, while remote repairs continue; its stopped instrument sums the unresolved wait and retains finite timeouts and conditional quantum outputs. Two such clocks on ports 0 and 72 of joined carriers read one continuing history with an observer-chosen field algebra. Exact elimination of the hidden factor gives a record-conditioned quantum memory recurrence valid for arbitrary untouched references. At completed events the experiment admits a four-mark measure-and-prepare description, while unresolved waits retain coherence that the completed instrument erases. C and E share the completed clock law and predict field-plus probabilities 35/54 and 14/27 after three native attempts. Without feedback each two-tick cycle of clock A carries B-count mean two and variance 16; instrument-valued clock conversion preserves that count law, and direct native elimination and elimination through the joint event instrument agree, including B-triggered field feedback [18].

An active event clock lets each native event incident to port 0, including a zero-current one, trigger one fixed compiled modular word $U(1) = \exp(iH)$ generated by the native three-state attempt transfer; the carrier pauses during probe services, which are charged separately. For caps $c \in \{1, 2\}$ the execution retains all 3,661 joint output labels, the first timeout has probability $(5/6)^c$ and both ticks occur with probability $[1 - (5/6)^c]^2$. Closed native histories derive a faithful thermal reference for that generator, and passive repair itself cannot act as a nontrivial modular automorphism on an invariant observer algebra [19]. On the complete movement-bit tape $B_t = |j_t|$, including zero currents, four consecutive covariances $c_n = \mathbb{E}[B_t B_{t+n+1}] - p^2$ invert to the two transfer roots $\lambda_{\pm} = (25 \pm \sqrt{3})/30$ with $p = 5/33$, and the same roots predict the coherent quadratures $q_X = (1 + \cos(sE_j))/2$ and $q_Y = (1 + \sin(sE_j))/2$, $E_{\pm} = -\log \lambda_{\pm}$, with no frequency fit. The one-exponential approximation matching $c_0, c_1$ misses $c_2$ by exactly $1/2673000$. The inference does not distinguish C from E, whose classical tapes agree [20].

### 4.3 Identification of the record law

Nine source-executed scalar measurements identify all nine real coordinates $(k, g, h, m, n)$ of the record-law table of Theorem 2.3, and with them the coarse definite-current instrument on all 78 total-two states with arbitrary spectator and reference coherences. The execution comprises 36 compiled probes and ten complete two-event histories, five with untouched quantum references, run through the canonical observer runtime with serialized checkpoint validation and continuation. An admissible collision law agrees with E on the whole binary shell, and this experiment separates it. A reference-uniform bound carries measurement uncertainty through the finite feedback histories, so narrowing the admissible probability tubes narrows the complete-history bound [21]. The probes are synthetic; they identify an instrument within the classified family and select no law for nature.

### 4.4 Federations and matched screens

Federations join 20, 80 or 320 identical twelve-port carriers through the declared seam wiring, and native Kraus maps preserve signed-difference operator blocks with individual outcomes and reference correlations. The endpoint instruments fix the local Liouville action $H_{\mathrm{native}} = \sum_e (I - \Phi_e)$, with $h_e = P_1 + \frac{1}{2}P_{1/2}$ for C and $h_e = P_1$ for E on orthogonal projectors; the discrete uniform transfer is $I - H_{\mathrm{native}}/M$ for $M$ seams. For an observed region $R$, maximum degree $\Delta$ and retained radius $r$, truncating distant ports changes the complete local record, quantum and reference output by at most $\min(1, |R|(\Delta s)^{r+1}/(r+1)!)$ at auxiliary per-seam Poisson parameter $s$ [22]. Actual source neighborhoods are identified at levels 1/3, 2/4 and 3/5. On the 15,360- and 245,760-port screens matched radius-nine initial restrictions give complete root-observer output distance below 0.000382 at $s = 1/4$; after four local incident ticks, with no auxiliary time coordinate, the pair allowance is below $6.656 \times 10^{-9}$, and a density-1/3 preparation keeps a certified two-time field covariance above 0.15332581621. Full-screen executions retain the actual seam scheduler, current records, feedback repair costs, timeouts and serialized continuation. Conditional on a conserved total, native connected exchange derives the unique diagonal equilibrium, so the screens at totals 5,120 and 81,920 keep density 1/3, the feedback reversal of the connected four-field statistic and their finite-number correlations without product preparation [23].

A coherent observer runs on the same matched screens. A logical qubit on ports 0 and 1, possibly entangled with an untouched reference, passes through two native incident ticks and a local phase $Z_a$ after first mark 1, while every other port is occupied with probability 1/3. A degree-two operator dual evaluates the common 222-port ball with 561 seams and all 21 capped record labels, and each full-screen completed output lies within $7.779 \times 10^{-10}$ of it, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [24]. The native continuum operator of the refining screens and its density, wave and response limits are stated in Section 7.

### 4.5 What the native counts supply

Let $C_{xy}$ count the sixty seam and coin branches taking a binary configuration $x$ to $y$ and put $P = C/60$. Every attempt counts, including zero currents, so this complete-attempt transfer differs from the incident transfer $F$ of Proposition 4.1. Its closed histories give the positive functional $\rho_L = P^L / \operatorname{Tr} P^L$ on the 66-state matrix algebra through a chosen linear extension, and the same path counts give ordered modular correlations with generator $H = -\log P$ and conventional Heisenberg time $\tau = -Ls$. The entries of $\rho_L$ are classical endpoint probabilities divided by the return probability, native C and E evolution converges to $I/66$ without preparing it, and the current-only

adapter uses at most 11,398 attempts per experiment; the observers paper proves these statements [6].

Binary repair is exact conditional resampling at constant energy under the declared fair coin. A stopped coin prefix with stored cost $E = n \log 2$ gives a bounded register refresh that inhabits the finite four-law core in internal record-cost units; that core holds for every full-support law with $E = -\log(\mathrm{ref})$ and formalizes no contact relation [6]. The transfer spectrum and compact cores of the same kernel are in the particle paper [118], and regional entropy on the reference $C^2/171720$ is in the Einstein paper [8].

# 5 Field detectors on supplied actions

Layer four begins with detectors that couple native records to fields whose actions are supplied. They provide complete-output and refinement machinery that a source-derived field description must reuse; they derive no field.

## 5.1 The scalar detector and its refinement transfer

The scalar detector joins the native observer to the free-scalar continuum family of the theorem library [58, 59]. The scalar action, quantization, vacuum, preparation, Born readout and model clock are supplied. The field is the massive scalar on the golden tensor grids $q \in \{13, 34, 89, 233\}$, with Dirichlet boundary and mass one; after mass whitening its one-particle operator is $\mathsf{A} = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B$, with $\Omega = \sqrt{\mathsf{A}}$ and $H = d\Gamma(\Omega)$. A preparation profile $f$ and a detector profile $g$ lie in compact left and right slabs.

One trial runs as follows. (1) Prepare one of the 66 binary configurations uniformly, repair $(0, 1)$ and retain $(c_1, j)$. (2) Prepare the field vacuum and apply the momentum displacement of amplitude $\gamma j f$, $\gamma = 4$. (3) Evolve to model time $t$ and read the centered binary Weyl instrument

$$K_y = \tfrac{1}{2}[e^{-iX/2} - iy\, e^{iX/2}], \qquad X = a^\dagger(v) + a(v), \qquad v = \Omega^{-1/2} g/\sqrt{2},$$

with $K_y^\dagger K_y = \frac{1}{2}[I + y \sin X]$. (4) Commit $y$, repair $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, and halt. Model times lie in $[19/20, 1]$, with primary setting $39/40$, and carry no laboratory calibration.

**Proposition 5.1** (Scalar feedback and refinement transfer)**.** *For the process above at model time $t$, let $r(t) = \Pr(y = +1 \mid j = 1) - \frac{1}{2}$. Then:*

*(i) the branch operators $M_{(r,y,s)} = (L_s L_r) \otimes (K_y e^{-iHt} V_j)$, with $V_j$ the controlled displacement, form one CPTP instrument on arbitrary joint source and field inputs, and each branch retains its conditional field state as a superposition of two coherent components with their relative phase;*

*(ii) $p(y \mid j) = \frac{1}{2} + yj\, r(t)$ with $r(t) = \frac{1}{2} e^{-\|v\|^2/2} \sin \mu(t)$ and $\mu(t) = \gamma \langle g, \Omega^{-1} \sin(\Omega t) f \rangle$;*

*(iii) over the 528 histories, $\mathbb{E}[j_2] = 5r/132$, $\mathbb{E}[N_{0,\mathrm{final}}] = \frac{1}{6} - 5r/132$, $\mathbb{E}[yj] = 10r/33$ and $\Pr(y = \pm 1) = \frac{1}{2}$; disconnecting the preparation or replacing $y$ by an independent fair bit gives $\mathbb{E}[j_2] = 0$ with the same seam frequencies, and reversing the kick reverses the sign;*

*(iv) if $|r_q - r_\infty| \le \varepsilon_q$ on the window, then the complete labelled histories satisfy the total-variation distance $\mathrm{TV} = \frac{5}{33}|r_q - r_\infty| \le 5\varepsilon_q/33$, and $|\Delta \mathbb{E}[j_2]| \le 5\varepsilon_q/132$.*

*Proof sketch.* The Weyl operators $e^{\pm iX/2}$ are displacements by $\pm iv/2$, which gives the branch states and (ii). Conditioned on the initial configuration and first coin, the two detector laws differ in total variation by $|j|\,|r_q - r_\infty|$, and the subsequent native update is common [59]. □

At $q = 233$ the parent certifies, uniformly for $19/20 \le t \le 1$,

$$\varepsilon \le \frac{12513633763}{250000000000}, \qquad -\frac{111322580247}{500000000000} \le r \le -\frac{34438518047}{250000000000},$$

so the finite-grid mean second current satisfies

$$-\frac{37107526749}{4400000000000} \le \mathbb{E}[j_2] \le -\frac{34438518047}{6600000000000},$$

negative throughout the window, with the continuum mean within $\frac{5}{132}\varepsilon \approx 0.001896$ [95, 60]. At $t = 39/40$ the plus probabilities for $j = +1$ are about 0.334515, 0.319421, 0.317093 and 0.316771 on $q = 13, 34, 89, 233$, and the finest-grid prediction is $\mathbb{E}[j_2] \approx -0.00694050$. A seeded execution with 200,000 fresh trials per grid tests Born outcomes against predictions computed before sampling, with reversal, donor and fixed-seam controls; at $q = 233$ all 12,487,168 field modes contribute. An independent diagonalization gives largest relative frequency errors 2.137%, 0.2875% and 0.04116% at $q = 13$, 34 and 89 against the supplied dispersion $\omega^2 = 1+\pi^2(n_x^2+n_y^2+n_z^2)$ [87]. A local kick-drift-kick extension through the six-neighbour stencil runs at $q = 13$ and 34 and transfers the combined spatial and temporal bound through the same factors 5/33 and 5/132; at $q = 233$ the temporal term is at most 0.000008071512 and the combined probability error at most 0.050062606564 [58].

### 5.2 The retained two-read continuation

The continued observer keeps the disturbed field after its first read [61]. The field is read at model times 19/20 and 1; the first bit $y_1$ selects the second native seam, the conditional field evolves to time 1, the second bit $y_2$ selects the third seam, and the observer halts with five records readable. This continuation uses exact continuous spectral evolution and carries its own error bounds.

**Proposition 5.2** (Two reads of one field)**.** *In the continued protocol:*

*(i) all 2,112 labelled histories are retained; the record-prefix counts at the six phases are* $1, 4, 8, 24, 48, 124$*, so the 209 reached prefixes have an eight-bit encoding and the 124 terminal words need seven bits;*

*(ii) with the Gram kernel* $G_{ab}$ *of the two detector vectors,* $V = G_{11} = G_{22}$*,* $C = \operatorname{Re} G_{12}$*,* $c = 2 \operatorname{Im} G_{12}$ *(so that* $[X_1, X_2] = ic$*),* $S_a(j) = e^{-V/2} \sin(j\mu_a)$*,* $D = \cos(c/2)$ *and* $R_j = \frac{1}{2}[e^{-(V-C)} \cos(j(\mu_1 - \mu_2)) - e^{-(V+C)} \cos(j(\mu_1 + \mu_2))]$*, the joint law is*

$$p_j(y_1, y_2) = \tfrac{1}{4}\big[1 + y_1 S_1(j) + y_2 D\, S_2(j) + y_1 y_2 R_j\big];$$

*(iii) on the positive-current preparation, performing the first read changes the second plus probability by* $(-r(t_2))[1 - \cos(c/2)]$*, certified greater than* 0.000022281162 *at* $q = 233$ *and greater than* 0.0000252366 *in the declared compact continuum; a commuting Gaussian control with the same means and real covariance has* $D = 1$ *and no such disturbance; the zero-current vacuum correlation exceeds* 0.045619388905 *at* $q = 233$*;*

*(iv) the total variation distance of the complete labelled history from the declared continuum is at most* 1*,* 0.451806*,* 0.052189606637 *and* 0.025709913643 *on* $q = 13, 34, 89, 233$*, including the zero-current histories.*

*Proof sketch.* A state is stored as $\sum_n c_n D(i \sum_a h_{na} v_a)|\alpha\rangle$ with exact half-integer shifts, its Gram matrix is explicit in $G$, and both terms of each Kraus operator are applied before the branch norm is taken. For four atoms with Walsh-moment differences $a, b, d$, TV $= \frac{1}{2}\max(|a|, |b|, |d|, \frac{1}{2}(|a| + |b| + |d|))$, bounded in directed rational arithmetic [61]. □

At $q = 233$ the execution gives $\mathbb{E}[j_3] \approx -0.00130126$ and $\mathbb{E}[y_1 y_2 \mid j = 0] \approx 0.0514838$; deleting the first field record from the forecast raises its Brier risk by about 0.00064677. The 50,000-trial runs per grid do not resolve the small disturbance statistically.

### 5.3 The source-compiled two-site apparatus

The two-site apparatus executes preparation, evolution and pointer as emitted twelve-port control words [62, 63]. Its action lies outside the golden refinement family, so its occupation limit carries no spatial refinement. The Hamiltonian is

$$H = \tfrac{1}{2}(P_0^2 + P_1^2 + Q_0^2 + Q_1^2) + \tfrac{\kappa}{2}(Q_0 - Q_1)^2, \qquad \kappa = \tfrac{9}{32},$$

with normal-mode frequencies 1 and 5/4 and $d \in \{4, 8, 16\}$ number states per site. The executed Hamiltonian uses the compressed squares $Q_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$ and $P_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$, so that $H_d = \Pi_d H \Pi_d$. The prepared state is the normalized projection of the coupled vacuum onto total occupation at most $d-2$; at $d = 16$ its binary amplitude tree costs 5,379 native pulses. The current impulse is $e^{ijQ_{0,d}}$, 64 or 128 symmetric split steps reach $t = 1$, and the pointer coupled by $e^{-iZQ_{1,d}/2}$ and read in the $Y$ basis gives $\Pr(y = +1 \mid j) = \frac{1}{2} + jr_d$ and $\mathbb{E}[j_2] = 5r_d/132$.

**Theorem 5.3** (Complete ideal error budget of the two-site instrument)**.** *For the declared two-site action, model time $0 \le t \le 1$ and native current $j \in \{-1, 0, 1\}$:*

*(i) with $A$ the onsite and $B$ the bond part of $H_d$ and $S(h) = e^{-ihA/2}e^{-ihB}e^{-ihA/2}$,*

$$\|S(t/n)^n - e^{-it(A+B)}\| \le \frac{t^3}{n^2}\Big(\frac{\|[A,[A,B]]\|}{24} + \frac{\|[B,[A,B]]\|}{12}\Big),$$

*uniformly over inputs and references;*

*(ii) the prepared state differs from the infinite coupled vacuum in trace distance by less than 0.007576, 0.000079899 and $1.0324 \times 10^{-8}$ at $d = 4, 8, 16$;*

*(iii) the trace distance between the complete ideal finite instrument, including its classical outcome and unnormalized terminal field branches, and the infinite two-oscillator instrument is below 0.799447, 0.077171 and 0.000196 at $d = 4, 8, 16$, and the bound tends to zero as $d$ grows;*

*(iv) at $d = 16$ and $n = 128$ the combined occupation and splitting error is below 0.001464436, and the ideal source-word instrument satisfies $0.514957066204 \le \Pr(y = +1 \mid j = 1) \le 0.517885937293$, an interval that excludes 1/2.*

*Proof sketch.* Item (i) is a defect bound for the symmetric product formula [116] with nested commutators bounded by exact rational Frobenius norms. The infinite state is a displaced squeezed vacuum throughout; the Gaussian generating function bounds the number tails along impulse, evolution and both pointer arms, and projected Duhamel comparison with the orthogonal leakage gives (iii) without dividing by a Born probability. Item (iv) adds a rational enclosure of the infinite reference probability [63, 96]. □

The executed plus probabilities for a positive unit current, with their rational splitting bounds, are 0.516680958 (0.000103957) and 0.516679563 (0.000025990) at $d = 4$ with 64 and 128 steps, 0.516424174 (0.000693013) and 0.516422803 (0.000173254) at $d = 8$, and 0.516423330 (0.005076875) and 0.516421959 (0.001269219) at $d = 16$. The infinite reference has detector variance 9/20 and mean $\mu(t) = \frac{1}{2}(\sin t - \frac{4}{5}\sin(5t/4))$, so $\Pr(y = +1 \mid j) = \frac{1}{2}(1 + e^{-9/40}\sin(j\mu(t)))$, equal to 0.516421501748 at $t = 1$ and $j = 1$ [62]. The $d = 16$, 128-step protocol uses nine work qubits, 18,612,125 native pulses and 37,224,250 proper-code load and export events, and retained 200,000-trial runs record the reversal, disconnected and donor controls.

# 6 The common interacting source

## 6.1 Construction

The common source places quantum electric flux, a charged fermion pair and a neutral scalar on one evolving work register. The sharp protocol measures it twice and feeds its records into the register and the declared native integer primitive [70, 71]; its four-read smooth extension joins compiled native quantum channels and encoded recovery (Section 6.5). The action, preparation, coupling and model-coordinate schedule are construction inputs.

Order four fermion modes as $L_0, R_0, L_1, R_1$. The two channels carry charges $q_0 = +1$ and $q_1 = -1$ and one fermion each; $b_j = 0$ means left and $b_j = 1$ right occupation of channel $j$. The electric link has basis $|\ell\rangle$, $\ell \in \mathbb{Z}$, with $\mathcal{E}|\ell\rangle = \ell|\ell\rangle$ and $U|\ell\rangle = |\ell+1\rangle$. The isometry

$$W|b_0, b_1\rangle = |\text{modes } b_0 \text{ and } 2+b_1 \text{ occupied}\rangle \otimes |-b_0+b_1\rangle$$

maps onto four orthonormal occupation and flux vectors.

**Proposition 6.1** (Flux-dressed matter code)**.** *On the image of $W$:*

*(i) both operator Gauss constraints hold, $G_L W = G_R W = 0$ with $G_L = \mathcal{E} - Q_L$ and $G_R = -\mathcal{E} - Q_R$, the supported flux set is $\{-1, 0, 1\}$ inside the infinite link, and fermion parity is even;*

*(ii) the dressed hopping terms $h_j = c_{Lj}^\dagger U^{q_j} c_{Rj} + c_{Rj}^\dagger U^{-q_j} c_{Lj}$ preserve both ambient Gauss values and the code, with $W^*(h_0 + h_1)W = X_0 + X_1 = H_{\text{hop}}$ and $W^*\mathcal{E}^2 W = \frac{1}{2}(I - Z_0 Z_1)$, and $\|[H_{\text{hop}}, \mathcal{E}^2]\| = 2$;*

*(iii) with $\theta = \operatorname{atan2}(4,3)$, $u = (3I - 4iX)/5$, $U_\theta = u \otimes u = e^{-i\theta H_{\text{hop}}}$ and $V_\theta = e^{-i\theta\mathcal{E}^2} = \operatorname{diag}(1, z, z, 1)$, $z = (3-4i)/5$, the echo $U_\theta^* V_\theta U_\theta$ applied to $|00\rangle$ gives right occupation of channel 0 with probability exactly $576/3125$, and without $V_\theta$ the probability is zero.*

*Proof sketch.* On every image column $Q_L = -b_0 + b_1 = \mathcal{E}$ and $Q_R = -\mathcal{E}$. The four-mode algebra of canonical anticommutation relations (CAR) is checked on all 16 occupation states with independent Jordan-Wigner signs, and all entries of the echo unitaries are Gaussian rationals [71]. □

The echo runs on the source controls with 22 native pulses and 44 coherent service events. Two neutral scalar sites join the same register, and with onsite quadratics compressed from the infinite oscillator the declared dimensionless action is

$$\begin{aligned} H &= H_{\text{hop}} + \mathcal{E}^2 + H_{\text{scalar}} + H_{\text{int}}, \\ H_{\text{scalar}} &= \sum_{j=0,1} \tfrac{1}{2}[P_j^2 + (1+\kappa)Q_j^2] - \kappa Q_0 Q_1, \qquad H_{\text{int}} = g\sqrt{2}\, Q_0(n_{L0} - n_{R0}), \end{aligned} \tag{4}$$

with $\kappa = 9/32$ and $g = 1/4$. The interaction commutes with both Gauss generators and with fermion parity and couples the sectors, $[H_{\text{int}}, H_{\text{hop}}] = 2ig\sqrt{2}\, Q_0 Y_0$. The experiment proceeds in four steps. (1) Native repair on $(0,1)$ records a current $j$, which controls the gauge pulse $V_j = e^{-ij\theta X_0}$. (2) The register evolves for interval 1 ($U_1$), and a source pointer commits the right occupation $a$ of the first charged species ($P_a$). (3) Only $a$ selects the hopping pulse $F_1 = (3I - 4iX_0)/5$, with $F_0 = I$, and the register evolves for interval $1/2$ ($U_2$). (4) A second pointer commits the sign $b$ of $Q_0$ ($R_b$), and only $b$ selects the second native seam, $(0,5)$ for $b = 1$ and $(0,1)$ otherwise.

**Proposition 6.2** (One joint instrument with retained histories)**.** *The operators* $K_{(a,b)} = R_b U_2 F_a P_a U_1$ *satisfy* $\sum_{a,b} K_{(a,b)}^\dagger K_{(a,b)} = I$*, and* $M_{(r,a,b,s)} = (L_s^{e(b)} L_r^{(0,1)}) \otimes (K_{(a,b)} V_j)$ *form one CPTP instrument on arbitrary joint native and quantum inputs. Under the uniform binary preparation and the declared quantum vector* $\psi$*, the* $66 \cdot 2 \cdot 4 \cdot 2 = 1{,}056$ *histories have probabilities* $p(a,b \mid j)/264$ *with* $p(a,b \mid j) = \|K_{(a,b)} V_j \psi\|^2$*, each bound to its terminal density* $|v_{j,a,b}\rangle\langle v_{j,a,b}|/264$*, and the first native record keeps the masses* 14/33*,* 14/33*,* 5/66*,* 5/66*. For a complete record-controlled instrument* $B_y = \sum_r |r\rangle\langle r| \otimes K_{(r,y)}$ *with channel* $\Lambda(X) = \sum_y B_y X B_y^\dagger$*, every old record projector* $Q_r$ *satisfies* $\Lambda^*(Q_r) = Q_r$*, and the block readback* $E_{\mathrm{block}}(X) = \sum_r Q_r X Q_r$ *and the public readback* $E_{\mathrm{public}}(X) = \sum_r \mathrm{Tr}(Q_r X Q_r) Q_r / \dim W$ *are distinct conditional expectations.*

*Proof sketch.* Summing $M^\dagger M$ over $s$, then $(a,b)$, then $r$ gives the identity. Each $B_y$ commutes with $Q_r$, so $\Lambda^*(Q_r) = Q_r \sum_y B_y^\dagger B_y = Q_r$; every reported probability is checked against the norm of the retained unnormalized vector [71, 97]. □

At 512 steps per interval the emitted words use 4 working qubits, 176,182 native pulses and 352,364 coherent load and export events at $d = 2$, and 6 qubits, 1,879,628 pulses and 3,759,256 events at $d = 4$. Disconnected, feedback-disabled, fixed-seam and independent-bit controls isolate the causal effect of the records [70].

## 6.2 The feedback contrast at finite and infinite occupation

**Theorem 6.3** (Finite feedback contrast)**.** *For the common source with intervals* 1 *and* 1/2*:*

*(i) every palindromic product* $F(h)$ *of the component exponentials whose generators sum to* $H$ *satisfies* $\|F(t/n)^n - e^{-itH}\| \le S^3 |t|^3/(3n^2)$ *with* $S \le 3 + (1 + 2\kappa)(2d-1) + |g|\sqrt{4d-2}$*, on all inputs and references; at* $n = 512$ *and* $t = 1$ *the half-diamond bounds are below* 0.000728 *for* $d = 2$ *and* 0.004184 *for* $d = 4$*;*

*(ii) at* $d = 2$ *and zero native preparation current, exact rational Taylor arithmetic of degree 64 with* $\|H\| \le 109/16$ *encloses the unconditional probability of a positive scalar coordinate and the feedback contrast for the continuous finite Hamiltonian:*

$$\begin{aligned} &\textit{feedback enabled:} && [0.370697726608684,\ 0.370697726608685],\\ &\textit{feedback disabled:} && [0.402359206973491,\ 0.402359206973492],\\ &\textit{contrast:} && [-0.031661480364808,\ -0.031661480364807]; \end{aligned}$$

*(iii) for the ideal source word at 512 steps per interval the contrast lies in*

$$[-0.033297300667839,\ -0.030025660061775],$$

*which excludes zero.*

*Proof sketch.* The multinomial Leibniz expansion bounds the third derivative of a product of unitaries by $S^3$, palindromic symmetry gives second-order agreement, and unitary telescoping gives (i). For (ii) the integral Taylor remainder gives $\|T_m(t) - e^{-itH}\| \le (St)^{m+1}/(m+1)!$, and at $m = 64$ the probability radius is below $7.091 \times 10^{-37}$; measurements and feedback contract trace distance, which gives (iii) [71]. □

With $g = 0$ the gauge-only feedback cannot change the scalar marginal, and a coarse splitting or inadequate Taylor degree yields an interval containing zero, which the evaluator reports as no

resolved signal. The finite pointer measures the sign projector of the truncated coordinate, which differs from the infinite half-line effect; at $d = 2$ the two differ in norm by $\frac{1}{2} - 1/\sqrt{2\pi} \approx 0.10105772$ [75], so the infinite-occupation statement needs the infinite detector.

**Theorem 6.4** (Infinite-occupation feedback contrast)**.** *Take the action (4) on two infinite oscillator Fock spaces, the infinite free coupled vacuum, proportional to* $\exp[-(a_0^\dagger - a_1^\dagger)^2/36]|0,0\rangle$ *with squared normalization* $4\sqrt{5}/9$*, the blank gauge code, zero native preparation current, the matter record and feedback of Section 6.1, and the half-line readout of* $Q_0$*. Then:*

*(i) the unconditional positive-coordinate probability with feedback minus that without lies in* $[-0.024045670143, -0.024043450955]$*, and without the interaction the contrast is exactly zero;*

*(ii) both matter records and both half-line outcomes retain their quantum states, including the post-read infinite occupation tail;*

*(iii) along* $d = 2^k$*, with exact palindromic factors and* $n = d^2$ *steps per interval, the complete finite sign instruments converge in trace distance to this infinite instrument, terminal states included, and the finite-family contrast is strictly negative for all sufficiently large* $d$*.*

*Proof sketch.* The linear force is relatively bounded by the coupled oscillator Hamiltonian, so $H$ is self-adjoint and finite Hermite polynomials lie in every $\mathcal{D}(H^k)$. Each of 24 steps of length $1/16$ applies a degree-16 Taylor polynomial of $H$, charges its remainder exactly, truncates above total occupation 16 and rounds to a $2^{-112}$ grid, charging both differences. Half-line probabilities are $r + s/\sqrt{2\pi}$ with exact rationals, and $\pi$ is enclosed by Machin's formula. For (iii), $P_d Q_0 P_d \to Q_0$ in the strong resolvent sense and the spectral measure of $Q_0$ has no atom at zero [72, 117, 114]. □

The $d = 2$ finite experiment has contrast about $-0.03166148$; replacing only its detector by the half-line effect gives about $-0.02526221$, and removing the cutoff gives the interval of Theorem 6.4. The signal survives both limits, which change its size; the interval is specific to zero native preparation current and the declared two-site action [72].

### 6.3 Occupation refinement and complete-output comparisons

Let $P_d$ project both oscillators onto occupations below $d$ and let $N = n_0 + n_1$ be the total oscillator occupation, a different object from the closure input $N$.

**Theorem 6.5** (Uniform pre-readout occupation bound)**.** *Let the matter start in an arbitrary state entangled with an untouched reference, let the scalar start in the fixed coupled vacuum, and let the finite preparation retain total occupations* $0, 2, \ldots, d-2$*. Then:*

*(i)* $M(s,t) = \langle e^{sN} \rangle$ *satisfies* $\partial_t M \le \frac{11}{16} \sinh(s)\, \partial_s M + \frac{25}{32} \sinh(s)\, M$ *through the matter measurement and feedback;*

*(ii) at total model time* $T = 3/2$ *the resulting moment bound holds with base* $\varrho = 113/64 \ge 7/4$ *and* $M < 13$*, where* $\varrho \le e^{s(T)}$*;*

*(iii)* $\|P_d H (1 - P_d)\| \le C_d = \frac{\kappa}{2}\sqrt{d(d+1)} + \frac{3\kappa}{2} d + \frac{1}{4}\sqrt{d}$*, and the retained-history trace distance up to the final scalar read is at most* $\left[(\sqrt{2v_d} + T C_d \sqrt{\tau_d})^2 + \tau_d\right]^{1/2}$ *with* $\tau_d = M/\varrho^d$ *and* $v_d$ *the omitted preparation probability, an* $O(d\, \varrho^{-d/2})$ *rate;*

*(iv) this bound is below* $0.011408$*,* $0.000002498$ *and* $6.175 \times 10^{-14}$ *at* $d = 32, 64, 128$*.*

*Proof sketch.* In common and difference modes only the squeezing term and the force change the number; weighted commutator, Cauchy and Young inequalities that hold for matter-entangled vectors give (i), and integration along $\tanh(s(t)/2) = e^{-11t/16}\tanh(s_0/2)$ from the initial value $\sqrt{80/17}$ at $\varrho_0 = 8$ gives (ii). The record dilation and feedback commute with $N$, and projected Duhamel comparison gives (iii) [73]. □

**Theorem 6.6** (Fourth moment and strong convergence of the sharp instrument)**.** *Let $w(N) = (N+2)^4$ and let $M(t) = \langle w(N)\rangle$ be summed over the unnormalized record branches. Then:*

*(i) $|\dot{M}| \le CM$ with $C = 8\kappa + 4|g| = 13/4$, and every finite preparation has $M(0) \le M_0 = 143748821/8192000$;*

*(ii) the record-dilation vectors immediately before the final read satisfy*

$$\|\Psi_d - \Psi\| \le \Big[(e_0 + I_T L_d)^2 + \frac{M_0 e^{CT}}{(d+2)^4}\Big]^{1/2} = O(1/d),$$

*with $I_T = \sqrt{M_0}\,(e^{CT/2}-1)/(C/2)$, $L_d = \kappa/(d+2) + |g|/(d+2)^{3/2}$ and $e_0$ the preparation error;*

*(iii) the sign projectors of $Q_{0,d}$ converge strongly to those of $Q_0$, and the complete unnormalized classical and quantum history, together with its native continuation, converges in trace norm for all three native currents and both feedback settings.*

*Proof sketch.* On occupation at most $m$, $\|H\psi\| \le 4(m+2)\|\psi\|$ and $H$ raises occupation by at most two, so finite-particle vectors are analytic and core convergence gives strong resolvent convergence [115, 117]. The moment coefficients are bounded by $8\kappa$ and, through a positive polynomial certificate, $4|g|$; the gauge measurement and rotations preserve $w(N)$ summed over branches [74]. □

Strong convergence supplies no finite-$d$ rate for the sharp readout. The complete-output certificate supplies fixed-state enclosures at $d \in \{2, 32, 64, 128\}$ for the zero-current preparation and the named rectangular occupation codes [75, 98].

**Theorem 6.7** (Complete outputs of the sharp detector)**.** *For zero native preparation current, with both matter records and both sign outcomes retained:*

*(i) for two complete post-read outputs with rank-one branches, the half trace norm of their difference is $\frac{1}{2}\sum_b \sqrt{(p_b+q_b)^2 - 4|z_b|^2}$, summed over earlier records, where $p_b$, $q_b$ are the branch weights and $z_b$ their overlap;*

*(ii) the complete continuous-output distances between the finite and the infinite sharp histories lie in the intervals of Table 3; at $d = 2$ the ideal 512-step source budget widens them to $[0.454336, 0.455973]$ with feedback and $[0.456824, 0.458461]$ without;*

*(iii) every output state supported in the rectangular occupation code of $d = 256$ lies at distance above $0.122$ from the true output of either prepared history;*

*(iv) for the vacuum control, whose two sign probabilities are both $1/2$, the best approximation by a state supported in the code with $d = 2m$ has distance between $\sqrt{3/(8\pi)}\,m^{-1/4}$ and $\sqrt{4/(3\pi\sqrt{3})}\,m^{-1/4}$, hence of order $d^{-1/4}$, and distance at most $\varepsilon$ requires $d \ge 9/(32\pi^2\varepsilon^4)$.*

*Proof sketch.* Item (i) is the trace norm of a difference of rank-one operators. For (iii), the operator exchanging the normalized inside and outside directions of each branch is a norm-one witness, giving distance at least $\sum_{a,b}\sqrt{a_{ab}t_{ab}} - \epsilon_{\mathrm{poly}}$; half-line Hermite matrix elements are rational up to $1/\sqrt{2\pi}$ [106]. For (iv) the omitted vacuum mass per sign branch obeys $1/(4\pi\sqrt{m}) \le T_{2m} \le 2/(3\pi\sqrt{3m})$ [75]. □

Table 3: Certified complete continuous-output distances between the finite sharp history and the infinite sharp history, zero native preparation current, rounded outward [70, 98].

| Onsite dimension $d$ | Feedback enabled | Feedback disabled |
|---|---|---|
| 2 | [0.455154, 0.455155] | [0.457642, 0.457643] |
| 32 | [0.203713, 0.226530] | [0.203935, 0.226752] |
| 64 | [0.180917, 0.180924] | [0.181090, 0.181097] |
| 128 | [0.152201, 0.152204] | [0.152341, 0.152344] |

At $d = 64$ the pre-readout error is below 0.000002498, so the sharp detector dominates the final error: accurate sign probabilities do not certify an accurate post-measurement state.

### 6.4 The smooth Ramsey instrument

A detector variant keeps the preparation, evolution, first record and feedback and replaces the final sharp read by a source Ramsey pointer with work-space Kraus maps

$$M_0 = e^{-i\pi/4}\cos(\tfrac{Q_0}{2} - \tfrac{\pi}{4}), \qquad M_1 = e^{i\pi/4}\cos(\tfrac{Q_0}{2} + \tfrac{\pi}{4}),$$

and effects $\frac{1}{2}(I \pm \sin Q_0)$ [76]. Bit zero denotes the plus effect, a detector meaning different from the sharp sign record, so the sharp-sign certificates do not carry over.

**Theorem 6.8** (Smooth instrument)**.** *For the declared initial vacuum family, uniformly over the three native current kicks and both feedback settings, with $T = 3/2$ and the constants of Theorem 6.6:*

*(i) the complete recorded instrument, including the pointer interaction and both unnormalized outcomes, lies within trace distance $\min(1, E_d)$ of its infinite counterpart, where $A_T = \sqrt{M_0}e^{CT/2}$ and*

$$E_d = \Big[\Big(e_0 + I_T L_d + \frac{A_T(e-1)}{2(d+2)^{3/2}}\Big)^2 + \frac{M_0 e^{CT+2}}{(d+2)^4}\Big]^{1/2} = O(1/d);$$

*(ii) with $n(d) = d^2$ symmetric steps per interval the combined ideal temporal and occupation error is $O(1/d)$;*

*(iii) on normalized inputs in the form domain of the infinite Hamiltonian the unconditional increment of the full energy is exactly $1/8$ in the declared dimensionless convention.*

*Proof sketch.* Along the pointer parameter the generator $\frac{1}{2}Z_{\mathrm{anc}}Q_0$ obeys $dM/ds \le 2M$, which gives the pointer term. For (iii), $f_0(q) = \cos(q/2 - \pi/4)$ and $f_1(q) = \cos(q/2 + \pi/4)$ satisfy $\sum_b f_b^2 = 1$, $\sum_b f_b f_b' = 0$ and $\sum_b (f_b')^2 = \frac{1}{4}$, so the kinetic term gains exactly $\frac{1}{8}\|\psi\|^2$ [76]. □

The bound is about 0.2063 at $d = 64$, 0.04096 at $d = 256$ and 0.008840 at $d = 1024$, and vacuous at the executed $d = 2$ and $d = 4$ [70]; the energy identity belongs to the infinite oscillator and fails at a finite cutoff [76, 87].

### 6.5 Continued instruments, compiled native repair and encoded memory

One execution engine carries the sharp and smooth protocols and the extension

$$U(1),\ P \to r_0,\ F(r_0),\ U(1/2),\ M \to r_1,\ U(1/2),\ M \to r_2,\ F(r_2),\ U(1/2),\ P \to r_3,$$

with $P$ the complete gauge instrument, $M$ the smooth instrument and $F$ the gauge feedback, model time $5/2$ and all sixteen unnormalized work/reference branches retained, with no intermediate work preparation [77]. Complete Kraus families preserve every old-prefix effect, and imported checkpoints are validated by prefix reexecution; this supplies neither physical quantum serialization nor cloning. Carrying a moment bound $A$ and an error $e$ from stage to stage, with $L_H = 9/[32(d+2)] + 1/[4(d+2)^{3/2}]$ and $L_S = 1/[2(d+2)^{3/2}]$, each unitary interval of length $t$ adds $A(e^{at}-1)L_H/a$ $(a = 13/8)$, each smooth read $A(\mathrm{e}-1)L_S$, and the final tail gives $E_d = [e^2 + A^2/(d+2)^4]^{1/2}$; with $n(d) = d^2$ the complete output distance is $O(1/d)$ for the scalar vacuum family and vacuous at the executed $d = 2, 4$; the classical native attachment enumerates 4,224 joint outcomes at each executed cutoff. A separate state-specific certificate retains all sixteen outputs for the blank-gauge and coupled vacuum preparation and each $j \in \{-1, 0, 1\}$ [78]; the rank-one trace-norm identity, complete-channel contractivity and directed rational evaluation at 512 splits per interval give

$$\max_{j\in\{-1,0,1\}} D_{d,j}^{\text{ideal product},\infty} < \begin{cases} 0.264092, & d = 2, \\ 0.053956, & d = 4. \end{cases}$$

This bound transfers through the native and protected-memory attachments for arbitrary admitted native and reference inputs with this independently prepared work state; it is not an arbitrary work-input bound and excludes floating execution and physical service errors.

**Coherent native attachment.** Let $L_h^{A,e}$ be the compiled native Kraus operator for law $A \in \{\mathsf{C}, \mathsf{E}\}$, including its fair coin, and $K_r^{j(h)}$ the work branch including the first record's current-controlled kick. Source routing implements seam $(0, 5)$ by exchanging port labels 1 and 5. With $e(0) = (0, 5)$ and $e(1) = (0, 1)$,

$$M_{h,r,s} = (L_s^{A,e(r_2)} L_h^{A,(0,1)}) \otimes K_r^{j(h)}, \qquad \sum_{h,r,s} M_{h,r,s}^\dagger M_{h,r,s} = I_{78} \otimes I_W,$$

complete on arbitrary entangled native, work and reference inputs [79]. The execution keeps all 512 native amplitudes, padding and abort outcomes, with 295,106 terminal basis outcomes for C and 820,166 for E. For input $(|\{2,3\}\rangle + e^{i\theta}|\{1,2\}\rangle)/\sqrt{2}$ and independent work, the native interference effect gives $p_C(\theta) = B + \frac{1+q}{8}\cos\theta$ and $p_E(\theta) = B$, with $q = \Pr(r_2 = 0 \mid j = 0)$; the C phase contrast is about 0.349153278 at $d = 2$ and 0.351351028 at $d = 4$, against zero for E. Cyclic Gray ordering of the shared nine-qubit controlled-flip macro reduces its source pulse count from 11,782 to 2,044 while preserving the full 512-dimensional operator, including global phase, inverse and padding [124, 79, 80].

**Active encoded memory.** The code $V|0\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $V|1\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ has the flagged Knill–Laflamme identity $V^\dagger L_h^\dagger L_h V = p_h I_2$, $p_h = k_h/4$, $k_h \in \{0, 1, 2\}$. A source encoder and record-controlled full-register unitary $R_h$ realize $R_h L_h V = \sqrt{p_h} V$ [80]. Correcting after each native event, with final seam $(2, 3)$ for $r_2 = 0$ and $(0, 1)$ otherwise,

$$M_{h,r,s}^{\text{corrected}}(V \otimes I_W) = \sqrt{p_h p_s^{e(r_2)}}(V \otimes K_r^{j(h)}),$$

which preserves the logical and reference marginal after summing all work and record outputs, including initially entangled inputs. The passive control without recovery has complete plus/minus output distance $1-\bar{q}$, with $q_j = \Pr(r_2 = 0 \mid j)$ and $\bar{q} = (q_{-1}+2q_0+q_1)/4$, so every later decoder errs by at least $\bar{q}/2$ on one input, while the corrected distance is one; for fixed seams $(0,1),(2,3)$ passive Bell fidelity is $1/2$ and corrected fidelity one. Encoding, blank registers, source precision, storage and record services are supplied. This is active protection of the stated qubit before readout, not arbitrary-carrier recovery, fault tolerance or an autonomous observer.

### 6.6 Observer readouts and the fixed-reference clock

The readouts below extract observer representations, modular flow and regional structure from the same executed history [81]. The quantum algebra, state pairing and access choices are premises, and the Born readout of these instruments is a declared service (Section 2.5).

**Proposition 6.9** (Observer representations of the executed history)**.** *Let $v_{ab}$ be the four unnormalized terminal vectors of the history at $d \in \{2,4\}$ and $\rho_{\mathrm{hist}} = \bigoplus_{a,b} |v_{ab}\rangle\langle v_{ab}|$ on $\bigoplus_{a,b} M_{4d^2}$.*

*(i) For a positive functional $\omega$ the Gelfand-Naimark-Segal (GNS) construction $\langle [A],[B]\rangle = \omega(A^*B)$, $\pi(X)[A] = [XA]$, $\Omega = [I]$ has dimension $dr$ on a block $M_d$ whose density has rank $r$. The record and work algebra has GNS dimension $4(4d^2)$, that is 64 and 256; the four committed record projectors give a four-dimensional classical GNS space; the site-0 observer that forgets both records has GNS dimension $d^2$, that is 4 and 16; and the regional maps $J_R[A] = [\mathrm{embed}_R(A)]$ are isometric and intertwine the regional actions.*

*(ii) For faithful $\rho_0$ the modular group $\sigma_s(A) = \rho_0^{is} A \rho_0^{-is}$ moves number observables; the commutative record algebra has identity flow; states conditioned on the sharp terminal bit are nonfaithful on the full scalar algebra, with rank at most $d/2$.*

*(iii) At $d = 2$, $\rho_0 = \frac{1}{2}(I - \delta X)$ with $\delta = P(b=0) - P(b=1)$ and modular frequency $\nu = \log(P(b=0)/P(b=1))$. A source-compiled continuation of every terminal branch by $Z(\pi/4)$, $X(s\nu/2)$ and a fresh number pointer preserves the old two-bit record marginal and has readout law $P(1 \mid s) = \frac{1}{2}(m + \delta \sin(\nu s))$, where $m$ is the total branch mass, equal to one for the normalized history.*

*(iv) Scalar regional restrictions are completely positive conditional expectations, normalized partial trace followed by identity embedding; they are unital and idempotent and in general not multiplicative. The centered components of the actual Hamiltonian give an interaction graph that is a path through scalar site 1, scalar site 0 and the shared charged-matter and link sector.*

*Proof sketch.* For $\rho = V\,\mathrm{diag}(\lambda)V^*$ of rank $r$ the map $[A] \mapsto AV\sqrt{\mathrm{diag}(\lambda)}$ is the GNS isometry [101]; the modular identities, including the finite Kubo-Martin-Schwinger (KMS) condition, are checked with this sign convention [102], and at $d = 2$ the probe implements the Schrödinger dual of $\sigma_s$ [81]. □

In the executed history $\nu \approx 0.5292248981159262$ and $\delta \approx 0.2586046272446501$ [99]; ensemble frequencies read a modular interval on a declared monotone window, and full-cycle aliases are indistinguishable. The post-measurement site-0 state taken as its own modular reference fails to reproduce the interacting dynamics: its modular flow preserves it while its density changes under $H$ with derivative norms about 0.2086 at $d = 2$ and 0.6603 at $d = 4$ [87]. A fixed reference on the full work algebra removes this obstruction [82].

**Proposition 6.10** (Fixed-reference modular relation)**.** *Let $H$ be the finite interacting Hamiltonian on the work space of dimension $4d^2$, $\beta > 0$ and $\sigma_\beta = e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$. Then*

$$\sigma_\beta^{is} A\, \sigma_\beta^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \alpha_{-\beta s}(A), \qquad \alpha_t(A) = e^{itH} A e^{-itH},$$

*so continuous Hamiltonian segments correspond to modular flow at $t = -\beta s$. For the full matrix algebra this identification of the two automorphism groups forces the Gibbs form of the reference. With record blocks $\Sigma_\beta = \bigoplus_r q_r \sigma_\beta$, $q_r > 0$, the weights cancel and every record projector is fixed. The actual state $\omega(t) = e^{-itH}\omega e^{itH}$ can change while $\sigma_\beta$ is stationary, and $\operatorname{Tr}(\omega(t)E) = \operatorname{Tr}(\omega\, \sigma_{-t/\beta}(E))$ for every effect $E$.*

*Proof sketch.* $\log \sigma_\beta = -\beta H - \log Z_\beta\, I$, and equality of the generators requires $[\log \sigma + \beta H, A] = 0$ for every matrix $A$, whose commutant is scalar [82, 105]. □

The reference, $\beta$ and full-work access are declared, and measurements and feedback act outside this one-parameter group. The receipt uses $\beta = 1$ and reference weights $(1/10, 2/10, 3/10, 4/10)$, with source continuations at model times 0, 1/2 and 1 appending a gauge-number pointer to every post-sharp branch and retaining all eight outcomes [82].

## 6.7 History repair of the completed record

**Theorem 6.11** (Local repair of the completed history)**.** *Fix $d \in \{2, 4\}$, the native current $j$ and the feedback setting. Let twelve event nodes carry unnormalized work vectors subject to the compatibility equations $x_{\mathrm{root}} = \mathrm{prepared}(j)$, $x_1 = U_1 x_{\mathrm{root}}$, $x_{(a)} = P_a x_1$, $y_{(a)} = F_a x_{(a)}$, $z_{(a)} = U_2 y_{(a)}$ and $w_{(a,b)} = B_b z_{(a)}$, of depths 0 to 5. Then:*

*(i) exactly one assignment satisfies all equations;*

*(ii) the common fixed set of the local updates $x_v \leftarrow A_v x_{\mathrm{parent}(v)}$ equals that compatibility set;*

*(iii) every node-fair schedule reaches the compatible assignment from arbitrary initial vectors after finitely many updates and stays there; five covering rounds suffice, a forward topological pass needs eleven updates, and a reverse order may need 55;*

*(iv) with subtree weights $w_v$, equal to 11, 5, 4, 3 and 1 at depths one to five, the potential $\Phi = \sum_{v \neq \mathrm{root}} w_v \|x_v - A_v x_{\mathrm{parent}(v)}\|$ decreases by at least the repaired residual at every active ideal repair;*

*(v) every deterministic extraction from the completed assignment, including branch probabilities, terminal densities, GNS representations and modular parameters, is independent of the schedule.*

*Proof sketch.* Agreement at depth zero forces agreement at each further depth. A node whose ancestors are correct becomes correct permanently at its next selection, and after the $k$th covering round every node of depth at most $k$ is correct. Updating $v$ zeroes its residual and changes each child's residual by at most $\|r_v\|$, since all maps are contractions [83]. □

The solved object is the complete branching instrument law for given apparatus inputs; its uniqueness selects no realized record pair, and the auxiliary solver is a host computation distinct from a native hardware process.

### 6.8 A same-action particle corner and regional interfaces

A finite attachment uses the supplied Weyl/contact CAR action $H = d\Gamma[-(1/8)\sigma\cdot\sin k]+\sum_x n_{x0}n_{x1}$ on its complete two-fermion sector. An isolated bound-pair band has an independently checked position-space wavepacket, number and energy currents, and complete contact/scattering spectral records. Its topology is trivial; it supplies neither a nontrivial native defect invariant nor a physical mass assignment. Its exact rational dispersion enclosure gives a kinetic coordinate mass different from the value inferred from its energy difference and the same speed parameter. All of these tests use this one action.

At total momentum $(\pi,\pi,\pi)$, an on-site singlet $d$ and a zero-contact singlet $r$ have energies one and zero, respectively. They have the same particle number and total momentum. A supplied isometry maps the native configurations $\{1,2\}$ and $\{7,2\}$ to $(r,d)$. The native product reference restricts to $(264/7505)\,\mathrm{diag}(25/33, 8/33)$, furnishing an exactly faithful Gibbs reference on this invariant $M_2$ corner, with $\beta = \log(25/8)$ and Heisenberg time $t = -\beta s$.

The executed native protocol retains an initial coarse corner herald, a balanced probe, every uniformly chosen C/E seam outcome, a second herald, the record-derived clock, an analyzer and the final current/coin energy label. Rejection, leakage, zero slots and export aborts remain outputs. Each coarse herald is a 2,068-pulse source word checked on all 512 original register inputs. Preparation error is charged before the first herald, and the complete average half-trace bound is below 0.018762 without success conditioning. The selected two levels do not prepare or encode the full spatial field, and this corner supplies no clock units, many-site isometry or full-field native reference.

The same local field action also has signed CAR restrictions retaining all regional number sectors zero, one and two. Their Kraus maps prove complete positivity, trace preservation, untouched-reference compatibility and composition. Number-preserving local observables have true algebra inclusions; compressed odd operators do not inherit a CAR homomorphism. The corresponding ambient conditional expectations satisfy tower laws but are not multiplicative. This instantiates the CP restriction interface; a scalar-bottom homomorphic retraction is impossible on this algebra.

Increasing open boxes admit exterior-square inclusions $J$ satisfying $J^* H_{\mathrm{large}} J = H_{\mathrm{small}}$. Boundary hopping makes $H_{\mathrm{large}}J - JH_{\mathrm{small}} \neq 0$, so this equality alone does not intertwine dynamics. The complete two-read feedback histories instead use the existing finite-boundary and product-formula bounds, followed by contractive regional restriction. Every quantum record and discarded local particle sector remains in that comparison. The growing boxes hold spacing fixed and do not reconstruct Lorentzian geometry.

On the same full finite action, local energy continuity and quantum Gibbs/entropy identities supply further finite premises. The regional modular generator fails the proposed affine identification with bare regional energy even after number and number-squared terms are allowed. The full interacting Weyl action also couples every proper momentum pocket to its complement. Consequently neither a regional clock nor a chiral sector follows by merely deleting terms or modes. The source-bound construction is recorded with the executable model [10].

**The separate continuing native memory.** The endpoint-biased 60-code process also continues through every forbidden antipodal flag by a supplied coherent decode/re-encode service. It preserves the unknown logical state and keeps boundary, zero and abort outcomes. Positive endpoint coins leave the exact flagged-code conditions unchanged. Every all-seam correctable sharp-$q = 1$ source qubit, without persistent ancillary storage of its logical state, has both core occupation compressions equal to $I_2/2$, so its core kernel is independent of the history-selected code. For this entire class, the exact absorption potential $h(c) = 1 + \sum_d Q(c,d)h(d)$ has a unique maximum $H \simeq 203.025378$

attempted repairs at core $(0,1)$. Every counted reset raises $h$ by at most $H$, so telescoping gives $\mathbb{E}N_n \geq \max\{0,(n-h(c_0))/H\}$. Resetting only at the boundary to $(0,1)$ attains asymptotic rate $1/H$. Exact Bellman checks through horizon 64 are recorded separately from this all-horizon asymptotic argument. The compiled family attains the bound; arbitrary exact code choices cannot improve it under the same law and reset contract. Active recovery is supplied after every repair attempt; neither reset counts nor source-pulse counts are physical energy. The complementary fair-coin mobile-memory construction has a different law and a mean edge excursion of 150 attempts. Its finite-horizon theorem is separate from the biased asymptotic result. Neither controlled memory is the interacting field pair or a native topological particle. The complete service and its bounds are retained with the executable model [10]. A separately certified variant omits 3,120 identity recoveries among 11,580 positive flags, saving about 48,391 supplied pulses per stationary native attempt. It preserves the admitted logical/reference channel and does not change the reset rate.

**Smaller exact controls and a separate clock.** At side four and zero total momentum, the contact-generated cyclic subspace has seven dimensions. Its free spectral weights are $(1,3,3,2,3,3,1)/16$ at energies $2c(-\sqrt{3},-\sqrt{2},-1,0,1,\sqrt{2},\sqrt{3})$. The resulting rank-one resolvent gives exact rational pole enclosures and intertwines complete contact/complement histories with arbitrary references. Other spatial observations require the full field sector.

**Complete Fock dynamics and the thermodynamic limit.** The same local action acts on all $2^{2V}$ occupations of a box with $V$ sites. For $E$ spatial bonds, each of norm at most $c$, its full-Fock stability bound is $-cEI \leq H \leq (cE+UV)I$, with $E \leq 3V$. The completed two-mode hopping, contact and fermionic-route words apply to all number sectors and arbitrary references. With $B = cE+UV$, a symmetric $n$-step split has ideal operator error at most $B^3|t|^3/(3n^2)$. The corresponding complete-history error is summed over segments without conditioning on outcomes. This bound depends on volume; a bound restricted to two particles is not substituted for it.

Removing the onsite contact terms by a local interaction picture leaves even, time-dependent two-site bonds of norm at most $c$. The commutator-chain estimate for disjoint, initially commuting observables is

$$\|[\tau_t(A_X), B_Y]\| \leq \|A_X\|\|B_Y\|\,|X|\,2^{-d(X,Y)}(e^{48c|t|}-1).$$

At least one of the observables is even; odd–odd locality uses the graded CAR relation. For a normalized observable in a radius-$r$ cube and strictly padded radius $R > r$, the comparison to any larger box is bounded by

$$\min\{2,\, 6c|X|(2R+1)^2|t|\,2^{-(R-r)}(e^{48c|t|}-1)\}.$$

The limit gives a strongly continuous automorphism group on the quasi-local CAR algebra at fixed spacing. The contact contribution $[H_U - H_0, a_0^* a_2] = U a_0^* a_2 (n_1 - n_3)$ remains nonzero in that limit. Thus the construction is an interacting lattice field with three spatial coordinates and continuous Hamiltonian model time. Neither an infinite-volume physical state nor a Lorentzian chiral continuum is selected. The executable full-Fock control and rational locality bounds are retained with the executable model [10].

A corrected memory branch and a clock instrument on a separate factor compose exactly, including correlated logical/clock/reference inputs. The clock requires its preparation, control, readout and event association; the tensor identity does not establish a stationary product reference, physical duration, dispersion relation or memory mass.

**Extension and obstruction for the full band.** The same native record-rate estimator drives counted local source words on the full 8,128-state field sector, with all 64 pair momenta and all four terminal occupation-pointer outputs. Its uniform complete average half-diamond bound is below 0.037991 for $|s| \le 1$. This requires a supplied globally normalized field Gibbs reference and a rate-transfer actuator; it is not an inherited native encoding. Every finite independent product of the canonical native reference has integer modular gaps in units $\log(25/8)$, whereas the same zero-momentum pair has $13/12 < E/U < 3/2$ relative to an eta dark state. The inherited intertwiner therefore fails, with generator error greater than $1/24$ in the stated units. A separate native coherent state has an exactly reference-matched exterior through every native record history and a computed finite contrast lifetime, but its coherent core is immobile and its modular-energy excess is zero.

### 6.9 One ordered spin state, continuum clock and spectral readout

A separate source trajectory gives a canonical field in three continuous spatial dimensions and time. Its starting point is the declared two-flavor field extension on the cubic lattice, with the complete colored repair instrument. On the single-occupancy subspace every repair acts as identity on spin and retains its fair coin record. For a box with $E$ bonds, the centered source with $c_E = JE^4$ and $U_E = JE^8$ converges, including its complete finite histories, to

$$H_J = J \sum_{\langle xy \rangle} (\mathbf{S}_x \cdot \mathbf{S}_y - 1/4), \qquad J > 0.$$

This is an interacting fixed-spacing spin dynamics. Its cubic lattice and coherent field encoding are specified source inputs, distinct from the one-bit occupation process on the spherical federation.

To select the ordered reference, prepare the unique finite ground of $H_B = H_J - B \sum_x \epsilon_x S^1_x$, where $B > 0$ and $\epsilon_x = (-1)^{x_1+x_2+x_3}$. The finite ground has a terminating source-control preparation algorithm. Taking volume to infinity before $B \downarrow 0$ gives the state used in the ordered-zero-field route. The matched infrared inequalities give staggered order $m > 1/1000$ and filtered excitation energies between positive multiples of momentum [127]. They do not assert a sharp spectral pole.

Let $K$ implement zero-field evolution in this ground representation. With $X = px$, $T = Jpt$, the fields

$$Q_p(F) = p^2 \sum_x \epsilon_x F(px) S^3_x, \qquad P_p(G) = -p \sum_x G(px) S^2_x$$

have a common bounded-history subsequential limit. The resulting regular Weyl field obeys the canonical relation with coefficient $m \int FG$, admits continuous spatial translations and positive time translations, and has a finite propagation cone. Both Stone fields create nonzero positive-energy weight at arbitrarily small momentum. These statements concern the same ordered state and clock $K/(Jp)$; they do not identify its limiting dynamics with the separate free Weyl branch.

**Selection-field and zero-field clocks.** The actual finite source admits a uniform positive-field preparation bound. In its Marshall frame, global Hadamard conjugation gives

$$\widetilde{H}_B = \frac{J}{4} \sum_{xy} (X_x X_y - Y_x Y_y - Z_x Z_y - I) - \frac{B}{2} \sum_x Z_x.$$

Its Lee–Yang coefficients satisfy $w_z = |w_x| = |w_y| = J/4$, so the full spectral gap is at least $g = B/8$ [125]. The same bound holds when all bonds are multiplied by $s \in [0, 1]$. This gives a gapped path from the staggered product ground to the actual selected ground. With $D = J|\mathcal{E}|$, the rank-one

adiabatic error in half trace norm is at most $(2D/g^2 + 7D^2/g^3)/T$ [126]. Time discretization, bond-product splitting and native port synthesis have explicit additional bounds. Their sum gives a polynomial finite-volume resource bound at fixed B, using the declared fresh blank input and coherent services. A complete two-site 140-pulse witness is executed; the large adiabatic tapes are specified (`model/spin_twist_closure/FINITE_FIELD_GAP.md`, [2]).

The gap also persists when only the bonds crossing a box boundary are interpolated. Gaussian approximate spectral flow and the source locality bound then compare an interior marginal with the cut-box ground. For m observation sites at distance d from M crossing bonds, set $u = d/(144J)$. The half trace distance is at most

$$\min\left\{1, JM\left[\frac{mu}{2}e^{-d/3} + \left(\sqrt{\frac{2u}{\pi g}} + \frac{1}{g}\right)e^{-gu/2}\right]\right\}.$$

This is uniform in exterior volume and tends effectively to zero with the box radius at fixed B. It passes to complete local instruments and independent references. It supplies no B-to-zero marginal rate (`model/spin_fixed_field_locality`, [2]).

A joint prescription instead retains the field in both state and clock: $p = n^{-2}$, $B = Jn^{-5}$ and $L_{B,p} = K_B/(Jp)$. The gap and local source commutators bound the order variance by $48N_0\|f\|_\infty^2/n$ and the normalized bond variance by $(792/n + 24/n^6)a^2V_0$. The same clock transports their scalar phases through bounded histories. Transverse infrared bounds and a staged mixed-commutator argument construct two regular canonical pairs $(Q_3, P_2)$ and $(Q_2, P_3)$, with full internal U(1), continuous spatial translations and positive time evolution. Fixed-scale native preparation, clocks and Pauli histories retain all records with a computable total error. This is a chosen joint prescription; continuum uniqueness and its effective convergence rate are unproved (`model/spin_joint_field_limit`, [2]).

For its actual source current $Z_p(G) = p^3 \sum_{x,i} D_i^p G\, C_{x,i}^2$,

$$i[L_{B,p}, P_2(G)] = -p^{-2}Z_p(G) - n^{-1}Q_3(G).$$

A contraction energy filter $\rho$ supported in $[0, M]$ gives

$$\|\rho(L_{B,p})Z_p(G)\Omega_B\| \le p^2(M\|P_2(G)\Omega_B\| + n^{-1}\|Q_3(G)\Omega_B\|).$$

The stronger source-norm cancellation $\|\rho(L_{B,p})[Z_p(G) + p^2 aQ_3(\Delta G)]\Omega_B\| = o(p^2)$ would identify the proposed linear force on these filtered vectors. It is not supplied by the $O(p^2)$ bound. Complete Pauli scores measure the unnormalized squared residual, with their full error divided by $p^4$ when testing this limit. The exact finite control includes the actual two-spin spectrum and all 64 labelled setting/pointer records; no continuum mass or pole is inferred (`model/spin_joint_field_limit/FORCE_READOUT.md`, [2]).

Both exchange-current equations also give a same-source susceptibility test. For a common positive-window filter, let $q = RQ\Omega$, $a = RP\Omega$, $c = \mathrm{Im}\langle q, a\rangle$ and $\chi_Q = \|K^{-1/2}q\|^2$, $h_Q = \|K^{1/2}q\|^2$, with analogous P quantities. The exact nonnegative defects are $h_Q - c^2/\chi_P$ and $h_P - c^2/\chi_Q$. Complete Q, P and $Q \pm iP$ preparations followed by the same clock measure these moments with bounded scores and retained failures. Their smallness would bound the original weight outside a source-determined energy tube; continuum smallness is not inferred from the finite readout. Positive-weight coherent time filters also construct a resolvent-moment hierarchy with explicit continuous-score errors, without assuming an executable response projection (`model/spin_joint_field_limit/WARD_DYNAMICS.md` and `model/spin_dynamical_memory/SPECTRAL_BRANCH.md`, [2]).

The first circle coefficient of $\arctan(X_\theta)/\pi$ gives bounded charged records. For real even tests in caps of radius $\delta|k_0|$ about $\pm k_0$, $0 < \delta \le 1/4$, inversion divides each charge's band weight equally. Every charge and cap has original probability at least $1/774400$ and energy in $[9|k_0|/774400, 580800|k_0|]$. Seven records retain all complements. Narrow caps control spatial phases, without a sharp time phase. A six-site periodic source resolves order-three translations in thirteen records, with positive energy width (`model/spin_joint_field_limit/DIRECTED_CHARGED_BANDS.md`, [2]).

An E seam scan and record-controlled inverse signed transport exactly measure one-hole position: $\rho \mapsto \sum_x P_x \rho P_x$. Internal spin, references and every fine record are retained; classical export dephases positions. Through identical sector-preserving controls, m uniform global repair attempts have complete source/reference/archive trace error $2m\sqrt{(1+\vartheta)d_{\max}/|E|}$ relative to identity repair with the same seam/law/coin records; $\vartheta$ is the E-law weight. This vanishes for $|E| \asymp p^{-3}$, $m = O(p^{-1})$. General Hubbard hopping does not preserve this sector (`model/source_family_compatibility/REPAIR_TRAJECTORY.md`, [2]).

On the entire Fock space, let $b_j = \langle D \rangle_j$ on each ideal prefix, $D = \sum_x 1_{n_x \neq 1}$. Complete repair trace error is at most $\min\{2, 2\sum_j \sqrt{2d_{\max} b_j/|E|}\}$. For $H = T + (U/2)D$, energy at most C and $a = U/2 - cd_{\max} > 0$, $b_j \le 2C/a + 4c^2|E|d_{\max}/a^2$; interventions must count injected work (`model/source_family_compatibility/DEFECT_BUDGET.md`, [2]).

At fixed c,U, local readouts instead have half-trace error $\min\{1, m[6(2R+1)^3/|\mathcal{E}| + \epsilon_R]\}$, where $\epsilon_R$ is the complete dual locality error for a radius-R buffer. Local records and references remain. Distant raw records stay in the archive and are unavailable to feedback.

For $\|f\| = 1$, $A_+ = a^*(f)$ and $A_- = a(f)$, supplied coherent services implement the complete centered-source interference instrument

$$J_{q,b}(t) = \tfrac{1}{2}[A_q + b\tau_{-t}(A_q)], \qquad q = \pm,\ b = \pm 1, \quad \sum_{q,b} J_{q,b}^* J_{q,b} = I.$$

Geometric times with ratio $s = (1+\eta\Delta)^{-1}$ and dummy records approximate $Q_\eta = \int \eta^2/(\eta^2 + E^2)\, d\nu(E)$ with error $\Delta\sqrt{M_2}/2 + s^m$. Collar, marginal and service errors add. Pressure secants certify infinite-ground scores under a strict common margin. On the full dimer, 8194 proper native terminal records and every first-failure record give original weight above 0.595 in $[0,1]$. Rational spectral intervals and complete-channel error below $10^{-5}$ certify this finite-ground result; the infinite collar is unexecuted (`model/source_family_compatibility/NATIVE_LORENTZIAN_READOUT.md`, [2]). Odd-field locality and matrix Bernstein bounds give $\|Q_\eta(k) - Q_\eta(k_*)\| \le Cd[1 + \log(1/d) + c/\eta]$, $0 < d = \|k - k_*\|_1 \le 1$. On one fixed normalized ground, a positive nodal atom is equivalent to bounded physical-window survival for packets with $d \le aK$ under $c = v/a$. Its value is $m_0/(\alpha^2 + m_0)$, where $m_0 I = R^* 1_{\{0\}}(L_\perp) R$ and $\alpha = U/2$; its positivity is unproved. Fixed-reference source homogeneity makes every full joint ballistic limit dilation covariant, excluding positive original weight on nonzero relativistic mass shells. A whole geometric-refinement limit gives only discrete covariance. This identity supplies neither convergence nor a running reconstruction (`model/source_family_compatibility/MASS_SCALE_SELECTION.md`, [2]).

On an interior cubic edge, two-fermion calibration gives $p(t) = c^2t^2/2 - c^2(U^2 + 25c^2)t^4/24 + O(t^6)$. Exact signed walks and a volume-independent remainder identify positive $U/c$ from two times. A finite empty collar controls arbitrary exterior matter; its preparation and clock are supplied (`model/source_family_compatibility/BULK_ACTION_IDENTIFICATION.md`, [2]).

Aligned-hole controls use a separate nonrelativistic reference [118].

For the same $N$-site input and maximum branch duration $T_{\mathrm{abs}}$, Duhamel and complete-channel contraction bound the half-trace change under $H_B \mapsto H_J$ by $\min\{1, BNT_{\mathrm{abs}}/2\}$. Locality removes

unrelated exterior volume. This complete-record comparison requires separate ground preparation and certified energy origins.

For $\varepsilon_x = \frac{1}{2}\sum_{y\sim x} h_{xy} - B\epsilon_x S^1_x$ and $j_{xy} = i[\varepsilon_x, \varepsilon_y]$, exact finite evolution gives

$$\dot{\varepsilon}_x + \sum_y j_{xy} = 0, \qquad \frac{d}{dt}\sum_x f_x\varepsilon_x = \sum_{x<y}(f_y - f_x)j_{xy}.$$

The Hermitian currents are antisymmetric and vanish beyond graph distance two. Adjacent bond currents contain $i[h_{xy}, h_{yz}] = J^2\mathbf{S}_x \cdot (\mathbf{S}_y \times \mathbf{S}_z)$. For the bipartite staggered field the site-pair currents agree with their zero-field values, although the energy densities and evolution differ. An intervention has a separately counted source-energy jump. These identities supply energy continuity, not a symmetric Lorentz stress tensor or a responsive geometry.

**Relaxed response in the same continuum.** Let $L$ denote the continuum positive generator obtained from $K/(Jp)$. For either actual continuum field vector $a = A\Omega$, its inverse-energy susceptibility $\chi_A = \|L^{-1/2}a\|^2$ is positive and finite on the specified momentum annulus. The exact variational identity is

$$\inf_{2\,\mathrm{Re}\langle a,v\rangle = b} \|L^{1/2}v\|^2 = \frac{b^2}{4\chi_A}.$$

The vectors are perpendicular to $\ker L$. For $\eta > 0$, writing $\chi_\eta = \langle a, (L+\eta)^{-1}a\rangle$, $v_\eta = b(L+\eta)^{-1}a/(2\chi_\eta)$ gives a feasible sequence whose energy tends to this infimum. Attainment at $\eta = 0$ requires $L^{-1}a$ to belong to the Hilbert space. Positive Fourier averages of the same Heisenberg evolution implement the regularized filter, retaining complementary outcomes. Finite spectral certificates bound its response without treating bare twist energy as the relaxed stiffness. The bounded source probes $\sin(sA_p)/s$ give its regularized susceptibility through the same bounded-history limit at fixed $s$ and $\eta$. Taking $s \to 0$ and then $\eta \downarrow 0$ recovers the actual continuum coefficient, without a finite-state convergence rate. The quantity above is the cost of an infinitesimal response, not an excitation mass.

**Testing a particle branch.** On an even periodic box, set $K_B = H_B - E_0(H_B)$. The commuting projectors $Q_k$ for the combined translation and internal rotation, and the complete energy-clock operators

$$B_j(K_B) = \frac{1}{M}\sum_{r=0}^{M-1} e^{2\pi ijr/M} e^{-ir\tau K_B}$$

give joint effects $Q_k B_j^* B_j$ summing to the identity. Every unsuccessful preparation and source-control export retains its own outcome. For a phase margin $\delta$ measured in turns, the summed clock leakage is bounded by $\min\{1, [2(M\delta)^2]^{-1} + (2M\delta)^{-1}\}$. Inner and outer bin sets therefore bound the original, unnormalized spectral weight in a prescribed energy–momentum tube, with source, input and sampling errors added. The clock step must also resolve aliases across the full finite spectrum.

The unchanged positive-field eight-site cube supplies a finite control: its complete 256-vector certificate and 128-bin clock bound the weight in one reflection character and excitation interval $[4/5, 7/5]$ below by $1/5$. This interval contains the first excitation cluster near 1.0922846777 in the specified source units. The cube's characters are reflections, not small translation momenta; the larger native 128-bin tape is specified rather than reported executed. A positive continuum particle component additionally needs a uniform nonzero fraction of the original response in shrinking tubes about one common dispersion, with readout error small relative to that response. For width $Jp\delta_p$, the clock requires $M\tau Jp\delta_p \to \infty$. Fixed-history convergence alone supplies no such growing-time

estimate. Lorentz covariance, surviving connected interactions, massive species and the corresponding gravitational stress are separate conclusions requiring proof. The complete constructions and finite certificates are in the executable model [10].

### 6.10 Compiled instruments on interacting sources

Three executed components act on specified interacting sources. This subsection gives their resources, retained records and error bounds; the owning papers give the mathematics.

**The composite probe.** On the authenticated hidden-frame ground preparation of the open eight-site Weyl/contact source, whose inherited 262,137 pulses are not reexecuted, a selector word of 1,008 source pulses in 56 contraction blocks with five work bits implements the success Kraus operator $C_0/3$ of the composite charged probe of the particle paper [118]. It retains all 7,776 raw selector records, native export failures, inherited preparation aborts and 2,097,152 coherent joint amplitudes over 256 reference columns. Its complete operator error is below $1.6 \times 10^{-13}$ on every field input and untouched reference, and its success probability is approximately 0.08261623248916 [37]. It detects the total response, with no executed clock resolving the 138 spectral lines.

**Intervening observations.** On the compact 180-mode core the complete-history comparison of the particle paper [118] retains every native record, detector result, feedback choice and the final field and reference state, with the memory continuing by its actual transition law. Its rational schedule for three intervals of duration 1/3 and complete accuracy 1/100 requires 46,370,150,349 native steps, a finite resource bound. An independent execution of the 128-record law with binary filters of probability 3/8 retains all 12 old memory inputs and 512 fine labels on a four-mode probe, reaching 52,488 class-history rows at four steps per interval, and obeys the bound [36].

**The regional modular actuator.** On the prepared open plaquette, a palindromic source word for the regional modular polynomial at dimensionless parameter 1/2 executes all 39 Pauli terms on all 16 regional input columns in 37,568 pulses, with the same number of proper-code loads and exports. It retains the eight preparation record bits, every earlier abort, every first-failure output, the exterior field bits and any untouched reference, and its complete half-diamond error is below $6.4 \times 10^{-9}$, with no postselection on source success [38]. The Einstein paper states the theorem [8]; this operational clock identifies no geometric flow or physical duration.

## 7 One repair history: geometry, continuum, clocks and fields

This section derives what one native repair history supplies on the carrier and on the refining spherical source. Its statements are dimensionless carrier displacements, attempt counts and event counts; physical position, length scale and propagation speed need their own operational identification.

### 7.1 Native geometry, causal cones and field relations

**Proposition 7.1** (Native geometric readout)**.** *(i) Exact enumeration of all 4,680 state, seam and coin branches gives the one-attempt mean $T = I - L/60$ under uniform seam selection.*

*(ii) With $Q$ removing the constant mode and $C_n = QT^{2n}Q$, the normalized response $12C_n/\operatorname{Tr} C_n$ tends to $G = 4P_{\mathrm{slow}}$, where $P_{\mathrm{slow}}$ projects onto the rank-three band $5 - \sqrt{5}$; all eleven nonconstant modes are present at every finite attempt count.*

*(iii)* *For the readout $x(n) = \sum_p n_p u_p$ with frame directions $u_p$, a committed current $j$ on the seam $(a, b)$ gives the relative displacement $x(n') - x(n) = j(u_b - u_a)$, computable from the record without hidden loads.*

*(iv)* *The two-event controller spans two displacement directions. A declared continuation that sends the final scalar bit to $(0, 5)$ or $(0, 7)$ after the first event re-executes all 1,056 second events, and its three displacement directions $(2, 0, 0)$, $(1, 1-\varphi, \varphi)$, $(1, 1-\varphi, -\varphi)$, with $\varphi = (1+\sqrt{5})/2$, have determinant 4.*

**Proposition 7.2** (Native causal cones)**.** *For a fixed seam schedule and shared coins, the load at a port $v$ after the schedule is a function of the initial loads and coins in its backward dependency set. An initial difference supported on $D$ influences $v$ only along a time-ordered path, so $\mathrm{dist}(D, v) \leq$ edge crossings on that path $\leq$ elapsed seam events, and at most one edge is crossed per causal layer of disjoint simultaneous events.*

Exhaustive checks cover all 78 states for 27 schedule and coin pairs, and two executed examples show influence at graph distance two arriving only after two properly ordered seams [85]. One edge per causal layer is a native signal speed of at most one seam per ordered repair event; remote repairs can carry a difference to a receiver before its first local tick [14]. Units are observer conventions [87]: if a common operational propagation law fixed a dimensionless speed $c_{\mathrm{model}}$, choosing the SI speed of light would impose only $L/T = c_{\mathrm{SI}}/c_{\mathrm{model}}$ on the model's length and time units, an operational action constant would add $ET = \hbar_{\mathrm{SI}}/\hbar_{\mathrm{model}}$, and one overall scale would stay free. The construction supplies no $c_{\mathrm{model}}$ relating its native distance readout to its modular clock.

**Proposition 7.3** (Native field relations)**.** *(i) Every event satisfies the discrete continuity law $x' - x = -b_e j$ with $b_e = \mathbf{e}_a - \mathbf{e}_b$.*

*(ii)* *Conditional on the past and the selected seam, $\mathbb{E}[j] = d/2$ and $\mathrm{Var}(j) = (|d| \bmod 2)/4$.*

*(iii)* *Under the uniform calibration the mean load obeys $\mu_{k+1} - \mu_k = -L\mu_k/60$, exactly, by $\mathcal{K}X = XT$ for the 78-state kernel $\mathcal{K}$ and load matrix $X$; the deviation $V = \frac{1}{2}\|\mu - \frac{1}{6}\mathbf{1}\|^2$ decreases strictly unless the mean is constant, while every binary trajectory keeps $\|x - \frac{1}{6}\mathbf{1}\|^2 = 5/3$.*

*(iv)* *The slow band has mean multiplier $q = (55+\sqrt{5})/60 \approx 0.953934466291663$, every nonconstant multiplier lies in $(0, 1)$, and the second-difference coefficient $q + 1/q - 2 > 0$ has the opposite sign to the coefficient $-4\sin^2(\omega\Delta t/2) \leq 0$ of an undamped oscillator; no positive rescaling of the attempt count turns this closed mean law into a wave law.*

*Proof sketch of Propositions 7.1 to 7.3.* The fair coin averages a seam to its endpoint mean, which gives the mean operator and the conditional current moments; the other two positive response eigenvalues have strictly smaller magnitude than $1 - (5 - \sqrt{5})/60$, which gives the limit in Proposition 7.1(ii). Conservation gives the displacement identity. For the causal cones, scanning the schedule backwards and adding both endpoints of every seam that touches the dependency set proves the dependency claim by induction [84, 85, 86]. □

The rank-three readout is a local load-weighted coordinate. An absolute value needs an initial reference, while every current fixes its displacement. On the native quantum carrier, each complete fine branch $K_h$ is a scaled partial injection. Applying the classical increment identity to each supported basis column proves, for $d(h)$ the recorded net displacement,

$$[X_i, K_h] = d_i(h)K_h, \qquad e^{iq\cdot X}K_h e^{-iq\cdot X} = e^{iq\cdot d(h)}K_h.$$

These are operator identities and retain arbitrary untouched references. The branch acceptance depends on the initial load. A unitary on the whole finite carrier cannot instead satisfy $U^*X_iU = X_i + a_iI$ for nonzero $a_i$: taking the trace gives a contradiction and the operator-norm residual is at least $|a_i|$.

The protected spin source makes the distinction concrete. Its inclusion $P$ has one fermion at each site, so $N_xP = P$. For any finite real profile $f$, set $X_f = \sum_x f_xN_x$ and $c_f = \sum_x f_x$. Then $(X_f - c_f)P = 0$. The low-band isometry $W$ with $\|W - P\| \le 2\tau/U$ consequently obeys

$$\|W^*X_fW - c_fI\| \le 4\Big(\sum_x |f_x|\Big)(\tau/U)^2.$$

Indeed only $(W - P)^*(X_f - c_f)(W - P)$ survives after expansion. Thus fixed local total-load coordinates become scalar on the stated strong-coupling trajectory. A spatial detector of a moving spin excitation must read its internal field response.

There is a conjugate coordinate constructed from that same continuum field. Write $q(F) = Q(F)\Omega$ and use Fourier measure $d\nu = d^3k/(2\pi)^3$. The Weyl relation and the second-moment bounds imply $2\,\mathrm{Im}\langle q(F), P(G)\Omega\rangle = m\int FG$. Cauchy–Schwarz, optimized with $\widehat{G} = \widehat{F}/|k|$ on Fourier supports away from zero, gives

$$m^2\int \frac{|\widehat{F}|^2}{|k|}\,d\nu \le \|q(F)\|^2 \le 3\int \frac{|\widehat{F}|^2}{|k|}\,d\nu.$$

Strong commutativity of the equal-time $Q$ fields makes their Gram form real on real tests, so these bounds extend complex linearly. The map $T(\widehat{F}/\sqrt{|k|}) = q(F)$ is bounded below onto the closed subspace $\mathcal{H}_Q$ of the original field representation. It intertwines the actual spatial translations with multiplication by $e^{-ik\cdot a}$. Its positive Gram operator commutes with these multipliers, and the polar factor $V = T(T^*T)^{-1/2}$ is a unitary onto $\mathcal{H}_Q$. Define

$$X_i = Vi\partial_{k_i}V^*, \qquad P_j^{\mathrm{space}} = V\hbar k_jV^*.$$

The second operator is the same field translation generator in the chosen action units. On the common invariant core $V\mathcal{S}(\mathbb{R}^3)$,

$$U(a)^*X_iU(a) = X_i + a_iI, \qquad [X_i, P_j^{\mathrm{space}}] = i\hbar\delta_{ij}I, \qquad \Delta X_i\Delta P_i^{\mathrm{space}} \ge \hbar/2.$$

The coordinate spectral measure extends to the full field space with the no-position effect $I - \Pi_Q$. A normalized, translation-covariant measure on all of space without this outcome would give the invariant vacuum a translation-invariant probability measure on $\mathbb{R}^3$, which is impossible. This source-defined coordinate is generally nonlocal and differs from the carrier load coordinate and the field quadrature $Q(F)$.

Complete finite detectors use the declared coherent load, pointer and refinable six-level control services. For $q$ spins and $b$ fine Kraus branches, a computable Stinespring completion has dimension $D = 2^{q+\lceil \log_2 b\rceil}$. Complex rank-one elimination and rational angle synthesis give at most $20D^2+4D+10q$ native pulses. Their precision and complete load/export failures enter the channel bound. The eight-site coordinate detector executes a 49,228-pulse decoded core with an untouched reference; its 49,324-pulse encoded completion has analytic half-diamond error below $1.53 \times 10^{-10}$. The full encoded tape is not claimed executed. This construction consumes the coherent service specification; bare occupation repair does not supply it.

Let $R_n$ exhaust a computable Fourier basis and write $C = T^*T$. The compressed polar maps

$$V_n = TR_n(R_nCR_n)^{-1/2}, \qquad m^2I \le C \le 3I,$$

use the inverse on $\operatorname{Ran} R_n$, extended by zero. They converge strongly to $V$, with their adjoints converging on the field representation. For a fixed computable spatial partition, compress its ordinary coordinate effects by $R_n$, conjugate by $V_n$, and retain $I - V_n V_n^*$. These complete effects and their positive square roots converge strongly. The associated Lueders instruments therefore converge in trace norm on each fixed state and reference. Source vectors $\sin(sQ_p(F))\Omega/s$ realize each finite Gram through bounded histories, first taking the source limit at fixed $s, n$, then $s \downarrow 0$, then $n \to \infty$. Positive finite Gram gaps and the compiler control each fixed stage. No effective box-size modulus, uniform infinite-dimensional channel norm or second-moment tail bound follows from this ordered limit.

A localization space invariant under the actual time generator is

$$\mathcal{H}_{\mathrm{resp}} = \overline{\mathrm{span}}\{e^{-itL} V f : t \in \mathbb{R},\ f \in L^2(\mathbb{R}^3)\}.$$

Its spatial spectral measure is absolutely continuous: a momentum-null projector kills $\mathcal{H}_Q$ and commutes with every $e^{-itL}$. The countable rational-time history Gram consequently constructs an isometry $J : \mathcal{H}_{\mathrm{resp}} \to L^2(\mathbb{R}^3; \ell^2)$ that intertwines the same spatial translations and satisfies $JVf = f \otimes e_0$. Compressing ordinary position gives $E(B) = J^* E_0(B) J$, with the invariant dark effect $I - \Pi_{\mathrm{resp}}$. For normalized states in $\mathcal{H}_{\mathrm{resp}}$, its position-outcome and momentum spreads obey $\Delta_E x_i \Delta P_i^{\mathrm{space}} \geq \hbar/2$ whenever both moments are finite. The measure restricts to the preceding sharp coordinate on $\mathcal{H}_Q$. Its general outcomes need not be projections, and its effects alone do not determine measurement back-action. No time invariance of $\mathcal{H}_Q$, preservation of finite moments, sharp particle shell or operational continuum realization of the entire response measure is inferred. Proofs and finite certificates are in `model/spin_dynamical_localization`, `model/spin_operational_position` and `model/spin_position_limit` [2].

A complete local creation experiment uses the actual selected eight-site ground at $J = 1, B = 1/4$. A one-site logical $Z$ preserves the initial exterior and flips the global internal $X$ parity. Every full recorded C/E branch and the centered source Hamiltonian preserve that parity, including the fermionic determinant sign on double occupation. The reference and created programs execute 1,090 and 1,092 native pulses on the full 131,072-row work register, retaining all first-failure records and pointer outcomes. Their complete pointer contrast exceeds 0.67. For degree $d$, each pointer probability changes at rate at most $dJ/2$ under the exchange clock. Combining both inputs with the unhalved source-history error $D = 8m\tau/U + 6T\tau^3/U^2$ gives contrast at least $d_X(0) - dJT - D - 2\eta$, where $\eta$ bounds each protocol's additional half-trace error. This exceeds $2/5$ for ten repairs, $T \leq 1/16$, $c = 12^4, U = 12^8$ and $\eta \leq 10^{-8}$. The lifetime tape is analytic; the creation and detector programs are executed. This finite symmetry sector supplies no topological invariant or relativistic particle shell (`model/defect_localization_readout`, [2]).

The continuous source charge is $Q = \sum_x \epsilon_x X_x/2$, with $Q\Omega_B = 0$. The complete local instrument $A_q = \Pi_{q\epsilon_0 X_0} Z_0$ prepares coherent $q = \pm 1$ branches. A separate coherent Fourier pointer reads charge modulo $N$; its class projectors preserve coherence within each class. The native two-spin $N = 4$ implementation executes 55 pulses with every failure retained. The eight-spin $N = 16$ construction has 222 pulses and exact alias-free charge projectors on its code; its complete tape is not claimed executed. On the actual continuum configuration subnet, the source U(1) gives a charge second-moment bound $6s^2$ for $\sin(sQ_3(F))\Omega$ with a unit annular test. The absolute unwanted-charge weight in the $\pm 1$ modulo-$N$ bins is at most $6s^2/(N-1)^2$. This is an original probability bound, before any herald normalization (`model/spin_dynamical_memory`, [2]).

A bounded angular charge projection of $f(x) = x/(1+x^2)$ gives exact opposite-charge field preparations and a retained positive complement. The source infrared proof certifies $m > 1/10$ and original positive joint-band weight at least $1/1757600$ at every momentum scale. Spatial mean averaging and positivity give separated preparations of every fixed finite charge-sign pattern, retaining

all outcomes without assuming independent particles. Compactly supported charge creators have original joint-band probability at least 9/200000000. A source-clock filter gives positive energy transfer $[ar/2, 2br]$ and a complete three-record instrument. Here $r$ is the momentum scale, $d = 3/10000$, $a = 3d^2/2$ and $b = 2/d^2$. Analytic Fourier bounds fix its normalization $M$ and retained probability $p = 9/(200000000M^2)$. Almost-local commutator bounds and finite spatial averaging give separated equal-sign $n$-charge preparations with original weight at least $p^n/4$, with all $3^n$ records retained. The selected n-charge branch has entire energy in $[nar/2, 2nbr]$; the interval is not imposed on its complementary records. Every nonzero integer internal-charge sector has positive spectrum approaching zero. These are source charge labels, not elementary particle counts. Finite computable instruments have complete native compilations; the selected ground limit has no effective source-size modulus (`model/spin_charge_dispersion` and `model/spin_multipacket_comparison`, [2]).

On the same selected cube, a complete 64-setting Pauli instrument measures $H_B$ with all zero-coefficient settings and source failures retained. Its charge-resolved mean excitation energies are approximately 1.84463098607 and 1.73054620199 for $q = -1, +1$. An independent 4,907-pulse controlled-clock measurement recovers their unconditioned creation work with error below $3.59 \times 10^{-4}$. The signed generator action and endpoint system work differ from pulse count and repair-record cost; bare protected repair effects are scalar (`model/source_energy_effort`, [2]).

The cube's two-site ground marginal has spectral floor greater than 1/41. Its certified modular generator and an actual source-current pulse produce a nonzero regional entropy response. Complete 3,441-pulse histories retain all failures; an informationally complete 32-record detector supplies the regional density-response Fisher metric. For half-trace perturbation $\epsilon < \mu$, the bound $0 \le D(\rho\|\sigma) \le \epsilon^2/(\mu-\epsilon)$ supplies the first-law remainder. A fixed graph-cut area does not vary with this pulse while the entropy does. Neither that fixed area nor the response Fisher metric supplies responsive spacetime geometry or a boost-normalized stress (`model/spin_modular_geometry`, [2]).

The recovered frame also supplies an unbounded three-component *axial circulation*. With oriented incidence matrix $B$, seam current $J_n$ and columns $W_{(a,b)} = \hat{u}_a \times \hat{u}_b$ of the unit frame vectors, exact source identities give $WB^T = 0$, $WW^T = 8I_3$, $\mathbb{E}[J_n \mid x_n] = -B^T x_n/60$ and $\mathbb{E}[J_n J_n^T] = I_{30}/198$ in the uniform binary equilibrium. Hence $Z_N = \sum_{n<N} W J_n$ is a record-readable martingale with $\mathrm{Cov}(Z_N) = (4N/99)I_3$, and $N^{-1/2} Z_{\lfloor Nt \rfloor}$ converges to three-dimensional Brownian motion with covariance $(4/99)\min(t,s)I_3$ and Cameron–Martin form $(99/8)\int |\dot{z}(t)|^2\,dt$ in a rescaled attempt count. A three-event face loop restores the carrier state while adding a nonzero counter increment. The counter's memory grows with the observation horizon; it supplies neither a radial field nor a physical trajectory [11].

### 7.2 The common continuum operator

On the declared spherical source family, the native cell Green operators converge in norm along one common subsequence to a positive compact injective operator $G$ on the mean-zero observation space. Its inverse $A = G^{-1}$ is densely defined and self-adjoint, and restoring the constant zero mode gives a conservative strongly local regular Dirichlet form with domain $H^1$ on the supplied flat-polyhedron comparison surface, energy bounded above and below by positive multiples of the gradient energy, and heat-trace spectral dimension two. A boundary-action test excludes a boundedly invertible $L^2$ energy identification of $A$ with the static boundary action of a smooth compact three-dimensional Dirichlet bulk with smooth boundary, whose Steklov eigenvalues grow like $\sqrt{n}$; bounded-energy internal rank-three modes vanish at this scaling [11]. An exact terminal reduction gives local dual-edge conductances in $[5/8, 15/19]$ with Green-operator error at most $182\,2^{-l}$ relative to native cell means, so this weighted graph has the same continuum subsequences.

A positive native lag readout realizes the Poisson extension. For $K_l = I - L_l/(2M_l)$, Sibuya jump probabilities $p_n = \binom{2n}{n}/[4^n(2n-1)]$ and a Poisson number of jumps with mean $R$ give $\mathbb{E}K_l^N = e^{-R\sqrt{I-K_l}}$. Restricting jumps to $b$ and their count to $j$, with overflow retained, has maximum native lag $bj$ and operator error at most $\min(1, R\binom{2b}{b}4^{-b} + R^{j+1}/(j+1)!)$; at $R = 1/2$, $b = 64$, $j = 8$ the full 66-state source gives 0.034581 within the bound 0.036. Taking $R_l = r\sqrt{2M_l 4^l}$ yields the compressed limit $e^{-r\sqrt{A}}$; here $r$ labels a positive mixture of native lags, not a physical radius.

Full native density correlations converge to $e^{-\tau A}$, and the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. The full-space extension $\bar{A} = A \oplus 0$ satisfies $\|\mathbf{1}_F e^{-\tau\bar{A}}\mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)}$, and its cosine propagator vanishes between these supports when $|t| < \sqrt{2}\, d(E,F)$ in the declared flat comparison distance. Two consecutive long observation blocks of one history supply a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$, with observation count taken to infinity before spatial refinement [11]. A declared static filter $A_l^{1/4}$ changes it to $A_l^{-1/2}\cos((t-u)\sqrt{A_l})$, twice the real part of the free-field vacuum covariance; the stationary increments obey $2A_l^{-1/2}(I - \cos(\Delta t\sqrt{A_l})) \le (\Delta t)^2\sqrt{A_l}$ and, with $\sum_m \langle h_{jm,l}, \sqrt{A_l} h_{jm,l}\rangle \le 12j^2$ for centered cell averages of spherical harmonics, the geometric $H^{-s}$ increment moment is at most $12[1 + 1/(2s-3)](\Delta t)^2$ for $s > 3/2$, giving convergence in $C([0,T], H^{-s}(S^2))$. The wave readout is declared classical processing, distinct from passive density waiting and from a quantum vacuum.

### 7.3 Native density histories and their limits

At level $l$, let $V_l = 240\,4^l$, $q_l = V_l/3$ and $\chi_l = q_l(V_l - q_l)/[V_l(V_l-1)]$. In the uniform canonical equilibrium, with independent uniform seams and fair repair, the actual snapshot readout of a mean-zero test function repeated on its twelve native ports,

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\,16^l\tau\rfloor}(p) - 1/3]}{\sqrt{V_l\chi_l}},$$

converges in joint distributions at finitely many times as space and the native attempt count are refined together. The limiting family is stationary and Gaussian, with covariance $\langle f, e^{-|\tau-u|A} g\rangle$. Representing each unread repair as a fair identity-or-swap, the conditional profile overlaps concentrate with variance at most $4\|g\|_\infty^2\|f\|_{\ell^2}^2/V_l^2$ uniformly in attempt count, and a conditional finite-population central limit theorem applies. The declared filter $G_l^{1/2}$ gives covariance $Ge^{-|\tau-u|A}$. The law is reflection positive with one-particle semigroup $e^{-\tau A}$, generator $A$ and not $\sqrt{A}$. The interpolated density converges in $C([0,T], H^{-3}(S^2))$ and its Green-filtered readout in $C([0,T], H^{-2}(S^2))$, by a forward–backward martingale decomposition with $g = 2(I+P_l)^{-1}f$ and harmonic tail bounds; the step paths have the same Skorohod limits [11]. For $\phi = \Phi/\sqrt{2}$ a mode of eigenvalue $\lambda$ has covariance $e^{-\lambda|\tau-u|}/(2\lambda)$, so the Gaussian precision is the Euclidean form

$$S_E(\phi) = \frac{1}{2}\int_{\mathbb{R}} (\|\partial_\tau\phi\|^2 + \|A\phi\|^2)\, d\tau,$$

with reconstructed Hamiltonian $d\Gamma(A)$ and spatial precision $A^2$. The action follows from the native readout law and has diffusive dispersion.

Two restrictions bar identifying this sector with a relativistic field. On the shorter attempt scale $b_l = 1260\,8^l$ both the density and Green-filtered paths freeze to their initial field, with expected squared supremum increments at most $50(T2^{-l} + a_l^{-1})$, $a_l = 1260\,16^l$. For a complete linear

readout with bounded injective dense-range composite map $B = TC^{1/2}$, $C = I$ or $G$, the left polar decomposition $B = D_0^{1/2}U$ gives

$$D_0^{-1/2} D_\tau D_0^{-1/2} = U e^{-\tau A} U^*,$$

so normalization preserves the ordered decay rates $\lambda_n(A) \asymp n$. No common positive clock factor turns these into the $n^{1/3}$ Euclidean decay rates of a complete stationary free scalar on a smooth compact three-dimensional manifold without boundary, and the same argument excludes every spatial dimension $d \geq 2$, whose rates grow like $n^{1/d}$; a continuous nonconstant time change preserving the semigroup is a scalar factor. The excluded object is the complete linear Gaussian identification; restricted boundary observations, nonlinear dictionaries and interacting fields need their own analysis, and a causal microscopic theory may have this diffusive sector as an effective description [11].

### 7.4 Incident clocks and active waves

Let $a_l = 1260\,16^l$, $M_l = 630\,4^l$ and let a fixed event set contain $d_l > 0$ seams. Its incident counter $N_l(n)$ includes zero-current repairs, with $p_l = d_l/M_l$; write $T_l(k)$ for the attempt of its $k$th tick. For $n_* = \lceil a_l(T+\delta) \rceil$,

$$\Pr\Big[\max_{n \leq n_*} |N_l(n) - n p_l| > a_l p_l \delta\Big] \leq \frac{n_*(1-p_l)}{a_l^2 p_l \delta^2},$$

and on the complementary event the normalized count $C_l(t) = N_l(\lfloor a_l t \rfloor)/(a_l p_l)$ and the inverse clock $U_l(t) = T_l(\lfloor a_l p_l t \rfloor)/a_l$ differ from $t$ by at most $\delta + 1/a_l$ and $\delta + 1/(a_l p_l)$ on $[0,T]$. A primitive port has $d_l = 5$ or 6 and a whole carrier $d_l = 33$. The density path theorem transfers to these local count parametrizations without independence between clock marks and fields [13]. Passive port access chosen before each seam draw from earlier records also works: integer credits $R_n = \sum_{j \leq n} (30/d_j) I_j$ have conditional mean $30/M_l$ per attempt, deviation probability at most $n_*(M_l - 5)/(5a_l^2\delta^2)$, and inverse clock error at most $\delta + M_l/(5a_l)$.

For fixed incidence counts $d_i$ and shared-seam counts $c_{ij}$, centered clock counts divided by $\sqrt{a_l/M_l}$ converge to Gaussian increments with covariance $\min(t,u)c_{ij}$, asymptotically independent of the density snapshots. Adjacent degree-five port clocks have limiting correlation $1/5$, the endpoints of an external seam $1/6$, and nonadjacent clocks zero. These are native model predictions under the stated schedule and preparation, without an atomic-clock or proper-time dictionary [13].

The admitted coherent walk can use the same tick. A protected wave register receives one fixed step $W_l$ per incident event at a degree-$d$ port, including zero current, while the timing source pauses during coherent services. With $Q_l = 630\,16^l$, $D_l = 4^l L_l$, $\beta = 2d/\sqrt{630}$, $\eta = \delta + 1/a_l$ and $e_l = \Lambda/[6(4Q_l - \Lambda)]$, on the band $0 \leq D_l \leq \Lambda < 4Q_l$ and the count good event,

$$\sup_{\tau \leq T} \Big\| \cos(N_l(\lfloor a_l \tau \rfloor) \arccos K_l) - \cos(\beta\tau\sqrt{D_l}) \Big\| \leq \beta\sqrt{\Lambda}\,[\eta + (T+\eta)e_l],$$

so the compressed wave converges to $\cos(\beta\tau\sqrt{A})$. Step error $\epsilon_l$ accumulates at most $N_{\max}\epsilon_l$ with $N_{\max} = O(\sqrt{Q_l})$, so $\sqrt{Q_l}\epsilon_l \to 0$ suffices. An exact control retains 237,600 two-attempt clock histories; the continuing Ramsey-plus probability is 1297/1320 against 7901/7920 when zero-current triggers are dropped. The wave step is compiled apparatus; passive repair does not supply it [13].

### 7.5 An encoded quantum field and its modular clock

The classical covariance does not determine commutators. $N$ independent proper-code registers in $|0\rangle$ give $Q_N = \sum_j X_j/\sqrt{2N}$ and $P_N = \sum_j Y_j/\sqrt{2N}$, with commutator $i\sum_j Z_j/N$, $\mathbb{E}Q_N P_N = i/2$

and $\mathbb{E}Q_N^4 = 3/4 - 1/(2N)$. For Weyl factors $W_N(v) = e^{i(sQ_N+tP_N)}$ the quantum central limit theorem [1] gives

$$\lim_{N\to\infty} \mathbb{E}\prod_j W_N(v_j) = \exp\left[-\frac{1}{4}\left|\sum_j v_j\right|^2 - \frac{i}{2}\sum_{j<k}\sigma(v_j,v_k)\right], \qquad \sigma((s,t),(s',t')) = st' - ts',$$

the CCR vacuum with its ordered-product phase. Source pulses execute four codes and one Ramsey pointer with all 32 outcomes retained; the pointer quadratures give $0.7452257133 - 0.3013609949i$ for the declared three-factor word, reversal conjugates the phase and the tracial preparation removes it. The native density bands fix the mode rotation $e^{i\theta_l Z/2}$ per incident trigger, $\theta_l = \arccos(1-\lambda/(2Q_l))$, with encoded frequency $\Omega_l = \sqrt{Q_l}\theta_l \to \sqrt{\lambda}$, ordered covariance $e^{-i\Omega_l(t-u)}/(2\Omega_l)$ and kick response $\sin(\Omega_l(t-u))/\Omega_l$, executed for both kick signs at three native bands [11]. Polarized preparation, mode encoding, storage, pulses and readout are specified services. The sine kernel $A^{-1/2}\sin(t\sqrt{A})$ vanishes outside the comparison cone for mean-zero tests; unrestricted smearings add $tP_{\text{const}}$, and restoring the compact massless zero mode gives a free particle with no normalizable ground state, so the mean-zero field is no complete local net.

For $\rho_r = (I + rZ)/2$, $0 < r < 1$, and $H = \Omega(I - Z)/2$, the finite Gibbs and Kubo–Martin–Schwinger identity gives $\sigma_s = \alpha_{-bs}$ for every real $s$ exactly when $\log\frac{1+r}{1-r} = b\Omega$, that is $r = \tanh(b\Omega/2)$; equal polarization at unequal frequencies fails a common $b$, the pure endpoint is not faithful and the tracial endpoint has trivial flow. With $Q_N = \sum X_j/\sqrt{2Nr}$ and $P_N = \sum Y_j/\sqrt{2Nr}$ the ordered copy limit has thermal covariance $C(\delta) = [\cos(\Omega\delta) - ir\sin(\Omega\delta)]/(2r\Omega)$ with $C(t - ib) = C(-t)$, executed on two codes with retained purifying references. The native bound $\lambda_n(A) \geq n/49920$ gives $\operatorname{Tr} e^{-b\sqrt{A}} \leq 99840/b^2$, so the mean-zero Fock space has the faithful normal state $Z_b^{-1}e^{-b\,d\Gamma(\sqrt{A})}$ with the same modular flow, and under $t = \kappa\tau$, $\kappa = 2d/\sqrt{630}$, its modular parameter is $s = -\kappa\tau/b$. The common $b$ is a measurable preparation condition, not a repair-selected temperature.

### 7.6 The product field, its clock and its Lorentzian density

Independent long blocks of one native history supply radial innovations. Their mean-zero covariance is $B_l = G_l - P/(2520\,16^l)$, and a second block adds a normalized constant-direction readout, giving $B_l + P_{\text{const}}$. On a declared interval $(0, L)$ with Dirichlet ends, a mode with $h = \sqrt{\lambda}$ has covariance

$$C_\lambda(r,s) = \frac{\sinh(h\min(r,s))\sinh(h(L-\max(r,s)))}{h\sinh(hL)}, \qquad C_0(r,s) = \frac{\min(r,s)(L-\max(r,s))}{L},$$

realized by applying $Q_{rs}^{1/2}(B_l + P_{\text{const}})^{-1/2}$ to successive blocks, with $Q_{rs}$ the conditional innovation. The whitening means the radial direction carries no native information beyond the spectrum of $A$. Complete 96-event current tapes check the finite decoder. The spatial subsequence gives the product operator $\mathcal{B} = -\partial_r^2 + \bar{A}$ with action $\frac{1}{2}\int_0^L(\|\partial_r\phi\|^2 + \mathcal{E}(\phi))\,dr$; for $L = \pi$ its count satisfies $\Lambda^{3/2}/(12\sqrt{2}) \leq N_{\mathcal{B}}(\Lambda) \leq 49921\Lambda^{3/2}$ for $\Lambda \geq 3$, and its tensor form gives comparison wave speed one in $d^2_{\text{product}} = |r-s|^2 + 2d^2_{\text{screen}}$. On a radial grid the sampled bridge has an explicit sine-mode covariance, and a spectral filter realizes the thermal covariance $\coth(b\sqrt{\mathcal{B}}/2)/(2\sqrt{\mathcal{B}})$ from the same records; the full-port generator and the cell Schur generator have compressed resolvents within $1/(8\,4^l)$ at each positive shift, so both give the same product target. Radial labels, filters, boundaries and block access are specified operations.

For a full-port angular mode set $a_l = \sqrt{Q_l}\arccos(1 - \lambda/(2Q_l))$ and $c = \sqrt{\rho_{k,h}}$. Each incident trigger applies the source $Z$ pulse of angle $a_l/(2\sqrt{Q_l})$ and the radial $X$ pulse of angle $c/(2\sqrt{Q_l})$;

anticommutation gives $\omega_{l,k} = \sqrt{a_l^2 + c^2}$ with no fitted product frequency, and the split word differs from the joint Hamiltonian by at most $na_l c/(4Q_l)$ after $n$ triggers. The faithful reference is polarized along $(cX + a_l Z)/\omega_{l,k}$ with $r = \tanh(b\omega_{l,k}/2)$; ordered two-time measurements retain all 32 outputs. A deterministic packet $v \propto (12I - L)^{1024}(\mathbf{1}_{\text{cell }0} - \mathbf{1}_{\text{cell }1})$ on the 240-port carrier has Rayleigh quotient $\lambda \simeq 0.044442251696$, residual $\|(L - \lambda)v\| < 0.000304$ and squared cell compression above 0.996458773594; Householder encoders and a Bell reference read it after incident-triggered phase, with 25-trigger Chebyshev discrepancy about 0.000149026 at $M = 630$, at a cost of 1,747,068 source pulses. Copy and spectral limits followed by radial refinement give

$$C_b(t) = \frac{\coth(b\sqrt{\mathcal{B}}/2)\cos(t\sqrt{\mathcal{B}}) - i\sin(t\sqrt{\mathcal{B}})}{2\sqrt{\mathcal{B}}}, \qquad D(t) = \frac{\sin(t\sqrt{\mathcal{B}})}{\sqrt{\mathcal{B}}}, \qquad \sigma_s = \alpha_{-bs},$$

and with $C_L = L(49920 + L^2/9)/3$, $\operatorname{Tr} e^{-b\sqrt{\mathcal{B}}} \le 6C_L/b^3$, so the encoded Fock algebra has the faithful normal state $Z_b^{-1} e^{-b\, d\Gamma(\sqrt{\mathcal{B}})}$ with $s = -\kappa\tau/b$. Vanishing count tolerances require $4^l \eta_l^2 \to \infty$ and complete-step errors $\sqrt{Q_l}\epsilon_l \to 0$. Radial design, common $b$, preparation, mode access and source precision are declared services [11].

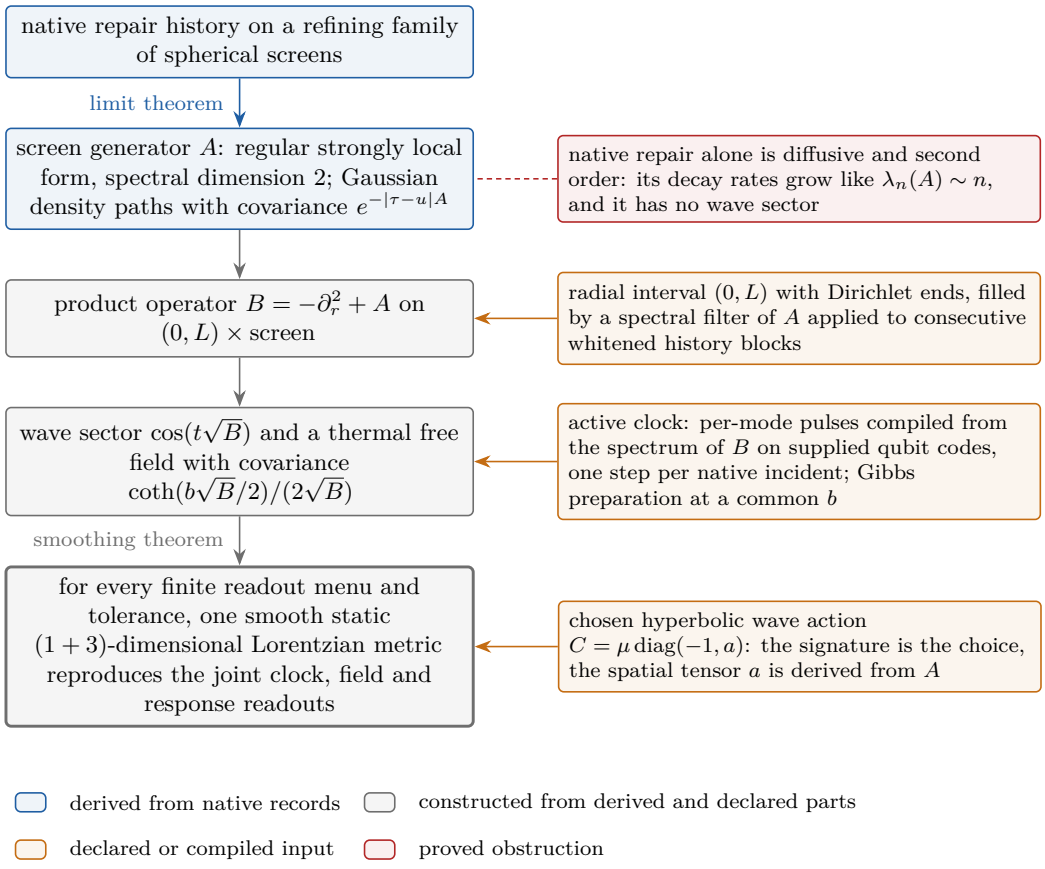



Figure 3: What the effective $(1 + 3)$ scalar description derives from native records and what it declares. Blue: derived from one native repair history by a limit theorem. Grey: constructed from derived and declared parts. Orange: declared or compiled inputs. Red: the proved obstruction that rules out a wave sector from passive native repair. The smoothing theorem is an existence result for fixed finite readout menus.

**Lorentzian density.** The regular strongly local angular form with its $H^1$ domain admits a measurable uniformly elliptic tensor in each chart [120, Theorem 7.2 and Corollary 7.3], recoverable from complete native records: with the disclosed total-one kernel $K_l = I - L_l/(2M_l)$, including every zero-current hold, the ordered limits

$$\lim_{t\downarrow 0}\lim_{l}\int \phi\,\frac{\bar{P}_l(t)(fg) - f\bar{P}_l(t)g - g\bar{P}_l(t)f + fg}{2t}\,d\mu = \int \phi\, a^{ij}\partial_i f\partial_j g\,d\mu$$

recover weighted tensor averages. In barycentric face coordinates the one-attempt increment tensor obeys $\frac{3}{7}a_{\mathrm{bare}} \le a_{\mathrm{screen}} \le \frac{15}{19}a_{\mathrm{bare}}$ with $a_{\mathrm{bare}} = \begin{pmatrix} 1/36 & -1/72 \\ -1/72 & 1/36 \end{pmatrix}$, so the one-attempt-first limit reads a different tensor. The radial product adds $a^{rr} = 1$. With coordinate measure $\mu(x)\,d^3x$ the chosen wave action has coefficient density $C = \mu\,\mathrm{diag}(-1, a)$, and

$$g = \sqrt{-\det C}\,C^{-1} = \mu\sqrt{\det a}\;\mathrm{diag}(-1, a^{-1}), \qquad \sqrt{-\det g}\,g^{-1} = C$$

has signature $(-,+,+,+)$ almost everywhere, with weak wave equation $\partial_\alpha(C^{\alpha\beta}\partial_\beta\phi) = 0$ [123, Equations (23)–(26)]. The signature is that of the chosen action; the derived content is the measurable tensor $a$. The native measure is the radial pushforward of flat face area, with density $\rho_F(u) = c/(n_F\cdot u)^3$, $c = \pi r^2/(5a_0)$, on a face of unit normal $n_F$ and plane distance $r$. Adjacent faces agree on their common edge but their spherical gradients differ by $-3c(n_F - n_G)/(n_F\cdot u)^4 \neq 0$, so the density is not $C^1$, and since $C^{tt} = -\mu$ the exact dictionary in this atlas fails the smooth-density premise. An alternative smooth dictionary would need a separate argument.

The density is globally Lipschitz relative to smooth sphere volume. Smoothing only the angular variables and the tensor, with the unitaries $U_\varepsilon f = \sqrt{\rho_\varepsilon/\rho}\,f$ between the changing $L^2$ spaces, gives transported Mosco convergence and then norm-resolvent convergence with gap $(\pi/L)^2$. Thermal covariance and sine response converge in operator norm, and ordered Weyl laws, finite adaptive Ramsey histories and finite detector outputs converge. Hence for any fixed finite admissible menu on bounded wave times and any positive tolerance one smooth static $1+3$ metric approximates the joint readouts. Both sides are free fields on the same supplied codes; the theorem removes the roughness of the native coefficients, not the declared radial product, clock or state, and gives no computable rate. In the smooth comparison $g = N^2\,\mathrm{diag}(-1, a^{-1})$, $N^2 = \mu\sqrt{\det a}$, the canonical momentum is $\mu\partial_t\phi$ and the thermal covariance and response have the standard static CCR/KMS normalization with a reflecting radial boundary.

Angular observations do not select the radial law. For the 240-port $L$, three 720-port products $B_R = I_3 \otimes L + R \otimes I$ with the unit path, fourfold path and unit triangle Laplacians satisfy $J^*F(B_R)J = F(L)$ for the radial average $J$ and every spectral function, while a radial-contrast probe has cosine second derivatives $-5/2, -10, -3$.

For positive finite-measure regions $E, F$ of the product, let $d(E,F) = \sup_u(\operatorname{ess\,inf}_F u - \operatorname{ess\,sup}_E u)$ over bounded local-form functions with $\Gamma_{\mathcal{B}}(u) \le 1$. Davies–Gaffney estimates and the short-time theorem of Ariyoshi and Hino [119, Theorem 2.7], $\lim_{\tau\downarrow 0} -4\tau\log\langle \mathbf{1}_F, e^{-\tau\mathcal{B}}\mathbf{1}_E\rangle = d(E,F)^2$, combined with heat-cosine transmutation give

$$\inf\{t>0 : \|\mathbf{1}_F\cos(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = \inf\{t>0 : \|\mathbf{1}_F\mathcal{B}^{-1/2}\sin(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = d(E,F).$$

This is the sharp finite-propagation front of the chosen wave equation, fixed by its intrinsic metric; it supplies no independent propagation speed, and same-field modular time has onset $|s| = d/b$.

### 7.7 Interacting readouts on the product

Retain the angular constant and radial profiles $\sqrt{2}\sin r$ and $\sqrt{2}\sin(2r)$ on $(0,\pi)$. The projected local quartic polynomial is

$$W = \tfrac{3}{2}(q_1^4 + q_2^4) + 6q_1^2 q_2^2, \qquad H = \tfrac{1}{2}(p_1^2 + q_1^2 + p_2^2 + 4q_2^2) + \frac{g_{\mathrm{eff}}}{24} W,$$

with $g_{\mathrm{eff}} = (\lambda/\pi)\int \mu_{\mathrm{screen}}^2 \sqrt{\det a_{\mathrm{screen}}}\, dx$ for a constant covariant coupling; $g_{\mathrm{eff}} = 1$ is an input. At occupation cutoff $4 \times 4$, native source pulses prepare the purification of $e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$ and execute two reads with feedback, retaining all work, reference and record outputs. For $\beta = 1/4$, intervals $1/16$ and 64 split steps, the complete-output distance from an independent exponential is below $1.97 \times 10^{-8}$; deleting the interaction or feedback changes the record law by about $7.31 \times 10^{-4}$ or $1.95 \times 10^{-3}$. The worst branch uses 217,110 source pulses, 434,220 load/export events and two native repairs. A separate protocol advances this Hamiltonian on every incident event of two native timing attempts, retaining all 3,600 tapes and 4,900 field-prefix outputs, with timeout probability $35/36$ and complete error below $3.44 \times 10^{-9}$. On the refined clock family, incident rate $R_l = 2d4^l$ gives a controlled inverse-count comparison; each metric readout uses its own clock pullback, and no universal physical conversion is inferred.

For $g \geq 0$ the positive quartic form has a Friedrichs Hamiltonian $H \geq H_{\mathrm{harm}}$, Hermite compressions form a core, and

$$Z_d \longrightarrow Z, \qquad 0 < Z \leq \frac{e^{-3\beta/2}}{(1-e^{-\beta})(1-e^{-2\beta})}, \qquad \rho_d \longrightarrow \rho \quad \text{in trace norm},$$

so every fixed finite complete adaptive history converges and the limiting Gibbs state has $\sigma_s = \alpha_{-\beta s}$. This proves occupation existence for the two-mode theory, without a small-code certificate: the complete $d = 4$ versus $d = 8$ output distance is about 0.56045, and $d = 8$ versus $d = 12$ about 0.24501. An inverse control recovers $(g_{\mathrm{eff}}, \beta, c) = (1, 1/4, 1)$ within $10^{-4}$ from six noiseless source probabilities and predicts a withheld interval within $10^{-8}$, with local Jacobian singular values about $0.264713, 0.001489, 0.000110$. For fixed covariant $\lambda$ the smoothed metrics give convergent quartic projections. The quartic excites omitted modes; no invariant two-mode band, interacting ultraviolet limit or finite-cutoff microcausality follows [39].

**The reference-state boundary.** Increasing native-return caps match modular phase magnitudes to the cosine limit, and a fixed cap freezes the low-band phase. On the full configuration algebra the uncapped reference approaches normalized trace with half-trace error at most $3/(8\,2^l)$, its density-wave sector has vanishing weight, and fixed-parameter modular evolution becomes trivial in normalized-trace $L^2$. If the relevant fluctuation moments converge, any limiting scalar canonical commutator in this bounded-overlap regime is zero. Refinement-rescaled time, other states and other fluctuation regimes are not excluded [11].

## 8 A common matter metric and its gravitational response

The conditional scalar description supplies a concrete dictionary for comparing matter and geometry. With coefficient density $C = \mu\,\mathrm{diag}(-1, 1, a)$ and metric $g = \sqrt{-\det C}\, C^{-1}$, the scalar wave symbol, both Weyl principal determinants and the Maxwell symbol share one characteristic cone, and the product has induced measures $dA = \mu\, d^2x$ and $dV = \mu N\, dr\, d^2x$ with $N^2 = \mu\sqrt{\det a}$. The same dictionary yields a control. Both radial null Einstein contractions of a smooth static product vanish,

while a coherent scalar excitation has positive relative null stress, so that geometry cannot satisfy the Einstein balances for the reference and the excitation at positive coupling with other sources fixed. The spacetime paper [8] proves these statements, gives the finite field operators and entropy variations, and derives the necessary area-response equations.

# 9 Prediction protocols

**A response without a fitted coefficient.** In the stationary binary number sector, disclose local occupations and apply one extra fair native repair on seam $(a, b)$ exactly when $n_a = 1$. Detailed balance gives, for any fixed future record event $F$ whose continuation does not consult the earlier pulse mark,

$$\Pr_{\text{feedback}}(F) - \Pr_{\text{passive}}(F) = \tfrac{1}{2}\mathbb{E}_{\text{passive}}[(n_b - n_a)\mathbf{1}_F].$$

Complete capped clock and field forecasts and the native spatial Green response satisfy this identity exactly, with every trigger, coin and current label retained [11]. No matched public intervention dataset exists for these forecasts.

**A controlled continuum response.** For a disclosed port profile $h$, a uniform seam $(a, b)$ executes its unchanged fair repair with probability $[1 + \epsilon(h_b - h_a)(n_a - n_b)]/2$, and the clock counts executed repairs, including zero currents. The exact linear occupation response is $2\chi K(I - K^r)h$, and on the refining source

$$\partial_\epsilon \mathbb{E}\eta_l(g)|_0 \longrightarrow 2\langle g, (I - e^{-\tau A})f\rangle.$$

At $\tau = 1$ with four selection slots per target repair, stopped likelihood bounds control the nonlinear mean error by $36\epsilon^2\|g\|$, and for a unit-axis flat coordinate at $\epsilon = 10^{-6}$, $\liminf_l \mathbb{E}_\epsilon \eta_l(f) \geq \epsilon\, 9/32792 - 36\epsilon^2 > 0$ [11]. In the selection-slot clock the derivative converges to $\langle g, R_\tau f\rangle$ with $R_\tau = 2(I - e^{-\tau\bar{A}})$, and positivity improvement of the heat semigroup gives, for nonzero nonnegative Lipschitz profiles at distance $r > 0$,

$$0 < -\langle g, R_\tau f\rangle \leq 2e^{-r^2/(2\tau)}\|f\|\|g\|, \qquad \tau > 0.$$

This derivative has no exact finite support cone. Reading its control, support and slot parameter as physical local interventions and time would contradict an exact relativistic cone; that identification is not derived, and a causal microscopic theory with diffusive effective observables is consistent with it.

## 9.1 Complete finite observation laws

The following statements connect the declared finite sources to their complete observer outputs. Each instrument includes its original failures, timeouts and terminal quantum states. The coherent services, preparations and schedulers are construction inputs; the conclusions do not identify a physical clock, apparatus accuracy or a particle dictionary.

**Periodic readout.** On the sixty-six-state binary carrier, extend the probe Pauli operator $Y$ by zero and write $P = Y^2$, $Q = I - P$. The full-carrier Kraus operators $K_\pm = (P \pm Y)/2 + Q/\sqrt{2}$ have effects $(I \pm Y)/2$, whose sum is the identity, including every untouched reference. The declared modular intervention has phase $\theta = s\log(5/4)$. Exact quarter-turn evaluations give decoded conditional plus probabilities

$$p_{\mathsf{C}}(\theta) = \frac{1}{2} - \frac{\sin\theta}{140}, \qquad p_{\mathsf{E}}(\theta) = \frac{1}{2} - \frac{\sin\theta}{220}.$$

The original herald probability is $5/33$; its complementary failure is retained. If $T_0$ is the no-current map, the literal source gives $T_0^*(I) \le 13I/15$ on every input. Thus capping the first event at $h$ attempts changes the original completed-plus weight by at most the retained timeout probability, which is bounded by $(5/33)(13/15)^h$. All $4h+3$ original records are executed at caps $4, 32, 128$, and the stopped-map trace identity is checked on all 4356 matrix units. Native attempts do not generate the declared modular phase.

**Joint event clocks.** The full sixteen-dimensional work-operator space carries the marked four-port native instrument. For no-event map $M_0$ and marked maps $M_A, M_B$, put $U_j = M_j(I - M_0)^{-1}$. The noncommuting resolvent identity

$$U_A(I - zU_B)^{-1} = M_A(I - M_0 - zM_B)^{-1}$$

holds on every operator coordinate. Its coefficients preserve the conditional work state with any reference, including record feedback. The passive two-$A$-tick cycle has $B$-count generating function $(2-z^2)/(3-2z^2)$, mean 2 and variance 16. Complete three-attempt field reads give plus probabilities $35/54$ and $14/27$ for $\mathsf{C}$ and $\mathsf{E}$. Execution retains all 250 histories per law and all 50 histories of the separate reference-feedback control. This is a stochastic relative clock law, with no supplied affine conversion to physical time.

**Correlations at every lag.** For the full sixty-six-state incident-clock kernel $K = B(I - A)^{-1}$, let $D = I - A$, $p = 3/10$, $M = pD^{-1}/66$ and $f = n_0 - 1/6$. The actual source has $Df = pf$, so $c_n = f^T K^n f/66 = f^T M K^n f$. Its literal cyclic inclusion $C = [f, Kf, \dots, K^6 f]$ satisfies

$$C^T MC = G > 0, \qquad KC = CR, \qquad GR = R^T G, \qquad 0 < GR < G.$$

Consequently $K^n C = CR^n$ for every $n \ge 0$: the degree-seven recurrence inferred from exact correlations holds on the original source at every unused lag. Independent native powers verify lags 15 through 32. Finite sampled correlations alone cannot certify this exact rank or closure. The three constructions above, including their complete output laws, are reproduced by `model/finite_observer_readout_closure` [2].

**Joint screen-reference interference.** On the eight-coordinate logical-pair/reference carrier, set $O = |2\rangle\langle 5| + |5\rangle\langle 2|$ and $E_\pm = (I \pm O)/2$. These effects include the six-dimensional complement. For a retained stopped word with probability $w_r$ and record-corrected coherence $c_r$, the two terminal probabilities are $(w_r \pm c_r)/2$. The actual seam-mask counts $(23, 5, 4, 1)$ out of 33 give completed two-tick probabilities

$$P_{\mathsf{C}}(+) = \frac{61}{72}, \qquad P_{\mathsf{E}}(+) = \frac{809}{1089}, \qquad P_{\mathsf{C}}(+) - P_{\mathsf{E}}(+) = \frac{101}{968}.$$

All 21 capped words retain both terminal outcomes; a separate five-port native recursion checks all 42 probabilities. The untouched reference marginal is $I_2/2$ under both laws, as required by trace preservation. This distinction is a joint observable; the untouched reference alone cannot distinguish the laws.

**Two-particle incident-clock statistics.** The twenty-cell source has 240 ports and 28680 fixed-two configurations. Four alternating field reads, each capped at one attempt, give 124 original

records per feedback policy, including 60 timeout records. Literal native transfer and feedback matrices give nonnegative integer terminal vectors with common denominator

$$28680 \cdot 2520^4 = 1156594998988800000.$$

Their total equals this denominator exactly, with no rounded remainder. The completed-read probability is 28561/252047376; the complementary timeout probability and the changed source states are retained. This exact cap-one statement supplies no transfer of its unrounded-law guarantee to the separate cap-256 or density-1/3 comparisons.

**Measurement within an interacting source.** The sixteen-state common source executes its actual primitive words, retaining the four original intermediate/final measurement records for each feedback setting. An independently constructed Gaussian-rational Taylor enclosure compares the continuous Hermitian generator with fresh host outputs, whose binary components are converted to exact rationals. For branch vectors $u_r, v_r$, the difference $A_r = v_r v_r^\dagger - u_r u_r^\dagger$ has rank at most two, so $\|A_r\|_1/2 \le \sqrt{\|A_r\|_F^2/2}$. Exact squared norms and an outward Taylor remainder certify complete half-trace error below $3 \times 10^{-7}$ per setting. The trace defect is included in the $6.01 \times 10^{-7}$ unconditional event-contrast allowance; the resulting contrast interval lies strictly below $-0.0316608$. There is no branch renormalization or hardware-precision assertion. The joint, counting and interacting instruments are reproduced by `model/finite_observable_closures` [2].

**Axial counter convergence.** For the axial counter defined above, exact enumeration retains all 32400 two-attempt seam/coin/current words. The actual four-step transition matrix has uniform minorization $\delta = 11/90000$. Writing $C(x)$ for the conditional increment covariance and $\Sigma = (4/99)I_3$, the source gives, for every initial classical law,

$$\mathbb{E}\left\| \frac{1}{N} \sum_{i<N} (C(X_i) - \Sigma) \right\|_F^2 \le \min\left\{1, \frac{1440000}{11N}\right\}.$$

Indeed $\|C(x) - \Sigma\|_F < 1$ and the lag-$k$ cross term is at most $2(1-\delta)^{\lfloor k/4 \rfloor}$; summing this geometric bound gives the estimate. Bounded increments supply the martingale Lindeberg condition, and a finite time grid extends bracket convergence uniformly on compact intervals. The martingale functional central-limit theorem therefore yields covariance $(4/99)\min(t,u)I_3$ in scaled native count [11]. Physical position and laboratory time are not identified by this counter limit.

**Integer face lifts.** For an integer face/seam incidence matrix $B$ whose columns sum to zero, $m = Bn$ obeys $\sum_f m_f = 0$ for every integer link field $n$. A seam twist by $j$ adds $jB_{\cdot s}$; reduction modulo six commutes with this update. The generic identities in `Lean/Dynamics/IntegerFaceLift.lean` apply by induction to every history, including rejected and no-operation branches. The executable checks the actual 20-, 80- and 320-carrier incidence matrices, all 2592 local load/coin/class contexts, and the 80-carrier, 200-sweep reference. The imported 99.7% hypercharge-value census is a separate population diagnostic. Integer values differing by six have the same class, so these identities supply no conserved physical hypercharge label. The counter and lift certificates are reproduced by `model/finite_current_readout_closure` [2].

## 9.2 The phase experiment

The classical repair law fixes the diagonal of every admissible record law and none of its coherences (Theorem 2.3). A fixed interference experiment reads the multiplier $g$ that separates the members,

and its count protocol, comparing the executed laws $\mathsf{C}$ and $\mathsf{E}$, is fixed before any data [66, 67]. All members are ordinary quantum instruments, so the test separates record laws and makes no claim to separate OPH from quantum mechanics.

Write $x = \{2,3\}$, $z = \{1,2\}$ and $w = \{0,2\}$. Prepare $|\phi_\theta\rangle = (|x\rangle + e^{i\theta}|z\rangle)/\sqrt{2}$ for $\theta \in \{0, \pi, \pi/2\}$, or the incoherent mixture $\frac{1}{2}(|x\rangle\langle x| + |z\rangle\langle z|)$, apply one repair on the seam $(0,1)$, retain its record, and measure $P_+ = |\phi_0\rangle\langle\phi_0|$ and $P_- = I - P_+$ inside each committed record block.

**Proposition 9.1** (Phase law of the two record instruments)**.** *For the preparations above:*

*(i) the nonzero recorded source densities of* $\mathsf{C}$ *are* $\frac{1}{2}|\phi_\theta\rangle\langle\phi_\theta|$ *for* $(c,j) = (0,0)$, $\frac{1}{4}|x\rangle\langle x|$ *for* $(1,0)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1,-1)$*; those of* $\mathsf{E}$ *are* $\frac{1}{4}|x\rangle\langle x|$ *for* $(c,d) = (0,0)$ *and* $(1,0)$, $\frac{1}{4}|z\rangle\langle z|$ *for* $(0,-1)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1,-1)$*;*

*(ii) the plus probabilities are* $p_{\mathsf{C}}(\theta) = \frac{3}{8} + \frac{1}{4}\cos\theta$ *and* $p_{\mathsf{E}}(\theta) = \frac{3}{8}$*, and the incoherent preparation gives* $3/8$ *for both; the phase-reversal contrast is* $1/2$ *for* $\mathsf{C}$ *and zero for* $\mathsf{E}$*;*

*(iii) all four preparations have identical classical load probabilities, the readout preserves the probability of every old record, and no record is erased and no outcome postselected.*

*Proof sketch.* On the seam $(0,1)$ the configuration $x$ has contrast zero for both coins, and $z$ has contrast $-1$, current 0 for coin zero and current $-1$ for coin one, which moves it to $w$. The coherent sum survives only in the $\mathsf{C}$ branch $(0,0)$, where both currents vanish; $\mathsf{E}$ separates $x$ from $z$ by their contrasts [67]. □

Table 4: Plus probabilities of the four fixed settings [66, 40].

| Preparation | $\mathsf{C}_0$ | $\mathsf{C}$ | $\mathsf{E}$ | $\mathsf{T}$ |
|---|---|---|---|---|
| Phase 0 | $\frac{3}{8} + \frac{\sqrt{2}}{4}$ | 5/8 | 3/8 | 3/8 |
| Phase $\pi$ | $\frac{3}{8} - \frac{\sqrt{2}}{4}$ | 1/8 | 3/8 | 3/8 |
| Phase $\pi/2$ | 3/8 | 3/8 | 3/8 | 3/8 |
| Incoherent mixture of $x$ and $z$ | 3/8 | 3/8 | 3/8 | 3/8 |

For every admissible record law the plus probability is $p_\Phi(\theta) = \frac{3}{8} + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$, since the empty seam of $x$ stays in place and $z$ waits with probability $\frac{1}{2}$ while keeping coherence $g$ with $x$. The settings 0, $\pi$ and $\pi/2$ therefore estimate $\mathrm{Re}\,g$ and $\mathrm{Im}\,g$ for whichever law is realized; laws with equal $g$, such as $\mathsf{E}$ and $\mathsf{T}$, are separated by the same construction with the full seam $\{0,1\}$ in place of $z$, which gives $\frac{1}{2} + \frac{1}{2}\,\mathrm{Re}(k\,e^{-i\theta})$ [40].

The count protocol fixes four settings with 8,192 fresh independent attempted trials each and no optional stopping. Each attempt falls in one of twelve common bins of coin, current and outcome, or aborts. At each setting it scores the twelve bin indicators, the plus indicator and the abort indicator, $m = 4(12+2) = 56$ statistics, with sampling half-width $b = 1/32$, independently established apparatus probability budget $\eta = 1/64$ and familywise sampling requirement $\delta = 0.001$.

**Proposition 9.2** (Sampling certificate)**.** *If the attempted trials of each setting are independent, then with* $n = 8192$ *and* $b = 1/32$ *the probability that any of the 56 observation intervals* $[\hat{p}-b, \hat{p}+b]$ *misses its actual probability is at most* $112e^{-16} < 112/S_{32} < 0.000012606 < \delta$*, where* $S_{32} = \sum_{k=0}^{32} 16^k/k!$*.*

*Proof.* Hoeffding's inequality [108] gives $2e^{-2nb^2} = 2e^{-16}$ for each statistic, a union bound over 56 statistics gives $112e^{-16}$, and $S_{32} < e^{16}$; the final comparison is exact rational arithmetic [67]. □

A candidate is rejected when some observation interval is disjoint from its ideal probability widened by $\eta$. Under the apparatus contract, rejection occurs only when an observation interval fails, and both candidates use the same intervals; if the calibration guarantee fails with probability at most $\beta$, the combined bound is $112/S_{32} + \beta$. An abort has ideal probability zero, and a resolved excess returns the verdict that the apparatus budget is exceeded. Without supplied counts the verdict is "no data", and software demonstrations are labelled synthetic [66]. One survivor discriminates the conditional instruments, two survivors do not separate them, and two rejections mean neither specified apparatus model explains the counts; in every case the same counts estimate $g$.

### 9.3 The source-executed implementation

Both complete recorded instruments, the four preparations and the readout also run as emitted twelve-port control words [64, 65]. A configuration is written as the seam pair and the ten spectators: the six endpoint pairs use three bits and the spectators six, placing the 78 configurations injectively in a nine-qubit register with 434 unused addresses. A local permutation with blank record bits realizes each repair from phase-exact transpositions, and the fair coin is prepared by a source pulse and recorded. The encoding is a mathematical isometry, not a physical interface from occupation states.

**Proposition 9.3** (Exact source realization of the two instruments)**.** *In the nine-qubit encoding $J$:*

*(i) the encoded recorded channels intertwine the native ones,*

$$\mathcal{E}_{\mathrm{code}}(J\rho J^*) = (I_{\mathrm{record}} \otimes J)\, \mathcal{E}_{\mathrm{native}}(\rho)\, (I_{\mathrm{record}} \otimes J^*),$$

*on all 6,084 matrix units of the 78-state carrier and therefore with every untouched reference; the complete Gram identity has 156 nonzero Kraus entries;*

*(ii)* $\mathsf{E}$ *is* $\mathsf{C}$ *followed by a three-pulse source controlled-NOT (CNOT) that copies the seam-total parity $p$, with contrast decoded as $d = 2j + (2c-1)p$; relative to the retained coin and current labels one additional binary record is necessary and sufficient, and forgetting it leaves parity dephasing without restoring the coherence of* $\mathsf{C}$*;*

*(iii) the readout is the Lüders instrument [112] of the rank-one plus projector and its complement, realized by coupling the plus projector to a blank pointer and measuring only the pointer; the minus outcome is one rank-77 projector whose internal coherences survive, and reading the full computational basis would implement a different instrument.*

*Proof sketch.* For fixed current the endpoint map is an injective translation, and for fixed contrast the spectators and endpoint sum reconstruct the input, so every input column of the joint coin, record and source isometry has two entries $2^{-1/2}$ with disjoint supports; an independent enumeration checks all 6,084 matrix units. For (ii), $\{2,3\}$ and $\{1,2\}$ share the $\mathsf{C}$ record $(0,0)$ and have contrasts 0 and $-1$ [65]. □

The complete attempted trial costs 32,006 to 32,010 native pulse requests, with one proper-code load and one export per pulse; a readout costs 26,632 pulses, a local $\mathsf{C}$ repair 5,370 and a local $\mathsf{E}$ repair 5,373, with peak live width 11 qubits for $\mathsf{C}$ and 12 for $\mathsf{E}$ [64]. Executed terminal states reproduce Table 4, and on synthetic counts drawn from executed branch norms the evaluator keeps the generating candidate and rejects the other. The compositional error contract

$$N_p u + N_l\, l + N_e\, e + p + z + c + a + r + s + m + g + n \le \frac{1}{64},$$

with pulse, load and export counts and their channel errors and whole-trial allowances for preparation, initialization, coin, randomizer, records, storage, measurement, coefficient compilation and numerical execution, gives sufficient conditions for $\eta = 1/64$; an equal allocation at the largest schedule is 1/6146496 per term. The same source alphabet, code services and classical transitions admit two quantum instruments whose phase-zero observable differs by 1/4, so these ingredients alone do not select the record law.

### 9.4 Calibration and electromagnetic response

The detector in Section 3.4 evaluates its declared coupling map. The complete electromagnetic identification additionally requires the same matter coefficients, current normalization, thresholds and physical low-energy readout. The particle construction states its certified conditional feedback roots and the omitted response terms [118]. Numerical root closure alone supplies no Thomson-limit identification. The source response archive includes exact rational tests of the direct-endpoint hypothesis [66, 67]; this hypothesis must be tested separately from existence of a mathematical fixed point.

### 9.5 A sequential design on one carrier

The phase test above assumes fresh preparations. A sequential design runs it on one continuing carrier in its binary total-two sector without resets or independent trials. Each trial applies 379 unread native attempts to the state the carrier holds, performs the fixed-seam native current herald, whose useful yield is 5/33, and on either useful branch executes the source mixer, three unread native attempts, a second mixer and the native coin and current pointer; a failed herald counts as a trial. After exactly 3,927 trials the two locked likelihoods are compared on all six counts, and an impossible record returns a verdict outside both models. Conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most 3/400 under the exact-source model; a uniform service allowance of 1/1570800 per complete trial would suffice for a one-percent bound, and that allowance is not authenticated [25]. Two consecutive trials per candidate and their reference controls are executed.

### 9.6 Native preparation and calibration of a complete readout

The clocked phase and work readout is a complete instrument at cap $N$: an independent work preparation, a fixed-seam native herald of useful yield 5/33, stopping slots, a supplied balanced mixer, the native phase pointer, all sixteen work records and their feedback, the failed herald and two unmeasured timeouts, with $128N + 3$ joint suffix records [33]. Nineteen fixed-order sweeps of the thirty seams, 570 native attempts with the tape ignored and no conditioning, replace its exact input $I/66$ and initial binary support: every density on $\mathcal{H}_{78}$, with any finite untouched reference, reaches the stationary reference within half trace norm $3.7187 \times 10^{-8}$, and at cap 64 all 8,195 suffix records enter a complete comparison below $3.03719 \times 10^{-6}$ [34]. Joint preparation-and-return tests on a trusted basis and on one uniform coherent vector per outcome block of rank above one determine a complete Lüders instrument, with a reference-uniform error bound [35]. The observers paper proves both theorems [6]. The total-two sector, the quantum instrument semantics, the coherent mixer, the work action, the trusted tests and the model clock are inputs.

### 9.7 Native records against public data

Published records test precise attachments of native records without a laboratory realization of the carrier; each test is retrospective and states its readout hypotheses.

The binary endpoint map is an identity or a local exchange, so diagonal preparations stay diagonal under recorded operations and local classical feedback. Under independent settings and local factorized responses the diagonal branch obeys $P(\text{win}) \leq 3/4$ in the CHSH game, that is $S \leq 2$; a control that lets one side read both settings wins every trial, and the native example on the 66-state carrier, with all 1,056 histories, gives $S = 26/33$. Replaying the deposited loophole-free records of Hensen et al. [28] reproduces 245 eligible trials, 196 wins and $S = 16036094/6619647 \approx 2.42250$, with a published fixed-count tail bound near 0.03917 [26]. This rejects the local classical reading of the diagonal branch under that experimental contract; it does not test the quantum C/E instruments.

Two native ports, or two regions of one carrier, counting incident repairs under the uniform scheduler have independent interval increments with exact count laws. Attaching the published 300-second blocks of the $Ar^{13+}/Yb^{+}$ frequency ratios of King et al. [29] to the normalized count ratio, through a gain of at least one, a declared bracket on the $Yb^{+}$ frequency, an absolute error of $10^{-12}$ per block and direct phase-count readback, gives conditional probability bounds below $10^{-104}$ for two ports and below $10^{-22}$ over all ordered pairs of the 3,433 regional event sets, after charging every choice of subwindow, gain and retained blocks [27]. This rejects the direct attachment with its readout assumptions; the native event clocks are unaffected as clocks of the model, and a clock calibration needs an explicit model of the measuring apparatus.

Published trapped-ion records test a supplied experimental model executed through the same source interface: two calibrated preparation parameters predict 100 reserved complex readouts with per-spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876427868 in two-point position-estimator units [10]. They test the supplied oscillator model, whose action and apparatus do not follow from native repair.

## 10 Finite premises and physical interpretation

A conclusion supplies a subsequent theorem's premise only on the same state, algebra, operation, clock and refinement domain, with compatible error bounds. The model–theorem crosswalk records these maps [43, 44], and Table 5 states the division of supply for the main results. The empirical electron anchor converts specified source energies to GeV. The hadron and neutral-family actions also use disclosed calibration data; their agreements with those inputs are not independent mass predictions. Complete finite realization and unit calibration do not establish a physical species or pole interpretation. The following entropy bounds and countermodels constrain that interpretation without excluding other specified designs.

**Finite entropy and the gravitational premise.** For positive sector counts $d_i$, normalized probabilities $p_i$ and fixed $M = \sum_i d_i$, the counting entropy obeys

$$S_{\text{count}} = H(p) + \sum_i p_i \log d_i = \log M - D(p\|d/M) \leq \log M.$$

The maximizer is $p_i = d_i/M$. In the positive-real relaxation, joint fixed-$M$ variations have zero first derivative there and second derivative $-\sum_i (p_i' - d_i'/M)^2/p_i$. This is no operation varying integer dimensions, and with bulk factors full saturation also requires maximally mixed bulk blocks.

Table 5: What native repair and its counts supply and what each result declares.

| **Result** | **From native repair and counts** | **Declared input** |
|---|---|---|
| Seam rule and record family | Current law, coherence table, uniform stationary density, gap $(5-\sqrt{5})/60$ | Tie-break and covariance premises, uniform schedule, fair coin |
| Binary repair [6] | Conditional resampling at constant energy, entropy identities, complete tape accounting | Total-two sector, fair coin, exogenous schedule |
| Record costs [6] | Prefix weights, Gibbs register reference, bounded refresh kernels | Fair coin, stopping rule, multiplier $b$, comparison controller |
| Closed-history functional [6] | Count moments, $H=-\log P$, modular correlations from path counts | Linear extension, complexification, census settings; $\rho_L$ is unprepared |
| Measurement | Diagonal acceptance effects, complete classical record laws | Complex tomography: coherent services and quantum semantics |
| Reversible seam repair (Section 3.8) | Native carrier and one-event population kernel | Coherent record reuse, address reflection, stationary filter, optional phase probes; internal code and links for gauge transport |
| Preparation (Section 9.6) | Finite warm-up to $I/66$ with a reference-uniform bound | Total-two sector, fair coin, instrument semantics |
| Compiled instruments (Section 6.10) | Source pulses, raw records, aborts, complete-output bounds | Source generators, load and export services, target action, prepared reference |
| Strain and pressure [6] | Charge weights from a complete two-attempt E census | Volume control, energy meter, response class, calibrated sector forces |
| Interacting fields and mass law [118] | None | Weyl/contact action, texture, memory law, couplings, links, mass-law architecture |
| Maxwell form [9] | Incidence tables and their exact ranks | Locality, Faraday, conservation, reversal, field definitions, coefficients |
| Carrier | Rank-three response band of the given incidence | Incidence and response layer (premise A1); the readout selector takes a supplied three-dimensional quadratic readout |
| Physical units and fits | Dimensionless gaps and complete finite energy records | Electron mass anchor; explicit hadron and oscillation calibration; no common laboratory-clock or thermal-history calibration |

Uniform native preparation realizes counting weights. Fixed count is not identified with fixed physical volume, nor the logarithmic sector term with a geometric area term of universal coefficient [12]. For every cap of the wave reference's return-length law, an absolute bound holds on the full $D = \binom{V}{V/3}$ configuration space:

$$\Delta = \log D - S(\sigma) \le \frac{2025}{64V}, \qquad I_\sigma(R : R^c) \le \log J + \Delta,$$

where $J$ counts the allowed regional particle numbers. A complete base-face patch has $V/20$ ports and $3\,2^l$ crossing seams. Its stationary reduced entropy is extensive in port count, while the return reference's mutual information divided by crossing-seam count vanishes. Neither observable therefore has a nonzero universal coefficient times that boundary count. This does not exclude interpreting a whole spherical screen as a bulk boundary; that physical region map is a premise [12].

Four countermodels fix boundaries of the construction [39]. Each rejects a named identification or an overly broad inference; none rejects the theorem premises.

**Complete-carrier center.** The centers of $M_{66}$ and $M_{78}$ consist of scalars, so the unextended working algebra cannot supply twelve nonzero primitive central ports. Binary occupations are noncentral, overlap and sum to $2I$, and total-two occupations also admit the eigenvalue two. The native matrix algebra is kept as a sector; a separate twelve-block central-address extension supplies a central algebra with port-controlled operations, and occupation measurement is a separate contract.

**Marked triplets.** The slow Gram matrix satisfies $G^2 = 4G$, $\operatorname{Tr} G = 12$ and $(5I - A)G = (5-\sqrt{5})G$. One fixed order-five symmetry has trace $(1-\sqrt{5})/2$ on the slow frame and $(1+\sqrt{5})/2$ on the lower response triplet, which excludes an invertible intertwiner for the same action, so spatial and internal response domains are kept distinct. The comparison separates these two specified triplets only; it neither shows every internal triplet inequivalent to the frame nor identifies carrier rechartings with physical rotations. The local rank-three carrier band and the refining global low band are distinct. The conditional Spin–matter construction has independent commuting factors, and its locked diagonal action fails to give a uniform fermionic central sign. A physical separation requires the spatial and internal actions to be identified on the same clocked interacting response and preserved by its dynamics.

**Feedback class.** With the two endpoint loads in the uniform mixture of their one-particle configurations, a rational number-preserving exchange gives endpoint probabilities 769/1250 and 481/1250 for the two private preparations, a difference of 144/625. Four Kraus operators certify that it is CPTP, and its private commutant has dimension one, which excludes every single-product exponential representation, alternate logarithm branches included. The no-feedback theorems for reduced loads and single products hold within their stated classes, and this channel marks the class boundary.

**Configuration equilibrium.** For $A = I + L_{\text{path3}}$ the controls $\varepsilon = \pm 1/16$ keep the normalized configuration Gibbs marginal and change the full reference and the vacuum. For a stiffness response $Y = \gamma(A + \varepsilon A^2)$ one has $Y_{13} = \gamma\varepsilon$ and a third-mode fit residual $6\gamma\varepsilon$, while the actual acceleration is $-Yq$. The theorem identifying inertia with the same capacity keeps that premise, and equal equilibrium configurations do not select the kinetic law.

### 10.1 A recorded separation observable with finite relaxation

On the binary two-load carrier, a known preparation and the complete addressed coin/current records determine every later load by

$$n'_a = n_a - j, \qquad n'_b = n_b + j.$$

A bounded classical controller can therefore track the 66-state source using a seven-bit index and report the graph distance between the two occupied ports. It receives no working source loads after preparation. Exhaustive comparison of the 3,960 native input branches checks the update and readout identities; induction extends them to every finite complete history. Two controllers supplied the same preparation and records agree. Unknown initial states are a different inference problem: the same first current record can arise from preparations with different separations.

For independent uniform choice of the thirty seams and the fair native coin, distances $1, 2, 3$ form a strongly lumpable quotient with

$$K = \begin{pmatrix} 14/15 & 1/15 & 0 \\ 1/15 & 9/10 & 1/30 \\ 0 & 1/6 & 5/6 \end{pmatrix}, \qquad \pi = (5/11, 5/11, 1/11).$$

Representative independence follows by comparing the outgoing probability into each distance fibre for all 66 source states. The uniform source measure pushes forward to $\pi$. The one-step total-variation contraction coefficient is $14/15$. Direct rational powers improve the finite budget: the maximum of $\mathrm{TV}(K^n(i,\cdot),\pi)$ over initial classes first falls below $1/100$ at $n = 39$, where it is approximately 0.009811214413. Convexity supplies the same bound for every initial distribution on the admitted binary sector. No equilibrium fit is needed.

The addressed record alphabet has 120 letters. A capacity of $T$ attempts uses at most $7T$ tape bits. Including retained preparation, tracked state, counter, status and distance gives 298 logical register bits at $T = 39$, or 311 with the incoming packet workspace. A separate tape retaining every distance would add 78 bits. Missing, duplicate, incompatible and out-of-order records halt inference. The source protocol halts at tape capacity, so the last reported separation never stands for unrecorded later motion. Immutable program storage, thermodynamic costs and physical implementation are separate. The known preparation and addressed uniform scheduler are declared interfaces; the fixed two-event observer has its own scheduler. This result constructs a classical record readout, without a measurement claim for arbitrary unknown quantum states or an identification of attempts with physical duration. The executable controller and exact bounds are in `model/repair_sector_bridge`.

### 10.2 Native relaxation and static-geometry obstructions

Three results exclude specified mechanisms of the tested law. Each is a fact about that law and its stated class.

**The native double-load defect is transient.** On the twelve-port carrier, the only configuration functions invariant under every admitted seam and coin are constants, and the binary shell is the unique closed sector. A collision state carries one double load above that shell. Under the uniform seam probe, the complete C and E instruments both satisfy $\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$ for the projector onto the twelve collisions, so after $n$ attempts the collision probability is $(5/6)^n \operatorname{Tr}[(P_{\mathrm{coll}} \otimes I)\rho]$ for arbitrary quantum inputs and untouched references. The mean number of attempts to split is

six, and the double load never hops before splitting. Load differences telescope around all twenty faces for all 78 states, so this gradient map carries no holonomy. Holding one port fixed creates sectors only by disabling its five repairs, which changes the law. The record costs of the corrected memory code have expectations independent of the logical state, so they cannot split logical-state masses [31]. No persistent quantum subspace supported on the collisions exists under this evolution, and the particle paper classifies the compact amplitude subspaces of the binary transfer [118]; the theorem covers no other record-encoded sector.

**The static radial product fails null balance.** For the smooth static product metrics of the conditional scalar description, both radial null Einstein contractions vanish, while a coherent scalar excitation in the restored angular-constant Dirichlet sector has positive relative null stress. That geometry cannot satisfy the Einstein balances for the reference and the excitation at positive coupling with other sources fixed [8]. A responsive geometry is required.

**Passive repair has no wave sector.** The record-discarded native channels are Hilbert–Schmidt self-adjoint with spectrum in $[0, 1]$ and scalar fixed algebra (Theorem 3.4). The executed quantum channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier, and the half-step map of its principal logarithm is not positive, so it generates no completely positive semigroup [10]. The closed mean law is damped (Proposition 7.3(iv)), ballistic rescaling freezes the density paths, and the native decay rates grow like Laplacian eigenvalues, so no common clock factor gives a wave dispersion (Section 7.3). Every wave in this paper is declared record processing or a compiled apparatus step on supplied codes; other dilations or generators are not excluded.

## 11 Particles, interacting fields and gravity

The physical target is one continuing source whose compatible observer records admit an interacting chiral field description in $3 + 1$ dimensions, its particle spectrum and a common classical-gravity limit. Carrier, protocol and memory choices are permitted design variables. For each admitted protocol $p$, a controlled identification requires

$$\sup_p \frac{1}{2} \left\| D_{\text{native}}(\rho_{\text{native}}(p)) - D_{\text{eff}}(\rho_{\text{eff}}(p)) \right\|_1 \leq \epsilon,$$

with the same declared menu, all failure and wait outcomes, untouched references and an error that vanishes in the claimed limit. The decoding channels map to one common output algebra, preserve every claimed observation and intervention response, and are fixed independently of the comparison data. Matching separate marginals or isolated eigenvalues does not prove this statement.

The companion papers state the results on each side of that target. The gauge paper derives the compact Lie type and the conditional matter module from complete-response premises and constructs finite local Gauss constraints and a gauge-covariant band on classical links [7]. The particle paper gives the protected centre-class dictionary and the charged-lepton ratios of the declared generation register (Section 3.9). Positive native-current actions supply calibrated hadron states, complete weak-dressed lines, neutral oscillation instruments and bounded energy records on a common semileptonic source. Their calibration ancestry, failed held-out comparisons and complete observation domains are explicit [118]. The Einstein paper gives regional entropy on native and prepared references, null tomography and causal counts, and the conditional Einstein implication, which requires geometrically normalized modular flow, the same conserved matter stress, a universal positive area coefficient and fixed-proper-volume entropy stationarity with controlled

small-ball remainders and sufficient independent directions [8]. These conditions must hold together for the same source, state, clocks and refinement maps. A chiral continuum also requires controlled spatial refinement, locality, positivity and a complete accounting of tastes and residues; the photon-containing theory must retain massless sectors, and massive poles, unstable resonances and confined observables require their appropriate spectral conventions.

For a coherent rest angular frequency $\omega_0$, a physical clock calibration $T$ seconds per model unit and the physical relation $E = \hbar\omega$, the conditional rest mass is $m = \hbar\omega_0/(Tc^2)$. Assigning the model signal speed $v$ to $c$ sets the length unit $L = cT/v$ and determines neither $T$ nor $\omega_0$ [30], and a necessary small-momentum test of a massive relativistic branch is $\omega_0\omega''(0)/v^2 = 1$. Host work, relaxation rates and maintenance counts acquire no mass interpretation from a unit choice alone. Independent physical comparisons then test the complete dictionary: a failed candidate rejects the stated realization and interpretation, and a whole-class exclusion requires coverage of every admitted realization.

### 11.1 Prepared charge localization and relative spin phase

The complete charged energy measurement also controls position for its specified preparation. Write $H = H_0 + \lambda T + 4N_{\mathrm{ref}}$ for the eight-cell interacting source, with $\lambda = 2^{-34}$ and $\|\lambda T\| \le v = 360\,2^{-34}$. The reference is a retained static charge at the origin. Let $\chi = P_I(H)B_r\Psi$ be the unnormalized prepared charged line, and let $\phi$ be its atomic comparison vector. The complete spectral-projector and preparation bounds give $\|\chi - \phi\| \le \epsilon$, $H_0\phi + 4N_{\mathrm{ref}}\phi = a\phi$, and $\Pi_0\phi = \phi$. In the physical one-reference sector, $\Pi_y$ is the single-charge-at-$y$ projector: it requires every other matter cell to be a gauge singlet and retains all link states. These projectors are pairwise orthogonal on that sector. If $w = \|\phi\|^2 \in [w_-, w_+]$ and $p_- = (\sqrt{w_-} - \epsilon)^2 > 0$, Duhamel's identity gives

$$\Pr(\Pi_0 \mid \mathrm{line}, t) \ge 1 - \min\left\{1, \frac{(\epsilon + |t|v\sqrt{w_+})^2}{p_-}\right\}.$$

The complete cloud effect is $\Pi_{\mathrm{cloud}} = I - \sum_y \Pi_y$, so $I - \Pi_0 = \Pi_{\mathrm{cloud}} + \sum_{y\neq 0}\Pi_y$ retains every remote single-charge outcome and every multi-charged-cell cloud outcome, including screening configurations. Neutral local excitations and link excitations need not lie outside $\Pi_0$. For the native $e^c$ removal channel the origin probability exceeds 0.9999999127 at preparation and 0.9995481684 at $t = 10^6$ in source units; the latter interval exceeds 110,036 atomic source-gap cycles. These are bounds on the complete prepared state under its actual Hamiltonian. They do not apply to every state in the same energy bin, and the fixed compensator is part of the construction.

The source's central spin rotation has an exact finite readout on these same states. Its native generator is $S_z = (N_\uparrow - N_\downarrow)/2$, hence $e^{-2\pi i S_z} = (-1)^N$ on the full source Fock space. The prepared neutral cube is even, each successful elementary removal is odd, and the full Hamiltonian and energy filter preserve that parity. Controlled source rotation and a retained coherent pointer therefore distinguish the minus sign at $2\pi$ from the identity at $4\pi$. Rotating the odd compensating reference as well cancels the sign and supplies a distinct control. All preparation, pump and energy-bin failures are retained. This is a finite source and apparatus result; it does not identify a continuum electron angular momentum or derive an apolar tensor field.

The same energy-selected excitation has a complete finite transport experiment. After the energy record, measure the projector $S_0$ onto blank links and the true local ground doublet at every cell except the origin, retaining the full origin and reference carriers. Its failure is recorded. Since $S_0\phi = \phi$, the unnormalized successful vector $\xi = S_0\chi$ satisfies $\|\xi - \phi\| \le \epsilon$ and $\|\xi\|^2 \ge p_-$. The position failure has probability at most $\epsilon^2$, conditional only on the physical preparation. The

fraction outside the recorded energy interval after position success is at most $\epsilon^2/p_-$. This bound is preserved by the unchanged source clock; the position intervention does not preserve exact energy-bin support.

For each of the six incident oriented links, an arrival projector $P_e$ requires that link excited, every other link blank, the true local ground at every nonarrival cell, and the same complete local charged energy atom at the neighboring cell. The six effects are orthogonal, and their full complement is retained. The compensating reference stays at the origin. The exact no-hopping kernel $P_e e^{-itH_0} S_0 = 0$ suppresses the preparation error in the arrival amplitude by the hopping coefficient. The selected arrival atom differs in energy from $\phi$ by exactly one, the declared nonblank-link cost. With $c_e = \|P_e T\phi\|^2 > 0$ and $0 \le t \le 6$, the first Dyson term and the full remainder give

$$\left| \|P_e e^{-itH}\xi\| - 2\lambda \sin(t/2)\sqrt{c_e} \right| \le 15\lambda t\epsilon + \frac{(360\lambda t)^2}{2}.$$

The exact local positive and spin residues determine $c_e$ on each charged channel; no charged single-particle Hamiltonian replaces $H$. Outward rational bounds give strictly positive original-input arrival probabilities at source times one and three, and a lower bound at time three greater than the upper bound at time six. Every preparation, pump, energy, position and nonarrival outcome is included. This is a coherent off-resonant flux excursion of an energy-selected preparation with a quantified position disturbance. At time three the summed six-route original probabilities lie in $[2.78, 3.67]\,10^{-22}$ for $e^c$ and $[6.08, 7.37]\,10^{-22}$ for $L$; their time-six upper bounds are $2.51\,10^{-23}$ and $3.77\,10^{-23}$. The energy-bin leakage after position success is below $8.73\,10^{-8}$ and $1.15\,10^{-7}$, respectively. No charged dispersion relation or physical electron identification follows from these finite records.

On the invariant one-$e^c$-reference sector populated by the pump, each $e^c$ arrival also refines the complete native charge and flux measurement. The source endpoint has $Q = Y = -1$, its retained reference has $Q = Y = +1$, and the oriented flux of any cell region $S$ is $\mathcal{F}_S = \mathbf{1}_{0\in S} - \mathbf{1}_{y\in S}$ for arrival at $y$. The same original-input intervals therefore apply to these joint records, including every one of the 256 regions. They do not assign a mean flux to the unconditioned prepared source.

These probabilities imply large sampling costs. With independent complete source and apparatus preparations, sufficient original-input budgets to observe at least one time-three arrival with failure probability below $1/20$ are $1.08 \times 10^{22}$ for $e^c$ and $4.94 \times 10^{21}$ for $L$. Sufficient budgets to obtain a positive sampled time-three minus time-six contrast with error probability at most $1/20$ are $1.23 \times 10^{23}$ and $4.77 \times 10^{22}$ per time setting, respectively. Every failed preparation or measurement counts as a trial with no arrival. These are sufficient sampling bounds for the declared independent experiment, without a practical execution claim.

## 11.2 Finite electromagnetic and matter-wave readouts

For a finite incidence complex, let $D$ be the gradient and $C$ the curl, with $CD = 0$. Positive electric and magnetic energy, Faraday evolution and the declared time reversal give

$$\dot{E} = c^2 C^* B, \qquad \dot{B} = -CE.$$

A total-occupation cutoff preserves the linear Heisenberg equations when the positive boundary completion is retained. The energy differences of $H = \sum_j \omega_j a_j^* a_j$ establish $[H, a_j] = -\omega_j a_j$ and hence the linear Heisenberg equations on every retained state, including the cutoff boundary. This does not assert canonical coordinate commutators on the finite carrier. Propagation therefore reduces to the coordinate matrices, without diagonalizing the full Fock carrier.

For the declared periodic cubic extension, the two transverse polarizations obey

$$\omega_a^2 = \frac{4c^2}{a^2}\sum_j \sin^2\frac{ak_j}{2}, \qquad 0 \le c^2|k|^2 - \omega_a^2 \le \frac{c^2a^2}{12}\sum_j k_j^4.$$

Independent real-space evolution and exact modal propagation test this bound. The same gradient supplies the minimum-energy neutral electrostatic solution

$$E_L = D(D^*D)^+\rho, \qquad U = \frac{g^2}{2}\rho^*(D^*D)^+\rho.$$

Potential differences and forces approach the Coulomb form under the stated volume and spacing controls. The coefficient $g^2/(4\pi)$ retains the declared field normalization. A Dirichlet consumer with supplied nonrelativistic kinetic energy has negative ground and excited levels; exact reflection sectors reduce the largest $95^3$-site source to 110,592 even-sector rows. The ground-to-odd spacing decreases from approximately 0.686 to 0.4204 under the retained refinement, compared with the hydrogenic reference 0.375 in its declared units. This is a conditional bound-state calculation with unresolved continuum and finite-volume errors, not a precision hydrogen spectrum.

A separate complete sixteen-row native mass-channel calculation retains all eight valleys and verifies the massive second-order wave law through direct time evolution. Its finite-spacing boost defect is bounded and decreases under refinement. The supplied mass coefficients, electromagnetic normalization and physical particle identification are separate inputs; these finite calculations do not identify a common interacting relativistic source.

### 11.3 Metric response and conditional cosmological consequences

For the reciprocal scalar action density $C(r) = \mathrm{diag}(-1/r, 1/r, r^2, r^2)$, the metric dictionary $g = \sqrt{-\det C}\, C^{-1}$ gives unit proper spatial volume, two-face area $A = 2/r$, and lapse $L = r$. On the two branches $r = 9/10, 11/10$, a complete matter-randomization intervention followed by the unchanged thermal update therefore obeys

$$\Delta\langle L\rangle = -\frac{99}{200}\Delta\langle A\rangle.$$

The lapse response per unit randomization is enclosed about 0.0003103930363, retaining all sixteen joint levels and every rejected proposal. This is a finite metric readout; no spatial potential or frequency comparison between separated clocks is inferred.

The conditional regional Einstein bound has a direct weak-field consumer. For a static spherical pressureless source, with negligible or subtracted cosmological background, let every component of the Einstein residual be bounded by $\epsilon_E$ and vanish outside radius $R$. In $ds^2 = -(1+2\Phi)dt^2 + (1-2\Psi)d\mathbf{x}^2$, direct curvature differentiation gives $\nabla^2\Phi = 4\pi G_N\rho + (E_{00} + \sum_i E_{ii})/2$. Regularity at the origin and vanishing potential at infinity imply

$$|\delta\Phi'(r)| \le \frac{2\epsilon_E R^3}{3r^2}, \qquad |\delta\Phi(r)| \le \frac{2\epsilon_E R^3}{3r}, \qquad r \ge R.$$

Thus the matched regional entropy and area errors bound the exterior force directly. The geometric and stress identifications required by that regional theorem, together with the weak-field and compact-support assumptions, are explicit premises of this implication.

For the separately specified capacity-density branch $\rho_{\mathrm{DE}} = \kappa/N(a)$, Friedmann continuity yields $w = -1 + \frac{1}{3}d\ln N/d\ln a$. A flat comparison history with $E = H/H_0$ satisfies

$$\frac{N(a)}{N(1)} = \frac{\Omega_{\mathrm{DE}}}{E(a)^2 - \Omega_m a^{-3} - \Omega_r a^{-4}}.$$

This normalization-free identity tests capacity evolution while leaving its amplitude unselected. A numerical cosmological constant requires the additional global-capacity and horizon identifications; the displayed identity does not determine them.

### 11.4 Magnetic response of the native kinetic family

The complete sixteen-row native kinetic matrices obey the Clifford relations. In a declared Peierls gauge extension with scalar mass $m$, their finite Hamiltonian satisfies

$$H^2 = m^2 + \Pi_x^2 + \Pi_y^2 + \Sigma_z i[\Pi_x, \Pi_y], \qquad \Sigma_z = -i\gamma_x\gamma_y, \quad S_z = \Sigma_z/2.$$

A smooth uniform external field therefore gives the nonrelativistic spin term $qBS_z/m$, corresponding to tree-level $g = 2$ without an inserted Pauli coefficient. Exact internal projectors retain all eight valley labels and reduce the largest 266,256-row example to two 16,641-row spatial operators. Tapered finite-packet energy responses give $g_{\mathrm{eff}} = 1.95802, 1.98942, 1.99734$ at spacings $1, 1/2, 1/4$; coherent spin-phase measurements agree, with an explicit square-root truncation bound. The gauge extension, mass and external field are supplied. No self-sourced dipole, physical electron identification or radiative anomalous moment is inferred from this external-field calculation.

The interacting charged source has a different response to a declared central hypercharge background. On each complete 119-state quantum link, put $K_e = -G^Y_{e,\mathrm{left}}$. Its actual coordinates obey $[K_e, U^{(r)}_{ab}] = Y_r U^{(r)}_{ab}$. Multiplying the hopping paths by their hypercharge phases therefore gives the exact full-source identity

$$H(A) = V(A)H(0)V(A)^\dagger, \qquad V(A) = \exp\left(i\sum_e A_e K_e\right).$$

The link penalties commute with $V$, and the source contains no bare plaquette term. Hypercharge is central, so this conjugation preserves every Gauss constraint. Independent link angles are allowed, including a profile with a nontrivial square-loop phase. Every energy eigenvalue and multiplicity is consequently unchanged. For a profile generator $K$, the nonzero current $J = i[K, H]$ and second derivative $D = -[K, [K, H]]$ obey, for every complete energy projector $P_E$,

$$P_E J P_E = 0, \qquad P_E D P_E + 2P_E J(1 - P_E)(E - H)^{-1}(1 - P_E)J P_E = 0.$$

The inverse is restricted to the complementary energies. This exact cancellation includes the diamagnetic term and the full spectrum.

Conjugating the complete preparation and apparatus gives equal energy, spin and location record probabilities with all failures retained. Blank-link position preparations and the displayed arrival effects are themselves invariant under $V$, so their complete arrival laws have this background independence as well. A fixed coherent preparation can have a nonzero change in its mean energy when the background is quenched; spectral invariance does not equate that experiment with paired preparations. A fixed added plaquette term breaks the conjugation identity. The result establishes zero static hypercharge energy splitting for this interacting construction. It supplies no electromagnetic gyromagnetic factor and does not transfer the separate free-source value $g = 2$ to these charged excitations.

### 11.5 Conserved charge without a winding dictionary

The native internal current $Q = T_3 + Y$ and colour Cartan generator $C$, induced by $\mathrm{diag}(1, 1, -2)/3$, obey $3Q_i \in \mathbb{Z}$ and $Q_i + C_i \in \mathbb{Z}$ on all fifteen internal weights. Exterior products lift these identities

to every fermion number and any number of spatial or family copies:

$$\operatorname{spec}(d\Gamma(Q)) \subset \tfrac{1}{3}\mathbb{Z}, \qquad e^{2\pi i d\Gamma(Q)}\Gamma(Z_{\text{colour}}) = I.$$

Colour-singlet charge eigenvalues are integral, and a half-unit charge eigenvalue is absent. Every specified guarded texture-record map is $\sqrt{p_r}\Gamma(U_r \otimes I)$ and preserves these charge sectors. A mixture can have mean charge one-half while its retained outcomes are zero and one. This is finite spectral quantization and branchwise charge conservation, conditional on the declared current, normalization and record law; it does not exclude charge-coherent states or other external interventions.

The compact texture independently has integer shell degrees $-1, +1$. Every accepted guard-preserving update is a nonsingular homotopy of its closed shell; rejected updates are identity maps. Unit degree therefore cannot split into two equal integer degrees under that law. The same fixed texture nevertheless admits neutral and charged fermionic occupations, so the current cannot be a function of winding alone on the unchanged carrier. The direct current proof does not require that false identification. An observed electromagnetic dictionary and a local physical dressing require their own source attachment.

## 11.6 Native scalar attraction and its universality obstruction

In units of the local factor $\sqrt{14}$, the three-oscillator reciprocal source has

$$M(r) = a(r)D - \frac{C}{7r}, \qquad a(r) = \frac{r}{4} + \frac{1}{7r} + \frac{3r^2}{28}, \qquad J = \left.\frac{dM}{dr}\right|_{r=1} = \frac{9}{28}D + \frac{1}{7}C.$$

Here $D_{ss} = 2\,\mathrm{popcount}(s) + 3$ and $C = XXI + IXX$. A specified matter-state contrast supplies charge $j = \mathrm{Tr}(\delta\rho J)$. On a declared spatial extension with positive elastic energy $\kappa q^T L q/2 + j^T q$, completing the square yields the mutual interaction $-j_x j_y (L^{-1})_{xy}/\kappa$. The local coupling comes from the source derivative; spatial gluing, stiffness, zero boundary and a massless geometry mode are separate architecture inputs.

A finite dyadic geometry carrier admits exact minimum-energy enclosures from a trial residual and a lower bound on the Dirichlet Laplacian. For the occupation contrast, the native nonlinear remainder is exactly

$$\frac{q^2(7+3q)}{14(1+q)}.$$

Bounding it on the entire finite window controls the nonlinear minimum without solving a nonlinear optimization. The retained separation-six force interval is strictly attractive. However, two certified preparations above the actual local ground have positive excess energies and opposite $J$ contrasts. This scalar extension consequently does not supply universal positive-mass gravity or a derived Newton constant.

## 11.7 Charge, flux and filtered arrival on the interacting source

The central hypercharge current supplies an exact field attachment on the eight-cell source. Orient each stored link from its left to its right endpoint and set $b_e = L^Y_{e,0}$. The complete 119-state link satisfies $L^Y_{e,1} = -b_e$. With $(Df)_e = f_{\text{right}} - f_{\text{left}}$, the physical Gauss constraints give

$$D^* b = \rho, \qquad \rho_x = \rho_x^{\text{matter}} + \rho_x^{\text{scalar}} + \delta_{x0} Y_{\text{ref}}.$$

On the invariant one-$e^c$-reference sector populated by the native removal pump, each of the six elementary $e^c$ arrival routes has $Q = Y = -1$ and the retained reference has $Y = +1$. Summing

the actual endpoint currents over any region gives its total enclosed charge, including that reference. All $6 \cdot 256$ regional identities and all 24,840 native factor-column Gauss comparisons are checked. The full flux spectra include every link state. The noncentral weak component of $Q = T_3 + Y$ does not admit the same endpoint cancellation on an arbitrary link; the stated central-current identity does not omit its gauge charge.

An energy-resolved experiment uses the actual full-$H$ bin projector $E$ of radius $1/1024$ about the visible $e^c$ pole and an arrival projector $F_e$ refined by its local atomic pole. If $a$ is the atomic pole energy, then $H_{\mathrm{at}}F_e = (a+1)F_e$, where $H_{\mathrm{at}} = H_0 + 4N_{\mathrm{ref}}$, and $F_eTF_e = 0$. The latter follows from the singlet compression of the other matter endpoint and holds with all link states retained. The resolvent identity controls $F_eE$ without assuming that the complete charged band is isolated. Together with the exact native pole residue, it yields original-input probabilities between $1.04092024\,10^{-23}$ and $1.68685307\,10^{-23}$ for each tagged arrival at $t = 0$. Their change at $t = 2^{-24}$ is at most $2.24579\,10^{-34}$. These records measure virtual dressing present in the filtered state; they do not establish propagation from the origin. All seventeen outcomes, including preparation, pump, bin, untagged and nonarrival outcomes, are retained. The local tag can change the energy expectation and is an active intervention.

This experiment records the atomic ground and blank-link projector $P_0$ before pumping and filtering, so its successful initial vector is $EBP_0\Psi$. The finite excursion above instead measures $S_0$ after energy selection, giving $S_0EB\Psi$ and exactly zero initial arrival. Its positive later arrivals and quantified energy disturbance concern that complete measurement sequence. The virtual initial tail and the position-prepared excursion have distinct preparations and are not interchangeable transport laws.

The reference also has a quantitative binding effect. Let $\Pi_y$ require one unscreened charge at $y$ and gauge-singlet matter at every other cell, retaining all links. Exact Gauss strings and local endpoint embeddings give the constrained energy differences

$$\mathcal{E}_L(y) - \mathcal{E}_L(0) = d(0,y), \qquad d(0,y) - w \le \mathcal{E}_{e^c}(y) - \mathcal{E}_{e^c}(0) \le d(0,y) + w, \quad w \simeq 0.591071386691.$$

The additive reference cost cancels from these differences. The compressed subspaces need not be invariant under $H$, and the complete complement contains screening clouds. Thus this is a result about the specified single-unscreened-charge family, with no assertion of unrestricted confinement or free-electron identification.

## 11.8 Native free beats with complete coherent heralding

Let $P = P_0 \otimes |0_{\mathrm{ref}}\rangle\langle 0_{\mathrm{ref}}|$ restrict the atomic source ground projector to the reference vacuum. The local pole residue defines a partial isometry $A_r = S_rB_rP/\sqrt{w_r}$ with $A_r^*A_r = P$ and $A_r^2 = 0$. One declared coherent preparation makes an equal superposition of its two source subspaces. Thereafter the unchanged full Hamiltonian evolves the fixed quadratures $X = A_r + A_r^*$ and $Y = i(A_r - A_r^*)$. In the atomic comparison their expectations are $\cos \Omega t$ and $-\sin \Omega t$. For the ground-to-$e^c$ transition,

$$\Omega = 4 + \Delta_{e^c} \simeq 4.691656680938395.$$

The static reference energy is part of the measured beat. A common herald onto the union of its two actual full-$H$ bins preserves coherence. If $\eta$ bounds the projector error, the waveform error is at most $2\eta + 2v|t|$, uniformly over localized spins and untouched references. At $t = 10^6$ it is below 0.041953.

The distinct reference families also support $A_{e^c}A_L^*$, whose arms each contain one reference particle. Its frequency is $\Delta_{e^c} - \Delta_L \simeq 0.594768183046114$, with exact cancellation of the common

reference cost. Resolving the two herald arms separately destroys the relevant coherence, as does reference-family dephasing. Seven complete records retain the preparation and herald failures and the quadrature outcomes. The initial coherent preparation and its work are explicit apparatus inputs; these free beats supply neither passive self-starting nor a physical Compton-clock identification.

### 11.9 Coherent compilation of the lepton rest Hamiltonian

The full native pairing admits a finite-group reduction that preserves the entire physical Fock carrier. If $N_A$ counts ancillary fermions, put $R = \exp(i\pi N_A/2)$ and

$$\overline{V} = \frac{1}{4}\sum_{r=0}^{3} R^{-r} V R^{r}.$$

The average removes every monomial creating or annihilating one or two ancillary fermions. For the vacuum-ancilla embedding $J$ it obeys $\overline{V}J = JH_Y$, where $H_Y$ is the analyzer-derived mass pairing. Two-phase parity averaging fails on ancillary pairs. Parity of the complete charged-lepton leg changes the sign of $V$ and implements negative interaction segments without discarding any Fock state.

Let $S_2$ be the symmetric seven-factor formula for the four conjugates $R^{-r}VR^{r}/4$. With $p = (4 - \sqrt[3]{4})^{-1}$, the composition

$$S_4(h) = S_2(ph)^2 S_2((1-4p)h) S_2(ph)^2$$

matches the exponential through degree four. Exact noncommuting-word coefficients verify the cancellation. Since $\|V\| \le 6s$ on $s$ rest sites, a Taylor-tail bound and unitary telescoping give complete, unhalved output trace error

$$\epsilon \le \frac{(14s)^5 |t|^5}{10 n^4}, \qquad n \ge 14 s|t|,$$

uniformly over all inputs and retained quantum references. At $s = t = 1$, $n = 15{,}229$ cycles, comprising 533,015 interaction segments, suffice for $\epsilon \le 10^{-12}$. The repeated-reset implementation has a conservative budget of 108,000,000,000,000 rounds at that tolerance. The coherent construction uses one initial ancillary vacuum and retains every final outcome. Its interaction time is $(8p-1)|t|$, and its phase and parity controls are declared services. The proof assumes zero kinetic Hamiltonian; it does not reverse a noncommuting kinetic term.

The three coefficients, approximately $1 : 206.7703 : 3477.4728$, are unchanged by this compilation. With the empirical electron anchor $0.00051099895069\,\mathrm{GeV}$, the guarded numerical coefficient intervals are

$$m_\mu \in [0.105659408862295, 0.105659420129515]\,\mathrm{GeV},$$
$$m_\tau \in [1.77698487649602, 1.77698506513085]\,\mathrm{GeV}.$$

These intervals enclose the compiled coefficients under the declared register and calibration. They are not physical uncertainty intervals; the retrospective register choice, approximately 9.83 parts-per-million muon discrepancy, interacting-particle identity and physical clock remain distinct scientific conditions.

### 11.10 A neutral protected sector and its boundary dependence

The actual Higgs-lepton composite $J_s = \phi_0 c_{s,11} - \phi_1 c_{s,10}$ commutes with all twelve native gauge currents. It is odd under fermion parity and changes raw lepton number by one. Raw lepton number and baryon number commute with the unchanged local Yukawa and hopping Hamiltonian. The full atomic gap $g$, hopping bound $v = 360\,2^{-34}$ and an admissible local neutral trial give

$$g - v \le E_{\mathrm{sector,min}} - E_{\mathrm{ground}} \le \Delta_\nu + v, \qquad \Delta_\nu \simeq 0.089555330476.$$

The numerical enclosure is $[0.04117056754057018, 0.08955535143083537]$ in source units. The lowest-sector projector is stationary and cannot decay to the source ground under the specified lepton-number-preserving dynamics. This ideal spectral projector is defined mathematically; a full interacting eigenvector or a primitive preparation of that projector is not computed here.

The original antiperiodic translations square to fermion parity and exclude zero momentum in this odd sector. A separately declared periodic member reverses twelve seam signs. The second-order ground operator and quartic remainder bounds are unchanged. Eight translated normalized neutral trials are orthogonal, so their zero-momentum average has squared norm $1/8$ and supplies the same positive threshold bound. Using the existing electron energy calibration gives a conditional interval $[3.0416993564\,10^{-5}, 6.6163881404\,10^{-5}]$ GeV. This is a neutral finite-sector energy, with no inferred observed neutrino identity, relativistic mass shell or cosmological abundance.

### 11.11 Stationary intrinsic magnetic response

An exact unitary decomposes the complete sixteen native Clifford rows into four Dirac blocks while retaining every valley label. A declared spatial construction adds a scalar confining well and Wilson regulator to give a Hermitian operator $D_W(A)$ with positive mass lower bound $m$. Its positive matter kernel is $K(A) = |D_W(A)|$. For a normalized one-particle state and a mean-zero transverse field, use

$$\mathcal{E}(\psi, A) = \langle \psi, K(A)\psi \rangle + \frac{a^3}{2\kappa} \|CA\|^2.$$

Compactness of the finite state sphere and field coercivity prove that a minimum exists. The derivative includes both signed matter branches: if $X = \delta K$, then $KX + XK = D_W \delta D_W + \delta D_W D_W$. The Peierls derivative, including the Wilson contribution, supplies the actual bond current. The stationary solve obeys $D^* J = 0$ and $C^* C A = \kappa J / a^3$; its current is recomputed from its own field.

With $m = 8$, scalar-well strength 0.01, regulator $1/2$ and coupling $\kappa = 0.1$, three fixed-extent refinements give bound-energy $g$ values 1.47516960, 1.68416493 and 1.78875065. The convention is $g = 2E_{\mathrm{bound}}\mu_z/(qS_z)$ with $q = -1$; replacing the measured bound energy by the supplied $m$ gives different numbers. The computed unregulated central-difference control has intrinsic current below $10^{-10}$ on its selected spatial parity branch. The nonzero stationary dipole, its Maxwell residual and its static weak-field energy derivative are numerical results. The displayed branch is not certified to be the global minimum, and three spacings do not establish a continuum value of two. The construction includes external mechanical confinement and a semiclassical transverse field; the stationary calculation supplies no longitudinal electromagnetic completion or time-dependent experiment. The confined magnetic member's evolution is described in Section 11.15. Its classical connection and transverse Maxwell energy belong to this distinct construction. They do not satisfy the independent quantum-link conjugation hypotheses of the interacting charged source.

## 11.12 Longitudinal field energy on the native charge carrier

The same native Gauss identity permits the readout

$$G = (D^*D)^+, \qquad E_{\mathrm{long}} = DG\rho, \qquad \tfrac{1}{2}\|E_{\mathrm{long}}\|^2 = \tfrac{1}{2}\rho^* G\rho.$$

It is the minimum-norm field with divergence $\rho$. The complete native flux retains its divergence-free part; this readout does not project a quantum state onto a different carrier. Each of the six unit-charge arrivals on the side-two cube has quadratic value $7/48$. On the original source this is an observable, rather than its link energy. Adding this positive energy to the unit-link penalty cannot cancel the constrained linear binding cost.

A separate declared Hamiltonian replaces that penalty:

$$H' = H_{\mathrm{local}} + \lambda T + 4N_{\mathrm{ref}} + \frac{\kappa}{2}\rho^* G\rho.$$

All matter, scalar, reference and link states are retained. The central density commutes with the local gauge action, so all Gauss constraints and the native continuity equation are preserved. Choose the native $e^c$ local endpoint eigenstate at $y$, a local ground state at every other cell, a fixed one-particle reference at zero and a normalized physical flux path connecting the endpoints. This trial has exact expectation

$$(V-1)E_0 + e_{e^c} + 4 + \kappa(G_{00} - G_{0y}),$$

independent of its path length. The same local lower bound gives a constrained energy enclosure of distance-independent width. The computed Green interaction approaches $-\kappa/(4\pi r)$ in counting distance; finite-box potential-difference and force amplitude ratios at side 128 and radius eight are approximately 1.00407 and 1.02842. These are static trial and field energies. At nonzero hopping the strings need not be eigenstates. Unpenalized flux loops retain a degeneracy, and the original ground preparation and clock do not transfer to $H'$. The coupling and physical length normalization are declared; these identities determine no physical fine-structure constant.

The static hypercharge obstruction also holds for this alternative Hamiltonian. Its density acts on matter, scalar and reference factors, so $[K_e, \rho_x] = 0$ and the longitudinal energy commutes with every $V(A)$. Consequently $H'(A) = V(A)H'(0)V(A)^\dagger$: replacing the linear link penalty by the Coulomb energy does not produce static hypercharge spectral splitting. This operator identity imports no ground preparation, charged energy bin or transport law from $H$ into $H'$.

## 11.13 Separated metric observations and atomic refinement

The reciprocal scalar geometry supports an explicit test of its clock dictionary. On the computed weak geometry preparation, all seven ground-to-excited ratios of the complete eight-level local Hamiltonians differ from one another and from the metric lapse ratio. The devices have fixed coordinate extent and changing proper dimensions, so this does not constitute an equivalence-principle refutation. It establishes that this device choice supplies no universal metric clock attachment. A declared transverse scalar-wave extension uses the same density $C(r)$, retaining all 31 normal modes and the vacuum. Its complete detector laws have total-variation distances from the flat construction of approximately 4.0 to 7.7 parts per million at the executed times. The associated piecewise metric has transverse ray delay $6.4197332251\,10^{-5}$ in source time and radial null speed one. The dispersive wave arrival is not identified with the ray. A harmonic scalar tail also has nonzero linearized Einstein residual under the tested vacuum or pressureless interpretation; the regional Einstein theorem therefore cannot supply that interpretation without its stress and geometry premises.

For the separate Dirichlet atom with supplied $m = 1$ and $g^2 = 4\pi$, volume and spacing effects are tested independently. At fixed spacing $a = 1/3$, boxes of side 24, 32 and 40 give first gaps 0.420953171076, 0.420395862457 and 0.420371341048. At fixed side 24, spacings 1/3, 1/4, 3/16 and 3/20 give 0.420953171076, 0.399138718909, 0.388151976665 and 0.383407843476, compared with the hydrogenic reference 3/8. The second-shell splitting decreases from 0.005500452345 to 0.000410646363. Exact reflection reduction represents even standing waves over all $159^3 = 4{,}019{,}679$ sites through 512,000 even-sector rows; the other parity sectors are separate. The discrete maximum principle, zero extension and min–max theorem prove monotonicity of each ordered parity eigenvalue with volume at fixed spacing. They do not prove monotonicity of gaps or spacing convergence. Residuals bound distance to some spectrum; the reported floating residuals and successive-grid differences are not certified continuum uncertainties or level-index proofs.

### 11.14 An isolated stationary neutral band

Consider the periodic eight-cell fermion, quantum-scalar and 119-state-link source with nonblank-link coefficient two and hopping coefficient $\lambda = 2^{-34}$. Every local scalar and Yukawa coefficient is retained. In the physical sector with raw lepton number seven and baryon number zero, the atomic band has one local neutral singlet of energy $e_\nu$ and seven local ground doublets of energy $e_0$. It has rank 1024, with rank 128 at each of eight momenta. Its complement gap exceeds $d = 0.0110299637715905$ and the hopping perturbation has norm at most $v = 360\lambda$. Blank-link selection gives $PTP = 0$. A Schur-complement bound therefore confines the perturbed band to

$$[a - \epsilon, a], \qquad a = 7e_0 + e_\nu, \qquad \epsilon = \frac{v^2}{d - v} \le 3.9810\,10^{-14}.$$

The spectral projector satisfies $\|E - P\| \le v/d < 1.900\,10^{-6}$. The same source's positive second-order exchange and bounded quartic remainder prove a unique physical ground at zero momentum; the certified ground gap exceeds $4.4542\,10^{-25}$. Relative to that ground, the neutral band is contained in $[0.0895553304760346, 0.0895553304760794]$. This argument uses the actual finite exchange operator and a uniform remainder, avoiding diagonalization of the entire interacting carrier.

The local neutral rank-one projector is an explicit algebraic polynomial of the sixteen-dimensional native local block and is annihilated by all twelve gauge currents. On the full carrier, a Chebyshev filter with repeated rejection gives a degree-167608 polynomial $M(H)$ approximating the neutral band projector, with operator error below $10^{-69}$. Together with the commuting gauge, charge and momentum projectors, it defines the complete accept/fail instrument. From the maximally mixed input, the accepted source density commutes exactly with $H$ and is within $7.02\,10^{-13}$ in unhalved trace norm of the uniform zero-momentum band. The original-input success probability is approximately $2.85\,10^{-126}$. The polynomial degree counts Hamiltonian evaluations; it is not a primitive gate count or a claim of full-matrix execution. Stationarity concerns the reduced source density and its time-independent source readouts, without asserting stationarity of an arbitrary purification.

A complete 51-record instrument retains preparation failure, energy and profile complements, and 48 position/occupation outcomes. The stationary conditional profile is uniform. A separate localized filtered preparation has leakage bounded, up to the specified filter error, by

$$\left( \frac{\eta}{\sqrt{1 - \eta^2}} + \epsilon|t| \right)^2, \qquad \eta = v/d.$$

At $t = 10^{12}$ this bound is below 0.001585. Position and momentum projectors do not commute; their probabilities are separate instruments. The exact plane/link symmetry in Section 11.28 strengthens

the momentum statement: every momentum fiber has precisely the same full band spectrum, and its dressed location projectors commute with the Hamiltonian. This periodic side-two source supplies no positive propagation speed. A moving relativistic assignment is incompatible with its exact momentum flatness; larger or antiperiodic members require their own analysis.

Calibrating a separate native charged local gap by the empirical electron energy yields $6.616386592241\,10^{-5}$ GeV for this neutral band. It is a calibrated zero-momentum excitation energy. Neither a relativistic mass shell nor a dark-matter or neutrino identification is inferred from its stability and neutrality.

### 11.15 Self-consistent magnetic precession and refinement

For the confined positive kernel $K(A) = |D_W(A)|$, use the semiclassical energy $\langle\psi, K(A+A_{\mathrm{ext}})\psi\rangle + a^3\|CA\|^2/(2\kappa)$ on mean-zero transverse link fields. The full absolute-kernel derivative retains both signs of the Wilson operator. In its eigenbasis the density entering the current is

$$W_{ij} = \frac{\lambda_i + \lambda_j}{|\lambda_i| + |\lambda_j|}\rho_{ij}, \qquad J_e = -\operatorname{Tr}(W\partial_e D_W).$$

Only the transverse projection of $J$ sources the quasistatic magnetic field; the full current obeys the source continuity law. A uniform Hessian estimate on the executed coarse cube makes the fixed-state field map a contraction with factor below 0.608135. It proves a unique field minimizer for each normalized matter state, without asserting contraction of a coupled time-stepping algorithm.

Evolution under this same kernel gives a complete weak-field precession cycle. The spin and self-sourced dipole frequencies are approximately 0.002848096149 and 0.002848053578, while an independently optimized stationary energy split is 0.002848114846. Reversed, zero, halved and frozen-field probes and step refinement retain their respective errors. Complete energy, spin and moment instruments include all spectral branches. These dynamics are nonlinear quasistatic mean-field dynamics; they do not define a quantum channel on arbitrary entangled inputs. Their magnetic field has no propagating electromagnetic degrees of freedom.

At fixed matter extent six and field extent twenty-one, spacings $1, 3/5, 3/7, 1/3$ give static bound-energy factors approximately $1.47517, 1.78875, 1.88532, 1.92738$. Independent volume and symmetric-probe controls are retained. Fit-order-dependent intercepts ranging from 2.0002 to 2.0746 do not give a certified continuum uncertainty. On the finest matter grid a separately declared coupling $\kappa = 10^{-14}$ makes the self-field error uniformly small. A zero-probe stationary current witness on this same weak-coupling member has eigen-residual below $7.5\,10^{-15}$ and a nonzero self-field. Its static doublet normalization 1.92095809 agrees with the weak-field precession value to the stated finite-probe precision. Two actual full-operator eigenline witnesses prepare the precessing state. Exact rational link-phase Taylor bounds and tridiagonal Sturm counts certify their residuals and isolate the complete low quartet. Duhamel's identity bounds the full nonlinear trajectory through $T = 6000$, including every unprepared mode. The finite-duration ground-rest-energy normalization lies in

$$1.920947397728 \le g_T \le 1.920952163162.$$

The static convention divides by the measured canonical spin; the precession convention uses its rotation frequency. Their numerical values must not be equated when the projected spin is below one half. The regulator, confinement and weak-coupling choice are supplied. These finite results do not identify the precision physical electron magnetic moment.

## 11.16 Exact pinned Coulomb minima

Replace the unit-link penalty in the periodic gauge source by $(\kappa/2)\rho^Y G\rho^Y$, where $G$ is the mean-zero inverse of the same lattice incidence Laplacian. Pin a complete native hypercharge $-1$ weak/colour-singlet multiplicity space at $y$, a unit reference at zero, and complete gauge-singlet matter multiplicities elsewhere. Every link state and local multiplicity is retained. Each native hopping word changes a pinned local density, so its compression vanishes exactly. The constrained physical carrier factors into local multiplicities and a nonempty flux space. Actual native strings prove nonemptiness and saturate the tensor lower bound. Thus

$$E_{\min}(y) = (V-1)E_0 + e_\chi + 4 + \kappa(G_{00} - G_{0y}).$$

The unknown complete charged local minimum $e_\chi$ cancels from every separation difference. This removes the need to solve that local spectrum in order to calculate the static force. Flux-loop degeneracy and the full screening complement remain present; the pinning apparatus is explicit and its carrier is not invariant under unrestricted hopping.

Finite-box Green calculations give the attractive Coulomb force. On a side-128 box the force ratio to its continuum form is approximately 1.00898 at separation twelve; a potential-difference ratio is 1.00407 at separation eight. The supplied $\kappa$, physical length conversion and original source's different link energy are distinguished. No autonomous pair motion or empirical fine-structure constant follows from this constrained minimum identity.

## 11.17 Kinetic evolution in the coherent lepton compiler

The coherent phase cycle extends to a homogeneous periodic kinetic law without switching that law off. Let $K$ have the native nearest-neighbour Weyl symbol $-\boldsymbol{\sigma}\cdot\sin(a\mathbf{p})/a$ on every internal, character and address copy. Ancillary-number phase rotations $R = \exp(i\pi N_A/2)$ commute with $K$, so the four conjugated generators $H_r = K + R^{-r}VR^r$ average to $K + \overline{V}$. The ancillary-vacuum embedding therefore obeys

$$(K + \overline{V})J = J(K_{\text{physical}} + H_Y)$$

on the complete physical Fock carrier, including external quantum references. For each character, the physical address and its image under the rank-one analyzer span an invariant two-dimensional space. The noncommuting calculation acts on $24s$ fermionic modes at $s$ sites; every other enlarged mode evolves as an exact free kinetic spectator. This is an invariant-factor identity, rather than a discarded subspace.

Spatial inversion $I : x \mapsto -x$ reverses $K$ and preserves uniform onsite pairing. Parity $S$ of the complete charged-lepton doublet leg reverses $V$ and preserves $K$. The exterior lift of the site permutation includes every fermionic occupation sign. Thus $C = IS$ satisfies $CH_rC^* = -H_r$. This implements negative full-generator segments through declared coherent permutation and parity services; leg parity alone cannot do so. Exact finite Fock tests include that failing control.

On the invariant active factor, $\|H_r\| \le B = 36s/a + 6s$. The same fourth-order word identity applies with $R_0 = 7B/3$ and gives complete unhalved output trace error

$$\epsilon \le \min\{2, R_0^5|t|^5/(10n^4)\}, \qquad n \ge R_0|t|.$$

The free spectators accumulate exactly the same signed target time in both evolutions. The controls and their time/work costs remain apparatus inputs. The result retains the register's three coefficients and its phase, scalar-background and calibration ancestry; it does not select an interacting Higgs vacuum or eliminate massless spectators.

## 11.18 Charged branch masses and complete spatial records

The analyzer-derived pairing admits three explicit intertwiners from the complete native Nambu source to charged Dirac frames. The source Hamiltonian, electric current and all three spin generators restrict respectively to $h_k = -\tau_3 \otimes \sigma \cdot \pi + e_k \tau_1 \otimes I$, $Q = -I$ and $S_j = I \otimes \sigma_j/2$. Thus $E_k^2 = |\pi|^2 + e_k^2$, with strict branch minima $e_k$ at each lattice valley. Complete reservoir rest records and spatial positive-band probe records evolve under that same quadratic Hamiltonian; time-zero comparisons separate motion from the initial projected tail.

The side-four realization retains all 5760 physical CAR modes and eight valleys. Its three distinct positive masses therefore describe 24 family/valley copies, with 39 additional massless internal coefficients. An explicit three-quark colour-singlet zero-mode state has unit electric charge and zero energy. It rules out a positive global charged-sector mass floor in this charged-lepton-only action. A single electron energy anchor converts the other two branch values to approximately 0.1056594144959 and 1.7769849708134 GeV; the retrospective phase choice, interacting particle identities and physical clock are separate scientific conditions.

## 11.19 Native spin, field and frame return periods

On the periodic native kinetic and pairing construction, a literal spatial quarter-turn combined with its spin rotation preserves every Hamiltonian pulse. Its fourth power is fermion parity on the full carrier, including analyzer coordinates. An odd charged packet therefore acquires an interferometrically measurable minus sign at $2\pi$ and returns at $4\pi$, relative to an idle pointer and charge-compensating reference. The filled negative-band reference is invariant under proper cube rotations. Rotating the opposite reference too cancels the sign, and pointer dephasing removes the interference.

For an axial positive-band packet, the native spin current on its origin plane is $s = (p/2)(\cos\theta, \sin\theta, 0)$ under actual cube rotations, with $p > 0$ at every source time. An oriented frame constructed from this current has period $2\pi$. The apolar tensor $Q_{ij} = s_i s_j - \delta_{ij}|s|^2/3$, measured using two independent source copies, has period $\pi$. The complete classical tensor law includes all preparation failures and detector complements; its shorter period is not a period of the complete quantum source state. For a spin-up axial preparation, the same native excess density integrates to $1/2$ throughout autonomous evolution. The unfiltered occupied sea retains a nonzero spin variance, so this integral does not imply a total-spin eigenstate or a continuous spatial Noether angular momentum. The field is a declared two-copy observable, and the rotation theorem concerns exact cube symmetries of the finite source.

## 11.20 Fractional charge with native colour strings

The complete particle source also supports a direct quark-family charge/flux experiment. The specified preparation fills all 240 source fermion modes, leaves the scalar and link registers blank, and prepares the thirty-mode reference vacuum. It is a supplied highly occupied gauge singlet, with no assertion that it is the interacting ground. For either weak-singlet colour triplet, the reference-hole pump

$$B = \frac{1}{\sqrt{3}} \sum_i a^*_{\mathrm{ref},i} c_{0,i}$$

has $B^*B = N_{\mathrm{family,spin}\,0}/3 \le I$ on reference-vacuum inputs and succeeds with probability one on this preparation. Its full failure complement is retained.

Contracting the actual 119-state link coordinates along a path of $\ell$ edges gives normalized coefficients $3^{-(\ell+1)/2}$. The endpoint is the literal CAR vector $c_i|\mathrm{filled}\rangle$, including its occupation

sign. The reference, link and dual endpoint generators cancel at every vertex. The source-hole charges are $+2/3$ and $-1/3$; their weak charge is zero and their colour flux continues along the string. The calculation checks all 6,648 shortest-path constraints and 1,536 constraints using the original full CAR/scalar/link Gauss compiler.

For the native colour Cartan $C$ induced by $\operatorname{diag}(1,1,-2)/3$, the complete Fock identity is $e^{2\pi i d\Gamma(Q)}e^{2\pi i d\Gamma(C)} = I$. Both selected holes have electromagnetic two-pi phase $-1/2 - i\sqrt{3}/2$, completed by the inverse colour-centre phase. A controlled endpoint hypercharge rotation measures this phase without choosing a gauge-variant colour component. Its complete randomly chosen pointer quadratures have probabilities

$$\left(\frac{1}{8}, \frac{3}{8}, \frac{2-\sqrt{3}}{8}, \frac{2+\sqrt{3}}{8}\right).$$

The three-turn electromagnetic control gives $(1/2, 0, 1/4, 1/4)$. This is an internal gauge-path holonomy; the electromagnetic path does not close in one turn and supplies no fractional spatial texture winding.

The original full hopping operator places squared amplitude $1/16$ on each of six one-edge arrival subspaces, retaining both endpoint spins, and has total squared norm $3/8$ on the prepared vector. Each arrival projector annihilates the no-hopping evolution because the atomic Hamiltonian preserves blank links. With $\|H_0\| \le 480$, $\|T\| \le 360$, $\lambda = 2^{-34}$ and $t = 2^{-24}$, Duhamel's identity encloses each original-input arrival probability in approximately $[6.91604\,10^{-37}, 8.15583\,10^{-37}]$. The full complement, reference and pump failures are retained. The positive finite-time signal does not identify a low-energy quark, its rest mass, a hadron or an unrestricted confinement phase.

### 11.21 A measured scale-dependent colour coupling

The colour restriction of the five-level seam cost is exactly

$$1 - \tfrac{1}{5}\operatorname{Re}\operatorname{tr}_5 \operatorname{diag}(U, I_2) = \tfrac{3}{5}\left(1 - \tfrac{1}{3}\operatorname{Re}\operatorname{tr}_3 U\right).$$

A declared four-dimensional periodic thermal construction with cost multiplier ten therefore gives Wilson weight $\beta = 6$, or bare $g_0^2 = 1$ in the fundamental trace convention. The native generators give $\operatorname{tr}(T_a T_b) = \delta_{ab}/2$ and $C_F = 4/3$. This colour ensemble and its thermal state are declared separately from the finite particle Hamiltonian; no observed strong coupling is an input.

Let $W(R,T)$ be the ensemble mean of the normalized real trace of a rectangular Wilson loop, averaged over all sites and coordinate plane orientations. Define

$$\chi_R = -\log \frac{W(R,R)W(R-1,R-1)}{W(R,R-1)^2}, \qquad g_C^2(R;N) = \frac{\chi_R}{k_R(N)}.$$

The coefficient $k_R$ removes the free finite-box geometry. If $J_C$ is the conserved oriented link current of the rectangle and $\widehat{k}^2 = 4\sum_\mu \sin^2(k_\mu/2)$, the exact finite Gaussian covariance gives

$$f(R,T) = \frac{C_F}{2N^4}\sum_{k\neq 0}\frac{\sum_\mu |J_{C,\mu}(k)|^2}{\widehat{k}^2}, \qquad k_R = f(R,R) + f(R-1,R-1) - 2f(R,R-1).$$

In particular $k_1 = (1 - N^{-4})/3$. An independent position-space inverse verifies the coefficient. The Gaussian control $W = e^{-g^2 f}$ gives the same $g_C^2 = g^2$ at every scale; the unnormalized loop ratio would produce a geometric false trend. This defines the normalization scheme without assuming an interacting finite-volume zero-mode expansion.

Four cold/hot productions at the same $\beta = 6$ retain 400 records each on sides six and eight. Common-block deletions preserve loop and scale correlations. At side eight, the combined coupling squared is approximately 1.56186, 2.73542, 4.13672 at lengths 0.5, 1.5, 2.5 in lattice units; twenty-record block errors are 0.00050, 0.00444, 0.03837. Side-six values are 1.55976, 2.75157, 5.43421. The longest-loop discrepancy exposes substantial finite-volume sensitivity. Multiple block sizes, independent starts, split histories, all raw records and gauge controls are retained. These errors are sampling diagnostics, not rigorous mixing or confidence bounds. No imported perturbative running curve enters the calculation. The finite coupling weakens toward shorter measured lengths; a continuum beta function, physical energy scale and onset at a particle core are not identified.

### 11.22 Localized near-rest charged packets

A particle rest edge and a localized detector signal concern different preparations. They can be joined on the native kinetic and pairing source by an explicit smooth compact packet, without introducing a spatial trap. Let $L \geq 4$, let the periodic side be $N \geq L + 5$, and define

$$g_j = \sqrt{\frac{8}{3(L+1)}} \sin^2 \frac{\pi j}{L+1} \quad (1 \leq j \leq L), \qquad g_j = 0 \quad \text{otherwise.}$$

The product $f(x) = g_{x_1} g_{x_2} g_{x_3} |\beta+, +z\rangle$ is normalized and supported in a cube $C$. With the same native $h = K + m\beta$, $\{K, \beta\} = 0$ and $E = \sqrt{m^2 + K^2}$, source fermion addition on the filled reference succeeds with vector $P_+ f$, where $P_+ = (1 + h/E)/2$. The charged reservoir, idle opposite reference and full failure branch are retained.

Put $q = \pi/(L+1)$ and $D = (T - T^*)/(2i)$. Direct shifts, including both boundary collars, give

$$d_2 = \|Dg\|^2 = \frac{\sin^2(2q)}{3} + \frac{4 \sin^4 q}{3(L+1)},$$
$$d_4 = \|D^2 g\|^2 = \frac{8}{3(L+1)} \left[ \frac{L}{8} \sin^4(2q) - \frac{1}{2} \sin^2(2q) \cos(2q) \sin^2 q + \frac{1}{4} \sin^4 q \right].$$

Product structure gives $a_2 = \langle K^2 \rangle_f = 3d_2$ and $a_4 = \langle K^4 \rangle_f = 3d_4 + 6d_2^2$. Their exact finite formulas supply rational upper bounds by enclosing $\pi$ and the elementary trigonometric inequalities. No large spectral diagonalization is required.

The success weight is $w = (1 + m\langle E^{-1} \rangle_f)/2$ and $1 - w \leq \eta = a_2/(4m^2)$. For $\eta < 1$, the conditional packet $\psi = P_+ f/\sqrt{w}$ obeys

$$0 \leq \langle E \rangle_\psi - m \leq d = \frac{a_2}{2m(1-\eta)}, \qquad \langle (E-m)^2 \rangle_\psi \leq b^2 = \frac{a_4}{4m^2(1-\eta)}.$$

Although $P_+$ gives the compact seed a spatial tail, its complete outside-cube probability satisfies

$$\|(1 - Q_C) e^{-iht} \psi\|^2 \leq \left( \sqrt{\frac{\eta}{1-\eta}} + |t| b \right)^2.$$

With $L = 2^{30}$, $N = 4(L+1)$ and the light native charged coefficient, the conditional mean energy is at most $3.014\,10^{-9}$ relatively above its rest edge. The electron anchor therefore gives a localized mean energy immediately above 0.00051099895069 GeV, within about 0.003014 parts per million. Through source time $10^9$, the conditional outside probability is below $1.103\,10^{-7}$. A preparation-vector error $\epsilon_0$ and a complete-source Hermitian perturbation of norm $\delta$ add at most $\epsilon_0 + |t|\delta$ to the amplitude bound. A per-site perturbation bound cannot replace this total norm condition.

The exact local moments are unchanged when the volume doubles at fixed support. Complete smaller boxes execute all native position, energy and failure records. The large member is an analytic finite-carrier certificate, with every valley and massless spectator retained. Its full-volume evolution is not numerically enumerated. The complete compiler includes all ancillary outputs and arbitrary references. Exact decomposition into commuting four-CAR factors and a prepared-state energy estimate, derived below, control both the complete record law and the physical-energy marginal. With $0 \le H_{\mathrm{phys}} - E_G \le 282N^3$ and target mean at most $m_{\mathrm{upper}} + d$, the required cycle count is at most $6.564\,10^{27}$ for the stated volume and time. The energy error is at most $10^{-6}d$ and the unhalved trace error at most $10^{-12}$. A separate lower kinetic-moment bound exceeds that error, so the compiled mean also lies above the rest edge. This is the physical marginal energy, not the complete enlarged compiler Hamiltonian. An efficient local preparation, interacting electron or stationary soliton is not inferred.

## 11.23 The same packet's autonomous phase and drive response

A single coherent preparation entangles the packet and its compensating reference with a pointer, relative to the source ground and reference vacuum. The idle reference Hamiltonian is zero by the declared source contract. The success and failure operators form a complete contraction instrument on the full carrier. Let $C(t) = \langle \psi, e^{-iht}\psi \rangle$. Two coherent return quadratures, their orthogonal complements and preparation failure have seven original-input probabilities

$$p_{a,\pm} = \frac{w}{8}(1 + |C|^2 \pm 2c_a), \quad p_{a,\perp} = \frac{w}{4}(1 - |C|^2), \quad p_{\mathrm{fail}} = 1 - w, \qquad c_X = \Re C, \quad c_Y = -\Im C.$$

Since the same spectral law has $E - m \ge 0$, it gives

$$|C(t) - e^{-imt}| \le |t|d, \qquad |C(t)| \ge 1 - |t|d, \qquad |-\partial_t \arg C(t) - m| \le \frac{d}{1 - |t|d}$$

when $|t|d < 1$. The displayed packet preserves visibility above 0.9997728 for about twelve thousand rest cycles. Its phase and energy share one Hamiltonian, preparation, ground and reference.

A nonzero reference cost $\mu$ multiplies the return by $e^{-i\mu t}$. Dephasing the initial arms removes their quadrature contrast. A differential native scalar drive $v(t)Q_{\mathrm{source}}$ multiplies it by $\exp(i\int v)$, whereas a common source/reference gauge change cancels. The actual full-generator inverse returns the complete state. These controls establish prepared autonomous phase coherence of the declared quadratic target Hamiltonian. Passive onset is not established, and the differential scalar drive shifts its frequency. The finite phase-cycle compiler uses continuing controls; its sampled complete-record bound does not establish an uncontrolled microscopic oscillation or instantaneous frequency rigidity between pulses.

## 11.24 Native-current conversion into transverse radiation

A declared finite quantum extension couples the actual charged Dirac current $-\tau_3 \otimes \sigma_r$ to four transverse cochain radiation modes. All twelve source CAR coordinates and complete finite oscillator occupations are retained. Total charge and lattice momentum are conserved. For a scalar rest particle–hole pair on the filled free reference, both crossed-polarization two-photon matrix elements of $(H/m)^2$ have magnitude $2\sqrt{2/5}\,\eta^2$, with opposite phases. The source derivation sums all intermediate modes and both emission orders; no annihilation projector is inserted. A full-sector norm bound gives positive original radiation probability above $4.3451063\,10^{-38}$ at dimensionless time $10^{-8}$.

At time fifty, complete sparse evolutions give approximately 0.03544 and 0.02865 for caps one and two, retaining a material cutoff difference. Vacuum, zero-coupling and frequency-detuning controls and the interaction energy are retained. Number-readout work is distinguished from conserved premeasurement energy. The supplied box resonance, coupling and free preparation define this finite experiment; physical particle identity, continuum annihilation rates and asymptotic outgoing radiation are not inferred from the conversion probability.

## 11.25 Angular momentum and the localized field torque

A rotational regulator separates spatial angular momentum from an assigned spin label. Let $P_n$ retain every three-dimensional oscillator state of total excitation at most $n$, with $n \geq 2$. It commutes with rotations, and its coordinate wavefunctions are localized Gaussian polynomials. Define

$$p_i = P_n(-i\partial_i)P_n, \qquad L_i = P_n(-i\epsilon_{ijk}x_j\partial_k)P_n, \qquad J_i = L_i + \Sigma_i/2.$$

The differential $L_i$ obey the exact rotation algebra and rotate $p_i$ as a vector. Products of separately truncated coordinates and momenta have a different boundary term. The native sixteen-row Clifford representation retains four complete Dirac blocks, both signed branches and every spatial shell. A continuous nontrivial spatial rotation cannot act by permutations on a finite cubic site set, so exact cubic covariance alone does not supply this continuous generator.

Retain the three real transverse field amplitudes $b$ in $A_b(x) = e^{-|x|^2/(2R^2)}(b \times x)/2$ and their conjugate momenta $\pi$. Gaussian integration of the actual Maxwell energy gives

$$c_0 = \int |A_{\hat{b}}|^2 = \frac{\pi^{3/2}R^5}{4}, \qquad c_1 = \int |\nabla \times A_{\hat{b}}|^2 = \frac{5\pi^{3/2}R^3}{8}, \qquad H_f = \frac{\kappa|\pi|^2}{2c_0} + \frac{c_1|b|^2}{2\kappa}.$$

For the declared positive kernel $K(b) = |\beta m + \alpha \cdot (p - qA_b)| + \Omega\mathcal{N}$, the oscillator energy trap is distinct from the cubic scalar mass well. Its actual current is $-\partial_b\langle K\rangle$, evaluated by the complete signed spectral derivative. Rotational covariance and the Maxwell mode equations give

$$\frac{d\langle J\rangle}{dt} = b \times \partial_b\langle K\rangle, \qquad \frac{d(b \times \pi)}{dt} = -b \times \partial_b\langle K\rangle.$$

Thus $\langle J\rangle + b \times \pi$ and the total energy are conserved. The finite executable checks these identities, complete angular records and two integration tolerances. The tighter trajectory has angular drift below $4.9\,10^{-10}$ and energy drift below $3.9\,10^{-9}$; these are numerical diagnostics separate from the conservation theorem. This is a declared rotational source with semiclassical transverse fields, without longitudinal closure or a transfer of the cubic or interacting ground.

## 11.26 A common static and precession magnetic limit

Use the same complete-shell energy at $n = 2$, $\Omega = 1$, $q = -1$, $R = 6/5$ and $\kappa = m^{-6}$, with quasistatic minimization of the field. The field treatment differs from evolving its canonical momenta, while the matter kernel, current and energy are identical. Write $D = \beta m + A$. The native anticommutation relation gives $|D| = \sqrt{m^2 + A^2}$ on both signs. At zero probe its full forty-row kinetic square $T = A^2$ obeys

$$T(T - 5I/2) = 0.$$

For the uniform magnetic probe derivative $E_B = \partial_B A$, $Q = E_B^2$ obeys $Q(Q - I/4)(Q - I/2) = 0$. These exact finite identities give $\|A\| < 8/5$, $\|E_B\| < 4/5$ and $\|\partial_B T\| < 3$. On the oscillator vacuum

the two native spin states have $\langle \partial_B T \rangle = \pm 1$. The magnetic coefficient follows from the squared coupled Dirac operator, without an inserted Pauli term.

The number gap isolates the lowest axial eigenline. A positive Sylvester inverse bounds the magnetic derivative's difference from $(\partial_B T)/(2m)$, while the field-current Hessian makes the axial self-consistency map a contraction. The stationary branch has exact $J_z = 1/2$ and field momentum zero. At zero self-field its representation also has $J^2 = 3/4$. The fixed axial self-field can mix different $j$ values, so the dressed state's complete $J^2$ probabilities are retained without asserting that same Casimir. The separate canonical-spin expectation is not substituted for the angular generator.

For $B = 1/10$ prepare the equal superposition of the two actual lowest axial eigenlines and evolve through $T_{\mathrm{obs}} = 64m$. All other modes remain in the source. A full complementary Schur bound controls their energy split; a uniform Duhamel estimate controls the state-dependent self-field and the observed angular phase. Static and precession normalizations use the same empty-vacuum-relative stationary energy, including its field cost. At $m = 16384$ the rational bounds give

$$|g_{\mathrm{static}} - 2| < 0.001832, \qquad |g_{\mathrm{precession}} - 2| < 0.004863.$$

The bounds tend to zero along the declared mass/coupling route. Numerical values near 1.99999999884 and 2.00000000061 are evaluations, not narrow spectral or physical uncertainty intervals. This finite low-momentum theorem supplies no physical electron mass, fine-structure constant, anomalous moment or cubic-source continuum limit. Its quasistatic dynamics is semiclassical, without a linear quantum-channel claim for arbitrary references.

## 11.27 Autonomous native-current attraction with flux circulations

An explicit specialization supplies a complete autonomous charge-pair experiment without position pinning. Retain the full native periodic cube carrier of side four, but enable only the electron-singlet nearest axial terms on the line $(0, 0, z)$. Their literal kinetic coefficients are three $i\sqrt{2}/8$ entries and the periodic-boundary $-i\sqrt{2}/8$. Other hopping and Yukawa coefficients are zero by declaration. Add positive matter number energy $2N_{\mathrm{matter}}$, scalar excitation energy, reference energy $4N_{\mathrm{ref}}$ and the three-dimensional Green energy $(\kappa/2)\rho_Y^T G \rho_Y$ of the actual native current. The unit nonblank-link penalty is absent.

The complete 119-state electron-singlet matrix coordinate and its adjoint preserve three states with outgoing hypercharges $0, -1, +1$. The selected reference fills all thirty modes except one electron-singlet mode, giving hypercharge $-1$ and number energy 116. One active particle has charge $+1$. All 30 matter modes per cell, scalar factors, reference occupations and link complements remain in the defined source. Enumerating the native Gauss equation

$$\delta_{z,y} - \delta_{z,0} + f_z - f_{z-1} = 0$$

on the four-link ring gives nine position/circulation states for each spin. The full native column action closes this eighteen-dimensional physical sector exactly; its complement is retained. The interacting unit-Yukawa Hamiltonian does not preserve it, so no interacting ground or clock is transferred.

With $J_z$ the actual link-phase derivative, the autonomous clock obeys

$$i[H, N_z] = J_{z-1} - J_z, \qquad \left(\frac{dJ_z}{dt}\right)_{\mathrm{electric}} = i[V, J_z].$$

Here $z$ labels a bond and $J_z$ is its charge current, distinct from the angular generator above. The Green energies at unit coupling are $0, 21/128, 31/160, 21/128$ around the ring. A specified coherent

input on positions one and two gives the exact inward electric contribution $-19\sqrt{2}/5120$ to the outward current acceleration. Complete position, spin, circulation and energy records follow from the same Hamiltonian. At time eight the origin probability is about 0.47953, compared with 0.41818 in the zero-electric-coupling control. Larger coupling need not increase arrival monotonically because interference and the finite flux cutoff are retained. Removing circulation states changes the time-twenty position law by total variation about 0.25986. This is a finite quantum current response, without a classical trajectory, physical coupling, radiation field or far-field continuum inference.

## 11.28 Exact neutral momentum flatness and local preparation

For the periodic side-two fermion/scalar/link source, let $Q_i$ be fermion parity on the coordinate-one plane and let $R_i$ exchange and invert the paired axis-$i$ links. For single-link inversion $\mathcal{I}$, the full native coordinate identity $\mathcal{I}U_{ab}\mathcal{I} = U_{ba}^{\dagger}$ and endpoint-current exchange give $[Q_iR_i, H] = 0$. Translation satisfies

$$T_j(Q_iR_i)T_j^{*} = (-1)^{N_f\delta_{ij}}Q_iR_i.$$

In every odd-fermion sector these symmetries exchange translation characters. Every energy is therefore unitarily identical at all eight momenta, to all orders in the hopping. This includes the isolated neutral band. Its dressed location projectors commute exactly with $H$. A local band preparation has zero probability at other atomic locations for every time, with at most $(v/d)^2$ in its retained virtual-profile complement. This is an exact periodic side-two obstruction to motion, without a transfer to larger or antiperiodic members or a moving-particle mass interpretation.

The same neutral band has a useful preparation probability when its input is built from known local eigenspaces. A uniform choice of neutral cell and seven ground spins, with eight explicit local unitary services, blank links and ten retained classical bits, prepares the reduced atomic density $P/1024$. The local lines are algebraic projectors of actual sixteen-dimensional physical blocks. Supplied reservoirs and local services are declared inputs, rather than a free reset from an arbitrary state. The recipe archive and parity-balancing reference are retained.

A finite degree-10182 polynomial $M(H)$ has $\|M - E\| \le 10^{-16}$ for the actual interacting band projector $E$. With $\eta = \|E - P\| \le v/d$, put

$$A = PM^2P, \qquad q_0 = (1-\eta^2)(1-10^{-16})^2, \qquad L = \sqrt{q_0}\,MP\frac{3P-A}{2}.$$

The positive inverse-square-root series proves $L^*L \le q_0P \le I$ and success probability above 0.99999999999639, with failure square root retained. The conditional reduced source density is within $2.000001\,10^{-16}$ in unhalved trace norm of $E/1024$, giving an all-time stationarity defect below $4.000002\,10^{-16}$. It does not assert stationarity of individual recipes or source/archive correlations. Three polynomial evaluations require 30546 formal source-Hamiltonian applications; their finite Stinespring services are declared, without a primitive gate compiler. All eight momentum outcomes have probability one eighth. The complete factorized preparation, energy, position and local-profile instrument has 410624 records, including every failure and profile complement. This preparation does not remove the exact flatness of the same source.

## 11.29 Volume and sampling controls for the colour coupling

The same native-seam-derived pure-colour Wilson action at $\beta = 6$ is sampled on periodic four-dimensional sides $6, 8, 10$. Each volume retains independent hot and cold histories of four hundred records, with its declared warm-up, stride and final configuration. The two larger-box chains add eight hundred records; all twenty-four hundred original records enter the common analysis. The

five rectangle means and finite Gaussian normalization define the same three coupling readouts at lattice lengths $1/2, 3/2, 5/2$.

Common-block nonlinear jackknives retain cross-scale covariance. Block lengths $5, 10, 20, 40, 80$, split histories and hot/cold differences expose sampling dependence. The exact derivative of each logarithmic loop ratio gives its influence history; an initial-positive monotone covariance-pair sum supplies an autocorrelation diagnostic. Independent-box contrast covariances are sums of the two estimates. These finite histories do not certify mixing or rigorous confidence coverage. The receipt retains all volume contrasts, including effects visible at the shortest loop.

At side ten the three estimates are 1.564128, 2.759170 and 4.163746, with twenty-record-block standard errors 0.000337, 0.003190 and 0.026478. The side-ten minus side-eight contrasts are 0.002268, 0.023749 and 0.027022, with independent-box standard errors 0.000602, 0.005469 and 0.046618. The first two contrasts remain visible relative to these sampling diagnostics; agreement of the longest-loop estimate does not certify the absence of volume bias. All three measured coupling values increase with the loop scale on side ten.

The comparison changes volume at fixed bare action and spacing. It tests the finite-scale coupling trend without inserting perturbative running, while supplying no continuum beta function, matched step-scaling limit, particle-core radius or physical GeV conversion.

### 11.30 Complete compiler bounds from commuting native factors

The homogeneous kinetic and analyzer source decomposes into $6s$ disjoint four-mode fermionic factors on $s$ sites. Fourier transformation pairs opposite momenta, and invariance of the native spin epsilon tensor gives two helicity pairings per momentum and character. Every complementary mode has identical free evolution in the actual and target laws. On each factor the rank-one pairing has norm one. Centering its kinetic number operator gives $\|H_r - 2\kappa I\| \le 2|\kappa| + 1 \le 4/a + 1$. The identical scalar phases cancel in the unitary comparison without changing the recorded energy or reference.

Even operators on disjoint fermionic factors commute. Applying the same exact fourth-order word to each factor and telescoping the tensor product gives the complete unhalved trace bound

$$\epsilon \le \min\left(2, \frac{6sR^5|t|^5}{10n^4}\right), \qquad R = \frac{7}{3}(4/a + 1), \qquad n \ge R|t|.$$

It includes all ancillary outputs and arbitrary quantum references. A coupling between factors invalidates the factorization and is retained as a failing control. The transformation proves an identity of the existing pulses; it adds no Fourier apparatus operation.

For a nonnegative physical excitation observable with norm at most $E_{\max}$, a target mean at most $\mu$ and an isometry error $\delta$, purification gives

$$|\langle A\rangle_{\text{actual}} - \langle A\rangle_{\text{target}}| \le 2\sqrt{E_{\max}\mu}\,\delta + E_{\max}\delta^2.$$

The compact near-rest packet has $E_{\max} = 282s$ and $\mu \le m_{\text{upper}} + d$. Retaining its energy error budget $d/10^6$ and complete-record trace tolerance $10^{-12}$ requires at most $6.564\,10^{27}$ cycles under these bounds, a reduction by approximately $1.966\,10^{34}$ from the unseparated global-norm estimate. This is an analytic resource bound on the same huge carrier. Coherent control services, preparation costs and the physical-energy-marginal domain are explicit; an efficient local or autonomous implementation is not supplied.

### 11.31 All-channel native charged gap

**Complete native Higgs action and the charged floor.** Retain the fifteen-coordinate matter module, all three family characters, all momenta and the full spin-epsilon pairing kernel. With the unit neutral Higgs background, the three native invariant tensors define a rank-seven directed matrix $D$ satisfying

$$D^2 = 0, \qquad D^\dagger D + DD^\dagger = P_{\mathrm{ch}}, \qquad \operatorname{rank} P_{\mathrm{ch}} = 14, \qquad Q(1 - P_{\mathrm{ch}}) = 0.$$

The declared common family choice $Y_u = Y_d = Y_e = \operatorname{diag}(e_k)$ gives seven charged Dirac channels per character and valley, with energies $\sqrt{\sum_i \sin^2(ap_i)/a^2 + e_k^2}$. The remaining internal coordinate is neutral. On the complete quasiparticle Fock space,

$$H - E_{\mathrm{ground}} \;\geq\; m_{\min}|Q|, \qquad m_{\min} = \min_k e_k > 0.$$

Indeed every charged occupation has energy at least $m_{\min}$ and charge magnitude at most one, and every zero-energy occupation is neutral. The opposite-leg pairing ground has zero charge. A charged lepton rest mode attains the charge-one floor. The side-four census retains all 64 momenta and 11520 Nambu rows; its 48 zero-energy CAR modes are neutral. Removing either quark Higgs channel restores charged zero modes.

The complete controlled implementation has $42s$ disjoint four-CAR factors and retains all $324000s$ enlarged CAR coordinates and all ancillary outcomes. Its unhalved trace error obeys $\min\{2, 42sR^5|t|^5/(10n^4)\}$ for $R = (7/3)(4/a + 1)$ and $n \geq R|t|$. Fixed native leg creation gives success probability $1/2$, with failed creation and all four occupation outcomes retained. The source choice removes the massless charged counterexample of the charged-lepton-only action; it does not change that counterexample's validity for that earlier action. Equal up, down and lepton family coefficients are explicit architecture and produce equal family spectra. They do not reproduce unequal physical quark masses, determine the family selection, or identify interacting particles.

### 11.32 Neutral transport and weak-current records

**Neutral spin-half transport on the complete finite source.** On the native antiperiodic four-cell torus, the normalized local $e^cLL$ gauge singlet $(2|10,11,24\rangle - |9,10,26\rangle + |9,11,25\rangle)/\sqrt{6}$ has spin Casimir $3/4$ and charge zero. The full Hamiltonian retains 1920 matter modes, 192 links of dimension 119, and all five scalar states at each of 64 sites. Between adjacent spin-up packets its first three moments vanish and its third moment is $-3i\sqrt{2}/64$. The two native antiperiodic paths each have oriented coefficient $-i\sqrt{2}/8$; mixed path assignments vanish by Haar integration. For $t = 10^{-20}$, the full-source norm bound $\|H\| \leq 12937$ gives original arrival probability between $1.2181254471 \times 10^{-124}$ and $1.2232835274 \times 10^{-124}$. All other packet and complement outcomes are retained. This proves finite transport on its declared preparation and clock; no interacting-ground, relativistic-dispersion, or observed-neutrino identification is inferred. The exact columns and complete remainder are reproduced by `model/openwave_neutral_transfer`.

**A source-derived weak-current family instrument.** Let $Y$ be the positive native record-memory family matrix, $F$ its real ordered eigenframe, and $V = I + (-i - 1)P_2$ one declared native port pulse. The symmetric neutral coefficient $M_\nu = V^* Y V^\dagger$ has Takagi frame $VF$. Transforming the actual native charged current gives $C = F^T VF$, without imported neutrino angles or mass splittings. For momentum $p > 0$, $E_i = \sqrt{p^2 + \lambda_i^2}$ and $w_i = (E_i + p)/(2E_i)$ are the actual positive-band chirality weights. The contraction $K_p = \operatorname{diag}(\sqrt{w_i})C^\dagger$ and its adjoint return instrument give

three family records plus preparation and return-complement records. Their total variation distance from the ideal unitary oscillation law is at most $2\max_i(1-w_i)$. All native internal components and failed quantum outputs are retained. The chosen neutral tensor and port pulse define this conditional model; no observed-neutrino identification or absolute mass calibration follows. Exact definitions and complete records are in `model/openwave_neutrino_oscillation`.

**A native-current three-body channel.** On the complete 270-mode finite matter carrier, contract all three actual weak currents as $V = \sum_{a,\mu,p-r+q-u=0} \eta_\mu \, : j_a^\mu(p,r) j_a^\mu(q,u) :$, with $j_a^\mu = c^\dagger(\bar\sigma^\mu \otimes T_a/2 \otimes I_3)c$. The full contraction includes quark and lepton doublets and has 31464 canonical quartic words. Native analyzer masses are normalized to the middle charged branch. At the declared box momentum $q = (1 - m_{\text{light}})/2$, a middle charged rest excitation and the receiver containing a light charged excitation plus a neutral and antineutral have equal free energy and momentum. Their exact matrix element is $\langle f|V|i\rangle = -1$, including both charged chirality weights. For the declared coefficient $G = 1/64$, the full clock at $t = 10^{-14}$ has receiver probability in $[2.4413587178, 2.4414537827] \times 10^{-32}$; its complete complement is retained. Interacting energy expectations and measurement work are recorded separately from free resonance. This establishes a finite current-driven transition, without deriving a measured lifetime, the Fermi constant, or a continuum decay width. The construction and complete bound are reproduced by `model/openwave_weak_decay`.

## 11.33 Quantum angular momentum of a localized charged state

A complete oscillator-shell matter regulator can be coupled to a quantum Maxwell field while preserving exact rotations. Retain the three transverse modes $A_b(x) = e^{-|x|^2/5}(b \times x)/2$ and every field occupation of total number at most $C$. Their radius satisfies $R^2 = 5/2$, hence their electric and magnetic Maxwell metrics coincide. With $\kappa = c_0/32$, the field frequency is one and $b_i = (a_i + a_i^*)/8$. The projected field energy above vacuum is $N_f$. Its expression in truncated coordinates requires the exact boundary completion $(C+3)P_{\text{top}}/2$.

The native sixteen-row Clifford module decomposes into four complete Dirac blocks. The uniform source charge $q = -1$ is a declared assignment; it is not inherited from the separate chiral Higgs current. On each block let $D = \tau_3 m + \tau_1 B$, where $B$ is the actual minimally coupled spin/spatial operator and $m = 16$. Then $|D| = \sqrt{m^2 + B^2}$ on both signed branches. Complete matter shells through number two contain ten spatial modes. The ordered full fermion lift is

$$H = \sum_{ab} c_a^* K_{ab} c_b + d\Gamma(N_m) + N_f, \qquad K = |D|.$$

The blocks $K_{ab}$ are quantum field operators. If $K - mI = B_0^* B_0$, then $d_r = \sum_a (B_0)_{ra} c_a$ gives $\sum_{ab} c_a^*(K - mI)_{ab} c_b = \sum_r d_r^* d_r \geq 0$. Consequently $H \geq mN_{\text{ferm}} + d\Gamma(N_m) + N_f$ on every retained fermion sector. This is a linear quantum Hamiltonian with arbitrary external references; no state-dependent field substitution is used.

The actual angular generator is $J = d\Gamma(L + \Sigma/2) - i\epsilon_{ijk} a_j^* a_k$. Complete shells make the spatial differential and Maxwell mode actions covariant, so $[H, J_i] = 0$. The separate matter and field torques cancel as operators. Since

$$\|B\| \leq \frac{8}{5} + \frac{3}{5}\sqrt{C} = a_C, \qquad 0 \leq K - mI \leq \frac{a_C^2}{2m} I,$$

the two lowest levels of each reduced block are isolated below the number gap one. Projection onto the unperturbed oscillator/photon vacuum is an invertible rotation intertwiner. The full dressed

band therefore carries exactly spin one half, with $J^2 = 3/4$, including its quantum field dressing. The full native multiplicity retains eight identical doublets, forming a sixteen-dimensional charged ground band. The preparation selects one $J_z = 1/2$ line in one native copy. At cap two the up state has Maxwell angular expectation approximately $1.2056062\,10^{-5}$, with the matter contribution completing $J_z = 1/2$. The Maxwell field alone has integer angular spectra.

### 11.34 Common energy, current and finite field-cap records

A specified compensated fermion pulse and complete ground-band filter prepare the dressed state. Radial position, photon number and total angular records include the filter failure and retained idle reference. The exact oscillator kernel bounds the outside-radius-five probability by $1.1\,10^{-7}$ for every state of this spatial carrier. This localization uses the declared trap and complete-shell regulator.

A sixth-order square-root polynomial, exact algebraic matrix entries, normalized rational trial vectors and outward interval arithmetic enclose the ground energy. Its positive remainder is bounded by $(33/2048)a_C^{14}/m^{13}$. The complementary gap supplies a Temple bound; all ordered noncommuting terms of the polynomial derivative supply a current interval. The same current is the derivative of the joint source energy with respect to a uniform probe. At cap two the finite values obey

$$16.04615746140468 < E_2 < 16.04615746140603, \qquad 1.9982753534 < g_2 < 1.9982753546.$$

The moment convention uses this same rest energy and total angular momentum. The intervals are finite-source bounds, without a physical magnetic-moment uncertainty interpretation.

One empirical electron anchor, $0.00051099895069\,\mathrm{GeV}$, fixes the unit using the cap-two state. Caps one and three use that identical unit. Complete finite-cap comparisons retain field occupations, the Maxwell boundary term and photon-number readout work. They supply neither an infinite-photon or spatial-continuum theorem nor an observed electron, physical fine-structure constant or three-lepton-spectrum identification.

### 11.35 A native energy-lapse extension

Let $M_0$ be the complete native eight-level Hamiltonian at unit geometry, $E_0$ its actual ground energy and $K = M_0 - E_0 I \geq 0$. Declare the finite geometry-controlled energy $H(q) = \kappa q^T L q/2 + \sum_x e^{q_x} K_x$, with a positive Dirichlet graph Laplacian and a retained dyadic geometry register. The local response is exactly $\partial_{q_x} H_x = H_x \geq 0$ on every native level and reference. This lapse coupling and its ground counterterm are architectural inputs; they are distinct from the reciprocal scalar derivative.

For prepared native energy lines $E_x$, the geometry minimum of $F = \kappa q^T L q/2 + \sum_x E_x(e^{q_x} - 1)$ has positive Hessian $A = \kappa L + \mathrm{diag}(E_x e^{q_x})$. Inverse positivity gives

$$\partial_{E_x}\partial_{E_y} \min F = -e^{q_x}(A^{-1})_{xy} e^{q_y} < 0$$

on the continuous extension. A dyadic witness and its rational residual bound enclose the finite minimum by

$$F(q) - \|\nabla F(q)\|^2/(2\mu) \leq \min F \leq F(q),$$

where $\mu = 12\kappa/(n+1)^2$. Two first-excited native sources on a 31-cube at $\kappa = 10^4$ have certified negative mutual energies and an attractive separation-six energy difference of about $1.53313\,10^{-7}$. The model Green coefficient is $1/(4\pi\kappa)$; no physical Newton strength or autonomous geometry-minimization controller is inferred.

## 11.36 Common native clock and wave observations

On the same prepared energy-lapse geometry, every native transition has frequency $\omega_k(x) = e^{q_x}(E_k - E_0)$. Complete eight-output Ramsey instruments at two sites give 64 joint records and the common ratio $\omega_k(a)/\omega_k(b) = e^{q_a - q_b}$ for all seven excited levels. The source reference and probe preparation work are retained. Geometry is fixed during the probes, without an autonomous backreaction assertion.

Declare the scalar coefficient density $C = \mathrm{diag}(-e^{-4q}, 1, 1, 1)$. Its metric dictionary gives $g = \mathrm{diag}(-e^{2q}, e^{-2q}, e^{-2q}, e^{-2q})$, with that same lapse. All vacuum and one-quantum modes of the finite field have $K_{\mathrm{wave}} = e^{2q} D^T D e^{2q}$; complete site-frame observations retain measurement work and independent evolution controls. The metric ray time $\sum_j e^{-2q_j}$ has a certified positive delay, distinct from a claimed lattice group-delay limit.

Independent first-order curvature gives $G^{(1)}_{00} = 2\Delta q$ and $G^{(1)}_{ij} = 0$. Native energy stationarity therefore bounds $G^{(1)}_{00} - 2\rho_q/\kappa$ by the actual geometry residual, with $\rho_q = e^q E$ defined from matter energy. This uses no installed Einstein source equation. The isotropic spatial gluing is a declared choice: a changed spatial metric can preserve the clock and force laws while violating the exterior curvature comparison. Physical units, full covariant stress and nonlinear Einstein dynamics are separate conditions.

## 11.37 One hyperfine spectrum, charge profiles and complete gauge records

**One native hyperfine action.** A positive current interaction $h[\frac{3}{2} N_q N_{\bar{q}} + 2\mathbf{S}_q \cdot \mathbf{S}_{\bar{q}}]$ is the sum of quark–antiquark pair triplet squares. It vanishes on every pure-quark state and shifts a spin-one one-pair meson by $2h$, leaving its spin-zero partner unchanged. Together with a calibrated coefficient of the existing local spin square, one declared action gives a neutral vector energy 0.77526 GeV and an actual $S = I = 3/2$, charge-one baryon energy 1.232 GeV, while preserving the five specified proton, neutron, pion, axial and nuclear-binding calibration agreements. These are seven empirical inputs. The two added mass conventions are Breit–Wigner numbers, not derived complex resonance poles or widths. The complete 120, 64 and 2520 baryon, meson and nuclear singlet spectra, their coloured preparation failures and all magnetic partners are retained. Exact ground-line bounds test transfer of the native current and charge records, without identifying states from fitted labels alone. For the complete local $2^{24}$ matter Fock space, exact SU(3) fusion gives $\sum_x C_x^2 \le (22/7) \sum_e E_e^2$ on the physical zero-boundary-flux Gauss carrier. An independent quark/antiquark-count bound also retains the positive pair and baryon-density terms. Both apply to the same conditional quadrupole/string-length attachment; neither is a four-dimensional pure-gauge mass-gap assertion. The complete source and instruments are specified in `model/openwave_common_hyperfine_spectrum`.

**Native charge profiles and their independent radius test.** On the same seven-input strong source, the literal native isospin raising operator transports the complete neutron ground-ray certificate to the proton ray. Their mass difference is 0.00129333251 GeV. The intrinsic neutron charge is positive at graph radius 1/3 and negative at 2/3. A maximally mixed input on all 120 physical colour singlets gives a ground-ray filter probability 1/120; all 2024 occupation effects and their failure records are retained through a compressed positive-operator measurement. Its nonselective mean source work on that input is zero.

With the shared quadrupole/string length near 0.7559087287 fm, the orientation-averaged charge records give a proton root-mean-square radius near 0.3345681563 fm and neutron charge second moment near $-0.03731193543$ fm$^2$. The common length cancels in $-\langle r_n^2 \rangle / \langle r_p^2 \rangle$. Its certified interval is disjoint from the comparison box formed by the CODATA2022 proton radius and

PDG2025 neutron mean-square radius. No single length therefore fits both radii on this selected source. The graph orientation average is specified; no relativistic Sachs-current or continuum nucleon-pole identification follows from these records (`model/openwave_baryon_form_factors`). The complete source and instruments are specified in `model/openwave_baryon_form_factors`.

**Complete strong-source reduction and a linear potential.** On the same seven-input action, the complete native baryon singlet carriers have dimensions $120, 364, 816, 1540$ on two through five cells. Eliminating the full long-span complement yields the exact Schur action $H_{\mathrm{eff}}(E) = A + V(E - D)^{-1}V^{\dagger}$, together with its normalization metric and native current vertex $R_f^{\dagger} J R_i$. Rational complement bounds control their variation across a stated energy window; full finite-time records retain the complement and all failures. Graph-dependent masses and axial responses are diagnostics, without a continuum QCD running-coupling claim. For a pinned quark and diquark at nonadjacent interior sites, all 40 internal singlet states obey $H_{\mathrm{pin}}(r+1) - H_{\mathrm{pin}}(r) = (4\kappa/3)I$. This exact total-source identity includes local, spin, isospin, hopping and Gauss-electric terms. Its common conditional length gives a constrained inter-charge slope $1\,\mathrm{GeV/fm}$, explicitly used as an attachment target. The result concerns prescribed positions and does not identify the unpinned ground energy or a four-dimensional pure-gauge mass gap. The complete source and instruments are specified in `model/openwave_strong_step_scaling`.

**Calibrated interacting source poles.** The complete native three-quark Gauss source has five compact cluster fields per translation cell, with matrix spectral measure of zeroth moment $I_5$. Exact bounds retain every relative configuration outside this frame; the squared compact-to-complement coupling norm is $5/64$. Two declared counterterms vanish on the entire calibrated two-cell operator. With the inherited neutron and axial inputs, the neutron rest pole lies in $[0.939565387072, 0.939565456806]$ GeV, and the native axial magnitude in $[1.274757, 1.275843]$; native isospin gives the proton shift and vector amplitude one. The fixed-coefficient control has disjoint axial interval $[1.291676, 1.292715]$, so preserving two-cell fits alone does not preserve spatial refinement.

The pole residue is rank one in each tested momentum and spin/isospin fibre. If $q$ bounds complementary probability, $s$ bounds the compact-ground angle and $\tau$ the actual rational preparation's angle, its pole weight obeys

$$Z \geq (1 - q)\left[1 - (s + \tau)^2\right] > 0.9999997$$

at rest. The complete five-coordinate filter succeeds with probability $1/5$; its $4/5$ failure, other spectral outcomes and work are retained. This certifies observable poles of an interacting one-dimensional source, without identifying physical four-dimensional Lorentz particles (`model/openwave_hadron_refinement_flow`, `model/openwave_baryon_poles`).

**Propagation with the same native rest operator.** A positive correlated kinetic operator covers every colour-singlet triquark shape of occupied span at most one. Its common rule is $J = \zeta M^{-1}$ for the complete compact rest matrix $M$ and $\zeta = (\hbar c/a)^2/(2u^2)$; the action vanishes on the complete three-quark rest fibre. Exact five-band and infinite-complement bounds give width $[0.1450650, 0.1450654]$ GeV and neutron inertial mass $[0.91037, 0.97069]$ GeV. The electron unit, strong length and physical speed are supplied. Full Fock positivity supplies neither higher-number rest cancellation nor exact Lorentz dispersion (`model/openwave_baryon_dispersion`).

**Uniform native particle bands.** The complete interacting baryon source has an isolated simple ground pole at every momentum on its Brillouin circle. An exact finite shortcut splits its five compact triquark coordinates into three nearby light coordinates and two high coordinates, retaining the latter with the complete infinite relative-position tail. Subtracting the common scalar part of inverse-rest propagation reduces the remaining momentum derivative bound to less than 0.178 source units. A rational interval cover then certifies a ground isolation gap above 0.127 source units and high-complement probability below $2.7 \times 10^{-6}$ for both charges. These are unchanged calibrated one-dimensional source poles, with all spectral complements retained. They supply a uniform input for normalizable packets; no physical four-dimensional particle identification follows. Exact source records are in `model/openwave_uniform_pole_bands`.

**Normalizable source particles and complete records.** The explicit momentum amplitude $f(k) = \sqrt{15/(16\kappa)}[1 - (k/\kappa)^2]$ on $|k| \le \kappa = 1/512$, zero outside, has unit norm on the infinite chain and finite position variance $5/(2\kappa^2) = 655360$ cell$^2$. Uniform full-source projector bounds give neutron and proton band probabilities above 0.9999614. After conditioning on the band outcome, more than 98.4025% lies within 8192 source cells of the origin. The same native weak-current packet response has vector amplitude above 0.983876558018 and axial interval $[1.226387005970, 1.324212994030]$. Preparation, band, current and position outcomes retain every failure, reference and work term. These are actual normalizable states, rather than fixed-momentum plane waves. The exact band filter remains a declared nonlocal spectral service; finite local compilation and physical four-dimensional particle identification are separate obligations. Exact records are in `model/openwave_normalizable_particle_records`.

**Native weak response on the same propagating poles.** Exact CAR contraction of the complete compact triquark source gives $V_P = I_5$ and $A_P = 5/3 \oplus \left(\begin{smallmatrix} 1 & 2/3 \\ 2/3 & 1 \end{smallmatrix}\right)^{\oplus 2}$. The full fixed-charge axial square satisfies $(M - I)(M - 4I) = 0$, retaining current-produced spin and isospin channels outside the pole frame. Complement and actual rational-preparation bounds transfer these currents to the isolated interacting source poles at six momenta. The rest vector amplitude is exactly one; the rest axial interval is $[1.274757331914, 1.275842668086]$, with its strong axial calibration retained. At lattice phase $\pi$ the vector amplitude remains above 0.994488237318. A complete charge-conserving current instrument records success, failure, every complementary spectral outcome and bounded source work. These source responses do not establish autonomous beta decay or a physical lifetime. Exact records are in `model/openwave_pole_weak_current`.

**Local strange families and an energy-conserving shell.** The complete two-cell native source uses two positive number/pair coefficients to calibrate charged kaon and Lambda energies to 0.493677 and 1.115683 GeV. Complete 816- and 256-state colour-singlet carriers retain the failed Sigma, Xi and neutral-kaon comparisons. The charge-conjugate strong action and nonzero native weak vertex are explicit. Its global number correction prevents separated-cluster additivity, and its full two-cell outgoing channel is closed even after the weak norm bound (`model/openwave_strange_composites`).

A finite-range extension of the complete native strange action restores additivity of separated colour-singlet clusters. With the same positive inverse-rest-operator propagation rule, full relative-position bounds give $m_\Lambda \in [1.11568462, 1.11568488]$ GeV and $m_{\pi^-} \in [0.13959620, 0.13959623]$ GeV in the supplied electron unit. These inherited-calibration bulk energies retain their shifts from the two-cell fits. The proton and pion ground-band energy sum crosses the Lambda rest interval between $\tan(k/2) = 1/6$ and $1/5$, proving an open strong-sector energy shell. The unfiltered weak

current requires an interacting scattering-state transfer; no strange lifetime follows from the shell crossing alone. The complete construction and records are specified in `model/openwave_cluster_bandwidth`.

**A native current identity for the bulk strange source.** The complete compact *uds* frame has 56 shapes and its $S = 1/2$, $I = 0$ Lambda sector has seven coordinates with Gram matrix $G$. Exact CAR contraction of the inherited strangeness-changing weak current into all 1,656 nonzero five-body occupation outputs gives

$$6c^2G \;\le\; W_P^\dagger W_P \;\le\; 7c^2G, \qquad c = \sin\theta\cos\theta.$$

Thus every compact Lambda state has nonzero response, independently of a fitted eigenvector. The current preserves occupied spatial span; a complete source-pole complementary probability bound $q$ therefore gives $\|P_{\mathrm{out}} W \psi_\Lambda\|^2 \ge 6c^2(1-q) > 0$. At fixed total momentum, the calibrated bulk Lambda source pole lies in $[1.115684628607, 1.115684878081]$ GeV. All current outputs and probe failure are retained. A separately enclosed proton–pion energy shell is present in the same source, but its interacting five-body scattering projector has not been identified with this compact current response. No physical Lambda lifetime follows. Exact records and the remaining transfer premise are specified in `model/openwave_strange_shell_current`.

**Native proton–pion fields and complete strong evolution.** Literal CAR contraction projects the complete native weak output onto 23 independent antisymmetrized proton–pion fields. Relative to the seven-coordinate $\Lambda$ Gram matrix, the projected-current square has eigenvalues $6/5$, $241/150$ and $(277 \pm \sqrt{54409})/200$, with multiplicities $1, 2, 2, 2$, before the inherited Cabibbo factor. The complete autonomous five-body Hamiltonian includes both correlated inverse-rest triquark and meson terms. Its exact separated-field response derivative has squared norm $0.02761018661589379\ldots$ times the Cabibbo factor squared for a specified localized $\Lambda$ coordinate. Direct separated contraction vanishes; the full derivative is nonzero. A spectral-theorem remainder on the complete carrier certifies positive finite-time response, retaining every orthogonal output and probe failure (`model/openwave_full_strong_current`).

**Normalizable strange-particle records.** The native $\Lambda$ trial has normalized momentum amplitude proportional to $1 - (k/\kappa)^2$ for $|k| \le \kappa = 1/512$. Its leftmost-cell variance is exactly 655360 and its interacting ground-band probability exceeds 0.9997. The seven-field residue theorem and a bounded-momentum perturbation estimate suffice; isolation over the whole Brillouin circle is unnecessary for this trial. All 72 native weak CAR words preserve the leftmost occupied cell. The complete current and 23-field channel therefore act as the identity on the spatial packet times their exact internal maps. Actual trial probabilities are Gram contractions, with all preparation, other-channel and probe failures retained. Every nonzero internal branch keeps the same packet. This principal experiment uses no exact band filter; optional full-pole conditioning is an explicitly nonlocal service. Source rest-pole GeV intervals retain their calibrations, and the packet mean is a distinct energy observable (`model/openwave_lambda_wavepacket`).

**The native breakup spectrum.** The proton–pion energy shell belongs to the essential spectrum of the complete interacting five-body strong Hamiltonian. Exact CAR/Gauss factorization on separated singlet supports includes both correlated cluster operators. The finite Schur theorem gives factorially controlled relative tails; normalized spatial boxes then produce a singular Weyl sequence at the native $\Lambda$ rest energy. Its coefficients use the isolated full-source spectral projections, and the

executable certifies quantitative residual bounds, without executing a finite observer preparation of these Weyl vectors. An outgoing wave operator and nonzero on-shell weak spectral weight are not established, so neither a physical lifetime nor a $3+1$-dimensional particle dictionary follows. The source is one dimensional and retains its inherited GeV calibrations (`model/openwave_cluster_weyl_sequence`).

**Nuclear isospin and native charge response.** An occupancy-gated three-cell spin interaction vanishes on every sector with at most six quarks, preserving all seven two-cell calibration domains. One triton mass input fixes its coefficient. Exact colour exterior characters reduce complete 572572- and 176729280-singlet carriers to 64 and 256 low coordinates with full complement bounds. The triton-assigned interval $[2.8089192030, 2.8089230704]$ GeV contains its 2.80892113668 GeV input (`model/openwave_light_nuclear_curve`).

The native isospin intertwiner fixes the strong triton–helion separation at 1.29333251 MeV; the supplied nuclear comparison is 0.52952556 MeV. Thus an isospin-scalar nuclear interaction cannot repair both masses. A charge-breaking contribution of 0.76380695 MeV is required. A declared native-charge Coulomb pair kernel, using the inherited electromagnetic coupling and strong length with no nuclear fit, gives full-source helion and alpha intervals $[2.80884329, 2.80884820]$ and $[3.74016768, 3.74019421]$ GeV. Both held-out comparisons fail. The finite collinear pair kernel carries no asserted dynamical Gauss-field or physical halo transfer. The complete construction and records are specified in `model/openwave_nuclear_isospin`.

The complete spin multiplets, preparation failures and high spectral complements are retained. The Coulomb comparison and the positive correlated-propagation source are distinct members: their higher-number spectra have no asserted transfer.

**Covariant matter and compact Maxwell refinement.** The declared compact-field matter source uses the actual native charged spin coordinates and the positive kinetic operator $\sqrt{m^2 + (\boldsymbol{\sigma} \cdot \boldsymbol{\Pi})^2} - m$. Its Pauli magnetic commutator is retained. Every integer electric flux, compensating charge reference and native measurement complement remains in the source. Complete position–flux records have rational tail and finite arithmetic bounds, with preparation and absorption work included. A positive Wilson energy lifts all seven extra free lattice corners. The free matter and quadratic Maxwell symbols share $c^2 = 1/128$ and have controlled spatial refinement. The construction preserves excitation number; no vacuum pair current, signed-Dirac transfer or interacting Lorentz theorem is supplied. Charged mass ratios and the GeV anchor retain their input ancestry. The complete source and instruments are specified in `model/openwave_dirac_maxwell_refinement`.

**The dynamical field connection.** A field-dependent signed-band transformation also transforms the quantum field momentum. For $E = -i\partial_\theta$ and a smooth periodic unitary $U(\theta)$, the exact identity is

$$U^* E U = E + \mathcal{A}, \qquad \mathcal{A} = -iU^* \partial_\theta U, \qquad U^* E^2 U = (E + \mathcal{A})^2.$$

A complete compact-rotor diagnostic executes both bands and every integer flux. Exact blocks of dimension two certify the full ground and show nonzero field-induced band conversion. Omitting the connection changes the spectrum and measurement work even though the fixed-field matter square is scalar. The native charged mass parameters are inherited inputs to this diagnostic. The result identifies a required transfer term for a signed-Dirac completion; it supplies no equivalence to the full native fermion theory or interacting Lorentz-covariant continuum. The construction and complete records are specified in `model/openwave_dirac_field_connection`.

**Resolved charge profile and native effective coupling.** A declared Wilson coefficient $1/2$ resolves the charged core on the positive Pauli–Maxwell member. A retained scan over $1/16, 1/4, 1/2, 1, 2$ selects this coefficient by finite spatial resolution, without empirical radius or running data. Full integer-flux ground and resolvent certificates give a radius of approximately $0.523$ cells for each charged family and prove a half-suppression crossing between $3\pi/4$ and $7\pi/8$ per cell. The literal native scalar-charge intervention gives a certified ground-energy derivative equal to the same form factor after separate reference subtraction. Executed positive and negative probes independently cross half coupling inside the spatial Brillouin interval. The same current satisfies the Ward identity and has four strictly ordered imaginary-frequency responses, each enclosed to absolute error $10^{-9}$. Both spin branches, preparation failure, the true field-vacuum subtraction and complete pointer/readout work are retained. The resulting GeV sector energies are reference dependent and are not observed particle poles; no physical running-coupling curve or interacting continuum is inferred. The construction and complete records are specified in `model/openwave_resolved_core`.

**Native clocks and the phase resource.** The phase boundary admits a branchwise proof. An energy-invariant joint source and battery state, an energy-conserving unitary, and a battery readout commuting with battery energy give an energy-invariant conditional source state for every outcome. Exact execution retains all $4 \times 81$ native input and battery-word records. A two-copy native exchange creates relative coherence in a degenerate energy block, whose free frequency is zero; its interaction-driven population frequency is $2\lambda$, with $\lambda$ a declared coefficient. An energy-mixing battery readout creates the missing phase and has explicit mean work $m(p-1/2)$. These results exclude the stated finite phase-free collision route without excluding other declared observer architectures. The construction and complete records are specified in `model/openwave_clock_phase_resource`.

**Continuous quotient winding and fractional charge.** The native fractional charges admit a continuous quotient-group lift. For $G = (SU(3) \times U(1))/\mathbb{Z}_3$, the universal-cover kernel is $\{(e^{2\pi ik/3}I, 2\pi k/3) : k \in \mathbb{Z}\}$, so $\pi_1(G) = \mathbb{Z}$. An explicit generator has electromagnetic phase lifts $2\pi(2/3)$ and $2\pi(-1/3)$ on the actual quark coordinates, compensated by colour. Complete 120- and 364-state Gauss carriers satisfy the same endpoint identity. Their electric eigenstates have uniform conjugate-angle density; the continuous group loop does not establish a localized classical texture or a quark rest mass. The construction and complete records are specified in `model/openwave_quotient_homotopy`.

**Full-Haar gauge records and spatial blocking.** An actual spatial cube retains all six plaquettes and every Haar representation. Complete native contractions and whole-complement spectral bounds give a positive source gap and strictly positive adjacent-face Wilson covariances at the declared coupling. All fifteen face pairs, all 64 joint outcomes and the apparatus-work bound remain. Separately, the exact bounded-Wilson identity for the shared-link source controls its coupling derivative without differentiating an unbounded truncated matrix. A fixed physical gap supplies only a necessary source-clock tangent, not a measured beta function.

Subdividing the cube gives an isometric coarse Haar/Gauss embedding. Complete parity enumeration and independent Haar contraction certify a positive coarse Wilson coefficient and a uniform remainder for the blocked magnetic factor. Matching the electric term in the unchanged clock requires $g_f^2 = g_c^2/4$ at half spacing; magnetic mismatch and induced operators remain. These are finite-source and spatial-blocking results, without a full-Hamiltonian renormalization, GeV glueball or four-dimensional continuum identification (`model/openwave_cubic_haar_source`,

`model/openwave_haar_coupling_flow`, `model/openwave_gauge_refinement_clock`).

**Interior-resolved full-Haar refinement.** The actual refined cube has 27 vertices, 54 links and 36 faces. Arbitrary-valence Clebsch–Gordan fusion trees retain every Gauss intertwiner, including the two and five singlets of four and six fundamental legs. The vacuum, 36 four-cycles and 164 six-cycles give the complete 201-mode frame through electric energy 9/2; every omitted state has electric energy at least 5. All 7,236 face columns and their exterior fusion multiplicities enter the certificates. The coarse $(a, g^2) = (1, 64)$ and fine $(1/2, 16)$ sources have the same electric clock under edge subdivision. Their full-complement gap and Wilson-variance bounds retain interior modes and the eight-link boundary field. The exact first spectral moment for a loop of length $L$ is

$$m_1 = \frac{\alpha L}{4}(1 - \langle W^2 \rangle), \qquad v e^{-t m_1/v} \le C(t) \le v e^{-t\gamma}.$$

Here $v$ is the connected variance and $\gamma$ a certified full-source gap. These are finite spatial-refinement comparisons, without an inferred running coupling or a physical continuum identification (`model/openwave_refined_haar_source`, `model/openwave_refined_haar_spectrum`).

**Controlled memory of omitted gauge modes.** For this complete frame, $QHQ \ge 80Q$. Full Wilson action computes the leading positive Schur correction

$$G = PHQ(16E_Q + 9)^{-1}QHP, \qquad \|K(t)\| \le \|QHP\|^2 e^{-80t}.$$

Its 3,260 nonzero entries include every output intertwiner. A Neumann ratio 9/89 bounds the infinite-channel zero-energy self-energy error in source-energy units by $9\|QHP\|^2/(89 \cdot 80) \le 0.00175460$. A separate exact control rejects autonomous boundary-only compression. At $t = 1/1024$, retaining interior modes improves the Haar-vacuum semigroup-defect bound by a factor greater than 225. This controls one specified probe and finite memory. Exact autonomy and a uniform continuum refinement are not inferred.

The full calculation is specified in `model/openwave_haar_memory_closure`.

### 11.38 Positive energy response and common gravitational records

An autonomous two-state geometry register couples to both complete native eight-level factors through their positive ground-referenced energies. For $H_g = \omega(I - \sigma_x)/2$, $Q = q\sigma_z$, and the positive three-dimensional harmonic profile $F$, the source $H = H_g + e^Q K_s + e^{FQ} K_t$ has an explicit square-root ground energy. Its mixed source/probe energy derivative is strictly negative. Rational bounds certify attraction for all 49 excited native pairs at three separations, with both self energies subtracted. Complete autonomous records conserve combined energy and retain measurement work. The positive energy coupling is a declared construction; the reciprocal scalar source's opposite-charge control is not transferred to it. Its dimensionless Newton coefficient follows from the mode stiffness, while physical strength and Einstein dynamics require separate attachment. The complete construction is `model/openwave_universal_energy_geometry`.

**Native lapse and finite curvature.** For the complete two-state geometry source, the weak-source derivative of the actual reference-normalized clock lapse is $-2\sinh(q)\tanh(qF)/\omega$. Its linear-profile error is bounded by $2\sinh(q)(qF)^3/(3\omega)$. Commuting lattice derivatives give exact linearized Bianchi identities on the declared metric variation, including every boundary halo. Their curvature Laplacian differs from the nearest-neighbour Poisson Laplacian by $(a^2/4)\sum_i \Delta_i^2$; that correction

is retained. This identifies a finite response and error budget, without deriving a physical metric assignment or nonlinear Einstein limit. The source and complete records are specified in `model/openwave_source_ward`.

**Joint transverse light and native clocks.** A complete three-dimensional Maxwell complex has 882 directed edges and 540 transverse coordinates, with Gauss law imposed at every vertex. The declared positive-frequency field, two native clocks and the quantum geometry share one autonomous Hamiltonian. Complete finite records give arrival delay $0.006354 \pm 0.000509$, transverse displacement $-0.003216 \pm 0.000050$, and Fourier-direction change $-0.004595 \pm 0.000057$, in source units. The joint native clock phase ratios are approximately 0.992837 and 0.992829. Photon vacuum, no-arrival, polarization complements and apparatus work are retained. These are finite constitutive Maxwell records, with no canonical metric quantum-field or Einstein-continuum transfer asserted. A separate attachment of the electron energy and strong length fails both the observed Newton strength and light speed; it supplies no fitted or derived physical $G$. The complete source and instruments are specified in `model/openwave_transverse_gravity`.

**Common-unit compatibility.** For the positive native-energy geometry, write $u$ for the electron-anchored energy unit, $a$ for the edge length, and $g = 1/(4\pi\kappa)$ for its weak-source coefficient. The two unit equations imply

$$\frac{G_{\mathrm{model}}}{c_{\mathrm{model}}/c} = \frac{g}{u^2}.$$

Thus a change of length alone cannot match both physical constants. The speed-matched length is about 386.159 fm. Exact rational stiffness and arctangent enclosures then give $G_{\mathrm{model}} \simeq 4.05075 \times 10^4\,\mathrm{GeV}^{-2}$. Retaining the electron anchor would require a stiffness factor about $6.04 \times 10^{42}$; that adjustment would be a calibration to $G$. This rejects the specified unit attachment, without excluding other declared geometry sources. Six Lean identities establish this compatibility condition, the source scaling relations and the inherited capacity law. The construction and complete records are specified in `model/openwave_gravity_unit_constraint`.

**Calibrated collective gravitational strength.** A distinct member fixes each register frequency, $H_g = (\omega/2)\sum_i (I - X_i)$, while retaining $Q_N = (q/N)\sum_i Z_i$. Its weak response decreases as $1/N$ and its free register gap stays $\omega$. With the declared $q = 1/8$, $\omega = 1$ and $g_0 = Sq^2/(2\pi\omega)$,

$$\frac{g_0}{N}\left(1 - \frac{q^2}{3N^2}\right) \le g_N \le \frac{g_0/N}{1 - q^2/(2N)}.$$

Choosing the integer $N \simeq 6.04 \times 10^{42}$ using the observed Newton coefficient makes weak gravity and the free-symbol light-speed unit equations compatible with the unchanged electron anchor. This is a calibration of the declared hardware to $G$, without an independent prediction of its value. Full coordinate readout costs $N\omega/2$ source units. No common strong/geometry spatial theory or dressed-photon identification is proved (`model/openwave_collective_gravity_strength`).

## 11.39 Native calibrated states, complete currents and common mass records

A family of declared native-current constructions supplies calibrated composite states, massive neutral families and complete observer records. Exact representation identities, finite spectral instruments and Gaussian contractions reduce the full admitted sources. The positive spin-gradient strong member fits five stated data and supports an actual three-dimensional beta-current transfer.

The weak decay member has controlled massive-neutral survival intervals and an unused tau-width-ratio comparison. Source members, reference states and empirical inputs are stated separately; the boundaries of each transfer are part of the result.

**Positive native spin gradients.** The positive interaction $\lambda \sum_a (S_0^a - S_1^a)^2$, with the declared rational coefficient $\lambda = 5/2$, changes the actual native action while preserving its colour, spin, isospin and cell-reflection symmetries. Five disclosed inputs calibrate the proton, neutron and charged-pion energies, the axial magnitude and the deuteron binding. Exact colour and angular-momentum reductions retain all 2,520 six-quark singlet states. The even spin-one, isospin-zero energy is approximately 1.87561294497 GeV, with its binding fixed by the input. The odd spin-zero, isospin-zero and odd spin-one, isospin-one channels lie approximately 0.33047 and 0.16430 MeV above their compatible two-nucleon thresholds. The rational coefficient is a retrospective architecture choice; the complete comparison menu and failed controls are retained. A spin-three, isospin-zero bound channel relative to its two-Delta threshold is explicitly unassigned. Positive native-colour bounds and complete baryon spectra control every independent-copy quark-fragment partition of six. Integer preparation amplitudes, complementary failures and a retained charge reference supply a complete finite instrument. These are finite mass and binding records with stated calibrations, without a continuum nuclear or orbital-parity identification.

**The native axial-state mechanism.** The four-input positive action without a spin-gradient term supplies a separate exact mechanism control. Its native vector current is one and its calibrated axial coefficient is enclosed in $[1.27529999995, 1.27530000005]$. Only about 3.58% of the baryon state is co-located. Split-site correlations evade the local multiplicity-one scalar-action value $g_A = 5/3$ without multiplying the weak current by an effective form factor. The positive colour square vanishes on every original two-cell singlet, including a separated quark–diquark state; that control supplies no spatial-binding theorem.

**Operational rest-energy spectroscopy.** The five-input spin-gradient strong action can be coupled to the full native light-family contact and its filled massless-neutral reference before measuring energy. The charge-zero and charge-one sectors retain 1,400 and 360 states. Exact spin and reflection reductions give complete 199- and 51-dimensional prepared blocks. The neutral preparation has substantial lines near 0.939547847, 0.939570731 and 0.939580102 GeV, with probabilities 0.53512, 0.33840 and 0.11574. The charged preparation has a dominant line near 0.938279942 GeV with probability 0.999978. These contact outputs use the fixed strong coefficients without refitting. A finite Fourier register supplies a complete energy instrument, including all pointer bins, preparation failures, posterior-state operators and the idle charge reference. Registers of dimensions 1024, 4096 and 16384 have controlled finite-source approximation bounds. The finest bin spacing is approximately 7.9844 keV. This is apparatus resolution, without a physical decay-width or infinite-volume pole interpretation. The electron unit is calibrated, and the massless-neutral finite reference is distinct from the fitted three-family massive Dirac source. The filled-neutral spectroscopy reference and the massive Gaussian reference of the angular calculation are specified separately. Their contact energy shifts are not transferred by matching particle labels.

**Common strong-force controls.** For the four-input control in the complete native three-quark singlet sector, the colour and baryon-number costs reduce to $A + (8U/3 - 4A/9)P_{\mathrm{split}}$. Preserving the coefficient of $P_{\mathrm{split}}$ preserves the calibrated axial state. A common positive total-spin square changes the Delta–proton splitting by exactly $3k_s$, while min–max bounds the rho–pion splitting

between $2k_s$ and $2k_s + 2J_s$. The required coefficients are incompatible: a Delta fit gives a rho candidate near 0.33530 GeV, and a rho fit gives a Delta candidate near 1.89194 GeV. A positive colourful electric-current square can instead calibrate a neutral scalar energy at 0.1349768 GeV with a disclosed fifth datum; its physical neutral-pion wavefunction is not identified. A local colour-spin alternative has the positive completion $h\sum_x[4N_x^2 - \sum_{a,k} d\Gamma(T_a \otimes \sigma_k)_x^2]$, proved by Cauchy on every particle-number sector. Complete native-current reductions and feasible four-input controls are retained, together with failed positive-coefficient searches. These results constrain the stated architectures without claiming a universal strong-interaction exclusion.

**Nuclear and spatial controls.** The exact native Casimir and cell-swap reductions resolve all 2520 six-quark singlets, or 240 at fixed zero spin and isospin projections. The four-input action without a positive spin gradient has its lowest even $(S, I) = (1, 0)$ and odd $(0, 0)$ lines about 0.19121 MeV above two nucleons. Fitting only its edge weak-current coefficient binds both channels, including an unassigned odd state. These are distinct controls of the five-input spin-gradient member described above. For three quarks the native colour isometry gives $C_X^2 = \frac{2}{3}n_X(3 - n_X)$. On a four-cycle the positive edge-colour law costs $8\kappa/3$ for an adjacent quark–diquark pair and $16\kappa/3$ for an opposite pair. Complete 120-, 364- and 816-state singlet spectra retain finite-range spatial attraction and geometry-dependent energies, without a continuum-confinement claim.

**Colour strings and fixed-source spatial records.** For an open chain with zero external colour flux, Gauss elimination gives $E_x = \sum_{y\le x} C_y$ and a positive electric action $\kappa\sum_x E_x^2$. The native three-quark singlet identity $C_X^2 = 2n_X(3 - n_X)/3$ implies the exact string cost $4\kappa(x_{\max} - x_{\min})/3$. On two cells, replacing $U$ by $U - \kappa/2$ preserves the entire colour-singlet strong and colour-preserving contact action. Integer checks retain all 120 baryon and 2,520 nuclear singlets. With the same coefficients on chains of two through five cells, the complete singlet dimensions are 120, 364, 816 and 1,540. The neutral lowest energies are approximately 0.93956542194, 0.9396642972, 0.939664299718 and 0.939664299718 GeV. The two-cell mass is calibrated; the larger-chain values are fixed-source outputs. Their axial response is approximately 1.6658, so the fitted two-cell axial value does not transfer under this spatial extension. A positive source bound controls relative-separation tails uniformly in chain length. For one shared physical length, the exact relation $\sigma^2 Q = (4\kappa m_e/3)^2 Q_{\text{graph}}$ tests the string tension and nuclear quadrupole together. Using the stated quadrupole comparison and the illustrative 1 GeV/fm tension benchmark, every $\kappa \le 2U$ preserving a nonnegative local colour cost fails the simultaneous match. The resulting upper tension is below 0.429 GeV/fm. This is a finite-source comparison, without a continuum confinement or string-breaking claim. Using $Q_d = 0.285699\,\text{fm}^2$ as the quadrupole comparison[1] under the stated graph-to-quadrupole attachment, the declared $\kappa = 4$ sets the tension to approximately 0.00360537 GeV/fm. The termwise nonnegative-colour restriction gives at most 0.4286771 GeV/fm under that same length attachment. It is sufficient rather than necessary for full-source positivity. The exact local Fock bound $C_x^2 \le 3N_x$ and Gauss incidence inequality allow the number term to control a negative local colour coefficient. On the five-input source this admits a shared length near 0.755908729 fm and the illustrative tension 1 GeV/fm. Both comparison values select the attachment; no independently predicted dimensional scale follows (`model/openwave_string_positive_cone`).

**Three-dimensional native beta angular records.** The complete native baryon current, on the calibrated strong action, gives $a = (1-g^2)/(1+3g^2) = -0.1065439613$, $A = 2g(1-g)/(1+3g^2) =$

[1] M. Puchalski, J. Komasa and K. Pachucki, Phys. Rev. Lett. 125, 253001 (2020), arXiv:2010.06888v2.

$-0.1194352515$, and $B = 2g(1+g)/(1+3g^2) = 0.9871087098$. These coefficients follow from an explicit Pauli contraction checked against all 120 native baryon coordinates. The axial magnitude is a calibration input; comparisons with neutron angular data share that input's experimental ancestry. An explicit quartet and current recontraction transfers these results to the positive spin-gradient strong action with its five disclosed calibrations. The finite three-dimensional Weyl sources retain all valleys, helicities, calibrated neutral families and the two-contact form factor. On cubes of sides four and eight, a negative momentum-parity record survives the entire Fock-space Duhamel remainder $\delta = (54Gt)^2/2$, with probability error at most $(8Gt + \delta)\delta$. Preparation failure, complementary quantum records and measurement work are retained. The point-contact spin coefficients do not identify the finite anisotropic source with a physical continuum angular distribution. The displayed GeV energies use the electron anchor; the strong free-channel endpoint is distinct from an interacting pole endpoint. The reproducible source and proof are in `model/openwave_beta_angular`. The executed five-input strong source has a certified negative positive-minus-negative electron momentum probability below $-2.6887 \times 10^{-14}$ at $t = 10^{-4}$ and $G = 1/64$, conditional on its spectral preparation. The scalar-kinetic control has exact zero parity contrast; a separate one-dimensional chiral-ring control also has a negative contrast. Momentum sign and group velocity are distinct instruments on the finite regulator. The PDG 2025 zero-recoil comparisons are $a_0 = -0.1044 \pm 0.0007$, $A = -0.11958 \pm 0.00021$ and $B = 0.9807 \pm 0.0030$.[2] The axial input shares experimental ancestry with these data. The deviations in $a_0$ and $B$ are retained; the finite calculation does not supply an independent simultaneous precision fit.

### 11.40 A calibrated neutral family tensor and its native oscillation records

A common neutral multiplier of the charged hierarchy fails a scale-independent test. The ratio of the solar to atmospheric squared-mass gaps lies below the normal-ordering range of the NuFIT 6.0 comparison, while the inverted assignment gives its complement, close to one. Exact analyzer intervals exclude both assignments for every positive common multiplier.

An explicit Dirac family tensor uses the six normal-ordering oscillation parameters in table 1 of NuFIT 6.0[3], including IceCube24 and Super-Kamiokande atmospheric data, as calibration inputs. With the declared lightest mass 0.001 eV, its masses are

$$(m_1, m_2, m_3) = (0.001,\ 0.008712060606,\ 0.050139804547)\,\text{eV}.$$

The same conditional charged analyzer and electron anchor fix the energy conversion. Gauge-neutral singlet coordinates complete the Dirac pairs. They contribute no gauge anomaly, and their sterile coordinates share the masses of the active coordinates. Conserved Dirac lepton number excludes a Majorana term.

The full 288-mode source retains the native weak-contact tensor and its filled reference. Direct native contractions produce an isometry into all 144 outgoing channels. Complete free-flight records retain active and sterile-coordinate outputs, the production complement and reference work. Over atmospheric phase magnitude at most $64\pi$, the exact complete record law differs from the calibrated relativistic vacuum oscillation law by total variation below $4.1 \times 10^{-9}$. This estimate follows from the positive source dispersion and complete current/readout isometries. Independent contraction of the charged-antiparent current gives the conjugate flavour protocol. At atmospheric phase $\pi$, the two complete protocols give electron-flavour neutral and antineutral probabilities 0.04794190585 and 0.03465026894, respectively. Their contrast follows from the calibrated CP phase. The six

[2] Particle Data Group, Neutron listing, 2025 edition.
[3] I. Esteban et al., JHEP 12 (2024) 216, arXiv:2410.05380.

empirical oscillation parameters are fitted inputs; physical beam preparation is outside this free-flight comparison. The matter and decay constructions below state their own sources and transfer domains.

**A native matter potential and complete medium records.** Direct and fermionic-exchange contractions of the literal weak current on a neutral and rest-electron port give a twelve-state operator with characteristic polynomial $x^9(x-1)(x+1)^2$. Its unpolarized electron mean is $(P_e/2 - I/4) \otimes I_{\text{spin}}$. The calibrated weak unit therefore gives $V_{\text{CC}} = \sqrt{2}G_F n_e$, with its common neutral-current term retained. Compressing by the actual massive active weights retains all three neutral masses, both helicities and sterile outputs. For $N$ fresh electron-spin ports, complete Kraus-word composition gives an exact finite instrument. Its neutral-record distance from the coherent mean is at most $2B^2e^{2B/N}/N$, where $B = 5|G_{\text{Weyl}}n_eT|/4$. A declared 2.5 GeV, 1300 km medium gives $P(\nu_\mu \to \nu_e) = 0.0697011$, versus 0.0491046 in vacuum; the antineutral values are 0.0197144 and 0.0351399. These are scenario outputs from the calibrated tensor and a forward-current port architecture, without a claim of equivalence to unfiltered scattering or a measured beam-event fit. All preparation, electron-spin and sterile records and apparatus work are retained. The source and proof are in `model/openwave_neutrino_matter`.

**Native leptonic widths and their physical units.** The actual family current and three-body phase space give $\Gamma = G^2M^5f(m/M)/(1536\pi^3)$. With the conventional weak normalization, $G = 2\sqrt{2}G_F$. All nine calibrated massive-neutral channels are retained for each charged transition; a rational bound places their total width within $1.7\times10^{-8}$ relative to the massless-neutral value. The widths are $3.00876260 \times 10^{-19}$ GeV for the muon-like electron channel and $4.04898950 \times 10^{-13}$ GeV, $3.93790076 \times 10^{-13}$ GeV for the two tau-like leptonic channels. Their ratio is 0.972563836, compared with the unused PDG 2025 fit $0.9730 \pm 0.0022$. The weak coefficient is a calibration input with muon-lifetime ancestry; the muon result is a normalization check and lacks radiative corrections. The positive-frequency source has an exact parent-plus-all-daughters invariant sector. Rational Volterra and finite-bin survival bounds transfer to its calibrated massive-neutral member: exact angular coarea gives $\|\sqrt{\rho_m} - \sqrt{\rho_0}\|_2 \le 2\mu_{\max}^2$, so the inclusive probability error is at most $4Gt\mu_{\max}^2$. This controls a finite weak-lifetime interval with all angular dark channels retained. Hadronic tau channels and a total physical tau lifetime are not derived. The source and proof are in `model/openwave_lepton_widths`. For the supplied weak normalization, the exact massive muon-like survival probability at 2.1876500221 $\mu$s differs from $e^{-1}$ by less than $1.223 \times 10^{-7}$; at four such intervals the bound is $4.892 \times 10^{-7}$. The finite energy-bin error is included. The measured muon lifetime is 2.1969811 $\mu$s, so this radiatively uncorrected normalization check has a $-0.4247\%$ discrepancy. The independent on-shell tree relation using $\alpha(0), M_W, M_Z$ instead gives 2.35418 $\mu$s and fails that precision comparison. The tau ratio is compared with the unused HFLAV fit in the PDG 2025 tau listing; the weak coefficient and unit constants use the named physical-constants table. Each prepared-parent decay member retains its full admitted sector. The tau member contains both leptonic continua; a subsequent muon-daughter decay and the hadronic tau vertices are outside that reducing-sector construction.

**A controlled outgoing meson interval.** For the complete native bound-meson current, the six parent coordinates couple to every admitted emitted channel through $C^\dagger C = 24I_6$. Each of the nine family blocks is one $4L^2$ operator plus its exact mass offset. With $\lambda = \sqrt{24}/64$, the reconstructed first-emission vector satisfies

$$\|b(t) - b_1(t)\| \le \lambda^3 t^3/6.$$

This follows from unitarity and two substitutions in the exact memory equation; it retains every return and subsequent emission. A 128-site calculation at source time $1/10$ certifies a strictly positive infinite-line light-channel probability above $7.8898788951\times10^{-8}$ (and original-input probability above $1.8935709348\times10^{-8}$) after local-coefficient, binary64, Taylor-tail, cubic-feedback and finite-volume errors are included. The original preparation succeeds with probability greater than $6/25$ and its failure is retained. Full-source energy is conserved; pair/no-pair readout work has magnitude at most $2\lambda^2 t$. The fixed resonance-selected box and current are unchanged. The volume-sensitive two-site peak supplies no irreversible rate or physical strange-particle identification. The source, explicit error bounds and complete records are in `model/openwave_meson_memory_window`.

**One positive semileptonic mass source.** A declared two-cell, three-family extension retains all 192 native fermion modes, the four-input strong action, the charged analyzer tensor and the oscillation-calibrated Dirac neutral tensor. Its actual semileptonic current has coefficient $G = 1/64$. Its lepton spatial kinetic coefficient is zero; the massive Dirac branches belong to the full on-site family operator. The complete one-quark/one-family local current has exact norm 6, giving $|V| \le 18N_Q$ and $H \ge (m_{\min}-18|G|)N_{\text{colourful}}+H_L$. The retained coefficients obey $m_{\min}-18|G| > 1$. Consequently this finite matter source has a unique massive Dirac reference and full charge $\pm1$ threshold exactly the electron unit. The three charged and three neutral Dirac branches coexist on its quark-vacuum sector; purely leptonic contacts and photons are absent.

The pure lepton reference has active covariance $I/2$ and zero active–active anomalous covariance. Exact Wick contraction gives $\langle V\rangle = 0$ and $\langle V^2\rangle = 3\sum_r \langle J_{Q,r}^2\rangle$. For the actual integer native baryon trial, the complete interacting energy variance is therefore its retained strong residual squared plus $G^2\kappa$, with $\kappa \simeq 36.091699949922$. The spectral theorem gives at least $0.964754199267$ conditional probability in a half-width $0.255499475345\,\text{MeV}$ window about each calibrated nucleon mean. The corresponding original native-occupation-seed probability exceeds $0.042440240225$; preparation failure and the outside-window outcome are retained. These exact rational bounds avoid enumeration of $2^{48}$ lepton states without truncating the Hamiltonian. They identify finite energy records, not physical pole masses or widths. The full operator polynomial, native intertwiner and complete instrument are in `model/openwave_common_mass_source`.

**Longitudinal charge on the common matter source.** The complete 192-mode native matter action admits an exact compact-rotor Gauss completion. On two cells, the integer flux and reference constraints are $E_x - 6Q_x - R = 0$ and $E_0 + E_1 = 0$. Every matter occupation has a unique flux/reference image, giving a unitary map onto the complete physical constraint kernel. All charged hopping and cross-cell weak-current terms are dressed by their literal charge shifts. The full native charge commutator vanishes, including the charged Dirac reference transformation.

With declared stiffness $\kappa = 1/8$, the longitudinal Green energy is $\kappa(Q_0 - Q_1)^2/8$. A specified diagonal self-energy subtraction $-(\kappa/8)\sum_a q_a^2 n_a$ preserves every single-lepton analyzer mass. Its coefficient is fixed by the graph Green function; it is not selected by a mass refit. The bare control raises each unit-charge rest energy by $7.98435860453125\times10^{-6}$ GeV. The full physical charge-one threshold of the normalized source is exactly the electron anchor: the colourful floor exceeds it and the two-charged-lepton floor is $63/32$ electron units.

Exact native baryon trials give rational changed energy means and full interacting energy variances, using the pure Dirac reference current covariance $3I/4$. Complete spectral-window records retain the preparation failure and the outside-window outcome. These means are not nucleon pole masses. The source includes a quantum compensating background and longitudinal energy, without independent transverse photon modes, a calculation of the physical electromagnetic coupling,

or a continuum gauge theorem. The source, integer Gauss controls and moment certificates are implemented in `model/openwave_common_electromagnetism`.

**Moving records on the longitudinal matter source.** Replicate the complete 192-mode longitudinal matter block and add the positive Dirac excitation kinetic energy $\sqrt{m_j^2 + L_a} - m_j$, with $L_a(p) = 4\sum_i \sin^2(ap_i/2)/a^2$. The entire operator is dressed by the product of the native Gauss isometries, including the quantum reference shifts between blocks. Full-Fock positivity and the exact one-electron threshold are preserved. In the exact colourful-vacuum, one-lepton reducing sector the dispersion is $E_a(p) = \sqrt{m^2 + L_a(p)}$, with group speed at most one and one minimum per massive branch.

A standalone Taichi kernel executes all 48 positive source coordinates and 96 native particle/hole probe rows on 16, 32 and 64-site periodic probes. Every position outcome and the one-half preparation failure are retained. On a common momentum ball $|p| \le P$, the complete record variation from the same-input continuum positive-energy source is at most $\min\{1, s|t|a^2P^4/(24m)\}$. The finite electron packet variations are approximately 0.0069404, 0.0017534 and 0.0004396 at the three spacings. An electron continuum four-vector $(E, p) = (5/4, 3/4)$ supplies an explicit boost-to-rest comparison with a bounded regulator defect.

The source retains a preferred frame and its square-root spatial kinetic is nonlocal. These results concern complete records of an exact reducing sector; they do not establish a unitary finite-grid Lorentz symmetry, interacting-object contraction, transverse photons or a common matter/light continuum. The source and portable executor are in `model/openwave_boosted_records`.

**Three charged minima of a finite-range native source.** The full native Higgs tensor admits a declared nearest-neighbour regulator with charged-branch dispersion

$$E_k(p)^2 = \frac{\sum_j \sin^2(ap_j)}{a^2} + \left[m_k + \frac{\sum_j (1 - \cos(ap_j))}{a}\right]^2 .$$

For $x_j = ap_j$, differentiation gives $\partial_{x_j} E_k^2 = 2\sin x_j [am_k + 1 + \sum_{i \neq j}(1 - \cos x_i)]/a^2$. Since $m_k > 0$, each charged branch has exactly one local and global minimum on the Brillouin torus, at zero momentum. The native colour and charge operators identify three colourless charged Dirac branches, with electron-anchored energies 0.00051099895069, 0.1056594144959 and 1.7769849708134 GeV. The complete side-four execution retains all 11 520 doubled rows, all colourful and neutral modes, and complete creation/arrival records. The charged corner states are retained eigenstates with higher energy and negative curvature in at least one direction. The neutral sine valleys and the degenerate neutral reference are retained. The regulator coefficient is declared, and the analyzer phase 2/9 has retrospective ancestry. No interacting particle identification or chiral gauge continuum is inferred. On momenta $\|p\| \le P$, the operator error from the continuum Dirac branch is at most $aP^2/2 + a^2P^3/6$; complete same-input record variation is at most $\min\{1, |t|(aP^2/2 + a^2P^3/6)\}$.

## 11.41 Two native clocks and a controlled common-clock limit

The actual eight-level native energy admits an exact Jordan–Wigner representation as three normal fermion occupations. Two occupations therefore supply independently prepared clock bits within the same source cell. Their joint Ramsey instrument retains all settings, both clock outputs and the spectator complement. They share one fixed quantum geometry and scalar-wave source of dimension 80000.

Complete records resolve a finite response difference. At event 16, the difference between the two background-induced phases after normalization by their respective bare gaps lies in

$$[3.06881,\ 3.21525] \times 10^{-7},$$

corresponding to 31.37–32.87 parts per million relative to the second clock response. An exact dyadic partial-transpose witness in the vacuum-field controls also certifies correlations that entangle the clocks through their common geometry. Mean wave-arrival delays measured by both clocks are positive and use the same receiver law. These finite source effects are distinguished from a measured equivalence-principle violation.

There is nevertheless a controlled common-clock limit. For the declared probe family $H_\lambda = H_{\mathrm{env}} + \lambda B(an_a + cn_c)$, with $B = \exp(Q_{\mathrm{right}})$, both normalized clock phases converge to the same integral of the Heisenberg-evolved $B$. For $0 < \lambda \le 10^{-3}$ and source parameter at most 8, their disagreement is bounded by $1500\lambda/(1 - 600\lambda^2)$, uniformly over arbitrary environment states and passive references initially independent of the two prepared $|+\rangle$ clocks. This follows from the exact finite Duhamel remainder and the complete joint instrument. No physical time or gravitational constant calibration is assumed.

**A complete native kinetic kernel in OpenWave.** The periodic native kinetic and analyzer-pairing source admits a portable Taichi implementation on all ninety canonical fermion modes per site. Its complete 180-row doubled symbol has $H(p) = M + \sum_j \Gamma_j \sin p_j$, with the native Clifford relations and $\|H\| < 2$. The fourth-order Taylor step is contractive for $2t/n \le 2$, and its exact-arithmetic evolution error is at most $2^5 t^5/(120n^4)$. Complete charged-addition records include preparation failure $1/2$ and every source-coordinate outcome. Cubes of sides four and six retain 11520 and 38880 doubled coordinates. A dense native stencil and an independent Fourier solution test the Taichi action; a reversed kinetic sign is detected. At twenty steps to $t = 1$, the largest recorded arrival-law discrepancy is below $6.85 \times 10^{-8}$, while the analytic operator bound is $1.667 \times 10^{-6}$. Floating norm drift is retained separately. The engine executes in the OpenWave namespace with its Taichi dependency; no upstream model award, physical clock or interacting continuum follows from this port. The complete replay is `model/openwave_platform_kernel`.

## 11.42 Native electric charge of a dressed quantum state

Fill all seven native weak-singlet coordinates in both spin states and leave the doublets empty. Every original gauge current annihilates this reference; its electric charge is exactly zero. For each actual signed Higgs pair define $d_L = c_L$ and $d_R = s\epsilon c_R^\dagger$. The canonical CAR transformation preserves both spin components and gives $Q_{\mathrm{pair}} = q(N_d - 2)$. The three up, three down and one lepton charges sum to zero, so the full native current obeys

$$Q_{\mathrm{native}} = \sum qN_d$$

without a reference-charge offset. The mass pairing becomes a Dirac bilinear. A constant chirality rotation identifies its positive absolute kernel with the corresponding quantum rotational block, including its actual current derivative and field coupling. This attaches one charged block, rather than identifying all sixteen Clifford coordinates with one native family.

The declared extension retains all native pairs, families and neutral coordinates, together with a charge-conjugate native copy. Its neutral confined Weyl kernel has energy at least $1/2$, as follows from $T^2 = (5/2)T$ and the exact positive decomposition of $N_{\mathrm{spatial}} + T/2 - I/2$. The ordered positive CAR lift gives $H \ge 16N_{\mathrm{charged}} + N_{\mathrm{neutral}}/2 + N_{\mathrm{photon}}$. The light charged ground energy

lies below 16.5, every additional neutral costs at least 1/2, and every two-charged sector costs at least 32. Thus each charge-one sign has a four-dimensional lowest band containing two spin-half doublets. The complete preparation selects one line with actual native charge of magnitude one and dressed $J^2 = 3/4$. The absolute kernel, trap, conjugate copy, fixed Higgs background and empirical electron unit are explicit; local longitudinal gauge completion and a physical particle dictionary do not follow from this attachment.

**Infinite-photon confined spin-half band.** On the complete finite matter shells, the transverse photon oscillator can be extended to its full Fock space. The positive form $H = \sqrt{16^2 + B_\infty^2} + N_{\mathrm{m}} + N_\gamma$ has two reduced levels below 17 and all other levels at least 17. Projection to the matter/photon vacuum identifies the low space equivariantly with spin 1/2; its exact Casimir is 3/4 and its energy is a scalar on that space. For a trial with photon number at most $C$, fourteen additional photon levels evaluate $B_\infty^{14}\phi$ exactly. The global estimate

$$0 \le \sqrt{16^2 + x} - p_6(x) \le \frac{33x^7}{2048\,16^{13}}, \qquad x \ge 0,$$

then gives a vector residual and Temple enclosure without an operator-norm bound on the infinite field. Three independent finite witnesses enclose the same infinite-source energy and certify photon tails. The spatial trap, field coupling and electron calibration are retained inputs; the result removes the photon cap and does not establish a spatial continuum or an observed-electron magnetic moment.

## 11.43 Quantum magnetic response

**Same-source quantum Larmor dynamics.** For the complete cap-two source, a uniform probe gives $H(b) = \sqrt{16^2 + (B + bE_z)^2} + N_{\mathrm{m}} + N_\gamma$. The exact static derivative satisfies $KX + XK = BE_z + E_zB$ and restricts to $PXP = \mu\sigma_z$ on the dressed spin-half band. Exact angular-sector interval witnesses enclose both finite-field branches. Their precession coefficient $g_{\mathrm{pre}}(b) = 2E_0|E_+(b) - E_-(b)|/|b|$ converges to the same static value $4E_0|\mu|$. A source-derived propagator estimate bounds all joint matter, photon and spectator-reference records over finite time; the complete executable instruments retain filter failures and photon outcomes. This result retains the finite regulator and declared field and mass parameters, without assigning a physical electron error budget.

## 11.44 Neutral production and inclusive weak conversion

**Production, neutral flight and recoil on one source.** On an odd periodic ring, retain the complete native charged and neutral Higgs kernel, its filled reference and every spectator. The actual charged-current frame is $C$, and its positive-band left weights are $w_i(n) = (E_i(n) + \sin(2\pi n/L))/(2E_i(n))$. A charged recoil register makes the source-current contraction

$$K_{ink';amk} = \frac{C^*_{ai}\sqrt{w_i(n)w_a(m)}}{L}\delta_{k',k+m-n \pmod L}.$$

Opposite momenta give $\sum_n w_i(n) = L/2$, hence $(K^\dagger K)^2 = K^\dagger K/4$ and $\sqrt{I - K^\dagger K} = I + (2\sqrt{3} - 4)K^\dagger K$. A rest-family input succeeds with probability $1/(4L)$ for every normalized recoil input. The same quadratic Hamiltonian then propagates every neutral momentum, family and recoil coordinate. The receiver uses the actual reverse current and its complete square-root failure operator. Source failure, detector failure and all received family/momentum/recoil records sum to one. Each detector

aperture is a separately chosen complete experiment; a uniform aperture choice retains that classical label on every outcome.

A sharp source recoil momentum makes the spatial response uniform, while a localized recoil register produces moving interference profiles. A separate retained mass record removes the time dependence of the spatially summed family-interference signal; it is distinct from a momentum recoil record. At side five the native untagged family contrast between source times zero and nine is approximately $1.17942\,10^{-4}$. Source and detector charge references each cost $1/4$ in the declared units, and their preparation and readout work are retained. These are current/spectral apparatus services on one free source, without a measured neutrino fit, an autonomous decay claim or an identification with the separate gauge-singlet composite.

**Inclusive charged-current decay with a finite survival bound.** Consider the positive-frequency part of the native charged-current contact interaction, its Hermitian reverse, and every spin and momentum channel of a middle charged particle, a lighter charged particle and two distinct massless neutral species. This is a declared contact member, with continuum relativistic dispersion and free complementary native fields. The three conserved combinations $N_p + N_j = 1$, one for each daughter species, restrict the admitted full Fock evolution exactly to one parent or three daughters. The native weak generators give unit cross-family coefficient. With $u^\dagger u = 2E$, the parent-spin average and complete daughter-spin sum give

$$\overline{|\mathcal{M}|^2} = 8G^2(P \cdot p_{\bar{\nu}})(p_e \cdot p_\nu).$$

The neutrals are distinct and contribute no identical-particle factor. The two-body phase-space volume is $\lambda^{1/2}(S, a, b)/(8\pi S)$, so its three-body factorization and the parent factor $1/(2M)$ give

$$\Gamma = \frac{G^2 M^5}{1536\pi^3} f(r), \qquad f(r) = 1 - 8r^2 + 8r^6 - r^8 - 12r^4 \log r^2, \quad r = m/M.$$

Here $G$ has dimension energy to power minus two, and $r$ is the actual native analyzer ratio. The numerical coefficient follows from the complete current and phase-space contraction.

In units $M = 1$, choose the ultraviolet amplitude $u(E) = (2 - E)^2$ for $m \le E \le 2$. The inclusive spectral density is $\rho(E) = E^5 f(m/E)(2 - E)^4/(3072\pi^4)$. Its exact Volterra survival equation has kernel $C(t) = \int \rho(E) e^{-i(E-1)t}\, dE$. Rational derivative bounds give $\int_0^\infty |C| < 1/2000$ and $\int_0^\infty t|C| < 1/250$. If $\alpha = \int_0^\infty C$, unitarity and the Volterra equation imply

$$|a(t) - e^{-G^2\alpha t}| \le G^2 B_1(1 + G^2 B_0 t), \qquad \Gamma = 2G^2 \operatorname{Re}\alpha.$$

At $G = 1/64$, survival at one and four nominal lifetimes differs from $e^{-1}$ and $e^{-4}$ by at most $0.000050782$ and $0.000197266$. Rounding the entire reservoir energy to bins of width $h$ adds at most $th$ to any survival probability; $h = 2^{-60}$ gives an additional error below $7.11\,10^{-10}$ through four lifetimes. This large finite-carrier statement is an analytical bound. Executed smaller carriers retain every emission bin, the survival complement, and the measurement energy work $W = -2G \operatorname{Re}(a^* \langle \sqrt{\rho}, b \rangle)$.

The finite realization retains its orthogonal source complement and arbitrary external references. The cutoff and contact strength are declared inputs; there is no transfer to the unfiltered contact source, measured weak lifetime or exact irreversible finite-volume decay. Electron anchoring supplies a conditional conversion of the calculated width to GeV.

## 11.45 Bound native composites and a common energy scale

**Bound native colour composites.** On a two-vertex graph retain the full fifteen-coordinate native matter module, two spin coordinates and all sixty fermionic modes. With the actual native colour currents $T_x^a$ and the spin current $S_x$ on the entire colourful subspace, declare

$$H = mN + t \sum_{\langle xy\rangle,r} (c_{xr} - c_{yr})^\dagger (c_{xr} - c_{yr}) + U \sum_{x,a} (T_x^a)^2 + J \sum_x S_x^2, \qquad (m,t,U,J) = (1, 1/8, 1, 1/4).$$

Every term after $mN$ is positive, so $H \geq mN$ controls the full Fock complement and the empty matter vacuum. The action preserves global native colour, weak and hypercharge symmetries. Its graph kinetic term, number energies and positive interaction strengths are declared architecture; no local gauge-link dynamics or equivalence to a Weyl/Higgs action is asserted. Idle opposite-charge or conjugate-colour apparatus references have declared zero energy and are retained in every preparation outcome.

The complete three-quark colour singlet is $\mathrm{Sym}^3(\mathbb{C}^8)$, dimension 120, inside the 2024-dimensional three-quark sector. The quark/conjugate-quark colour singlet has dimension 64 inside a 576-dimensional sector. Explicit exterior-colour contractions give isometric reducing embeddings into these full occupation sectors. On a graph of maximum degree $z$, a local spin-half three-quark trial has energy at most $3m + 3J/4 + 3zt$, whereas every two-plus-one fragment threshold is at least $3m + 8U/3 + 3J/4$. A local spin-zero meson trial has energy at most $2m + 2zt$, whereas its two-fragment threshold is $2m + 8U/3 + 3J/2$. Thus the executed binding margins are at least $55/24$ and $67/24$, respectively, without enumerating the $2^{60}$-dimensional carrier.

The lowest internal-spin-half baryons of charges zero and one are degenerate. For the spin-zero mesons, all three charges $-1, 0, 1$ share the same energy:

$$E_{\mathrm{baryon}} = 3.544903181009, \qquad E_{\mathrm{meson}} = 2.229589021818.$$

The probability that all constituents occupy the same cell exceeds 0.993. Complete spectral preparation includes its failure outcome and every original occupation record. Spin-one mesons, the neutral meson singlet and spin-three-halves baryons are retained. Spin is the internal rotation representation on this finite graph. These finite bound eigenstates establish neither relativistic hadron identity nor physical lifetimes or continuum confinement. The source, exact reductions and complete records are in `model/openwave_native_composites/`.

**A common-scale obstruction for the symmetric control.** Three conserved family coordinates with the native analyzer number energies give exact colourless constant-mode energies $m_f$. The empirical electron unit 0.00051099895069 GeV then preserves the conditional charged-lepton values 0.105659414496 and 1.776984970813 GeV. It puts the light bound meson below 0.00115 GeV and baryon below 0.00183 GeV, excluding the charged-pion and proton assignments on this action. Weak symmetry also forces the charge-zero and charge-one baryons to be degenerate for every $U, J, t$. For families $a, b$, $H_{a,b} = H_{1,1} + (m_a + m_b - 2m_1)I$, so the common binding correction cancels from family differences. Its light/second-family difference 0.105148416 GeV cannot equal the kaon-minus-pion comparison 0.35410661 GeV by changing that correction. These exact controls explain the need for distinct hadron coefficients in the calibrated construction. Complete source spectra and comparisons are retained in `model/openwave_composite_scale`.

## 11.46 Autonomous geometry and complete wave deflection

**Autonomous finite quantum geometry.** The native eight-level excitation operator $K$ admits the declared source $H = H_g + e^{Q_0} K_0 + e^{Q_1} K_1$ on all $64(C+1)^2$ states, with two product oscillator

caps and graph Laplacian $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Both native excitation populations and total source energy are conserved; geometry energy responds autonomously. Exact dyadic eigenframe residuals enclose every native conditional block. At $C = 4$ and source time $10^4$, a rational partial-transpose witness for two initially product native clocks lies in $[-0.054402654, -0.054401892]$. All nine product Pauli instruments retain every geometry and complementary native outcome. All 49 positive-energy source pairs have negative finite mutual ground interaction energy. Deleting the spatial edge factorizes the source and prevents this clock entanglement. The graph, stiffness $10^4$, cap and clock unit are declared; physical $G$, irreversible relaxation, continuum gravity and a transfer to the static wave profile are not inferred. The construction and certificates are in `model/openwave_quantum_geometry`.

**Complete finite wave bending.** On the central plane of the declared native $31^3$ lapse profile, the scalar source $H = \sqrt{SD^*\overline{AD}S}$ retains all 961 wave modes and the vacuum, where $S = e^{(1+3\gamma)q/2}$ and $A_e = e^{(1-\gamma)q_e}$. Complete position and bounded velocity instruments admit arbitrary references. At source time 16 and $\gamma = 1$, a finite packet has transverse displacement $-1.9417333 \times 10^{-5}$ and velocity change $-3.6701444 \times 10^{-6}$ relative to the identical flat preparation. An independent local second-order wave evolution agrees to numerical roundoff. A side 47 control retains the sign; reflection reverses it. The principal ray dispersion and complete wave records are explicitly compared: $\gamma = -1$ has zero principal-ray deflection and a nonzero finite-wave residual. The scalar planar extension, static source, $\gamma$ and units are declared. No physical $G$, three-dimensional photon limit, observed lens angle or transfer to the autonomous quantum-geometry source is inferred. The finite law, analytic polynomial tail bound and separate arithmetic residuals are in `model/openwave_wave_deflection`.

## 11.47 Connections to observed particle and field structure

The finite constructions reproduce several characteristic mathematical structures used in descriptions of observed particles and fields. Each comparison below states the generated record, its calibration and its physical domain. These results use declared source members; combining their conclusions into one interacting realization requires the corresponding source-transfer statements. A structural agreement, an empirically anchored energy and a held-out numerical prediction are different tests.

**Quantized electric charge.** The native Fock charge operator has an exact discrete spectrum, including the charged-lepton unit and quark charges $2/3$ and $-1/3$. Actual charged arrivals carry the corresponding central current and regional Gauss flux, with complete failure and reference outcomes. Charge cannot be continuously divided within a fixed native representation. This is a finite compact-current realization of charge quantization. It does not equate texture winding with electric current or by itself identify the observed electromagnetic field. The complete source calculations are in `model/openwave_charge_protection` and `model/openwave_native_charge_flux`.

**Charged rest energy.** The exact native CAR and local Gauss isometry attach a positive energy to a dressed charged state, including the root-reference field. Equal-depth tree dressing adds its explicit electric energy without changing the matter current on its finite-orbital member. Separately, the full spatial/photon charged block, without tree dressing, has a trap-release threshold 16, conditionally 0.000509529040252 GeV under its retained electron conversion. The empirical electron anchor and its source conventions are explicit; this energy is not an independent precision electron-mass prediction. The threshold has no normalizable minimizing state. The distinction between a

free threshold and a localized trapped state is proved in `model/openwave_spatial_completion`. Its enlarged spatial domain does not inherit the finite-orbital tree energy or neutral-sector bounds.

**Particle persistence.** The complete finite charged packet has a proved persistence domain. For the spatial/photon source $H_\Omega = \sqrt{16^2 + B^2} + \Omega N_{\mathrm{m}} + N_\gamma$, positivity and oscillator coercivity isolate a spin-half band when $\min(\Omega, 1) > 3/64 + (3/8192)(5/7)^{5/2}$. Shell convergence controls the full source in this trap domain. This is a quantitative stable-band construction. The result supplies neither a self-bound localized electron after trap removal nor a nonlinear classical soliton.

**Magnetic response near two.** The actual differentiated current, including the photon field and its complement, gives a static coefficient in $[1.99827530667, 1.99827540121]$. Global divided-difference estimates and finite moment witnesses control the infinite-photon source; finite-time precession uses the same current. The proximity to two is a quantitative structural comparison with a spin-half magnetic response. Its deviation does not reproduce the observed electron anomaly. Field normalization, trap and regulator are declared inputs, as specified in `model/openwave_infinite_current`.

**Spin one half.** The finite-orbital dressed band carries $J^2 = 3/4$ and the two spin projections $\pm 1/2$ in units of $\hbar$. The native CAR charge map and longitudinal Gauss isometry preserve this representation on that member. Independently, the spatial/photon charged block includes complete matter shells and photon angular momentum, with a spin-half band and full-space convergence in its stated trap domain. The tree dressing and magnetic response bounds are not transferred to this enlarged spatial source. These exact representations agree with the spin structure of a fermionic particle; physical species identification is an additional comparison.

**The double-cover rotation law.** Literal proper cube rotations act on the kinetic/pairing source and its filled reference. An odd state changes sign under $2\pi$ relative to an idle reference and returns under $4\pi$. Complete oriented spin-frame records return under $2\pi$, while a declared two-copy apolar tensor record returns under $\pi$. These distinct record periods are computed in `model/openwave_spin_frame_records`. They realize the spinorial double-cover structure and do not prove an exchange-statistics theorem or a fundamental nematic field.

**Particle–antiparticle conversion to radiation.** The actual native CAR current converts a neutral particle–hole pair into two transverse quantum modes. A positive full-Hamiltonian probability bound, complete observations and conservation of total energy, charge and momentum establish this finite conversion mechanism. The source uses a declared resonance, reference quench and field cap; measurement work is retained. The calculation in `model/openwave_radiative_pair` supplies no physical positronium branching ratio or infinite-volume annihilation rate.

**A stable neutral candidate.** The periodic link-penalty-two source contains an isolated stationary neutral band. Its finite polynomial/polar preparation has success above 0.99999999999639 and stationarity error below $4.000002 \times 10^{-16}$. This supplies a stable neutral sector with complete failure records. Exact location conservation makes its momentum band flat, so these properties do not identify a moving relativistic dark-matter particle or a cosmological abundance. The preparation and obstruction are both retained in `model/openwave_neutral_preparation` and `model/openwave_neutral_dispersion`.

**Electrostatic spatial dependence.** The declared Gauss Green operator gives the Coulomb $1/r$ potential and the associated static inverse-square force in its controlled spatial domain. A separate autonomous eighteen-state native-current experiment supplies actual force and displacement records. This connects a constraint law to finite dynamics; its coupling is supplied and is not a prediction of the observed fine-structure constant. The source and reference domains in `model/openwave_pinned_coulomb` and `model/openwave_autonomous_coulomb` are retained.

**Two transverse Maxwell waves.** A finite cochain field source has two transverse Maxwell modes, exact finite-occupation dynamics and independently computed real-space propagation. Discrete constraints remove longitudinal radiation in the stated sector. This reproduces the transverse wave structure of electromagnetism on the declared lattice. A shared physical particle, coupling and elapsed-time calibration is required for a common electromagnetic interpretation; see `model/openwave_field_observables`.

**A massive second-order wave law.** For the full native kinetic/pairing source, charged frames obey

$$h_k^2 = \left( \sum_i \frac{\sin^2(ap_i)}{a^2} + e_k^2 \right) I.$$

Complete spatial wave records and a controlled low-momentum limit give the massive Klein–Gordon form. Every finite-derivative valley and spectator is retained. This is a source-derived wave identity with stated momentum/spacing error, rather than an identification of all interacting particle masses or an elimination of lattice multiplicities.

**Discrete atomic levels.** The declared Gauss atom has negative standing-wave levels under independent volume and spacing variation. In its atomic units, the first excitation gap approaches $3/8$, and the tested second-shell splitting decreases to 0.000410646363. This is the hydrogen-like orbital pattern generated by the specified kinetic and Gauss source. Mass and coupling are supplied. Precision physical hydrogen, fine structure and radiative shifts are outside this calculation, whose refinement evidence is in `model/openwave_atomic_refinement`.

**Charged-lepton numerical comparison.** The separate analyzer-family comparison gives electron-anchored values 0.00051099895069, 0.1056594144959 and 1.7769849708134 GeV. The muon residual is approximately +9.83 parts per million. The phase $2/9$ has retrospective selection ancestry, and the electron fixes the common unit. These small residuals are useful quantitative comparisons, with those empirical dependencies retained; they are not three independent parameter-free mass predictions. The full physical particle dictionary and interacting mass-shell identification require further same-source tests.

## 11.48 Source domains, complete records and physical identification

The target is one executable OPH construction whose native observers recover all the physical observations of our Universe, including its effective fields, gravity and particle spectrum. Hardware and software may be reverse-engineered; neither architectural uniqueness nor an a priori derivation of the design is required. Architectural choices define a legitimate candidate. Its complete outputs must be derived and compared with observation, with empirical calibration distinguished from independent prediction. The starting premise is a self-referential mathematical structure, without an independently existing background spacetime or external global clock. Computational

events have a partial order; physical spacetime and mutually calibrated clocks must be recovered from compatible records. An operational observer has readback, record, feedback, prediction and continuation interfaces. A minimal controller can qualify when it implements those interfaces; no consciousness premise is imposed. Auxiliary generator times in the constructions below are calculation parameters whose identification with native observer clocks must be stated and verified. Exact invariant sectors and positivity bounds avoid enumerating entire Fock carriers while retaining their full complements. Each construction supplies its own complete preparation and observation law. Separate source results require an implementation or transfer theorem before they constitute a common interacting particle and gravitational model.

**Native compact Gauss completion.** The actual native charge identity $Q_i = \sum_a q_a N_{ia}$ permits a compact rotor tree with local constraints $E_i = 3Q_i$ and a retained root charge $R = -\sum_i E_i$. The exact physical isometry dresses every native Fock word with these fluxes. Wilson-linked positive kernels retain the full ordered CAR floor. Electric energy reduces to $H_E = (\kappa/2)\sum_i d_i Q_i^2$ for branch depths $d_i$. For an equal-depth star, the entire one-electron block differs from the quantum source by the constant $\kappa/2$; its current, spin and complete observer laws transfer through the isometry. This energy includes the charged reference's longitudinal field. The declared orbital graph does not establish a physical Euclidean Coulomb law or an isolated-electron rest mass.

**Infinite-photon magnetic current and precession.** For the fixed complete matter shells, the photon-coordinate Weyl operator $B(x)$ is unbounded globally but is a finite Hermitian matrix at each $x$. Divided differences give the uniform derivative bound $\|K'(x)\| \le \|E_z\|_{\mathrm{HS}} = \sqrt{5}$. A global differentiated Taylor remainder is controlled by a finite trial's twenty-sixth radial field moment. Fourteen padded photon degrees evaluate its current and energy witnesses against the full photon source. The resulting static coefficient is enclosed in $[1.99827530667, 1.99827540121]$ in the unchanged source conventions. The same uniform derivative and isolated-band gap give finite-time Larmor bounds with arbitrary references and every field outcome retained. Physical field calibration, electron anomaly and spatial trap removal are separate identifications.

**Spatial completion and trap-release threshold.** On the full spatial and photon carrier, the declared positive source $H_\Omega = \sqrt{16^2 + B^2} + \Omega N_{\mathrm{m}} + N_\gamma$ has an exact isolated spin-half doublet whenever $\min(\Omega, 1) > v_0 = 3/64 + (3/8192)(5/7)^{5/2} < 1/16$. Complete-shell core and form convergence, together with oscillator coercivity, give convergence of fixed low energies without equating a truncated square root with a compressed infinite operator. Gaussian width-$L$ trials give

$$E_\Omega \le 16 + \frac{3}{64L^2} + \frac{3L^2}{8192(1 + 2L^2/5)^{5/2}} + \frac{3\Omega}{4}(L^2 + L^{-2} - 2).$$

Hence trap release has threshold 16, with no normalizable state attaining it. This excludes a demonstrated self-bound localized ground, not an ordinary free-particle mass threshold. The retained electron unit assigns that threshold approximately $0.000509529040252\,\mathrm{GeV}$, conditionally.

**Native bound-meson charged-current conversion.** Retain the positive two-cell native colour/spin action, all three analyzer family energies and the complete 180-mode fermionic carrier. A declared native port word gives the quark family current $C = F^\dagger V F$. The lepton reference fills all twelve neutral coordinates. The autonomous semileptonic source uses every local spin and family current,

$$H = H_{\mathrm{hadron}} + H_{\mathrm{lepton}} + GV, \qquad V = \sum_{x,\mu} \eta_\mu (Q_+^\mu L_-^\mu + Q_-^\mu L_+^\mu + 2Q_3^\mu L_3^\mu).$$

Conserved charge forces at most one emitted charged lepton and neutral hole on the prepared one-quark/one-antiquark sector. Exact colour reduction, total spin and cell parity reduce its complete 888-state block to 150 states without removing a reachable outcome.

The heavy-family and light-family spin-zero mesons have identical orbital wavefunctions. Their vector-current form factor is one, and the complete bound-to-bound pair matrix element is $C_{01}/\sqrt{2}$. The native current expectations are $-3$ and $5/2$. With $G = 1/64$, a declared outgoing frequency $q = (m_2 - 2)/2 - 11G/4$ equates their interacting mean energies. Full evolution at source time 720 gives conditional pair probability approximately 0.994907948; the spectral-preparation probability is approximately 0.2483446023, and its failure is retained. Independent complete-block propagation checks the reduction. The finite evolution also returns toward the parent, so this is a coherent finite-box conversion.

Separate complete position and momentum instruments retain 2664 active colour/occupation records, the full source complement and preparation failure. Their different energy work is calculated from the full native occupation Hamiltonian. The port word, contact strength, filled reference and resonance are adjustable architectural inputs. No empirical hadron mass, lifetime or mixing datum enters, but the result is neither a parameter-free physical strange-decay prediction nor a repair of the rejected physical mass assignment. Numerical probabilities have independent residual checks without a certified all-arithmetic enclosure.

**The exact resonant orbit of the native weak current.** The unfiltered 270-mode contact has 31464 quartic words. Its complete free-mode transformation has 55575 ordered terms and 299 Bohr-frequency labels. Exact normal ordering and a positive nonzero-frequency gap give an invariant two-state resonant orbit whose current is

$$\overline{V} = \begin{pmatrix} 173/2 & 1 \\ 1 & 85 \end{pmatrix}, \qquad p_{\mathrm{receiver}}(t) = \frac{16}{25} \sin^2 \frac{5Gt}{4}.$$

The $3G/2$ detuning follows from the actual contact; no selected transition operator or occupation penalty defines the source. Prepared-domain bounds below control its approximation by the full Hamiltonian, retaining every nonresonant outcome and arbitrary references. The finite momentum box, weak coupling and source parameter have no physical lifetime calibration. The source orbit is reproduced by `model/openwave_current_averaging`.

**Native-charge mixing.** One conjugate and one same-representation copy of the actual colourful weak-singlet module form an anomaly-cancelling vectorlike extension. The gauge-invariant native-hypercharge mixing gives $M_q = \left(\begin{smallmatrix} e_f & 0 \\ GY_R & M \end{smallmatrix}\right)$. For $G, M > 0$, its light singular values obey $0 < s_{u,-} < s_{d,-} < e_f$, while all heavy partners remain. The declared example $G = M = 1$ retains13 charged Dirac channels per family, the neutral Weyl coordinate and all finite-derivative valleys. Exact Ward identities, cubic anomaly cancellation and a rational charged energy-floor certificate accompany complete256-outcome active CAR addition records, success and failure, and the opposite-charge reference. The unchanged electron calibration supplies conditional GeV branch gaps. These are consequences of supplied extra matter and mixing coefficients; physical quark masses, heavy-particle identities and an interacting pole interpretation are not established. See `model/openwave_vectorlike_flavour`.

**Balanced-current baryon control.** The equal-number-energy action with positive colour, spin and weak squares has a complete 120-dimensional colour-singlet baryon sector, with a spin-half/isospin-half ground band near 3.73359579. The exact integer and rational instruments

retain all 2024 original three-quark occupation outcomes and failure. A number-centred graph charge readout gives positive inner and negative outer neutral charge, about $\pm 0.000575083$ in graph units. Using the free vectorlike gaps as number coefficients gives $0 < E_n - E_p = m_d - m_u < m_e$. The conditional graph energies 0.001709996503 and 0.001769595845 GeV fail the nucleon mass comparisons and forbid ground-state beta conversion. These are specific failures of the balanced-current/free-gap attachment, not of the separately calibrated hadron construction. The exact source and complete instruments are `model/openwave_baryon_ordering`.

**Dynamic geometry and its source law.** A declared seven-by-seven scalar grid retains 64 native matter sectors, two harmonic geometry coordinates and all 49 wave modes plus vacuum. Harmonic extension induces geometry stiffness and inertia from the same graph. The positive source $H = H_g + e^{Q_0}K_0 + e^{Q_1}K_1 + h(FQ)$ includes wave backreaction. Complete wave, geometry and native-clock records are computed using its exact reducing blocks. For comparison of occupation caps, the fixed-master compression below keeps the action fixed; rebuilding $e^Q$ after compression is a different source. `model/openwave_dynamic_wave` gives the dynamic action, and the fixed-master experiment states its own complete-record bounds and event convention.

**Source stress and the test-clock domain.** The autonomous wave construction determines its field force through $hG_a + G_a h = 2(D_a K_q + K_q D_a)$, with $h^2 = K_q$ and $D_a = \mathrm{diag}(F_a)$. The complete finite Ehrenfest equation retains the oscillator boundary term; geometry, native and field energy rates sum to zero. Full stress and momentum instruments retain all complementary outcomes. Interval certificates for all64 native vacuum sectors show a frequency-ratio spread of [87.8991138, 87.8991463]ppm among the seven dressed native clocks. A gap/resolvent theorem gives $|(f(s)-f(0))/s - \langle e^{Q_1}\rangle_0| \le [(17/250)^2/(9/50)]s$ on the stated probe domain. Thus the common test-clock response has a quantitative backreaction bound, while exact finite-strength universality fails in this construction. The source choices and separate clock preparations are retained; physical $G$, laboratory clocks and an Einstein stress tensor are not inferred. See `model/openwave_dynamic_stress`.

**Complete native six-quark binding and graph quadrupole.** A declared spin/weak-balanced current action on the full native sixty-CAR carrier has an exact $N = 6$, colour-zero, $S = 1$, $I = 0$ reduction from11700 occupation words to27 states. Exact integer Casimir projectors and a positive Gram factorization certify completeness. The six-quark Rayleigh energy is approximately 7.22070970417 and lies more than 0.24648187228 below two baryons of the same balanced source; all other fragment partitions have threshold at least $6+8/3$. The member without weak-current cost is certifiably unbound in this target channel. For the number-centred graph charge moment, $\langle Q_{zz}\rangle = \langle N_0 N_1\rangle/18$ on the isospin singlet. Binding forces it strictly positive; the certified ground comparison gives approximately 0.00263818496 in squared graph-separation units. The equal spin/weak costs also give a retained degenerate $S = 0$, $I = 1$ partner. Complete preparation/failure records and their stationary-ground error are included. The coefficient equality is architectural; physical deuteron mass, length, shape and species identification are not inferred.

**The vectorlike mass comparison.** For the declared $G = M = 1$ mixing, the light up/down gaps in the three analyzer families are approximately $(0.00042518, 0.00048478)$, $(0.0879117, 0.1002362)$ and $(1.46748, 1.68031)$ GeV on the empirical electron unit. Every heavy branch is retained, with gaps between 7.14 and 8.21 GeV. These are source branch energies, not identified quark or extra-particle masses. The observed quark spectrum is not reproduced, and family labels alone do not

supply a physical assignment. Complete intervals and current records are in `model/openwave_vectorlike_flavour`.

**Reuse of finite instrument theorems.** For every exact Hermitian projector $P$, the outcome maps $\rho \mapsto P\rho P$ and $\rho \mapsto (I-P)\rho(I-P)$ form a completely positive, trace-preserving summed instrument. This general finite theorem is formalized in the finite event-algebra library by the theorem `luedersChannel_isCPTP`. Its repeatability and complement-exclusivity results use the same algebraic premises. The source is `Lean/EventAlgebra/LuedersPhaseInstrument.lean`. The baryon and six-quark rank-one filters satisfy those premises by their stored exact integer vectors. Thus their failure channels and quantum references use the general theorem, with the concrete matrix identities checked by the executable certificates. Those concrete instances are not additional Lean proofs. The same formal library also proves that an effect table alone does not determine its state-update instrument. A declared preparation service cannot therefore be inferred solely from matching its outcome probabilities.

## 11.49 Mass-only controls and finite nuclear reductions

The mass-only member of the positive native-current action fits $m_p = 0.93827208943$, $m_n = 0.93956542194$ and $m_{\pi^\pm} = 0.13957039\,\mathrm{GeV}$ with three empirical inputs. An exact min–max slope bound of $2/9$ proves uniqueness within its declared coefficient family. Its complete three-quark and meson singlet sectors have dimensions 120 and 64. It is distinct from the four-input axial state described in Section 11.39.

**The axial discriminator.** The local colour-singlet representation is $\mathrm{Sym}^3(\mathbb{C}^2 \otimes \mathbb{C}^2) = (\frac{1}{2},\frac{1}{2}) \oplus (\frac{3}{2},\frac{3}{2})$, with multiplicity one. An integer colour contraction gives $g_V = 1$, local $g_A = 5/3$ and squared excited-current leakage $8/9$. Schur's lemma prevents a local scalar preserving both symmetries from changing this coefficient. On the mass-only source the positive separated-colour floor

$$\gamma = \frac{8U}{3} - \frac{4A}{9} - \frac{3}{4}(J_s + J_w) - 3t$$

and $\|I_- J_+^z\| \le 6$ give $|g_A - 5/3| \le 36t/\gamma$, hence $g_A \in [1.665734201267, 1.667599132066]$. This excludes the neutron comparison $1.2753 \pm 0.0013$ on that source. The split-site four-input construction changes the state rather than the native current, so the local multiplicity-one restriction does not apply to its full ground state. The exact controls are in `model/openwave_baryon_weak_formfactors` and `model/openwave_axial_state`.

**Nuclear spectra of the mass-only member.** Exact integer projectors give 27 levels at fixed spin projection $+1$ in the six-quark colour-singlet $(S,I) = (1,0)$ sector; the full spin multiplet has dimension 81. Its binding exceeds 1.4999965731 electron units, about 0.766497 MeV, below every same-source fragment threshold. A positive edge current shifts the spin-zero isotriplet by $2\eta$; its three components are above their respective two-baryon thresholds on the two-cell source. The excluded deuteron binding comparison 2.22456640 MeV fails. These masses, binding signs and partner exclusions are specific to the mass-only action, rather than the axial-calibrated action.

For an $L$-cell graph, exact local three- and six-quark colour-singlet dimensions 20 and 50 give the full locally singlet six-quark dimension $50L + 400\binom{L}{2}$. If the coloured complement is above $C$, $a = \max(PHP) < C$ and off-block hopping has norm $b \le 6tz$, every low level lies within $b^2/(C-a)$ below its compression; the next level is at least $C$. This controls paths and cycles with two to four cells, retaining excited bands, failures and an additional bound $(S,I) = (0,0)$ channel. The two-cell

isotriplet exclusion does not transfer to all these graphs. See `model/openwave_nuclear_exchange` and `model/openwave_nuclear_refinement`.

**Finite conversion of the mass-only member.** The native semileptonic contact $G = 1/64$ retains all 70 four-occupied states of its eight-mode lepton sector. With the baryon sector this gives 8400 states, 1400 at charge zero and 398 at fixed twice-spin projection one. The identity $N_e = N_{\mathrm{up}} - 1$ retains virtual two-electron states. At source parameter 512 the complete floating propagation gives conditional one-electron probability 0.96855747 and original-input probability about 0.32285249, retaining preparation failure and alternative complete position and momentum instruments. The 0.78233355931 MeV strong/free-channel endpoint does not include the filled-reference shifts $\mp 7.984$ keV. Recurrence and the failed axial comparison preclude a physical neutron lifetime interpretation. The full weak-dressed energy instrument and the axial source's outgoing momentum record are separate calculations, given in Section 11.39; the finite conversion control is `model/openwave_baryon_beta`.

**Prepared-domain control of the unfiltered native current.** Exact normal ordering of all 31464 native contact words retains the scalar 84, quadratic contractions and every quartic channel. Complete current columns from the derived parent/receiver orbit have nonresonant gaps above 0.00481278. The CAR identity $\sum_{|J|=b} \|a_J \psi_n\|^2 = \binom{n}{b}$ bounds coefficient matrices on particle-number sectors without truncating the source evolution. For its two-dimensional prepared domain, integration by parts gives $\|(U_I(t) - e^{-iG\overline{V}t})P\| \le GA + G^2 t(K + 5A/4)$. The first-peak coefficient is below 3367930.443. Thus $G = 2^{-30}$ certifies a receiver probability within 0.00628 of $16/25$, while $G = 2^{-44}$ retains probability error below$4 \times 10^{-7}$. All full-source complements and arbitrary references are included. The $G = 1/64$ control is uncertified; no observed weak lifetime or irreversibility is inferred. See `model/openwave_reachable_averaging`.

**Outgoing volume and a native meson memory kernel.** Keep the two-site bound meson, native family word, full semileptonic current and contact coupling $G = 1/64$, and extend only the outgoing lepton ring to $L$ sites. The original fermionic carrier is retained with $12(L-2)$ added coordinates. The filled neutral reference and charge identity $Q = -w - N_e = -1$ restrict this preparation to zero or one emitted electron–hole pair without a particle cutoff. The exact charge and spin sector has $24{+}216L^2$ states; spin zero and reflection reduce the prepared invariant subspace to $6 + 36L^2$ states. In its parent/emission decomposition,

$$H = \begin{pmatrix} A & GC^\dagger \\ GC & D \end{pmatrix}, \qquad C^\dagger C = 24 I_6, \qquad P_{\mathrm{pair}}(t) = \frac{3}{512} t^2 + O(t^3).$$

The coefficient follows from the native Pauli contraction, three emitted families and family unitarity. Eliminating the complete emitted sector gives

$$\dot{a}(t) = -iAa(t) - G^2 \int_0^t C^\dagger e^{-iD(t-s)} C a(s)\, ds.$$

The operator $D$ retains its neutral-current interaction. This identity supplies a memory kernel rather than an assumed decay rate.

With the same resonance-selected box held fixed, the pair probabilities at source time 720 are 0.994908, 0.244041, 0.063225 and 0.023513 for $L = 2, 4, 6, 8$. The selected plane-wave form factor is $2|C_{01}|/(\sqrt{2}L)$ and its contact detuning is $G(1/2 - 1/L)$. Consequently the two-site peak does not

define a volume-stable irreversible lifetime. A same-source infinite-ring comparison uses outgoing hopping norm below 103 and first differing hop order $R = (L-2)/2$:

$$\|(U_L(t) - U_\infty(t))P_{\text{parent}}\| \le 2\sum_{k\ge R}\frac{(103|t|)^k}{k!}.$$

For $L = 32, t = 0.01$ this is below $2.546754 \times 10^{-12}$. The separate 36,870-state floating propagation gives pair probability approximately $2.183 \times 10^{-7}$; its roundoff is outside this volume bound. Complete position and alternative momentum records retain preparation failure, all matching-colour outcomes and full-source measurement work. The source clock is not identified with seconds, and neither a measured strange-meson lifetime nor physical particle identification is asserted.

**A rigid neutral-hierarchy control.** On the same 288-mode Dirac carrier, the declared tensor $\epsilon R\,\mathrm{diag}(m, 1, M)$ with $\epsilon = 2^{-32}$ gives three neutral masses proportional to the charged analyzer. It retains both helicities, the sterile coordinates, the filled reference and the native weak contact. The exact complete current isometry has 144 outgoing channels and a production complement; it is a declared apparatus, distinct from autonomous decay. The rigid hierarchy fails the scale-independent squared-gap test in Section 11.40 for every positive common multiplier. Its conditional absolute masses therefore carry no measured neutrino identification. The calibrated neutral tensor uses the same current identities with explicitly different family coefficients; `model/openwave_dirac_neutral_current` retains the rigid control.

## 11.50 A fixed quantum geometry source and controlled compression

On a seven-by-seven scalar grid, two harmonic geometry coordinates use exact induced stiffness and inertia matrices. Retaining occupations zero through four in each coordinate gives a fixed source with 64 native matter sectors, 25 geometry states and 50 field states including vacuum. Its total dimension is 80000. The source contains the native energy–lapse coupling and the positive scalar-wave square root. Its nested comparisons compress this fixed action; exponentiating a smaller compressed coordinate defines a different action.

Complete arrival-history isometries certify the comparison for every input and untouched reference on the span of native levels 00 and 10, geometry vacuum, and field vacuum or the specified packet. The cap-three compression has complete-record total-variation error below 0.000358 relative to the cap-four source. A finite Schur bound and an exact rational trial give the native 10 one-wave ground interval

$$1.35873578062 < E < 1.35873578145.$$

All 64 native vacuum spectra are certified. The same master source gives transverse displacement in $[-0.001776299, -0.001775262]$ grid sites after eight evolution events, and conditional arrival delay in $[0.0105311, 0.0105598]$ in the event-parameter convention. Every receiver failure and field vacuum outcome is retained. These finite bounds prove neither an infinite geometry-cutoff limit nor physical gravitational units. The evolution word defines an experimental event order, without asserting a fundamental global clock.

## 11.51 A native clock read at wave arrival

The same 80000-dimensional matter–geometry–wave source prepares the right native cell in an equal coherent superposition of levels 0 and 1. A retained setting bit selects either the $X$ or $Y$ clock instrument when the wave arrives. All native complements, geometry outcomes and timeouts

are included. The resulting four active setting/sign frequencies determine the clock coherence without assigning simultaneous sharp $X$ and $Y$ values. Independently prepared vacuum-clock controls provide a strictly increasing phase calibration for the sixteen event labels.

Using this measured event-to-cycle dictionary, the source-on minus source-off mean arrival difference lies in

$$[0.0013506,\ 0.0013547] \quad \text{native clock cycles.}$$

The phase of the actual aggregate arrival-clock coherence gives a separate positive response in $[0.0004679, 0.0004687]$ cycles. It is a circular phase statistic, rather than the mean arrival delay. Complete finite numerical bounds control both quantities. A finite-repetition theorem also certifies the sign of the phase response with retained failure records; its large repetition count is an analytic resource bound.

Within a terminal event, clock, field and geometry effects act on distinct tensor factors. Their products commute, and every permitted ordering has the same full joint record. Copying these orthogonal records supplies two observer readouts without copying an arbitrary quantum state. This closes an operational clock–wave comparison on one specified construction. No physical-second calibration, Einstein proper-time identification or fundamental global-time variable is assumed.

**Finite theorem interfaces.** The exact native colour projectors and their representation meanings are reused across the three- and six-quark sources. Their actual current intertwiners are checked before any sector reduction. The finite complete instrument theorem in `Lean/EventAlgebra/LuedersPhaseInstrument.lean` applies once the concrete projector identities hold, retaining failures and arbitrary quantum references. Positive-root identities and uniqueness are formalized in `Lean/SourceLaw/MatrixLimits.lean`; the geometry calculation adds its stated executable residual and operator-norm estimates. The numerical instances and quantitative bounds are not additional Lean theorem counts.

## 11.52 Complete source transport and quantitative field records

**Exact Fock and field transport.** The complete 192-coordinate native CAR carrier admits 24 even pair implementers on all 48 lepton coordinates, with identity on the 144 colourful coordinates. Their exact inverse transforms give every canonical coefficient of the 114-term source and all 24 weak currents. Pair terms are retained. Charge, Gauss constraints, preparation, arbitrary untouched references and measurement work transform through the same unitary. For a smooth periodic field-dependent implementer, the full electric operator transforms as

$$U^{*}EU = E + \mathcal{A}, \qquad \mathcal{A} = -iU^{*}\partial_{\theta}U, \qquad U^{*}E^{2}U = E^{2} + E\mathcal{A} + \mathcal{A}E + \mathcal{A}^{2}.$$

The connection and derivative term are necessary even when a fixed-field matter square is scalar. A complete four-state Fock example retains both odd occupations and every integer flux, with exact nonzero measurement work. The native longitudinal Gauss carrier has dimension $2^{192}$; adding an independent rotor gives an infinite-dimensional carrier. There is no onto unitary equivalence between those different architectures. The exact algebraic transport therefore does not identify a common interacting continuum or additional physical rest masses (`model/openwave_fock_field_transport`).

**Finite references and the complete clock experiment.** A finite energy-conserving exchange gives source coherence $i\sin\vartheta\, e^{imt}/2$ and energy $m\sin^{2}\vartheta/2$, retaining every success and failure. Coevolution of both finite phase references, followed by energy-compatible readout, cancels the free

source phase in every terminal probability. A fresh terminal phase is a supplied timing resource. A distinct static controller executes the actual calibrated electron, muon and tau masses at common scale $L = 4096$, common times and coupling $g = 2/3$. The exact reducing carrier has dimension $256 \cdot 4 \cdot 2 \cdot 2 = 4096$; it retains every occupation record. The full source has the reducing projector $R = (P_{\mathrm{vac}} + P_j) \otimes I_{\mathrm{memory}}$, with an explicit $R^{\perp}$ terminal outcome and lower bound $H_{\mathrm{total}} \geq -L \max(|g|, 2)$. Its source-pair phase obeys the exact local error bound

$$\left| \frac{d\langle A \rangle}{dt} - im\langle A \rangle \right| \leq L|g|P_{\mathrm{prep}} + 2LP_{\mathrm{read}}.$$

Complete component-energy and terminal-measurement work accompany the finite trajectories. Numerical propagation has convergence diagnostics, without an outward arithmetic certificate. The moving controller packet, phaseful references and final occupation instrument are declared resources; no autonomous detector compilation, self-starting limit cycle or physical seconds follows (`model/openwave_finite_phase_clock`).

**Gauge-invariant localized quark records.** The full $L^2(SU(3) \times U(1))$ link carrier and thirty native source and reference CAR coordinates support an open-string operator $B$ with $B^2 = 0$ and $B^*B = N_{\mathrm{family}} P_{\mathrm{ref,vac}}/3$. Its controlled seed/path superposition has fractional source charge and a compensating quantum reference. The contractions $X = B + B^*$, $Y = i(B - B^*)$ give the complete physical instrument

$$\left\{ \frac{1}{4}G(I+X)G, \frac{1}{4}G(I-X)G, \frac{1}{4}G(I+Y)G, \frac{1}{4}G(I-Y)G, I-G \right\}.$$

Seed preparation on Gauss success has outcomes $P_{\mathrm{seed}}$ and $G - P_{\mathrm{seed}}$, with failure absorbing. The path energy is $2 + \ell(q^2/32 + 1/12)$ for integer hypercharge labels $q = 2, -1$. The displayed complex quadrature mean is $\langle B^* \rangle$. Complete charge and flux profiles are alternative terminal measurements, with their distinct work. The electric idle Hamiltonian and coherent preparation are supplied. This establishes finite localized charge records, without autonomous hopping, a spontaneously wound texture, confinement or a physical quark mass (`model/openwave_localized_quark_record`).

**Literal weak-probe energy and the Lambda spectral measure.** The normalized native Lambda packet uses $\kappa = 1/512$ and a trial preparation of probability $1/7$. All 72 native weak words, 522 nonzero current outputs and 19140 full-Hamiltonian successors enter exact packet moments, including every translated-packet cross term. The successful probe's energy mean and variance are enclosed by

$$\langle H_5 \rangle \in [2.418046697462, 2.418046697463] \text{ GeV},$$
$$\mathrm{Var}(H_5) \in [0.085780468809, 0.085780468810] \text{ GeV}^2.$$

The mean energy increase is $[1.302361802725, 1.302361802726]$ GeV; this is probe-induced energy, not the calibrated Lambda rest energy. Nonselective work retains probe failure. For $D = (H_5 - E)^2 + \eta^2$ and every full-source trial $x$,

$$\langle u, \eta^2 D^{-1} u \rangle \geq \frac{\eta^2 |\langle u, x \rangle|^2}{\|(H_5 - E)x\|^2 + \eta^2 \|x\|^2}.$$

All 23 native proton–pion columns and the literal current supply such witnesses. Their complete effect and complement certify a broad-window probability above one half; the tested narrow windows have zero certified lower bound. Zero lower bounds do not establish zero spectral weight. A Weyl

sequence at the rest energy establishes spectral membership, without proving current overlap or a physical decay density. A controlled shrinking-window limit and outgoing-current identification are required (`model/openwave_lambda_spectral_measure`).

**Dynamical electromagnetic nuclear bounds.** A compact $U(1)$ cycle extends the actual nuclear source with electric coefficient $a = 1/16$ and magnetic coefficient $b = 1/8$. The exact Gauss isometry retains all matter occupations and every integer cycle flux. Its unique positive field vacuum is gauge invariant; hence $H_{\text{field}} - E_\gamma \geq 0$ on the full carrier. An $n = 0$ trial gives an upper bound using the actual native compression plus longitudinal $V_0$, without deleting field polarization. The complete GeV intervals are

| | |
|---|---|
| triton | $[2.808919203020, 2.809127749492]$ |
| helion | $[2.807625870510, 2.808017057294]$ |
| alpha | $[3.739123982204, 3.739741842803]$ |

The triton input lies in its interval; both held-out comparisons fail. The positive-addition class cannot lower the excessive alpha mass. For general prepared matter state $\rho$, the field preparation energy is $\mathrm{Tr}(\rho V_0) + b - E_\gamma$, in $[\mathrm{Tr}(\rho V_0), \mathrm{Tr}(\rho V_0) + b]$. Uniform conditional quartet states give longitudinal means $9/32$, $3/4$ and $39/32$ in source units. The interval $[0, b]$ applies only to the zero-charge vacuum service. These bounds preserve all spin multiplets, failures, thresholds and inherited calibration; they are not a fitted Coulomb potential or successful nuclear-spectrum prediction (`model/openwave_nuclear_em_transfer`).

**Energy-dependent full-complement spectral certificates.** For the actual refined Haar source, let $P$ be the complete 201-mode frame, $T = QHP$ and $C = QHQ = D + B$, where $D = 16E_Q + 9$, $C \geq 80Q$ and $\|B\| \leq 9$. Every omitted representation and intertwiner remains. For real $z < 80$, define

$$\Sigma(z) = T^*(C - z)^{-1}T, \quad G(z) = T^*(D - z)^{-1}T, \quad J(z) = T^*(D - z)^{-2}T.$$

Factoring the resolvent around $D - z$ gives

$$\|\Sigma(z) - G(z)\| \leq \frac{9\|J(z)\|}{1 - 9/(89 - z)}.$$

All 8143 off-block rows evaluate these shifted Gram matrices; exact finite-frame inertia transfers to the full source. In the common electric clock, the coarse and refined gap intervals are

$$[96.00000060968, 96.00000069820], \qquad [47.99403921427, 48.00529277681],$$

with widths reduced by factors above 361 and 5.36. These are finite-source energies, with no GeV or continuum identification. A general finite-word Neumann remainder is bounded by $b_T^2 M^{n+1}/[(d + M - z)^{n+1}(d - z)]$ when $\|T\|^2 \leq b_T^2$; only the zeroth matrix term is executed here. Conversely, plaquettes vertex-disjoint from a fixed probe give the exact volume obstruction

$$\|QHP\|^2 \geq \frac{\beta^2 N_{\text{far}}}{4}, \qquad N_{\text{far}} \geq N_{\text{faces}} - 48.$$

Orthogonal Haar output supports prevent cancellation. Local incidence and interaction-picture propagation estimates replace the unavailable global volume-uniform norm bound; they do not establish a continuum mass gap (`model/openwave_energy_schur`).

**Finite-strength control on the full geometry register.** For the declared $H_g = (\omega/2)\sum_i (I - X_i)$ and $Q = (q/N)\sum_i Z_i$, Perron positivity orders all total-spin blocks. The complete geometry-sector spectral gap is determined by the two highest-spin Jacobi matrices, with exact rational Sturm inertia. This is not a gap across different matter sectors. For finite nonnegative energies, write $H = H_g + E_s e^Q + E_p e^{FQ}$. Convexity gives a product-comparison Hamiltonian and a positive remainder. For $M \geq E_s + E_p$, its ground gap is at least $d = \omega - Me^q\mu_2/2$ when $d > 0$, and the ground-ray angle is at most $Me^q\sqrt{\mu_4}/(2d)$, where $\mu_2, \mu_4$ bound the second and fourth $Q$ moments of the comparison product ground uniformly over $E_s + E_p \leq M$. Centered observable bounds control the finite lapse; Hellmann integration controls mutual force for all 49 positive native-energy pairs. A full Duhamel comparison bounds the complete quench records. Independently prepared contexts at times zero and 355/113 recover the static normalized lapse coefficient with the stated finite error. They do not identify a single continuously operating observer or physical seconds (`model/openwave_collective_finite_strength`).

**Fluctuation–response normalization.** The same complete collective geometry gives $g_N = S\,\mathrm{Cov}(Q, e^Q)/(2\pi\omega)$ and $W\,\mathrm{Var}(Q) = q^2\omega u/2$, with $\mathrm{Var}(Q) = q^2/N$ and coordinate-readout work $W = N\omega u/2$. Thus an independently determined native covariance would fix the weak gravitational coefficient without fitting it again. For the continuous source parameter in $H_g + E_s e^Q$, at $E_s = 0$, the complete coordinate measurement attains the geometry ground state's quantum Fisher information. Exact spectral sums imply $4v(1-q^2/(3N^2))^2/\omega^2 \leq I \leq 4v/[\omega^2(1-v)^2]$, where $v = q^2/N$. The zero-source quench information is at most $4I$ at every time. These bounds quantify the measurement resources of this specified geometry family; they do not bound all observer architectures. The inherited register count uses measured $G$, so its predicted fluctuations and response retain that calibration ancestry.

For $q = 1/8$ and $\omega = 1$, the upper information bound satisfies $1/I_{\mathrm{upper}} = 16N - 1/2 + 1/(256N)$. The Cramér–Rao inequality therefore requires at least $16N$ independently prepared geometry copies for a locally unbiased estimate of the continuous source parameter with unit native variance. The finite ground-state preparation, coordinate readout cost and measured-$G$ ancestry are retained; this is a bound for the specified geometry family, not for every observer protocol (`model/openwave_gravity_noise_normalization`).

**Conditional cosmological normalization.** For the declared Einstein–dust–radiation–scalar comparison, the regular adiabatic mode reduces exactly to $T = (3/2)(H/a)\int_0^a da'/H(a')$ and $\Phi = 3/2 - T$. The full evolution equations and Hamiltonian constraint vanish under this substitution. With $t = -\log a$ and $F = e^{-t}/(H/H_0)$, the background gives $-2 \leq h \leq 0$ and $0 \leq h_N \leq 33/8$, hence $|F_{tt}| \leq 9F$. Outward background intervals, trapezoidal remainders and the complete improper-integral tail enclose the present normalized metric coefficient in $[0.734179279095, 0.734257299186]$, compared with $[0.709108311479, 0.709181069590]$ for the same-density constant-$\Lambda$ control. These are conditional regular long-wavelength coefficients; the strict zero-wavenumber mode has zero comoving density and no measured growth or microwave-background comparison is implied. Exact rescalings $H_0 \mapsto \lambda H_0$, $\rho_{i0} \mapsto \lambda^2\rho_{i0}$, $L \mapsto L/\lambda$ and $k \mapsto \lambda k$ preserve the dimensionless equations. Their homogeneous perturbations also admit arbitrary amplitude. An absolute cosmic density and primordial normalization therefore require additional native source relations.

`model/openwave_capacity_normalization_transfer`.

### 11.53 Complete source acquisition and local refinement

**A common matter and compact-field carrier.** The full 192-CAR native source keeps all 114 source terms and 24 even pair intertwiners. Two parallel $U(1)$ links and an integer quantum reference solve the Gauss constraints exactly as $E_1 = 3Q_1 - n$, $E_2 = n$, $R = -3(Q_0 + Q_1)$, $n \in \mathbb{Z}$. The entire physical carrier is $\mathcal{F}_{192} \otimes \ell^2(\mathbb{Z})$. For the inherited source $H_0$, define the declared positive extension

$$H = H_0 + \frac{(3Q_1 - n)^2 + n^2}{32} + \frac{1}{64}\sum_{j=1}^{96}(d_{0j} - d_{1j})^*(d_{0j} - d_{1j}) + \frac{1}{64}\left(I - \frac{S + S^*}{2}\right), \quad S|n\rangle = |n+1\rangle.$$

It is self-adjoint on $D(n^2)$, positive, and has compact resolvent on this finite spatial graph. All native cross-cell terms carry their exact integer dressing. Charge, local current, reference and complete-instrument work identities hold on every retained matter/flux sector. The constant pair maps have zero connection; any field-dependent replacement retains $E + \mathcal{A}$ and every term of its square.

The full one-lepton reducing operators have second eigenvalue bounded from below by $1/16$ after subtracting their inherited rest coefficient. A rational trial with exact full integer-tail residual gives $\mu - v/(1/16 - \mu) \le E_0 \le \mu$ by Temple's inequality. The inherited coloured and charged occupation floors exceed the cycle vacuum bound $1/64$, while neutral-only states share its field operator. This proves that the cycle vacuum is the full-source ground. Subtracting this vacuum yields a common charged gap increase approximately $7.535487 \times 10^{-6}$ GeV with the supplied electron unit. Neutral one-lepton shifts cancel exactly. This is a calculated architecture dependence, not a physical rest-mass prediction. For every native preparation at $n = 0$, the added mean is $9\alpha\langle Q_1^2\rangle + t\langle K\rangle + \beta$; the operator difference is unbounded on the full flux space. A native-process compiler and an independent colour gauge field are not supplied by this $U(1)$ extension (`model/openwave_matter_flux_source`).

A separate optional normalization uses the exact computed spectral number $\delta = E_{\text{charged}} - E_v$, with $0 < \delta < 1/64$, and defines $H_\delta = H - \delta N_{\text{ch}}$. The inherited inequality $H_0 \ge cN_{\text{colour}} + N_{\text{ch}} + H_{\text{neutral}}$, $c > 1$, gives $H_\delta \ge cN_{\text{colour}} + (1-\delta)N_{\text{ch}} + H_{\text{neutral}}$. Every coloured or charged sector remains above the cycle vacuum, and the neutral-only sector retains exactly $H_v$ plus positive terms. Thus the global vacuum is unchanged and the same coefficient restores all inherited charged one-lepton rest coefficients; neutral ones are unchanged. This is an optional source convention with its own generator, not a native-process derivation or a new physical mass prediction. Weak transitions can change $N_{\text{ch}}$, so current and work formulas must be evaluated for $H_\delta$.

**Repeated finite mass-clock records.** A finite phase reference supports two native mass-clock cycles when source reset is implemented as an energy-conserving exchange with an explicit empty battery. For preparation coordinates $c^2 + s^2 = 1$, the two isolated source coherences are $ise^{imt_1}/2$ and $isce^{imt_2}/2$; depletion reduces contrast without changing the ideal frequency. Complete source and resource occupations give exact gap-switching work $m(1 - c^4/2)$, with battery energy and correlations retained. A static 512-position controller executes the two cycles on a 32768-dimensional reducing carrier for all three charged rest energies and two preparation strengths. A rational Taylor remainder and explicit arithmetic bounds enclose propagation, reset failure, record overwrite and both flight phase increments. The second flight has a resolved finite controller bias; exact preparation-strength independence applies to the isolated collision identity, not to this complete finite detector. The supplied phase references, clock packet, terminal measurement work and orthogonal native outcomes are retained. Sustained replenishment, physical clock units and native twelve-port execution of this protocol remain separate requirements.

**Three-refinement local gauge memory.** On open full-Haar $SU(2)$ cubes subdivided by $L = 1, 2, 3$, take $\alpha = 32/L$, $\beta = L^3/32$ and a fixed physical unit-square perimeter of $4L$ links. The Haar probe is supplied, rather than an interacting vacuum. Fundamental and spin-one Wilson characters have variances $1/4$ and $1/9$ and exact electric frequencies 96 and 256 at every subdivision. Electric evolution preserves support; a local Duhamel comparison gives

$$\delta_L = \min\{2, 2\beta N_{\mathrm{inc}}|t|\}, \qquad N_{\mathrm{inc}} = 5 \ (L = 1), \quad N_{\mathrm{inc}} = 8L - 4 \ (L \geq 2).$$

Normalized real correlation errors are at most $2\delta_L$ and $3\delta_L$. Both correlations exceed 0.97 at common time $t = 1/4096$, without a global low-energy projector or representation truncation. The probe is nonstationary, so its normalized correlation is not confined to $[-1, 1]$ by a ground-state argument.

For a Hermitian contraction $A$, the complete instrument $K_\pm = (I \pm \eta A)/2$, $K_0 = \sqrt{(I - \eta^2 A^2)/2}$, $\eta = 1/2$, has all nine two-time records. Its weighted moment is exactly $\mathbb{E}[s_0 s_t] = \eta^2 \operatorname{Re} \operatorname{Tr}(\rho A(t) A)$, including backaction and arbitrary idle references. The Haar derivative identity bounds the two-readout work by $32/3$ and $512/27$ source units. Probe preparation and intervention times remain explicit resources. At fixed time the local error grows as $L^4$; controlled windows of order $L^{-4}$ supply no fixed-time continuum or Clay theorem (`model/openwave_local_gauge_memory`).

**Quantum acquisition of the finite tensor stencil.** All 60 columns of the native coordinate Wilson source carry ten source observables: energy, three momenta and six coframe action derivatives. Three compact declared metric profiles give a full 34-site staggered stress/curvature stencil. Ten same-state mean records suffice for every component. Complete three-outcome instruments retain thirty exact probabilities, the zero events and arbitrary quantum references. Their nonselective trace-norm disturbance is at most $3/8$, and source work is at most $3\|H\|/8$ per acquisition. Independent preparations prevent incompatible sequential readouts from changing the target state. A two-state control changes $\langle X \rangle$ from 1 to $3/4$ after a prior $Z$ acquisition, exposing that distinction.

If $C$ is the maximum coefficient norm and each native mean has error $\epsilon$, every null contraction has error at most $8C\epsilon$. The sharp inverse $2 + \sqrt{3}$ and measured variation budget yield

$$\delta = (2 + \sqrt{3})(\widehat{q} + 8C\epsilon) + \widehat{\tau} + 2C\epsilon, \qquad |\Delta\lambda| \leq \widehat{w} + 8C\epsilon + 8\delta$$

at unit spacing. The complete sufficient Hoeffding budget is 3,239,348,495 independently prepared copies for $\epsilon = 10^{-3}$ and joint confidence 0.99. Every support point and zero halo is retained. This closes the finite acquisition interface for the stated carrier and readout assignment; its actual nonzero Einstein residual and physical metric/stress identification remain explicit (`model/openwave_stress_curvature_instruments`).

**Narrow Lambda weight and autonomous quark transport.** For the actual native weak current applied to a normalizable Lambda packet, an approximate inverse of the complete interacting Hamiltonian gives $\|1_{[E-w,E+w]}(H)u\| \leq \|u - (H - E)x\| + w\|x\|$. The rational witness includes native proton and pion ground trials at eight relative separations. Every occupation successor and translation return contributes to the residual. The conditional weight in the window of half-width 1 MeV is below 0.026622, compared with the moment upper bound 0.048210. These are complete finite-resolution spectral bounds, with the inherited electron and strong calibration. They imply neither an on-shell density nor a physical decay rate.

The complete source and records are in `model/openwave_lambda_narrow_weight`.

Gauge-covariant native hopping transports a fractional-charge endpoint with its complete reference-supported string. A full-carrier intertwiner reduces the specified preparation to

$$H|x\rangle = (2 + \sigma x)|x\rangle - \kappa(|x-1\rangle + |x+1\rangle).$$

where $\kappa = 1/8$ and $\sigma = q^2/32 + 1/12$. Rational exponential remainders enclose all charge, flux and pointer records on five- and nine-vertex carriers. The entire fermion and rotor complements remain in the source. Energy conservation gives the all-time conditional bound $\langle x\rangle - 1 \le 2\kappa/\sigma$, equal to $6/5$ and $24/11$ for native charges $2/3$ and $-1/3$. This autonomous source-unit transport theorem supplies no physical quark rest-mass or continuum identification.

The source and complete records are in `model/openwave_quark_string_transport`.

**Stable nuclear attraction and full-input work.** A neutral harmonic field coupled to a gauge-invariant four-site occupation projector supplies a signed nuclear interaction while preserving every sector with fewer than twelve quarks. Completing the full operator square gives $H \ge H_{\mathrm{EM}} - g^2 \sum_b O_b/(2\omega)$, an extensive lower bound. Exact coherent-state trials provide the corresponding complete-source variational upper bounds; the oscillator is not truncated. With declared $g = 1/8$ and $\omega = 1/16$, the helion and alpha comparisons fail. No held-out mass selects either coefficient. For the original full singlet input, an independent Casimir trace gives electric preparation energies $8955/9152$ and $2949/1360$ source units for three and four cells, respectively, including all charge sectors.

The alpha interval is $[3.739060107335, 3.739677967934]$ GeV; helion remains in $[2.807625870510, 2.808017057294]$ GeV. The necessary $|g| \ge 1.694976066305$ at fixed oscillator frequency is a diagnostic obstruction, not a chosen coefficient or sufficient fit. All oscillator and integer-flux complements remain (`model/openwave_nuclear_field_attraction`).

**Complete covariance measurements and normalization.** For $Q = (q/N)\sum_i Z_i$ in the product $X$ vacuum, two complete coordinate records $x, y$ give the unbiased statistic $D = (x-y)(e^x - e^y)/2$ for $C = \mathrm{Cov}(Q, e^Q)$. With $v = q^2/N$, its second moment is bounded by $3e^{2q}v^2$ and $C \ge e^{-q}v$. A median of independent block means therefore has a certified relative-error sample bound independent of $N$. Its measurement work per pair is $N\omega u$, and its coordinate resolution is a supplied resource. The complete source gives the weak rest slope $M(1) = \langle e^Q\rangle$ and the photon slope $M(f_\gamma)$. Using one clock and length yields $G_{\mathrm{weak}}/(c_\gamma/c) = SC/[2\pi\omega u^2 M(1)M(f_\gamma)]$. For a far photon reference, $M(0) = 1$ and this reduces to $Sq\tanh(q/N)/(2\pi\omega u^2)$. The register count requires independent observational ancestry; the calibrated count does not predict Newton strength. Finite-source physical identification and the native-process implementation are not established by this covariance protocol.

The same pair data also estimate $M(1)$, and a union bound propagates both relative-error intervals through the complete unit map. No independence between the two estimators is assumed. Geometry-copy counts for covariance estimation and local-unbiased source-energy estimation concern distinct parameters. The former can be independent of $N$ while its detector resolution and work remain $N$ dependent (`model/openwave_geometry_covariance_protocol`).

**Finite-wavelength cosmological transfer.** For the supplied Einstein–dust–radiation–capacity-scalar model, put $U = U_m$, $v = U_r - U_m$, $R = \Phi + U$ and $w = y - sU$. These comoving variables remove the exact adiabatic zero mode. At $N = \log a = -1$, the declared correction $\Delta_m = -2Q\Phi/(3\Omega_m)$ enforces the complete Hamiltonian constraint. An eight-coordinate affine

interval equation transports the $k^2$ response to $N = 0$; the exact variation-of-constants equation retains the complete $k^4$ remainder. The certified metric and comoving-dust coefficients lie in

$$\Phi_1 \in [-0.008281806997, -0.005759370982], \qquad \Delta_{m,1} \in [-1.455879943233, -1.454618725225].$$

Full finite-wavelength records are enclosed for $k/H_0 = 1/100, 1/50$, with overlapping independent 1024- and 2048-step certificates. The supplied late preparation is distinct from a primordial continuation. Gravity, density fractions, horizon choice and the common physical unit remain input premises; no early-Universe abundance or CMB match follows (`model/openwave_capacity_finite_wavelength`).

**Native-process admission.** OpenWave lists OPH as M10 with all 31 criteria untested. Finite theorems, calibrated extensions and implementation replays are supporting results. They do not substitute for an in-platform native carrier process generating the criterion's observable without installing the target field law. The admission gate preserves this distinction (`model/openwave_m10_review_gate`).

### 11.54 Exact repair response and complete observer records

One literal finite repair grammar fixes the twelve-port carrier, fair C/E law, endpoint coin and record-memory rule. Every experiment declares its schedule and input: the response and event-clock tests use independent uniform seam selection; causal-order and interaction controls use fixed literal seam words. Its local instrument retains all 25 raw slots, including zero hold and register padding. On the port-pair federations, three-event experiments retain all 15 625 raw histories. Disjoint events commute after relabelling their recorded event addresses; events sharing a port retain their causal order. The execution ordinal is an addressing convention rather than a fundamental global clock. Source preparation, randomness, blank memory and the coherent apparatus services are distinct declared resources (`model/openwave_native_process`).

**The complete finite response.** In the full global one-load sector, with graph Laplacian $L$, adjacency $A$, port degrees $d_i$ and $M$ seams, the exact unread-source channel is

$$\operatorname{diag}(\Phi\rho) = T \operatorname{diag}\rho, \quad T = I - \frac{L}{2M}, \qquad (\Phi\rho)_{ij} = \left[1 - \frac{3(d_i + d_j) - 2A_{ij}}{4M}\right]\rho_{ij} \quad (i \neq j).$$

Every endpoint matrix unit and every global one-load column is covered on 240, 960 and 3840 ports. The identity tensors with an arbitrary idle reference. The complete observer history is retained by the underlying instrument; the displayed marginal is sufficient only for the fixed independent seam schedule. It is also the exact linear response at the maximally mixed stationary input.

Let $C$ sum each cell's twelve populations and $S$ lift a cell population uniformly. The natural cell projection has

$$\|CT - (CTS)C\|_{1\to 1} = \frac{11}{12M}, \qquad Q = I - SC.$$

The complete complement gives the exact memory correction $CTQ(QTQ)^{\tau}QTS$. A memoryless cell process therefore fails on explicit basis columns. This response has no supplied translation representation or wave-time intertwiner identifying it with the separately declared twelve-direction cosine dispersion (`model/openwave_native_response`).

**Fast propagation with complete native arrivals.** Since the maximum port degree is six, $P = I - L/6$ is stochastic and

$$T^t = \sum_{k=0}^{t} \binom{t}{k} q^k (1-q)^{t-k} P^k, \qquad q = \frac{6}{2M}.$$

Positive uniformization certifies up to 80 640 original attempts using 128 local powers and an outward rational binomial tail. No source carrier or virtual-power remainder is discarded. Three graph-ball widths, including a predeclared radius-two control, use the common event-intensity coordinate $\tau = t/(2M)$ on all three graphs. Native crossing-current records determine first arrival at graph distance four; an observer flag retains arrival and nonarrival while the physical source continues. For the radius-two preparation on 240 ports at $\tau = 4$, the certified first-arrival probability is approximately 0.547397035737486. All corresponding coherent offdiagonals decay at their initial port addresses, so occupation transport is not a coherent translating packet. The calculations supply neither a physical speed nor a continuum limit (`model/openwave_native_packet_propagation`).

**Conservation and interacting native records.** Every actual current obeys $n'_a - n_a = -j$ and $n'_b - n_b = j$; complete histories therefore give the exact continuity equation and all cut-current identities. With a supplied known initial occupation, records reconstruct the final positions and retain explicit invalid-record outcomes. On a connected fixed-load carrier the native branch star algebra is the full matrix algebra: E-effect signatures distinguish configurations, and occupied-vacant edge exchanges connect them. Its source-only commutant is scalar. Total load is conserved, but a modulo-six accumulated current record is not a protected source charge or an identified Wilson flux (`model/openwave_native_defect_records`).

Two actual native loads give a nonfactorizing interaction observable. Two overlapping repairs produce connected record-current covariance 1/8. After one and two two-event blocks, the complete native pair densities are 1/2 and 3/8; independent labelled walkers with the same one-point means differ by 1/4 and 9/64. A complete additional E repair acquires a zero-imbalance statistic equal to that pair density in the native two-load sector, retaining its post-measurement state. The equality does not extend to the comparison process's collision sectors. All local operator columns and the full spectator complement remain. This is an exclusion interaction of the executed source, not a compiled unitary matter field or QFT scattering law (`model/openwave_native_interaction`).

**Event modes, phase readout and energy attachment.** For a graph eigenvector $v$, the native current record supplies $\Delta(v \cdot n) = (v_b - v_a)j$. On the elementary H66 carrier, four-event record means recover three exact first-moment multipliers

$$\lambda_\pm = \frac{55 \pm \sqrt{5}}{60}, \qquad \lambda_0 = \frac{9}{10}.$$

The complete instrument is contracted for the requested moments; its expanded four-event tree is not claimed to have run. Stationary current covariances separately satisfy $\mathrm{Cov}(\Delta f_0, \Delta f_n) = -\mathrm{Var}(f)(1-\lambda)^2\lambda^{n-1}$ for $n \geq 1$. These are three decay sectors, without a particle-family or physical frequency assignment (`model/openwave_native_event_clock`).

Every repair-only history map is a scalar times a partial injection, so its effect is diagonal in occupation. Phase twins with the same diagonal have identical complete record laws. A constructive extension loads a two-dimensional H66 code into the declared six-level response, executes its actual coherent H pulse and follows it with a native E measurement. The complete instrument retains the orthogonal 64-dimensional source outcome and all four processor complement outputs; all

4356 source matrix-unit trace identities are checked. This gives a phase-sensitive finite readout conditional on the supplied coherent service. It neither generates that service by repair alone nor compiles the full stress and curvature stencil (`model/openwave_native_geometry_readout`).

The unread repair channel has positive real decay eigenvalues and zero event phase. Its pure-port output differs from any unitary output by trace distance at least $d_i/(2M)$. Identifying one-load coherence decay rates with rest masses is an additional prescription; its full rate ratio is at most 30/23 on these graphs and fails the inherited electron-anchored charged-lepton hierarchy. No GeV rest-mass interval is obtained from this member. A Euclidean transfer or coherent dilation remains a distinct possible reconstruction with its own algebra, clock and readout proof (`model/openwave_native_rest_energy`).

**Capacity and physical normalization.** For export-oldest memory and a controller reading only the last committed record, concatenating archive and FIFO is an exact complete-instrument intertwiner across FIFO capacities. Every exported record and terminal outcome remains. A halt-on-full controller gives a different process. Raw storage, classical record entropy and source information capacity are distinct; none is an identified horizon area. Native event laws are also invariant under a rescaling of an assigned physical clock, while $H_0$ and $\Lambda$ scale inversely and inversely quadratically. A homogeneous perturbation has an independent amplitude freedom. The finite-wavelength cosmology attachment therefore requires a physical clock, horizon map and preparation normalization (`model/openwave_native_capacity_history`).

An author-run headless endpoint calculation in the pinned OpenWave M10 layout reproduces every local operator column with an independent rational oracle and rejects an altered conservation rule. This authenticates the finite source calculation and its declared platform interface. It is not a maintainer criterion award: the official M10 column has zero validated criteria and all 31 cells untested (`model/openwave_native_platform_gate`).

**A compatible finite curvature assignment.** Place scalar, vector and symmetric-tensor records at parities 0, $e_b$ and $e_a+e_b$. Half-step derivatives then give the same linearized Ricci polynomial with exact Bianchi and pure-gauge identities, while its scalar Laplacian is the original nearest-neighbour operator. The actual derivative graph connects scalar neighbours, so the centered collocated checkerboard ambiguity is absent in this assignment. Complete boundary halos are retained. Physical curvature and stress acquisition require a quantum source instrument; the co-location of its classical tensor records has the explicit finite construction below (`model/openwave_staggered_ward`).

The staggered tensor records also admit an explicit co-location map: keep diagonal entries and average the four surrounding off-diagonal entries. If $d_i(ab)$ bounds actual neighbouring record variation, then

$$\tau = \max_{a\neq b} \frac{d_a(ab) + d_b(ab)}{2}, \qquad \delta = (2+\sqrt{3})\epsilon + \tau$$

bound transport and nine-null reconstruction errors. The metric-term variation along each connected edge is at most $aw + 8\delta$. This implements finite record transport; physical curvature acquisition and nonlinear Einstein dynamics require their separate source transfer (`model/openwave_tensor_record_transport`).

**A static classical limit on the complete geometry register.** A permutation-invariant register has

$$Q_N = \frac{q}{N}\sum_{i=1}^{N} Z_i, \qquad H_g = \frac{\omega_N}{2N}\sum_{i=1}^{N}(I - X_i), \qquad \omega_N = \frac{\omega N \cosh(q/N)^N \tanh(q/N)}{\sinh q}.$$

This normalization preserves the original weak far-field response; it does not repair its failed physical $G/c$ attachment. For $E, P \geq 0$ and $0 \leq F \leq 1$, write $H_N = H_g + Ee^{Q_N} + Pe^{FQ_N}$. The mean-Bloch bound and spectral Jensen inequality imply

$$\inf \operatorname{spec} H_N \geq \min_{-1 \leq z \leq 1} \left\{ \frac{\omega_N}{2}(1 - \sqrt{1 - z^2}) + Ee^{qz} + Pe^{qFz} \right\}.$$

A spin-product trial bounds the difference above that minimum by $(Ee^q + Pe^{qF})(e^{q^2/(2N)} - 1)$, retaining the entire register rather than discarding nonsymmetric sectors. The relative vacuum lapse variance is at most $e^{q^2F^2/N} - 1$. At $q = 1/8$, $N = 256$ and $0 \leq F \leq 1$, this upper bound is below $6.103702 \times 10^{-5}$. The geometry gap closes as $1/N$: this is a static single-mode classical limit, without a uniform dynamical gap, spatial Einstein evolution, a physical gravitational normalization or a zero-cost preparation claim (`model/openwave_replicated_geometry`).

### 11.55 Capacity work and horizon-dependent expansion

For the finite active-register coupling $H(V, k) = \epsilon V/k$, the operator pressure is $P = -\rho = -\epsilon/k$ at fixed capacity. Complete register/geometry observations retain preparation work and every capacity-change stroke; the four-stroke work cycle vanishes exactly. Capacity drift has exchange $Q = -\rho \dot{k}/k$, which distinguishes mechanical pressure from the separately conserved effective-fluid pressure. These identities and all finite records are specified in `model/openwave_capacity_stress`.

The proved capacity drift law gives $w = -1 + \frac{1}{3} d \log N / d \log a$. With an area capacity $N \propto R^2$, a comoving radius gives $w = -1/3$. A Hubble radius with separately conserved dust instead gives $w = 0$ and $H^2/H_0^2 = a^{-3}$. Retaining mechanical pressure $-\rho$ on that branch requires matter exchange and a constant dark fraction, with no transition from matter domination to acceleration.

For the declared future-event-horizon member with coefficient one, $x = \sqrt{\Omega_{\text{dark}}}$ obeys the exact algebraic equation

$$\frac{x^6(1+x)^3}{(1-x)(1+2x)^8} = a^3 \frac{x_0^6(1+x_0)^3}{(1-x_0)(1+2x_0)^8}.$$

Monotonicity and rational bisection replace an ordinary differential equation solver. The future limit has positive finite $H$, and the event-horizon boundary condition follows. With the supplied $H_0 = 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}}$ and $\Omega_m = 0.315$, all fifteen correlated primary chronometer rows give $\chi^2 \simeq 6.57717$. The same comparison gives 6.19134 for fixed capacity, 21.54903 for comoving area and 70.48705 for separately conserved Hubble area. The alternative stellar-population reductions use the same observations. These are conditional capacity-history comparisons; neither the cosmic amplitude nor a physical register-to-horizon map is derived. The construction and complete records are specified in `model/openwave_horizon_capacity`.

**Radiation-era capacity histories.** With conserved matter and ideal radiation, the gamma-one future-event branch obeys $d\Omega_x/d \log a = \Omega_x(1 - \Omega_x)(1 + r + 2\sqrt{\Omega_x})$, where $r = \rho_r/(\rho_m + \rho_r)$. An outward Taylor enclosure controls the branch through $z = 10^6$. Its early radiation-era fraction scales as $a^2$, whereas fixed capacity gives $a^4$ and the Hubble-area control retains a constant fraction. The present amplitudes and radiation content remain supplied; these are conditional expansion histories, not derived particle abundances. The defining source and outward enclosures are specified in `model/openwave_radiation_capacity`.

**Perturbations of a declared capacity background.** Put $x = \sqrt{\Omega_x}$ and $r = \rho_r/(\rho_m + \rho_r)$. The conserved event-area branch has

$$c_a^2 = \frac{-2 - 3x + 3x^2 + 2x^3 + rx(1+x)}{6}.$$

The supplied past-background enclosures imply $c_a^2 < -1/5$, rejecting an adiabatic barotropic-fluid attachment. This instability is consistent with the holographic-fluid analysis of Myung. The established holographic-quintessence reconstruction of Zhang instead supplies a canonical scalar with

$$\frac{d\phi}{d\log a} = M_{\rm Pl}\sqrt{2\Omega_x(1-x)}, \qquad V = \frac{\rho_x(2+x)}{3}, \qquad c_s^2 = 1.$$

For this radiation-inclusive past domain, exact positive Bernstein coefficients and the Lean certificate give

$$2 \le \frac{V''}{H^2} \le 6.$$

This is a stable local scalar completion of the declared background, not a source-derived particle mass or a cosmological likelihood fit. The supplied amplitudes and horizon choice remain, and native-source transfer and perturbation-data acceptance are unproved. The polynomial bounds are in `Lean/ObserverPatchHolography/CapacityScalarCurvature.lean`; the complete background and controls are in `model/openwave_capacity_perturbations`.

**Constrained scalar, matter and radiation transfer.** The canonical capacity scalar has a declared linear attachment to Einstein gravity, ideal dust and ideal radiation. With $N = \log a$ and $h = H_N/H$, the complete metric–fluid–scalar equations give the exact Hamiltonian-constraint law

$$C_N = -(3 + 2h)C$$

and the spatial Einstein trace equation. The metric evolves with the matter; a deliberately nonzero constraint obeys the same homogeneous law. Regular growing initial data, an earlier-start comparison, tolerance refinement and an independent Einstein–de Sitter solution check the numerical transfer. With the inherited background inputs, present comoving matter transfer relative to a same-density cosmological-constant control is

| $k/H_0$ | 0.1 | 1 | 10 | 100 |
|---|---|---|---|---|
| $T_m/T_{m,\Lambda}$ | 0.97585 | 0.97001 | 0.91494 | 0.90822 |

These conditional differences supply no observed match or cosmic microwave background fit. Native scalar and primordial-state derivations are absent; free-streaming species and recombination are omitted. The constraint identities are exact, while the numerical solutions have convergence diagnostics without interval error certificates (`model/openwave_capacity_transfer`).

## 12 Reproducibility

Every executable result has a replayable receipt. Package manifests bind source, document and receipt artifacts by SHA-256. The repository's verification entry point recomputes these hashes, replays retained receipts and compares canonical bytes. Exact certificates must match exactly, and floating diagnostics carry declared replay tolerances. Independent test suites check the executable statements against separately written oracles, including adversarial mutations. Analytic limit proofs state their domains, normalizations and order of limits; a finite receipt does not replace those proofs. Model sources and the premise crosswalk are available at the pinned revisions cited with their results [11, 43]; the flagship synthesis is cited on the main branch [3].

## Statements and declarations

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**Data availability.** The model documents, executable sources, retained receipts and claim records supporting this study are available in the public OPH repository at the baseline revision [2] and the revisions cited with individual results.

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